

The Structure of Pronouns and Allomorphy

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Abstract

This dissertation investigates the following central question: *What is the internal structure of pronouns?* I propose and investigate the hypothesis that contextual allomorphy is a key tool for (a) accessing the inventory of morphemes that make up a pronoun across languages and (b) tracking the patterns of their interactions. Namely, allomorphic alternations involve an interaction of two morphemes, the *trigger* and the *target* of allomorphy. Identifying the trigger and the target in pronominal forms yields an *immediate result* - pronouns involve two morphemes that occupy distinct structural positions in their internal representation in the narrow syntax. The dissertation presents the idea that inspecting the series of sequential interactions between morphemes as basic units of syntactic computations reveals the complete structure of pronouns within and across languages.

The present study represents a comprehensive investigation of allomorphy in pronouns, based on a sample of 335 languages covering 75 language families. The analysis is couched within the Distributed Morphology framework. It scrutinizes directionality patterns, locality considerations, and the inventory of morphemes that act as triggers and targets of allomorphy in the pronominal domain.

The focus of the inquiry is on the properties of the internal pronominal syntax. On the basis of a series of studies, I propose a **multi-morphemic** approach to pronouns. Pronouns consist of a constrained set of syntactic terminals drawn from the set of functional morphemes. The study identifies the patterns of universality and variation in the pronominal syntax.

The present dissertation represents a case study on investigating the ways in which syntactic complexity is implicated in morphological complexity. The computation of the form is informed both by the syntactic structure and the PF-unique considerations such as Impoverishment and Underspecification of individual Vocabulary Items. Pronouns provide a particularly suitable testing ground for a number of central questions in morphological theory and the theory of the morphology-syntax interface. Inspecting pronominal systems reveals regularities in exponent distribution that go beyond principles of realization and implicate the narrow syntax.

The results of this dissertation contribute to research on nominal and pronominal syntax and contextual allomorphy. Recruiting allomorphy as a test for the internal structure of pronouns yields a twofold result, as findings from the pronominal domain inform the theory of allomorphy in important ways.

Zusammenfassung

Diese Dissertation untersucht die folgende zentrale Frage: Was ist die interne Struktur von Pronomen? Ich entwickle und untersuche die Hypothese, dass Allomorphie ein Schlüsselinstrument ist, um (a) auf den Bestand an Morphemen zuzugreifen, aus denen ein Pronomen in verschiedenen Sprachen besteht, und (b) die Muster ihrer Interaktionen zu verfolgen. Allomorphe Alternationen beinhalten eine Interaktion zweier Morpheme, des Auslösers und des Ziels der Allomorphie. Die Identifizierung des Triggers und des Targets in Pronominalformen führt zu einem unmittelbaren Ergebnis: Pronomen beinhalten zwei Morpheme, die in ihrer syntaktischen Repräsentation unterschiedliche strukturelle Positionen einnehmen. Die Dissertation präsentiert die Idee, dass die Untersuchung sequentieller Interaktionen zwischen Morphemen als Grundeinheiten syntaktischer Berechnungen die vollständige Struktur von Pronomen innerhalb und zwischen Sprachen offenbart.

Die vorliegende Studie stellt eine umfassende Untersuchung der Allomorphie in Pronomen dar, basierend auf einer Stichprobe von 335 Sprachen aus 75 Sprachfamilien. Die Analyse ist im Rahmen der Distribuierten Morphologie angelegt. Sie untersucht Richtungsmuster, Lokalität-süberlegungen und das Inventar der Morpheme, die als Auslöser und Ziele der Allomorphie im pronominalen Bereich fungieren.

Der Schwerpunkt der Untersuchung liegt auf den Eigenschaften der internen Pronominalsyntax. Auf der Grundlage einer Reihe von Studien schlage ich einen multimorphemischen Ansatz für Pronomen vor. Pronomen bestehen aus einer eingeschränkten Menge syntaktischer Bausteine, die aus der Menge der funktionalen Morpheme stammen. Die Studie identifiziert die Muster der Universalität und Variation in der Pronominalsyntax.

Die vorliegende Dissertation stellt eine Fallstudie dar, in der untersucht wird, wie syntaktische Komplexität mit morphologischer Komplexität verbunden ist. Die Bildung der Form wird sowohl durch die syntaktische Struktur als auch durch die PF-spezifischen Besonderheiten wie Impoverishment und Unterspezifikation (Underspecification) einzelner Vokabulareinheiten beeinflusst. Pronomen bieten ein besonders geeignetes Testfeld für eine Reihe zentraler Fragen der morphologischen Theorie und der Theorie der Morphologie-Syntax-Schnittstelle. Die Untersuchung von Pronominalsystemen offenbart Regelmäßigkeiten in der Exponentenverteilung, die über die Prinzipien der Realisierung hinausgehen und die Syntax implizieren.

Die Ergebnisse dieser Dissertation tragen zur Forschung über nominale und pronominal Syntax und Allomorphie bei. Die Verwendung von Allomorphie als Test für die interne Struktur von Pronomen führt zu einem zweifachen Ergebnis, da Erkenntnisse aus dem pronominalen Bereich die Theorie der Allomorphie in wichtiger Weise beeinflussen.

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Chapter 1

Introduction

Personal pronouns in natural language represent an area of extensive research from the origin of the generative tradition. In the early transformational approach, pronouns were considered substitutes for identical noun phrases or ‘words used in place of a noun’ (Lees 1960, Lees & Klima 1963; cf. Kuroda 1965, Ross 1967). In that sense, a pronoun is a surface realization of an underlying noun used to avoid repetition. Pronouns do not exist as primitives, but are derived from ordinary NPs via transformations and the rule of ‘pronominalization’ in particular, which converts the noun phrase into a pronoun. Other relevant notions involve rules such as reflexivization (Chomsky 1965), definiteization (Kuroda 1965), as well as genitivization and definite article attachment (Postal 1969). The idea of strict parallelism between nouns and pronouns was criticized already in Chomsky (1965), who notices that pronominalization does not obligatorily apply when two phonetically identical NPs co-occur, as is the case with two nouns having different reference.

Roughly at the same time, a growing interest in pronouns arises in logic and philosophy (Quine 1960, Geach 1962, 1964, 1967, 1969), and semantically oriented work couched within the Montague grammar (Partee 1970, 1973a, 1973b, 1978; Bach 1969; 1970; Karttunen 1969). The two lines of thought converge on the idea that pronouns in natural language are counterparts of variables in symbolic logic. This view has its far-reaching reflexes in the dynamic semantics (Kamp 1981; Heim 1982) and formal semantics where pronouns are variables, bound or free, consisting of an index and a set of φ -features with presuppositional semantics attached to the index (Heim & Kratzer 1998). Extensive research in the semantic theory has led to a broad consensus on the presuppositional semantics of φ -features and features on pronouns in particular (see Cooper 1983; Heim & Kratzer 1998; Sauerland 2003, 2008a; Schlenker 1999, 2003b, 2003a; Heim 1994, 2008; Chemla 2009; Jacobson 2012; Sudo 2012).

With the advent of Government and Binding, pronouns play a vital role in the formulation of the Binding Theory (Chomsky 1980, 1982), with reflexives being tied to Condition A, ordinary personal pronouns to Condition B, and non-pronominal DPs to Condition C. The tight connection of clausal syntax and pronominal forms found its way in prominent syntactic classifications of pronouns, grounded on the properties of pronouns in particular syntactic environments. Cardinaletti & Starke (1999) develop a typology of pronouns taking distributional properties of

pronominal forms (e.g., availability to surface in coordination, ϑ -positions, dislocated peripheral positions) as the primary diagnostic for their classification, yielding the distinction between (i) strong pronouns, (ii) weak pronouns, and (iii) clitics. In contrast, the main criterion employed in Déchaine & Wiltschko's (2002) typology is Binding Theory, i.e., whether a particular pronoun is subject to Binding Condition B or C, distinguishing between (i) pro-DPs, (ii) pro- φ Ps, and (iii) pro-NPs. The two proposals paved the way for a fruitful discussion in the syntactic theory about the properties of pronouns. Crucially, in both approaches, inferences about the internal composition of pronouns are made on the basis of their properties in the clausal syntax. With respect to their categorization, pronouns are typically considered objects of category D (Postal 1969), a view fully developed in the approach to pronouns as definite descriptions (Elbourne 2005; 2013), where they receive a uniform analysis with definite articles.

At the level of clausal syntax, pronouns are crucial ingredients of some of the central phenomena of syntactic theory, such as the silent pronoun *pro* in the *pro*-drop parameter (Rizzi 1982, 1986a, 1986b; Alexiadou & Anagnostopoulou 1998, 2021; Ritter 1995; Borer 1986, 1989; Barbosa 1995, 2011; Torrego 1998; Neeleman & Szendrői 2007; Shlonsky 2009; Roberts 2009, 2010; Holmberg 2005; Holmberg & Roberts 2013; Camacho 2013; Kučerová 2014). Likewise, the Person Case Constraint (PCC) emerges with the co-occurrence of two pronominal forms (Béjar & Rezac 2003; Adger & Harbour 2007; Anagnostopoulou 2017; Pancheva & Zubizarreta 2018; Deal 2022). The same holds for Clitic Doubling (CD), which involves a co-occurrence of a clitic, i.e., a pronominal form, with an overt DP (Uriagereka 1988, 1995; Anagnostopoulou 1994, 2003, 2004; Alexiadou & Anagnostopoulou 1997, 2000; Roberts 2010; Nevins 2011b; Kramer 2014; Harizanov 2014). Particular focus in the syntactic theory has been on so-called *adnominal pronouns*, i.e., syntactic environments in which a pronoun co-occurs with a noun, as is the case in *we linguists* construals (Postal 1969; Pesetsky 1978; Abney 1987; Cardinaletti 1994; Longobardi 1994; Déchaine & Wiltschko 2002; Ackema & Neeleman 2013b; Höhn 2017; Ilić 2024; Jambrović 2024). Detailed examinations of these domains contributed enormous insights into pronouns with respect to their role in clausal syntax.

Recent years have brought an increased interest in the morphology of pronouns, which have been addressed from various perspectives (see Moskal 2015a, 2015b, 2018; Harbour 2016; Smith et al. 2019; Pertsova 2022a; Zompì 2023). In this dissertation, I examine the patterns of morphological realization of pronouns to uncover the ways in which syntactic complexity is implicated in morphological complexity. The analysis is based on 335 languages covering 75 language families and couched within the Distributed Morphology framework.

The chapter is organized as follows. In 1.1, I introduce the main proposal and the central hypothesis examined in this dissertation. Section 1.2 introduces the research questions. Section 1.3 presents the overview of the data and the principles of data collection and classification. Section 1.4 introduces the theoretical framework. In Section 1.5, I provide a gist of the decompositional approach to pronouns developed here. Section 1.6 presents an overview of the dissertation.

1.1 Allomorphy as a tool

This dissertation investigates the internal structure of pronouns and its morphological realization in the natural language. It focuses on intricate patterns of morphological realization of pronouns, with the central goal of uncovering the ways in which syntactic complexity is implicated in morphological complexity. I propose and explore the following central hypothesis (1).

- (1) Contextual allomorphy is a key tool for
- a. *accessing* the inventory of morphemes that make up a pronoun across languages
 - b. *tracking* the patterns of their interactions

Allomorphic alternations involve an interaction of two morphemes, the *trigger* and the *target* of allomorphy. Identifying the trigger and the target in pronominal forms yields an *immediate result* - pronouns involve two morphemes that occupy distinct structural positions in their internal representation in the narrow syntax. This idea is at the core of the multi-morphemic approach to pronouns proposed here. Before proceeding with the specifics of the analysis, a brief overview of the notion of allomorphy is provided.

1.1.1 Proposal: From Allomorphy to Structure

Contextual Allomorphy is understood as a realization of a morpheme $[\xi]$ with two different exponents, i.e., *allomorphs*, depending on the morpho-syntactic context. In a context $[\Omega]$, the exponent y is chosen over an elsewhere exponent x (2). The competition between the exponents x and y is resolved in terms of *specificity*. The exponent y is more specific than the exponent x because it involves the context specification in its rule of insertion. That is, the exponent y is chosen only if the morpheme $[\xi]$ appears in the context of the morpheme $[\Omega]$. Otherwise, the exponent x surfaces. Specificity-based competition is regulated by the Subset Principle and its Elsewhere condition (Kiparsky 1973; Halle 1997). Therefore, allomorphy signals the sensitivity of a particular morpheme ξ to a morpheme Ω . Typically, the context specification is restricted to a single morpheme, as is the case with Ω in (2). The focus of the present investigation is allomorphy conditioned by morpho-syntactic features, that is, *suppletion*.

- (2) *Contextual allomorphy*

- a. $[\xi] \leftrightarrow y / __ \Omega$
- b. $[\xi] \leftrightarrow x$

For example, the person morpheme surfacing with the second-person feature in narrow syntax is realized as $[s]$ in the singular and as $[t]$ in the plural in Finnish pronouns. In that sense, $[s]$ and $[t]$ are *contextual allomorphs* of the same morpheme. In this case, the number morpheme is a trigger, and the person morpheme is a target of allomorphy. The following set of rules for the insertion of Vocabulary Items (VIs) for person accounts for the Finnish data (3). Throughout

this dissertation, I indicate the material under consideration in bold. In the Vocabulary Insertion rules, I provide exponents relevant for the present discussion and not the entire system, as is the case with exponents of person in (3). Furthermore, for clarity, in the Vocabulary Insertion rules, I indicate the first, second, and third person with numbers [1], [2], and [3], respectively, rather than using the terms *AUTHOR*, *ADDRESSEE*, *PARTICIPANT*, unless employing these notions is necessary for the analysis. It is important to note that I do not make claims about the representation of features, that is, whether features are binary or privative, and whether this distinction applies both at the level of narrow syntax and at the level of Vocabulary Items. For ease of exposition, the notational convention corresponding to privative features will be used (e.g., *PL*, *AUTHOR*), with an exception of gender features, where a binary $[\pm \text{FEM}]$ feature will be employed, in line with the literature on gender features.

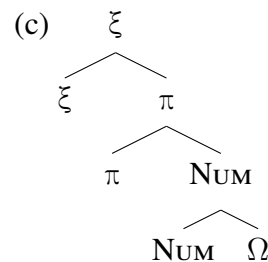
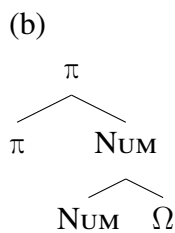
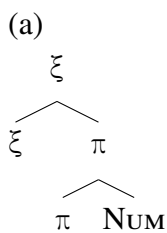
2		
	SG	PL
NOM	sinä	te
ACC	sinut	teidät
PART	sinua	teitä

(a) Finnish (White 2006)

(3) **VIs for PERSON in Finnish**[2] $\leftrightarrow$ t / $___ \#_{\text{PL}}$ [2] $\leftrightarrow$ s

Table 1.1: Allomorphy in Finnish pronouns

Crucially, Finnish data implicate the presence of two morphemes in the structure of pronouns: person and number. If we are able to identify the trigger and the target of allomorphy, these two features establish two morphemes in the syntax, yielding a minimal structure consisting of the two given morphemes. This pattern illustrates why allomorphy serves as a tool for uncovering the structure. Furthermore, in addition to *PERSON* and *NUMBER*, pronouns may involve more features merged in different positions (4). In (4-a), a morpheme ξ is merged on top of person $[\pi]$ and number $[\text{NUM}]$ morphemes, while in (4-b), a morpheme Ω is merged below the number morpheme. Combining ξ and Ω is an additional natural possibility (4-c). Several important questions arise. Is the merging of a new morpheme reflected in morphology? If the morphemes show an interaction, does ξ interact only with π , with both π and NUM simultaneously, or only with NUM to the exclusion of π in (4-a)? The same set of questions extends to structural representations in (4-b) and (4-c).

(4) **BUILDING THE STRUCTURE**

This dissertation presents the idea that inspecting the series of **sequential interactions** of ξ with π and NUM (4-a), Ω with NUM and π (4-b), ξ and Ω with π and/or NUM (7-c) is instrumental for understanding the structure of pronouns.

The **computation of the form** is informed and influenced both by the **syntactic structure** and the **PF-unique considerations** such as (i) allomorphy, (ii) Impoverishment, and (iii) underspecification of individual Vocabulary Items. Therefore, this dissertation demonstrates how alternations that are morphological in nature, such as allomorphy, inform the study of narrow syntax. Recruiting allomorphy as a test for internal compositionality of pronouns is motivated by a principled rationale. Allomorphy reflects narrow syntax, as it **directly operates on the output of syntax**. The competition of the two allomorphs applies to a morpheme, i.e., a basic unit of syntactic computations. Its realization provides a window into the structure as it indicates the syntactic environment in which it originates. Taking the general schema of allomorphy provided in (2) as an example, the insertion of the allomorph y directly implicates the presence of the morpheme Ω in the structure in addition to the morpheme ξ that is actually realized.

Furthermore, allomorphy is a window into the structure, as one can observe fundamental principles behind situations in natural languages where a morpheme of person comes in contact with a morpheme of number. As the data in this dissertation will demonstrate, to observe the behavior of a morpheme as a basic unit of syntactic computation, we often have to interact with it. This interaction in the present work is implemented by adding additional morphemes into perspective, which helps us determine the properties of the morpheme under consideration. For example, number-driven allomorphy of the person morpheme demonstrates that a pronoun involves at least two morphemes in its structure, thus being multi-morphemic and complex. Adding a new morpheme to this perspective, e.g., gender, sheds light on both person and number. This core principle underpins the idea presented here that the structure of pronouns is uncovered by inspecting the sequences of minimal interactions between morphemes.

This approach yields a **twofold result**, as findings from the pronominal domain inform the theory of allomorphy in important ways. Allomorphy is of central interest for the theory of morphology, syntax, and the morphology-syntax interface. It is among the phenomena that motivated the Distributed Morphology framework, based on the fact that meaning cannot be associated with sound in the syntax, as the same meaning can be mapped to several different phonological forms (Halle & Marantz 1994). Its ceaseless significance is evidenced by the ongoing vibrant discussion giving rise to the growing body of work, investigating larger sets of data (Veselinova 2006; Embick 2010; Bobaljik 2012; Moskal 2015a, 2015b, 2018; Smith et al. 2019; Kalin 2022a; Zompì 2023) or focused on individual case studies (Carstairs-McCarthy 1987, 2001; Bobaljik 2000; Embick 2003; Jenks & Sande 2017; Gribanova & Harizanov 2017; Winchester 2017; Christopoulos & Petrosino 2017; Ganenkov 2020; Banerjee 2020; Harðarson 2021; Trommer 2021; Paparounas 2022; Christopoulos & Zompì 2023; Dolatian & Guekguezian 2023; Caha 2024; Hein et al. 2024; Adamson 2024; Kampanarou 2024). The present study represents the most comprehensive investigation of allomorphy in pronouns up to date, based on a sample of 335 languages. It scrutinizes the directionality patterns, locality considerations,

and the inventory of morphemes that act as triggers and targets of allomorphy in the pronominal domain.

Pronouns came into focus in morpho-syntactically oriented work in recent years. In the study on the differences in patterns of allomorphy in nouns and pronouns, Moskal (2015b) examines 80 pronominal systems, demonstrating that pronouns undergo allomorphy for both number and case. In contrast, nouns exhibit allomorphy for number, but not for case. This asymmetry is proposed to stem from the structural difference between pronouns and nouns and the presence of the categorizing head in particular. Smith et al. (2019) examine allomorphy in pronouns, testing the CONTAINMENT HYPOTHESIS in allomorphic conditioning (Bobaljik 2012). The study focuses on number [SG-PL-DU] and case [NOM-ACC-DAT] triples, to observe whether the unattested *ABA patterns will arise. The discussion is based on the sample collected from the Free Personal Pronoun System database and the suppletion database at the Surrey Morphology Group (Brown et al. 2004). The patterns of number- and case-driven allomorphy were discussed in isolation. That is, whether the pronoun shows allomorphy for number was examined on a sample of 91 languages. Likewise, allomorphy triggered by case was independently examined on a sample of 89 languages. The study does not discuss an interaction of case and number, i.e., whether a pronoun undergoes allomorphy for both features and whether the two potential triggers mutually interact. Furthermore, the Containment hypothesis was examined in languages with clusivity in triples [1SG-1PL.EXCL-1PL.INCL] on the basis of 233 languages from the Free Personal Pronoun System database in Moskal (2018). Pertsova (2022a) revisits the proposal that [1SG-1PL.EXCL-1PL.INCL] should be considered a triple formalized in terms of containment, reporting several examples of *ABA. The study is based on 96 languages out of a total of 270 languages in the PRONOM database (Pertsova 2022b). This database was used as one of the sources for the present investigation.

In a recent study on the typological patterns of allomorphy couched within Optimality Theory, Zompì (2023) notes that the methodological strategy employed by Smith et al. (2019), where allomorphy is examined with respect to one trigger at a time, e.g., only number to the exclusion of case, provides insights into unidimensional paradigms, but leaves multidimensional paradigms out of consideration, where the members of the paradigm differ with respect to several features, e.g., both case and number. Notably, all previous work on the patterns of allomorphy in pronouns, including Zompì's (2023) study that takes into consideration several features simultaneously, considers **only one direction in allomorphic conditioning**. That is, a particular formant, analyzed as a pronominal base or stem, is observed with respect to number and case features. However, **allomorphic conditioning goes in both directions**.

The present study, based on the most extensive set of pronominal systems, investigates the complete set of features that make up pronouns across languages and the patterns of their interactions, all directions in allomorphic conditioning. It contributes a complete catalogue of the patterns of allomorphy in pronouns across languages. These empirical findings have implications for the structure of pronouns and the theory of allomorphy.

1.2 Questions

The fundamental question that this dissertation addresses is what is the structure of a pronoun. Which syntactic terminals make up pronouns within and across languages (1)? An answer to this question is not immediately obvious given their relatively simple morphological realization, especially compared to nouns, which is pushed to a complete non-segmentability in some languages. A natural question that arises in the context of syntactic theory is whether pronoun allows for drawing significant generalizations about universal patterns in their realization. In particular, do potential universal patterns in realization implicate universality in the syntactic structure (2)? The same questions can be raised for potential variation in the realizational component. If variation is attested in morphology, does it carry over to narrow syntax? An equally relevant concern for both the structure of pronouns and the theory of morphology-syntax interface is where to locate the source of variation (3). Finally, the big picture question that arises is to what extent does the close examination of the computation of the form reveal generalizations about the syntactic structure (4).

THE QUESTIONS

1. What is the internal structure of pronouns?
2. Are there universal patterns in the realization and the structure of pronouns?
3. Is there variation in the realization of pronouns? If there is, what is the source of variation?
4. Which broader generalizations about narrow syntax are derivable from observing the computation of the form?

Pronouns provide a particularly suitable testing ground for a number of central questions in morphological theory. Inspecting pronominal systems reveals regularities in exponent distribution that go beyond principles of realization and implicate the narrow syntax. As the discussion in upcoming chapters will demonstrate, considerations that emerge in studying the pronominal domain address several key themes central to the morphological theory.

1.3 Data

In this section, I will discuss the principles of data collection (1.3.1) and classification (1.3.2) in this study and provide an overview of the sources, language families, and languages investigated.

1.3.1 Data collection and overview

This dissertation is based on a large-scale, typologically balanced sample of 335 languages encompassing 75 language families, including 5 creoles and 18 isolates. The empirical coverage

enables drawing meaningful generalizations about universals and identifying points of variation in the realization of pronouns in natural languages. The reason for recruiting a high number of languages is that closely related languages exhibit different patterns of allomorphy, observed in the early stages of this study. For example, the pattern of allomorphy present in Icelandic is different from the one attested in German. In addition, Icelandic and Faroese exhibit gender distinctions in both singular and plural, while other Germanic languages typically exhibit gender neutralization in the plural. For this reason, it was not possible to identify one language as a representative of a family and count it as a single data point. Typologically different languages exhibit the same patterns of allomorphy. Hence, the result cannot be interpreted as an outcome of genetic relatedness.¹ As the upcoming discussion will demonstrate, within the Indo-European family, Greek exemplifies a language in which the person morpheme remains unaffected by allomorphy, while Germanic languages exhibit substantial allomorphy. In investigating the patterns of allomorphy on a large language sample, the present study contributes to the line of research entertained in recent years in which large samples of cross-linguistic data are taken into account in interpreting the results of allomorphy.

The investigated language sample is compiled from the following six sources (5). The complete collection of grammars published within the Mouton Grammar Library [MGL] was consulted, resulting in 78 languages. Mouton-CASL Grammar Series [MCASL] was consulted for four languages (Pashto, Bangla, Dhivehi, Hindko). Mouton Pacific Linguistics [PL] series served as a source for two languages: Nama (Papuan; Yam) and Kopar (Lower Sepik). Mouton Sinitic Languages of China [SLCH] contributed two languages: Yichun and Caijia. In addition to these series, I consulted individual reference grammars, grammatical descriptions, and studies, collecting the data from 110 additional languages, which resulted in 196 languages. This sample was supplemented with 139 languages from the recently published PRONOM database (Pertsova 2022b). These sources resulted in a total of 335 pronominal systems collected for this study. The collected sample provides a unique empirical base that underpins the present investigation, as the given combination of languages has not been examined before, although a certain number of languages naturally overlap with previous cross-linguistic studies on the morphological realization of pronouns.

(5) SOURCES

- Mouton Grammar Library [MGL]
- Mouton-CASL Grammar Series [MCASL]
- Mouton Pacific Linguistics [PL]
- Mouton Sinitic Languages of China [SLCH]
- Individual reference grammars, grammatical descriptions, and studies

¹I thank Elisabeth Verhoeven for helping me to clarify this important point.

- PRONOM database (Pertsova 2022b)

Detailed, comprehensive grammatical descriptions and studies consulted for this study, and the available database enabled data collection in a straightforward manner. Pronominal systems were manually extracted from the sources. All collected pronominal systems are distributed across Appendices A to C, corresponding to studies discussed in the upcoming chapters.

The collected sample of pronominal systems covering 75 language families provides a high level of both genetic and areal diversity. In addition to language families that involve a number of individual branches, as is the case with Indo-European and all its branches included in the present investigation (e.g., Germanic, Romance, Slavic, Indic, as well as individual languages such as Greek), that exhibit substantial geographic distribution, a number of indigenous families, including several endangered languages, located in a particular area was examined, as is the case with native American languages (Algonquian, Hokan, Penutian, Uto-Aztecan), indigenous languages of Central and South America (Araucanian, Arawakan, Chibchan, Mayan, Mixe-Zoque, Quechuan), and indigenous languages of Australia, to name a few. The same holds for language isolates, such as Trumai (Xingu Reserve, Brazil), Urarina (Peru), and Bangime (Mali).

The following table provides an overview of all language families with several examples of languages belonging to each family. When a single language family can be split into several sub-families, as is the case with Altaic, Australian, Niger-Congo, and Papuan languages, the respective name of the sub-family is provided.

FAMILY	LANGUAGE
Afro-Asiatic	Goemai, Tamashek, Amharic, Dahalo, Hdi, Somali, Pero, Hausa
Algonquian	Ojibwe, Nishnaabemwin, Passamaquoddy
Altaic Mongolic	Mangghuer, Buryat
Altaic Tungusic	Udihe, Evenki
Altaic Turkic	Uzbek, Kazakh, Chuvash, Khakas, Turkish, Tatar, Baskir
Araucanian	Mapuche
Arawakan	Alto Perené, Baure
Arawan	Kulina
Australian	Kunbarlang, Ngandi; Wardaman, Malakmalak, Maranungku
Australian Nyulnyulan	Bardi
Australian Bunuban	Gooniyandi
Australian Iwaidjan	Maung
Australian Pama-Nyungan	Pitjantjatjara, Warrongo, Kuuk Thaayorre
Australian Tangkic	Kayardild
Australian West Barkly	Wambaya
Australian Wororan	Worrorra, Ungarinjin
Austro-Asiatic	Pacoh, Sapuan, Ho, Semelai, Temiar
Austronesian	Madurese, Tukang Besi, Maranao, Daakaka, Malagasy
Baining	Qaqet
Barbacoan	Cayapa, Awa Pit
Cariban	Pemón
Caucasian	Kartvelian (Svan, Georgian), Nakh Daghestanian (Hinuk)
Chibchan	Bribri, Cabécar, Damana, Ika, Kuna, Tunebo
Choco	Embera
Chicham	Aguaruna
Chonan	Selknam
Dravidian	Malayalam, Tamil, Kannada, Gondi, Gadaba, Brahui, Telugu
Eskimo-Aleut	Aleut, Greenlandic, Central Alaskan Yupik
Guaicuruan	Pilagá, Toba
Hmong-Mien	Hmong Njua, Xong
Hokan	Jamul Tiipay, Chimariko, Salinan, Central Pomo

FAMILY	LANGUAGE
Indo-European	Greek, Armenian, Germanic, Slavic, Celtic, Romance, Indic
Japonic	Japanese
Korean	Korean
Kxa	Jul'hoan
Mayan	Modern Yucatec, Tojolabal
Misumalpan	Miskito, Sumu
Mixe-Zoque	Sierra Popoluca, Mixe (Ayutla)
Mosetenan	Mosetén
Muskogean	Choctaw
Nadahup	Hup
Na-Dene Athapaskan	Navajo
Na-Dene Tlingit	Tlingit
Nambikuaran	Nambikwara
Niger-Congo	Bantu (Chichewa, Eton, Kisi, Limbum), Kwa (Idoma), Fongbe
Niger-Congo Dogon	Yamsay, Tommo So
Niger-Congo Kordofanian	Orig
Nilo-Saharan	Ma'di, Koyraboro Senni, Koyra Chiini, Lango, Turkana, Nandi
Oto-Manguean	Chinantec
Páezan	Páez
Panoan	Cashinahua, Kakataibo
Papuan	Bunaq, Teiwa, Adang, Blagar, Kaera, Kamang, Sawila, Wersing
Papuan Eleman	Orokolo, Toaripi (Sepoe)
Papuan Engan	Kewa
Papuan Nimboran	Kemtuik
Papuan Sepik	Yimas, Kopar, Kwoma, Awtuw
Papuan Solomons East Papuan	Lavukaleve, Savosavo
Papuan Trans-New Guinea	Hua, Kuman, Coastal Marind, Asmat
Papuan Yam	Nama
Peba-Yaguan	Yagua
Penutian	Nisenan, Nez Perce, Takelma, Sierra Miwok

FAMILY	LANGUAGE
Quechuan	Ecuadorian Kichwa
Salish	Bella Coola, Musqueam
Sino-Tibetan	Nuosu, Qiang, Meithei, Jingpho, Apatani, Mandarin, Yichun
Siouan	Winnebago
Tacanan	Cavineña
Torricelli	Arapesh
Totonacan	Totonac
Tucanoan	Cubeo, Siriano
Tupian	(Xingu) Asurini, Sirionó
Uralic	Udmurt, Estonian, Finnish, Tundra Nenets, Pite Saami
Uto-Aztecan	Hopi, Luiseño, Western Tarahumara, Tübatulabal, Yaqui, Pipil
Witotoan	Ocaina
Yeniseian	Ket
Yukaghir	Kolyma Yukaghir
creoles	Fa d'Ambô, Berbice Dutch Creole, Saramaccan Creole
isolates	Basque, Chitimacha, Haida, Sandawe, Tonkawa, Yuracare, Zuni

Table 1.2: Languages investigated in this study: An overview

1.3.2 Data classification

A coarse inspection of the collected sample, consisting of 335 languages, suggested that pronominal systems of natural languages differ in the number of overt pronouns and morphologically realized features. This motivated the decision to classify pronominal systems with respect to three major features, (i) **PERSON**, (ii) **NUMBER**, and (iii) **GENDER**, as the realization of these features is directly traceable in pronouns. It is important to note that the focus of the study is on personal pronouns. Natural languages employ several types of pronominal forms, such as possessives and demonstratives. As these pronouns involve a set of additional syntactic and semantic considerations, they are outside the scope of this investigation.

The category of **PERSON** as a criterion resulted in a split between (i) languages without clusivity and (ii) languages with clusivity distinction. In the former group, the meaning of the *inclusive*, that is, a pronoun whose referent includes both the first [AUTHOR] and second person [ADDRESSEE] is morphologically realized in the form of the first person plural pronoun, as is the case with *we* in English and *wir* in German. In contrast, in a number of languages, the information whether the second person is included in the referent of a pronoun is overtly mor-

phologically encoded. These languages, such as Malayalam and Lavukaleve, surface with two distinct pronouns depending on whether the second person is included (*inclusive*) or not (*exclusive*) in the referent of a pronoun. In this study, 210 languages lack the clusivity distinction, while 115 languages mark it morphologically. This is consistent with the finding reported in Harley & Ritter (2002) that languages with clusivity are generally less frequent.

When it comes to the category of **NUMBER**, all languages investigated in this study involve the singular~plural distinction, which enabled close examination of the morphological realization of the person feature in the context of different number specifications. The presence of the dual will occasionally be discussed.

Some languages were recognized in the literature to surface with minimal~augmented systems (McKay 1978; Noyer 1992; Corbett 2000, 2005; Harbour 2016). The existence of a pronoun whose referent consists of *exactly* two individuals, the **AUTHOR** and the **ADDRESSEE**, [1+2], motivated the proposal of the **MINIMAL** feature in natural language. However, it is important to note that minimal~augmented system is not necessarily a different kind of a number system that should be sharply contrasted with the singular~plural. As the upcoming discussion will demonstrate, the variation emerges in the **PERSON** morpheme that can overtly morphologically mark first and second person to the exclusion of all other individuals. For this reason, languages involving minimal inclusive pronouns are discussed together with languages surfacing with the clusivity distinction. That is, in the same way languages with clusivity have more morphological possibilities to express the person feature specifications, the minimal inclusive pronoun simply involves a more fine-grained morphological distinction in this respect, being able to mark that the referent of a pronoun consists of the **AUTHOR** and a single **ADDRESSEE**.

In contrast to **NUMBER** and **PERSON** features, which are present in *all* languages investigated here, examination of **GENDER** features resulted in a significant split in the present dataset. Namely, gender marking on pronouns is attested in 80 out of 335 languages investigated in this study. As adding gender features to person and number leads to increasing the complexity, I developed a **DECISION TREE** (Roberts 2010; Kalin 2022a) for tracking the patterns of interaction of gender features with person and number to apply the same criteria across languages.

In addition to the three central features and the patterns of their interactions investigated in this thesis, in Chapter 6 (6.4.1), I will discuss the ways in which the present analysis can be extended to case features, presenting the empirical split between Greek, where exponents of person remain unaffected by the case, and Germanic languages, that exhibit substantial case-driven allomorphy.

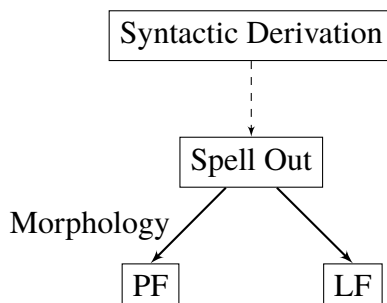
Finally, pronouns in some languages combine with extra pieces of morphology, spread across the entire paradigm, thus being insensitive to person, number, and gender considerations. These additional exponents are optional, but their overt realizational is typically followed by a set of syntactic and semantic effects, such as focus interpretation. In Chapter 6 (6.4.2), I present two case studies on Greek and Serbian complex pronouns, extending the proposal to complex forms.

This broad classification served as an input for identifying the exponents of person, number, and gender morphemes, and the analysis of allomorphy in pronouns.

1.4 Distributed Morphology: The Architecture

Distributed Morphology (DM) is a syntactic theory which posits syntax as a single generative engine building all complex objects (Halle & Marantz 1994). The same principles building syntactic objects in clausal syntax apply below the ‘word’ level in the traditional sense of the term. That is, the word formation is syntactic, meaning that the standard structure-building operation *Merge* (Chomsky 1995) applies across the board. For overviews of the theory, detailed discussions on its aspects, and the principles of building syntactic objects see Harley & Noyer (1999), Embick & Noyer (2007), Embick (2015), Alexiadou (2016), Bobaljik (2017), and the contributions in Alexiadou et al. (2025). The theory makes a principled division between narrow syntax and the post-syntactic realizational component of the grammar. Placing formation of all complex objects in the syntax makes Distributed Morphology compatible with the standard Minimalist Y model of grammar (6), where the output of the syntactic derivation is sent to Spell Out and consequently interpreted at the two interfaces: the Phonological Form (PF) and the Logical Form (LF) (Chomsky 2000, 2001). Furthermore, DM is compatible with the Meaning First approach to natural language (Sauerland & Alexiadou 2020) that places syntactic and semantic derivation within a single generative component that corresponds to the level of syntactic derivation in the representation in (6), i.e., there is no LF branch after Spell Out but only prior to it. The place of Morphology remains the same.

(6) THE ARCHITECTURE OF GRAMMAR: Y MODEL



Morphology interprets and reflects syntactic hierarchical representations without feeding them, i.e., it does not serve as an input for syntactic computations. Instead, Morphology operates on the raw syntactic structure sent to interfaces in one cycle of Spell Out. Under this architecture, a central ingredient of the theory is the existence of three lists (7). Each list is consulted at a distinct stage in the derivation, indicated after the arrow.

(7) THREE LISTS OF DISTRIBUTED MORPHOLOGY

- a. *The Syntactic Terminals*, a list which includes $\sqrt{\text{Roots}}$, categorizing heads (n, v, a), and functional morphemes ($\pi, \#, \gamma, D$) $\Rightarrow$ **Syntactic Derivation**
- b. *The Vocabulary*, a list consisting of Vocabulary Items (VIs), i.e., objects that supply syntactic terminals (morphemes) with the phonological material $\Rightarrow$ **Morphology**

- c. *The Encyclopedia*, a list which includes non-compositional semantic information

The list of *Syntactic terminals* (List 1), consists of *morphemes* as basic units of syntactic computations (see below) that are subject to Merge operation. This list comprises both $\sqrt{\text{Roots}}$ (e.g., $\sqrt{\text{dog}}$, $\sqrt{\text{flower}}$, $\sqrt{\text{home}}$) and functional morphemes, such as φ -features (π , $\#$, γ), definiteness (D), Tense, Aspect, and Voice, among others. As morphemes are primitive units the syntax operates with in building larger objects, this list is consulted during the syntactic derivation, i.e., it belongs to narrow syntax. *The Vocabulary* (List 2) contains all the Vocabulary Items (VIs) that pair phonological exponents with their respective rules of insertion. The role of the Vocabulary is to supply the morpheme that is devoid of any phonological material in narrow syntax with the phonological content in the post-syntactic component of the grammar. Therefore, this list is consulted in Morphology in the process of mapping from narrow syntax to the Phonological Form (PF). Finally, List 3, *The Encyclopedia*, supplies morphemes with special meanings (e.g., idiomatic interpretations) that are not available under regular compositionality. These interpretations have been connected with particular local contexts, demonstrated to be the first categorizing/phase head (Marantz 2001, 2007; Arad 2003; Alexiadou & Anagnostopoulou 2008b; for refinements and the discussion on the delimiter for idiomatic readings see Anagnostopoulou & Samioti 2014; Oikonomou & Alexiadou 2022; Alexiadou & Anagnostopoulou 2023). In the following sections, I will elaborate more on the properties of morphemes, exponents, Vocabulary Items, and the principles located in the Morphology component in (6), as these aspects of the theory will be crucial for the discussion throughout this dissertation.

1.4.1 Abstract Morphemes, Exponents, and Vocabulary Insertion

An abstract *morpheme* is a basic atomic unit of syntactic computations that carries a single feature or a feature bundle. In the syntactic structure, an abstract morpheme is a *terminal node*. When combined with other morphemes in the composition of complex objects, each morpheme represents a single terminal node, i.e., a head in narrow syntax. As such, a morpheme should crucially be distinguished from an *exponent*, an object that supplies the morpheme with the phonological content during the process of Vocabulary Insertion at PF. Therefore, abstract morphemes belong to the list of syntactic terminals (List 1), while exponents are part of a Vocabulary (List 2). Separating the (i) features of a morpheme from its (ii) phonological content is a crucial point that distinguishes the abstract morpheme of DM from a traditional morpheme as pairing of meaning and sound. In DM, these two components of the morpheme are distributed across two lists and accessed at two distinct levels in the natural language architecture, narrow syntax and PF, respectively. Therefore, during the syntactic computation, information about the phonological content of individual morphemes is not available to the generative engine.

- (8) a. **MORPHEME**: an atomic unit of syntactic computations, occupying a single terminal node (head) in narrow syntax
- b. **EXPONENT**: an object supplying a morpheme with a phonological content

- c. **VOCABULARY ITEM (VI)**: a pairing of an exponent with the condition on its insertion

Furthermore, the theory distinguishes between two kinds of abstract morphemes: (i) functional morphemes and (ii) $\sqrt{\text{Roots}}$ (9). $\sqrt{\text{Roots}}$ are morphemes that contribute ‘lexical’ meaning, such as $\sqrt{\text{dog}}$, $\sqrt{\text{walk}}$, $\sqrt{\text{purple}}$, and combine with categorizing heads and functional morphemes to compositionally build larger syntactic units (Halle & Marantz 1994; Marantz 1995, 1997; Embick & Marantz 2008; Alexiadou 2014, 2016; Harley 2014; Panagiotidis 2014, 2015; Alexiadou & Lohndal 2017b; Bobaljik 2017). A $\sqrt{\text{Root}}$ is typically considered the most deeply embedded morpheme in a particular complex head, and the first morpheme that undergoes Vocabulary Insertion, in a step-wise bottom-to-top realizational model (Bobaljik 2000; Myler 2017; Kalin 2024). The set of functional morphemes consists of categorizing heads (n , v , and a) and morphemes such as ϕ -features, Tense, Aspect, and Voice, among others.

(9) TYPES OF MORPHEMES

- a. FUNCTIONAL MORPHEMES: n , v , a , π , $\#$, Tense [$\pm\text{PST}$], Aspect [$\pm\text{PF}$], Voice, ...
- b. ROOTS: $\sqrt{\text{dog}}$, $\sqrt{\text{flower}}$, $\sqrt{\text{home}}$, $\sqrt{\text{sail}}$, $\sqrt{\text{dress}}$, $\sqrt{\dots}$

A property setting roots apart from other morphemes is the CATEGORIZATION ASSUMPTION (Embick & Marantz 2008), according to which a root cannot appear as part of a complex object without being categorized. Roots are categorized by combining with category-defining functional heads, n , v , and a , which turn these acategorial morphemes into nouns, verbs, and adjectives, respectively. As illustrated in the table below, an acategorial $\sqrt{\text{dog}}$ becomes a noun only after combining with a category-defining head n . The same logic extends to verbs and adjectives.

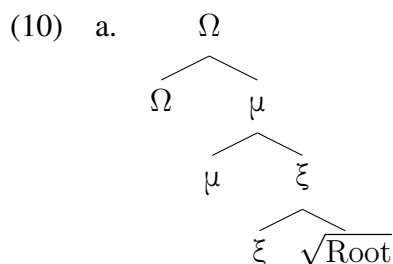
$n\text{P}$	$v\text{P}$	$a\text{P}$
$n \sqrt{\text{dog}}$	$v \sqrt{\text{walk}}$	$a \sqrt{\text{purple}}$

Table 1.3: Category-defining heads

An approach to roots described above takes into consideration both their structural and semantic properties. An alternative approach treats roots as pure objects of syntactic computations, lacking semantics altogether. Under this conception, roots are empty objects that carry indices [e.g., $\sqrt{\text{root}}_7$, $\sqrt{\text{root}}_{101}$] responsible for their interpretations at LF and PF (Harley 2014). A related understanding of roots formulated on purely structural grounds, as a radically empty/underspecified structural position subject to first Merge that surfaces with four properties (i) absence of grammatical features, (ii) absence of syntactic category, (iii) structural rather than lexical definition, and (iv) domination by functional material, has been put forth in De Belder & Van Craenenbroeck (2015). Therefore, under this conceptualization, roots are not morphemes but structural positions. As a consequence, $\sqrt{\text{dog}}$ and a feature [$+\text{PL}$], for instance, are treated alike by the syntactic computation.

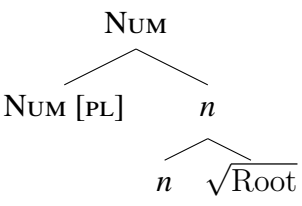
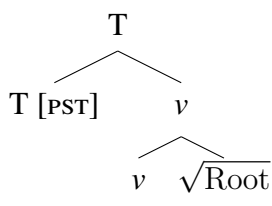
Alexiadou & Lohndal (2017a) demonstrate that roots are not uniform across languages, motivating the typology of root categorization based on the division of labor between the categorizing head v and roots. This idea is motivated by the empirical fact that in some languages, as is the case in Hebrew, the semantics of roots is highly general, being assigned arbitrarily in a given configuration. On the other hand, in English, roots are supplied with meaning independently of the environment in which they occur. This is best exemplified by the following contrast. In English, two morphologically unrelated roots ($\sqrt{\text{cream}}$, $\sqrt{\text{fat}}$) are employed to derive the two meanings (*cream*, *fat*), while Hebrew has at its disposal a single root ($\sqrt{\text{smn}}$). To obtain the two meanings, Hebrew employs an array of functional morphemes. This suggests that functional material has a ‘repair’ function in cases when the root itself does not carry substantial semantics. Thus, distinctions in the semantics of roots tightly correlate with the number of functional morphemes. In between these two poles is Greek, with a set of roots whose interpretation is highly dependent on the functional material, thus patterning with Hebrew in this respect, but also a set of roots that have substantial independent meaning on par with English.

With the distinction between different kinds of morphemes in place, we can now move to the distinction between **morphemes** and **exponents**. The following syntactic structure (10-a) consists of four morphemes: a $\sqrt{\text{Root}}$, ξ , μ , and Ω . In narrow syntax, these morphemes are compositionally combined into a complex object (10-a). These morphemes lack phonological content. In the flat representation in (10-b), each morpheme is enclosed within square brackets. The empty slot // indicates that morphemes lack phonological material and thus need to be supplied with it. I use flat representations in (10-b/c) rather than the syntactic structure in (10-a) to simply indicate that phonological material does not enter the level of narrow syntax. In that sense, the representation in (10-b) is a direct output of syntax, consisting of syntactic/semantic features and empty slots // for the phonological material. This representation shows morphemes prior to the Vocabulary Insertion operation. In the next step, the Vocabulary Insertion applies, targeting one morpheme at a time, starting from the most deeply embedded morpheme outwards. In the structural representation in (10-a), the most deeply embedded morpheme is the $\sqrt{\text{Root}}$, hence it is targeted for the insertion first, followed by the insertion targeting ξ , which is followed by the insertion into μ , and finally the insertion into Ω . Therefore, (10-c) illustrates the representation of a complex object after *all* morphemes have undergone Vocabulary Insertion. In this case, the slots within double slashes are not empty anymore, but filled. This material represents exponents. The exponent /a/ supplies the $\sqrt{\text{Root}}$ with the phonological content, /b/ supplies the morpheme ξ , /c/ is combined with μ , and /d/ with Ω . Put differently, the morphemes $\sqrt{\text{Root}}$, ξ , μ , and Ω are mapped to *a*, *b*, *c*, and *d*, respectively.



- b. $[\sqrt{\text{Root}}; / /] - [\xi; / /] - [\mu; / /] - [\Omega; / /]$ **morphemes *prior to* Vocabulary Insertion**
- c. $[\sqrt{\text{Root}}; /a/] - [\xi; /b/] - [\mu; /c/] - [\Omega; /d/]$ **morphemes *after* Vocabulary Insertion**
- d. $\sqrt{\text{Root}}, \xi, \mu, \Omega$ **morphemes**
- e. a, b, c, d **exponents**

Let us see how this abstract representation translates into the realization of nouns and verbs. The syntactic representation of the noun *books* and the verb *walked* is provided in examples (11) and (12), respectively. In narrow syntax, the noun *books* consists of at least three syntactic terminals: a $\sqrt{\text{Root}}$, a categorizing head *n*, and a number (NUM) morpheme carrying the plural feature (11-a). In (11-b), each morpheme is enclosed within square brackets. The inserted phonological material is provided within a double slash */ /*. The $\sqrt{\text{Root}}$ is realized as */book/*, the categorizing head *n* receives zero realization */∅/*, and the number node is realized as */s/*. After the Vocabulary Insertion operation, the form *books* is obtained (11-c). The same principle applies to the past tense verb *walk*. Three morphemes, $\sqrt{\text{Root}}$, verbalizing head *v*, and the Tense morpheme carrying the past tense feature, are combined in the syntax, giving rise to the structural representation in (12-a). At PF, the respective morphemes are paired with the phonological content (12-b), resulting in the past tense form *walked* (12-c).

- (11) a.  b. $[\sqrt{\text{Root}}; /book/] - [n; /∅/] - [PL; /s/]$ c. *books*
- (12) a.  b. $[\sqrt{\text{Root}}; /walk/] - [v; /∅/] - [PST; /ed/]$ c. *walked*

As abstract morphemes lack phonological content in the syntax, they are supplied with it in the process of Vocabulary Insertion that applies at PF (13), i.e., in Morphology, as illustrated in (6). The Vocabulary Insertion operation fills the empty slot in a morpheme with a phonological exponent. This empty slot corresponds to a Q position proposed in Halle (1992). In that sense, every morpheme involves two components: (i) the syntactic/semantic features and (ii) the empty Q slot. The latter is filled with the phonological exponent at PF.

(13) VOCABULARY INSERTION

An operation that adds phonological features to abstract morphemes in the realizational component of grammar.

The analysis is formalized in terms of Vocabulary Insertion rules. For example, the rule in (14) states that */s/* is the exponent of the given Vocabulary Item. On the left side of a bidirectional arrow ($\leftrightarrow$) is a feature or a feature bundle that undergoes insertion, in the case at hand, plural, while on the right side, the respective phonological exponent */s/*.

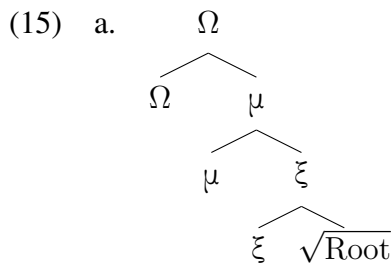
(14) **Vocabulary Insertion rule for plural**

$$\text{NUM} [\text{PL}] \leftrightarrow s$$

There is a debate in the theory as to whether $\sqrt{\text{Roots}}$ have phonology in narrow syntax or not. Approaches assuming that $\sqrt{\text{Roots}}$ are paired with the phonological content at PF, such as early DM (Halle & Marantz 1994; see also *Separation Hypothesis* of Beard 1966, 1995) and the Meaning First approach (Sauerland & Alexiadou 2020) propose *Universal Late Insertion*, as all morphemes without exceptions are devoid of phonological material in the syntax.

1.4.2 **Vocabulary Insertion targets terminal nodes**

In DM, the Vocabulary Insertion operation **targets only terminal nodes** (Embick & Marantz 2008; Embick 2016). That is, in the syntactic structure in (10), repeated here as (15), the Vocabulary Insertion targets only one terminal node at a time, e.g., ξ , but crucially not ξ and μ together. This point distinguishes DM from comparable realizational frameworks that propose insertion targeting larger chunks of structure, as is the case with phrasal spell-out of Nanosyntax (Starke 2009) and Spanning that allows insertion targeting contiguous sequences of structurally adjacent heads, i.e., ‘spans’ (Bye & Svenonius 2011; Merchant 2015; Svenonius 2016). The idea that the Vocabulary Insertion targets only terminal nodes will play a crucial part in this dissertation, and the pronominal domain will provide a strong additional argument in its support.



b. $[\sqrt{\text{Root}}; / \text{I}]/[\xi; / \text{I}]/[\mu; / \text{I}]/[\Omega; / \text{I}]$ **morphemes *prior to* Vocabulary Insertion**

c. $[\sqrt{\text{Root}}; /a/]/[\xi; /b/]/[\mu; /c/]/[\Omega; /d/]$ **morphemes *after* Vocabulary Insertion**

1.4.3 **Competition**

Vocabulary Items **compete** for insertion. Two kinds of competition can be distinguished: *context-free* and *context-dependent* (Halle & Marantz 1994). As the upcoming discussion will demonstrate, the kind of competition has direct consequences for the structure of pronouns. For example, the rule in (14) states that the elsewhere exponent of plural is inserted, that is, an exponent that does not require any context specification to be satisfied for its insertion. Notably, English employs several allomorphs of plural (16) that compete for insertion into the plural node. The competition is resolved in terms of the context specification, which states which allomorphs combine with which $\sqrt{\text{Roots}}$. The allomorph $[en]$ cannot appear in the context of the root

$\sqrt{\text{sheep}}$, producing an ungrammatical **sheepen*, as the given root does not appear on the list of roots specified for the allomorph *[en]*, as is the case with *oxen*. Instead, the root $\sqrt{\text{sheep}}$ is on the list that combines with the zero $[\emptyset]$ allomorph of the plural. Therefore, in the set of rules for realizing English plural morphology (16), **complexity** arises on the **right side** of the rule. That is, the context resolves the competition. Complexity on the right side of a rule will play a significant role in this dissertation, as an essential component of contextual allomorphy.

(16) ALLOMORPHS OF PLURAL IN ENGLISH (Gouskova & Bobaljik 2025)

[PL] $\leftrightarrow$ -en /]_N __, where $N \in \{\sqrt{\text{ox}}, \sqrt{\dots}\}$

[PL] $\leftrightarrow$ $\emptyset$ /]_N __, where $N \in \{\sqrt{\text{sheep}}, \sqrt{\text{fish}}, \sqrt{\dots}\}$

[PL] $\leftrightarrow$ [-back, +tense] /]_N __, where $N \in \{\sqrt{\text{foot}}, \sqrt{\text{goose}}, \sqrt{\dots}\}$

[PL] $\leftrightarrow$ -i /]_N __, where $N \in \{\sqrt{\text{alumn}}, \sqrt{\text{syllab}}, \sqrt{\dots}\}$

[PL] $\leftrightarrow$ -z /]_N __

The kind of competition just described relies on specificity, as all rules that involve context specification are more specific than the one inserting the elsewhere exponent /s/, thus taking precedence in the course of the Vocabulary Insertion operation. The competition grounded in specificity is resolved in terms of the Subset Principle and the Elsewhere condition (Kiparsky 1973; Halle 1997), also known as Pāṇinian ordering.

(17) **THE SUBSET PRINCIPLE** (Halle 1997)

The phonological exponent of a Vocabulary Item is inserted into a morpheme in the terminal string if the item matches all or a subset of the grammatical features specified in the terminal morpheme. Insertion does not take place if the Vocabulary Item contains features not present in the morpheme. Where several Vocabulary Items meet the conditions for insertion, the item matching the greatest number of features specified in the terminal morpheme must be chosen.

The approach to the structure of pronouns developed in this dissertation is couched within the outlined theoretical framework. As Morphology **interprets** hierarchical representations generated by the syntax, studying the patterns of universality and variation in the realization of pronouns provides direct insights into the underlying syntactic representations. Contextual allomorphy, proposed to serve as a tool for discovering the structure, is crucially informed by the syntax as it operates on the piece of structure sent to the post-syntactic component in one cycle of spell-out. In this overview of the theory, I outlined central architectural assumptions and fundamental notions. Other relevant phenomena such as **Underspecification** of Vocabulary Items and **Impoverishment** will be introduced in the course of the discussion in explaining the patterns in morphological realization of pronouns that necessitate their application.

1.5 Decomposition: A first look

The central proposal of this dissertation is that pronouns are multi-morphemic, multi-terminal, complex syntactic objects assembled in the syntax. Earlier in this chapter, I provided an analysis of Finnish pronouns as involving contextual allomorphy. Recall (3), repeated here as (18).

2			
	SG	PL	
NOM	<i>sinä</i>	<i>te</i>	(18) VIs for PERSON in Finnish [2] $\leftrightarrow$ t / ___ # _{PL} [2] $\leftrightarrow$ s
ACC	<i>sinut</i>	<i>teidät</i>	
PART	<i>sinua</i>	<i>teitä</i>	

(a) Finnish (White 2006)

Table 1.4: Allomorphy in Finnish pronouns

Why is an analysis of pronouns in terms of mono-morphemic feature bundles targeted by the Vocabulary Insertion not enough to explain this pattern? For example, the form *sinä* would be a bundle consisting of person, gender, and case features [2, SG, NOM], *sinut* would consist of [2, SG, ACC], and *te* of [2, PL, NOM]. This approach predicts that pronouns are stored in their entirety in memory. Hence, there are six exponents that realize the given six feature bundles. This approach **misses an important empirical generalization**. Namely, the six pronouns under consideration share not only substantial feature content, e.g., the second-person that repetitively occurs in each bundle, but also substantial material in the domain of their realization, as [s] is spread across all singular pronouns irrespectively of case specification, and [t] across the plural. The presence of a **shared morpho-phonological material** calls for a **decomposition**.

Let us now look at a different system. In Greek, [m], [s], and [t], the exponents of the first, second, and third person, respectively (Alexiadou & Stavrou 2000), consistently resist allomorphy for number and case, being preserved across both numbers and all case specifications.

	1		2		3					
	SG	PL	SG	PL	M.SG	M.PL	F.SG	F.PL	N.SG	N.PL
NOM	ego	e <u>m</u> is	e <u>s</u> i	e <u>s</u> is	aftos	afti	afti	aftes	afto	afta
GEN	e <u>m</u> ena	e <u>m</u> as	e <u>s</u> ena	e <u>s</u> as	aftu	afton	aftis	afton	aftu	afton
ACC	e <u>m</u> ena	e <u>m</u> as	e <u>s</u> ena	e <u>s</u> as	afton	aftus	aftin	aftes	afto	afta

Table 1.5: Personal pronouns in Greek

I propose that the presence of a shared morpho-phonological material is indicative of the presence of a person morpheme that occupies a terminal node in the internal composition of pronouns. This terminal is targeted by the Vocabulary Insertion operation. The *Vocabulary* of Greek contains a dedicated set of Vocabulary Items realizing person features (19). As noted

above, in the Vocabulary Insertion rules, I provide simple representations of person features [1, 2, 3] and include only those in focus of the present discussion. The Greek pronominal system consists of additional exponents realizing definiteness [*af*], as well as number, gender, and case features.

(19) **Vocabulary Items for PERSON Greek**

[1] $\leftrightarrow$ m

[2] $\leftrightarrow$ s

[3] $\leftrightarrow$ t

Recognizing the existence of Vocabulary Items specialized for person features **explains the shared morpho-phonological material** in the Greek system **in a principled manner**. Under the **decompositional** analysis I am developing in this dissertation, there are only 3 VIs for person in Greek. The Vocabulary Insertion targets terminal nodes, adding the phonological content of the VIs in (19). Under the **non-decompositional** approach, the pronominal system of Greek in Table 2 would result in 30 VIs, involving a significant morpho-phonological overlap, which, however, remains unexplained. In addition, an implication of the decompositional analysis is a *reduction* of the *Vocabulary*. Namely, the person feature is specified in only 3 Vocabulary Items when the person is individuated, in contrast to 30 otherwise. Furthermore, other features which make up Greek pronouns, such as number, gender, and definiteness, are realized via a limited number of VIs. In contrast, the non-decompositional approach would repeat all the features with every instantiation of a pronoun, on par with the specification of person. Therefore, the decompositional approach not only accounts for the morpho-phonological overlap but also brings economy to the post-syntactic component of the grammar.

This work is in line with decompositional approaches to natural language, such as Distributed Morphology (Halle & Marantz 1994, Embick 2015) and the Meaning First approach (Sauerland & Alexiadou 2020, To appear), where a single terminal in the narrow syntax corresponds to a single exponent at PF. In an implementation of the phenomenon of double plurality in terms of Maximize Exponence, Alexiadou (2024) demonstrates that overt exponents of the lexical and grammatical plural correspond to two nodes in the narrow syntax. Support for the decompositional account comes from much of the recent work on language acquisition (Guasti et al. 2023; Martin et al. 2022; Driemel et al. 2023; Sauerland et al. 2024; Martin et al. 2024). Furthermore, it contributes to the line of inquiry that proposes a complex internal structure of pronouns (Cardinaletti & Starke 1999; Déchaine & Wiltschko 2002; Moskal 2015a; Smith et al. 2019; Puškar-Gallien 2022, among others), by investigating minimal interactions of morphemes that build a pronoun.

If pronouns are separated into several morphemes in the syntax corresponding to several exponents at PF, this raises implications for memory storage and processing, as noted above. Namely, extensive research has provided significant evidence that words are not stored as such in memory, but decomposed. Evidence for this position comes from language-mixing varieties. As demonstrated in Alexiadou & Lohndal (2018), bilingual speakers of several mixed varieties

combine a root from one language with functional morphology from the other. Crucially, the observed mixing arises below the word level, thus implicating that words are not stored as such in the memory, but decomposed into a root and several pieces of functional material. In addition, an extensive line of experimental work provides evidence for the decompositional approach to words (Stockall & Marantz 2006; Solomyak & Marantz 2010; Fruchter & Marantz 2015; Flick et al. 2018; Gaston et al. 2021). In this dissertation, I argue that pronouns consist of functional morphemes only. The decomposition of the functional material reported in the existing experimental work is expected to carry over to pronouns. If pronouns, as syntactically smaller objects, are decomposed and their individual pieces are stored in memory, this would provide an additional argument that words are not stored and processed in their entirety, but decomposed.

1.6 Dissertation structure

The dissertation is structured as follows. In Chapter 2, I examine the realization of person and number exponents in pronouns, demonstrating that discrete exponents are indicative of person and number morphemes in the narrow syntax. This empirical finding motivates the multi-morphemic approach to pronouns. Chapters 3 to 5 present a series of studies on allomorphy. In Chapter 3, I focus on the interaction of person and number, presenting two studies: (i) Number-driven person allomorphy and (ii) Person-driven number allomorphy. Chapter 4 shifts the perspective to languages with the clusivity distinction. In Chapter 5, I introduce gender into perspective. Chapter 6 provides the main theoretical proposal about the structure of pronouns and the implications of the present study for the theory of allomorphy.

Chapter 2

Exponents of Person and Number in Pronouns

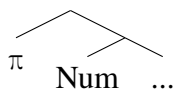
2.1 Introduction

Realization of person and number features and their interaction is a key component of the theory of pronouns that aims to uncover their internal structure. Before delving into the patterns of allomorphy as a tool for discovering the pronominal structure, I discuss the morphological realization of person and number exponents in pronouns, establishing a foundation for the series of studies on allomorphy introduced in the upcoming chapters.

The central empirical finding of the present discussion is that both person and number, as two core components of pronouns, are combined with distinct, discrete exponents in the realizational component of grammar. The discussion is based on illustrative examples that cover a range of typologically different languages. Morphological patterning, which gives rise to a number of empirical generalizations discussed and evaluated here, has direct implications both for the narrow syntax and the post-syntactic *Vocabulary*. Namely, an exponent of the person feature in the morphological realization of a pronoun is indicative of the person morpheme in the syntax. The same logic extends to the number. Importantly, exponents of both person and number freely co-occur, implicating a complex, multi-morphemic structural representation.

The obtained empirical facts give rise to a **multi-morphemic** formal account of pronouns. In the narrow syntax (1), person and number are morphemes, i.e., basic units of syntactic computations. They establish two distinct syntactic terminals in the internal structure of a pronoun (1). Being terminal nodes, both person and number can be targeted by the Vocabulary Insertion operation. This stance is directly supported here by the fact that they are mapped to clearly identifiable, distinct exponents at PF (2). The empty slots // in (2) indicate that the person and number morphemes do not have phonological features in the syntax, being supplied with them only at PF. $\mathbb{M}$ stands for a morpheme, and the square bracket includes information about its featural content (π , NUM) and the slot for the phonological content //.

(1) NARROW SYNTAX



(2) PF

$\mathbb{M} [\pi; / /] - \mathbb{M} [\text{NUM}; / /]$

The multi-morphemic approach yields a number of results. First, it explains the significant morpho-phonological overlap in pronominal systems across languages in a principled manner. The complex patterns of the realization of plurality in the nominal and pronominal domains are revealed once the two sources of plural allomorphs are recognized: a $\sqrt{\text{Root}}$ in nouns and a π morpheme in pronouns. These facts cannot be accounted for under a non-decompositional account that would consider all pronouns as single Vocabulary Items (VIs) inserted into bundles of φ -features. In a model of grammar where pronouns would be stored in their entirety, empirical generalizations would remain unaccounted for.

The chapter is organized as follows. In 2.2, I discuss the realization of person in pronouns, demonstrating that it exhibits regularities both within and across languages. Section 2.3 presents an examination of number-related exponents, proposed here to realize the number morpheme in the narrow syntax. Section 2.4 summarizes the chapter.

2.2 Realization of Person in pronouns: Germanic ~ Greek

Across languages, the morphological realization of the PERSON feature in pronouns gives rise to two patterns, which can be subsumed under a Germanic and a Greek pattern. Germanic languages are particularly illustrative of allomorphic processes in pronouns. First-person pronouns in English, German, and Icelandic (Neijmann 2021) undergo allomorphy triggered by both number and case, employing four different morphological realizations.

	English		German		Icelandic	
	SG	PL	SG	PL	SG	PL
NOM	I	we	ich	wir	ég	við
ACC	me	us	mich	uns	mig	okkur

Table 2.1: Allomorphy in first person pronouns in Germanic languages

Compared to Germanic, Greek shows a radically different behavior. As the table below demonstrates, $[m]$, $[s]$, and $[t]$, the exponents of the first, second and third person, respectively (Alexiadou & Stavrou 2000), consistently resist allomorphy for number and case, being preserved across both numbers and all case specifications.

	1		2		3					
	SG	PL	SG	PL	M.SG	M.PL	F.SG	F.PL	N.SG	N.PL
NOM	ego	<u>e</u> mi <u>s</u>	<u>e</u> si	<u>e</u> si <u>s</u>	aft <u>s</u>	aft <u>i</u>	aft <u>i</u>	aft <u>e</u> s	aft <u>o</u>	aft <u>a</u>
GEN	<u>e</u> me <u>na</u>	<u>e</u> ma <u>s</u>	<u>e</u> se <u>na</u>	<u>e</u> sa <u>s</u>	aft <u>u</u>	aft <u>on</u>	aft <u>i</u> s	aft <u>on</u>	aft <u>u</u>	aft <u>on</u>
ACC	<u>e</u> me <u>na</u>	<u>e</u> ma <u>s</u>	<u>e</u> se <u>na</u>	<u>e</u> sa <u>s</u>	aft <u>on</u>	aft <u>us</u>	aft <u>i</u> n	aft <u>e</u> s	aft <u>o</u>	aft <u>a</u>

Table 2.2: Pronouns in Greek

Resistance to allomorphy does not hold only for pronouns but extends to clitics. Dedicated person exponents are preserved in the genitive and accusative, with the absence of clitic forms in the nominative case in the language in general.¹ In the third person, clitics lack the additional morpheme [af], argued here to encode definiteness (Chapter 6), but realize the person morpheme [t]. Thus, Greek is a language where the person morpheme is an invariant building block in the morphological realization of pronouns.

	1		2		3					
	SG	PL	SG	PL	M.SG	M.PL	F.SG	F.PL	N.SG	N.PL
GEN	<u>m</u> u	<u>m</u> as	<u>s</u> u	<u>s</u> as	<u>t</u> u	<u>t</u> on	<u>t</u> is	<u>t</u> on	<u>t</u> u	<u>t</u> on
ACC	<u>m</u> e	<u>m</u> as	<u>s</u> e	<u>s</u> as	<u>t</u> on	<u>t</u> us	<u>t</u> in	<u>t</u> es	<u>t</u> o	<u>t</u> a

Table 2.3: Clitics in Greek

A closer look at Greek reveals that the PERSON morpheme and the morpheme encoding definiteness/demonstrativity [af] are inaccessible for allomorphy. In contrast, the component responsible for encoding number (and gender in the third person) is the one that undergoes changes. The relevant question that emerges is whether the observed pattern may receive an explanation in structural terms. I propose that the presence of a dedicated exponent that can be tracked across the paradigm in Greek is indicative of the presence of a person morpheme that occupies a terminal node in the internal composition of a pronoun. This terminal is targeted by the Vocabulary Insertion operation. The *Vocabulary* of the language contains a dedicated set of Vocabulary Items realizing person features (3).

(3) Vocabulary Items for PERSON Greek

[1] ↔ m

[2] ↔ s

[3] ↔ t

Recognizing the Vocabulary Items specialized for person features explains the shared morpho-phonological material in the Greek system provided in Tables 2.2 and 2.3 in a principled manner. Under the **decompositional** analysis that I am developing in this dissertation, there are only 3 VIs for person in Greek. The Vocabulary Insertion targets the terminal nodes, adding the phonological content of the VIs in (3). Under the **non-decompositional** approach, the pronominal system of Greek in Table 2.2 would result in 30 VIs, involving a significant morpho-phonological overlap, which, however, remains unexplained. In addition, an implication of the decompositional analysis is a *reduction* of the *Vocabulary*. Namely, the person feature is specified in only 3 Vocabulary Items when the person is individuated, in contrast to 30 otherwise. Furthermore, other features that make up Greek pronouns, such as number, gender, and definiteness, are realized

¹Clitics in the nominative case are possible in environments such as *Na tos!* ‘There he is!’ This construal is available in the accusative case: *Na ton!* with the same meaning (Schneider-Zioga 1994).

via a limited number of VIs. In contrast, the non-decompositional approach would repeat all the features with every instantiation of a pronoun, on par with the specification of person. Therefore, the decompositional approach not only accounts for the morpho-phonological overlap but also brings economy to the post-syntactic component of the grammar.

The comparison of Germanic and Greek reveals the split in the realization of the person morpheme across languages. While Germanic languages employ up to four exponents of person, Greek surfaces with only one exponent. There is no morpheme whose presence in the structure affects the exponence of person in Greek: gender, number, case, or demonstrativity. It is entirely protected from allomorphy. Germanic languages, with their substantial allomorphy, and Greek, with the complete absence of allomorphy, establish the two poles in the realization of person in pronouns. Across languages, pronominal systems follow Germanic or Greek to a notable degree, giving rise to a gradient scale. In the next section, I discuss data from several language families, including Turkic, different branches of Indo-European (Slavic, Romance, Uralic), Tacanan, Araucanian, Sino-Tibetan, and Papuan. I demonstrate that individual languages provide evidence for the decompositional approach to pronouns. In the next step, I will revisit Germanic data from a comparative perspective, demonstrating that even languages with substantial allomorphy may involve invariant exponents in their morphological realization.

2.2.1 A cross-linguistic perspective

Turkish (Turkic). In Turkish, the exponents of person remain inaccessible for person- and case-driven allomorphy. As illustrated in the table below, first-person pronouns in both numbers and all cases share the formant *[b]*. The same holds for second-person pronouns, which all surface with the formant *[s]*. I analyze *[b]* and *[s]* as exponents of the first and second person, respectively. They are stored as Vocabulary Items and do not involve any allomorphy.

	1		2		3	
	SG	PL	SG	PL	SG	PL
NOM	ben	biz	sen	siz	o	on-lar
ACC	ben-i	biz-i	sen-i	siz-i	on-u	on-lar-i
DAT	ban-a	biz-e	san-a	siz-e	on-a	on-lar-a

Table 2.4: Pronouns in Turkish (Csató & Johanson 2021)

There are two ways to account for the data. In terms of the absence of allomorphy (4), as I argue here, or in terms of allomorphy (5) (see, e.g., Paparounas & Akkuş 2023). Under the former approach, both person and number exponents are stored as separate Vocabulary Items that combine compositionally, giving rise to the obtained pronouns. The latter approach, *[ben]* is a Vocabulary Item which has *[biz]* as its allomorph in the presence of the plural number, realized as zero. The same holds for the *[sen~siz]* alternation.

(4) **VIs for Turkish**[1] $\leftrightarrow$ b[2] $\leftrightarrow$ s[PL] $\leftrightarrow$ iz / ___PARTICIPANT[PL] $\leftrightarrow$ lar[SG] $\leftrightarrow$ en(5) **VIs for Turkish: An alternative set**[1] $\leftrightarrow$ biz / ___#_{PL}[1] $\leftrightarrow$ ben[2] $\leftrightarrow$ siz / ___#_{PL}[2] $\leftrightarrow$ sen[PL] $\leftrightarrow$ $\emptyset$

Allowing for the analysis in (5) would have severe consequences for our understanding of allomorphy in general and in pronominal systems in particular. For instance, the second person pronoun in German in the nominative, accusative, and dative case [*du-dich-dir*] would be considered an ABC pattern in the sense of Bobaljik (2012), instead of an AAA (see Smith et al. (2019) for an AAA account). That is, the three pronouns would be allomorphs of each other. Such an approach entirely misses the fact that the exponent [*d*] is consistently present across the three forms. Thus, an analysis where [*dich*] is an allomorph of [*du*] in German, and [*biz*] is an allomorph of [*ben*] in Turkish, fails to capture significant empirical regularity in natural languages, which becomes immediately obvious once the person morpheme is individuated. Furthermore, in both cases, the forms are clearly phonologically related, which is not the case with allomorphs. An *allomorph*, by its definition, precludes any phonological relatedness. The fact that German [*du-dich-dir*] and Turkish pronouns (Table 2.4) share phonological material would have to be analyzed as a consequence of accidental homophony. I argue that if there is a shared phonological material, this calls for a decomposition (see also Caha et al. (2019) for the analysis of comparative exponents [š] and [ějš] in Czech, with the latter being further decomposed into [ěj] and [š], given that the two share the same material [š]). The possibility that neither (4) nor (5) entertains is an analysis in terms of a *feature bundle*, where person and number features would be located on a same morpheme (e.g., [1, PL] $\leftrightarrow$ biz²).

This point brings me back to the economy of the *Vocabulary*. Generalizing allomorphy, where pronouns would be allomorphs of each other, leads to an increase in Vocabulary Items specified for person. Under the decompositional analysis I am pursuing, there are two exponents for the first and second person (4). In the alternative analysis, the number of exponents is double, i.e., four (5).

In addition, it is not only person morphology that remains invariant across the paradigm. First and second person pronouns share the formant [en] in the singular [*ben*; *sen*] and [iz] in the plural [*biz*; *siz*]. If these pronouns were stored as single Vocabulary Items, the presence of [en; iz] would have to be considered mere accidental homophony. This point provides an additional

²This view has been pursued in Neeleman & Szendrői (2007), as person features are not individuated (6). In addition, this analysis does not involve any allomorphy, since it is based on the insertion into non-terminal nodes, realizing both person and number features simultaneously.

(6) [NP +p, -a, 1, SG] $\leftrightarrow$ ben
 [NP +p, -a, 2, SG] $\leftrightarrow$ sen
 [NP +p, -a, 1, PL] $\leftrightarrow$ biz
 [NP +p, -a, 2, PL] $\leftrightarrow$ siz

argument for the decompositional view. I analyze $[en]$, which transparently combines with the person exponents, an exponent of the singular number. Its phonological alternation with $[an]$ in the dative is a consequence of the vowel harmony, triggered by the dative exponent $[a]$. Furthermore, $[iz]$ is an exponent of plural in the context of the PARTICIPANT feature in the person node, as it surfaces only in the first and second person pronouns (4). In contrast, the third person combines with $[lar]$, an elsewhere exponent surfacing on nouns. The third person $[o]$ lacks $[n]$ only in the nominative singular. The presence of $[n]$ can receive a uniform analysis with $[en]$ as an exponent of singular, in which case it involves stacking of number morphology in the plural, which is not an unusual pattern cross-linguistically, as will be discussed in the next section.

The same morphological patterning is attested across Turkic. Azeri, Uzbek and Baskir minimally differ from Turkish in exhibiting number-driven person allomorphy, realized in the form of the $[m \sim b]$ alternation in the first person. Furthermore, these data demonstrate the variation in third-person pronouns, as Azeri behaves like Turkish in surfacing with $[n]$ in the plural. Uzbek differs in combining the plural exponent $[lar]$ directly with $[u]$. Baskir contrasts with all the data discussed here in surfacing with the consonant $[l]$ in the nominative singular, which is not the case in Turkish, Azeri, and Uzbek third-person pronouns that involve only a vowel.

	SG	PL		SG	PL		SG	PL
1	män	biz	1	men	biz	1	min	bïð
2	sän	siz	2	sen	siz	2	hin	hïð
3	o	onlar	3	u	ular	3	ul	ular

(a) Azeri (Ragagnin 2021)

	SG	PL
1	men	biz
2	sen	siz
3	u	ular

(b) Uzbek (Straughn 2011)

	SG	PL
1	min	bïð
2	hin	hïð
3	ul	ular

(c) Baskir (Berta 2021)

Table 2.5: Person exponents across Turkic

Serbian (Slavic). Individuation of the person exponent is transparently available in languages with substantial allomorphy. In Serbian, the first person undergoes both number- and case-driven allomorphy, the second person undergoes number-driven allomorphy, while the third person does not enter allomorphic alternations. Importantly, the case-induced allomorph of the first person in the plural $[n]$ remains intact across the system. Likewise, the number-triggered allomorph of the second person $[v]$ remains constant and does not interact with case. If pronouns in Table 2.6 were considered single Vocabulary Items inserted into bundles of number, person, case, and gender features, the phonological overlap would remain unaccounted for. Under the decompositional analysis, the exponents of person are stored as individual Vocabulary Items in the Vocabulary of Serbian. These data are representative of a larger Slavic pattern, which employs substantial allomorphy with transparently identifiable individual exponents.

	1		2		3			
	SG	PL	SG	PL	M.SG	M.PL	F.SG	F.PL
NOM	ja	mi	ti	vi	on	oni	ona	one
ACC	mene	nas	tebe	vas	njega	njih	nju	njih
DAT	meni	nama	tebi	vama	njemu	njima	njoj	njima
GEN	mene	nas	tebe	vas	njega	njih	nje	njih
INST	mnom	nama	tobom	vama	njim	njima	njom	njima
LOC	meni	nama	tebi	vama	njemu	njima	njoj	njima

Table 2.6: Person exponents in Serbian

French and Italian (Romance). The decompositional analysis of pronouns across Romance has been proposed in Kayne (2003). Namely, $[m]$ and $[t]$ in the singular and $[n]$ and $[v]$ in the plural are considered the exponents of the first and second person, respectively. In the present work, I argue for the same segmentation, explaining, in addition, the presence of $[n]$ and $[v]$ in terms of the number-driven allomorphy of person. The same holds for case-driven³ $[c]$ in the Italian first-person accusative plural pronoun. Furthermore, I consider $[l]$ the exponent of the third person, which does not undergo allomorphy, as evident from the data below. The same set of person exponents is present in both languages, with an additional allomorph $[c]$ in Italian. These data demonstrate how closely related languages may differ minimally in patterns of allomorphy.

	1		2		3			
	SG	PL	SG	PL	M.SG	F.SG	M.PL	F.PL
NOM	je	nous	tu	vous	il	elle	ils	elles
ACC	me	nous	te	vous	le	la	les	les

Table 2.7: Person exponents in French (Kayne 2003)

	1		2		3			
	SG	PL	SG	PL	M.SG	F.SG	M.PL	F.PL
NOM	io	noi	tu	voi	lui	loro	lei	loro
ACC	mi	ci	ti	vi	lo	la	li	le

Table 2.8: Person exponents in Italian

³Italian does not have morphological case marking. The form [ACC] is used for ease of exposition. Importantly, however, when a pronoun appears in the internal argument position, the observed morphological alternation emerges, on par with the accusative in languages that employ case morphology. Allomorphy induced by argument structure alternations will be discussed in Chapter 6.

Finnish (Uralic). The exponents of the first [*m*] and third person [*h*] do not undergo allomorphy triggered by number and case. This contrasts with the second person [*s*], which exhibits the number-driven person allomorphy, giving rise to the allomorph [*t*]. At the same time, the allomorph [*t*] remains unaffected by the case. I propose the following set of Vocabulary Items (7) for Finnish pronouns.

	1		2		3	
	SG	PL	SG	PL	SG	PL
NOM	minä	me	sinä	te	hän	he
ACC	minut	meidät	sinut	teidät	hänet	heidät
PART	minua	meitä	sinua	teitä	häntä	heitä

Table 2.9: Person exponents in Finnish (White 2006)

(7) **Vocabulary Items for PERSON in Finnish**[1] ↔ *m*[2] ↔ *t* / ___ #_{PL}[2] ↔ *s*[3] ↔ *h*

Pite Saami (Uralic). A related Uralic language, Pite Saami, departs from Finnish, as it surfaces with the complete absence of allomorphy, patterning in this respect with Greek. As illustrated in Table 2.10, [*m*], [*d*], and [*s*], the exponents of the first, second, and third person, respectively, are preserved in all number and case specifications. The set of Vocabulary Items in (8) accounts for the data.

	1			2			3		
	SG	PL	DU	SG	PL	DU	SG	PL	DU
NOM	mãn	mij	mäj	dãn	dij	däj	sãn	sij	säj
ACC	mu^v	mijáv	mãnov	du^v	dijáv	dãnov	suv	sijáv	sãnov
ILL	munje	mijjaj	munnuj	dunje	dijjaj	dunnuj	sunje	sijjaj	sunnuj
INESS	mu^vne	miján	munuon	du^vne	diján	dunuon	suvne	siján	sunuon

Table 2.10: Person exponents in Pite Saami (Wilbur 2014)

(8) **Vocabulary Items for Pite Saami**[1] ↔ *m*[2] ↔ *d*[3] ↔ *s*

Cavineña (Tacanan; Amazonian rainforest of Lowland Bolivia). The exponents of the first, second, and third person are transparently individuated in the pronominal system. As evident from Table 2.11, allomorphy is present only in the first-person absolutive. The exponent of the first person *[i]* alternates with the form *[e]* present in the plural number, realized as *[kwana]* and in ergative case, realized as *[ra]*. The first person is realized as *[i]* in the absolutive and as *[e]* in three other cells under consideration. Importantly, treating *[i]* as the default, as it appears in the default absolutive case, on par with the default nominative in languages with nominative-accusative alignment, would mean that both the plural and the ergative case trigger the allomorph *[e]*, which leads to accidental homophony, as two distinct features, plural number and ergative case, would trigger the same allomorph. Note that the ergative singular, that is, the form not involving the plural number in its structure, surfaces with *[e]*, which precludes a potential containment explanation where case would structurally contain plural, thus enabling syncretism. As the ergative singular and the absolutive plural share the same exponent *[e]*, together with the ergative plural, I analyze *[e]* as the default realization of the first person and *[i]* as the absolutive case-driven allomorphy. This explanation is incompatible with the containment-based approach to case-driven allomorphy in pronouns pursued in Smith et al. (2019) (for the case containment theory see Caha 2009), where absolutive is contained in the ergative as the more marked case. Under this analysis, the absolutive form *[i]* is predicted to appear in the ergative, contrary to fact. Therefore, the containment-based analysis must explain the distribution of *[e]* without resorting to accidental homophony. Note that in this language, the absolutive case receives an overt morphological realization *[ke]*, which disappears in the plural, across all three person specifications. The second person *[mi]* remains intact across both number and case specifications. The same holds for the person morpheme carrying the third-person feature in the narrow syntax, which is realized as *[tu]*. The exponent of the ergative case *[ra]* combines with person and number exponents without triggering any allomorphic alternations.

	1		2		3	
	SG	PL	SG	PL	SG	PL
ABS	i-ke	e-kwana	mi-ke	mi-kwana	tu-ke	tu-na
ERG	e-ra	e-kwana-ra	mi-ra	mi-kwana-ra	tu-ra	tu-na-ra

Table 2.11: Person exponents in Cavineña (Guillaume 2008)

I propose the following set of Vocabulary Items for person in Cavineña (9). On par with the data discussed above, if pronouns in Table 2.11 were single Vocabulary Items in their entirety, the shared morpho-phonological material would have to be considered accidental homophony. The fact that the entire pronominal system surfaces with clearly identifiable Vocabulary Items realizing person, number, and case, provides strong support for the compositional approach.

(9) **Vocabulary Items for Cavineña**

[1] $\leftrightarrow$ i / ___K_{ABS}

- [1] ↔ e
 [2] ↔ mi
 [3] ↔ tu

Mapuche (Araucanian). The exponents of the first [*iñch*], second [*eym*], and third person [*fey*], present in the singular forms of pronouns, are preserved in the plural. As evident from the data below and the set of Vocabulary Items in (10), the presence of the number morpheme does not trigger the allomorphy of person, which explains the shared material across both number specifications. The formant [*ng*] that appears in the third-person plural is an exponent of animacy (see also Chapter 5), as third-person plural pronouns can only be used for reference to animate referents (Smeets 2008), while the singular form [*fey*] for both animates and inanimates.

	SG	PL	
1	iñché	iñch-iñ	(10) Vocabulary Items for Mapuche
2	eymi	eym-ün	[1] ↔ iñch
3	fey	feye-ng-ün	[2] ↔ eym
			[3] ↔ fey

(a) Mapuche (Smeets 2008)

Table 2.12: Pronouns in Mapuche

Meithei (Sino-Tibetan). The exponents of person are preserved in plural forms, transparently combining with the plural exponent [*khoy*]. The presence of the plural exponent triggers morpho-phonological changes in the second and third person. For this reason, I consider only [*n*] and [*m*] Vocabulary Items for the second and third person (11), respectively. Morphological realization of this type, where both person and number are overtly realized, with different degrees of allomorphy or its complete absence, is a general property of Sino-Tibetan languages.

	SG	PL	
1	əy	əy-khoy	(11) Vocabulary Items for Meithei
2	nəŋ	nə-khoy	[1] ↔ əy
3	má	mə-khoy	[2] ↔ n
			[3] ↔ m
			[PL] ↔ khoy

(a) Meithei (Chelliah 2011)

Table 2.13: Person exponents in Meithei

Adang, Kaera, and Kamang (Papuan). Across Papuan languages, person morphemes are mapped to exponents transparently identifiable in the morphological composition of pronouns. As illustrated in the following data from Adang, Kaera, and Kamang, representative of a group of the so-called Timor-Alor-Pantar languages (Schapper 2014a), allomorphy is absent, and the singular and plural forms demonstrate a clear overlap. An exception in this respect are inclusive forms which involve the alternation. A few notes are in order when it comes to inspecting Papuan

pronominal systems. Namely, all pronominal forms in the singular share the formant $[a]$, while the plural pronouns share the formant $[i]$. I suggest that $[a]$ and $[i]$ should be considered as exponents of the singular and plural, respectively. In addition, pronouns in Kaera surface with an extra formant $[ng]$. The fact that it is spread across all person and number specifications suggests that it cannot be an exponent of person or number.

With these considerations in place, the individuation of person exponents in Papuan becomes straightforward. In the first-person pronouns, I analyze the formant $[n]$, an exponent of person. In the exclusive, it does not undergo allomorphy, being preserved in the pronoun $[ni]$. The second person receives zero realization $[\emptyset]$, consisting only of an overt number $[a/i]$ in Adang and Kamang or number and the formant $[ng]$ in Kaera. Given that the exponent of person is zero across both singular and plural, I analyze the second-person pronouns as the absence of allomorphy. Third-person pronouns also provide evidence for the presence of person exponents, with $[s]$ being the exponent of the third person in Adang and $[g]$ in Kaera and Kamang.

	SG	PL		SG	PL		SG	PL
1EXCL	na	ni	1EXCL	nang	ning	1EXCL	na	ni
1INCL		pi	1INCL		ping	1INCL		si
2	a	i	2	ang	ing	2	a	i
3	sa	supi	3	gang	ging	3	ga	ga

(a) Adang (Robinson & Haan 2014) (b) Kaera (Klamer 2014) (c) Kamang (Schapper 2014b)

Table 2.14: Person exponents across Papuan

It is important to note that the second person is not obligatorily realized as zero. As illustrated in Table 2.15 on the basis of Western Pantar data, that shares the same set of first- and third-person exponents as Kaera and Kamang, the singular $[a]$ and plural $[i]$ morphology, including the formant $[ng]$, present in Kaera, the second-person pronouns provide evidence for an overt realization of the person exponent, realized as $[h]$ in the singular and preserved in the plural. Therefore, the nullness of the second person is common, but not an obligatory property of Papuan.

	SG	PL
1EXCL	nang	ning
1INCL		ping
2	hang	hing
3	gang	ging

Table 2.15: Person exponents in Western Pantar (Holton 2014)

Nama (Papuan). The same availability of transparent person exponents is attested in other Papuan languages, as is the case in Nama, belonging to the Yam family. As illustrated in Table 2.16, pronouns in the singular and plural share significant morpho-phonological material, which

calls for a decomposition. I analyze [yènd], [fèm], and [yèm] as exponents of the first, second and third person, respectively. These exponents combine with the plural exponent [fem].

	SG	PL	(12) Vocabulary Items for Nama
1	yèndo(n)	yèndfem	
2	fèmo(n)	fèmofem/fèmafem	
3	yèmo(n)	yèmofem	

(a) Nama (Siegel 2023)

Table 2.16: Person exponents in Nama

Revisiting Germanic. At the beginning of this section, I contrasted Germanic languages, which employ up to four exponents of person, with Greek, surfacing with only one. A closer inspection from a comparative Germanic perspective reveals that all languages share the exponent [m] in first-person pronouns in the accusative case. In the nominative plural, it is the exponent [w, v]. This suggests that [m] and [w/v] may figure as Vocabulary Items (see also Chapter 6 for an argument that [m] is the elsewhere realization of the first person across Indo-European languages), although at the synchronic level nothing would change if Germanic pronouns are simply overt allomorphs of person with zero number morphology, as allomorphy applies in both segmentation options. Therefore, even languages with substantial allomorphy, such as Germanic, when observed from a comparative perspective, hint toward further segmentation and identifying individual Vocabulary Items.

	SG		PL	
	NOM	ACC	NOM	ACC
ENGLISH	I	me	we	us
FAROESE	eg	meg	vit	okkum
GERMAN	ich	mich	wir	uns
ICELANDIC	ég	mig	við	okkur
DUTCH	ik	mij	wij	ons
NORWEGIAN	jeg	meg	vi	oss
SWEDISH	jag	mig	vi	oss

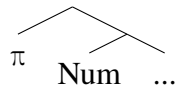
Table 2.17: The exponent of person across Germanic

2.2.2 Interim Summary

Close inspection of the morphological realization of pronouns in a wide range of typologically distinct languages reveals that PERSON feature exponence exhibits regularities within and across languages, giving rise to several significant generalizations. Pronominal systems demonstrate that an exponent realizing person should be individuated, as only in this way can substantial

morpho-phonological relatedness be explained. The **decompositional** approach developed here neatly relates the exponents at PF with the morphemes in the narrow syntax, accounting for the shared material across paradigms in a principled manner. To account for the regularities in the realization of person, I propose that it occupies a terminal in the structure of a pronoun in the narrow syntax (13). This terminal hosts only person features and is independent of number. In the realizational component of grammar, i.e., at PF, person and number morphemes ($\mathbb{M}$) linearize and undergo Vocabulary Insertion. The decomposition in the post-syntactic component provided in (14), where a Vocabulary Item $/a/$ is inserted into the person morpheme, illustrates the way exponents of person are mapped to morphemes in all the languages discussed above.

(13) NARROW SYNTAX



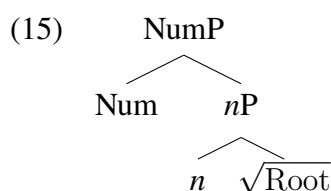
(14) PF

$\mathbb{M} [\pi; /a/] - \mathbb{M} [\text{NUM}; / /]$

In the next section (2.3), I focus on the realization of number, demonstrating that it provides further support for the decompositional approach. Namely, in the same way person exponents exhibit regularity in their morphological patterning, the exponents of number point toward the existence of the number terminal in the narrow syntax.

2.3 Realization of Number in pronouns

The representation of number is an indispensable component of a successful theory of pronouns. In the nominal domain, **NUMBER** is located on a dedicated layer in the extended projection of a noun (15), NumP (Ritter 1992, 1995; Borer 2005; Alexiadou 2011, 2019a, 2019b; Scontras 2013, 2022; Ouwayda 2014). In cases of double plurality, it is distributed across two nodes, NumP, which hosts the grammatical number, and the nominalizing head *n*, hosting lexical number (see Acquaviva (2008) and Kramer (2016) for double/split plurality, and Alexiadou (2024) for an analysis in terms of Multiple Exponence). The question that arises for the pronominal domain is whether a comparable syntactic terminal is present in the structure of pronouns.



In this section, I demonstrate that pronouns surface with discrete number morphology. Transparent exponents of plural, dual, paucal, trial, and augmented morphemes are detectable in the morphological realization of pronominal forms across languages. The fact that the number morpheme may receive a dedicated exponent suggests that pronouns are not person-number bundles, in which case a single Vocabulary Item would realize both features located on the same terminal. Instead, pronouns involve a syntactic terminal hosting number features, which receives its overt realization in morphology.

The logic of the present argument is straightforward. If there is an exponent of plural in pronouns, this exponent relates the form to a morpheme in the narrow syntax. If there is a plural morpheme in the syntax, this syntactic terminal occupies a particular position in the pronominal structure. Adding person to this perspective implies that the pronoun is multi-morphemic, involving at least two components in its structural representation. Although a morpheme can be a feature bundle, it is crucially mapped to a single exponent. Thus, several distinct, clearly identifiable exponents in the morphological realization of a pronoun are indicative of different morphemes in the syntax to which these exponents apply.

2.3.1 Singular exponents

Singulative morphology typically stays silent across languages (Alexiadou 2019a, 2024). If singular number receives an overt morphological realization in the nominal domain, it is argued to be hosted on the nominalizing head *n*, together with gender (Kramer 2015). In the pronominal domain, though sparsely attested, patterning in this respect with singulative morphology in the nominal domain, discrete exponents of the singular are identifiable. This is the case in Turkish (Table 2.18), as argued in the previous section. In first and second person pronouns, the formant [en] transparently combines with person exponents [b; s]. In accusative and dative, it intervenes

between person and case morphology. As noted above, [an] in the dative is a consequence of the vowel harmony, triggered by the dative exponent [a].

	1		2	
	SG	PL	SG	PL
NOM	b-en	biz	s-en	siz
ACC	b-en-i	biz-i	s-en-i	siz-i
DAT	b-an-a	biz-e	s-an-a	siz-e

Table 2.18: Overt singulative morphology in Turkish (Csató & Johanson 2021)

2.3.2 Plural exponents

Natural languages employ several allomorphs of PLURAL. This is exemplified on the basis of English data given in (16). Namely, English employs (at least) five Vocabulary Items for realizing plural in the nominal domain, with /-z/ [s] being the elsewhere form. The choice of the plural exponent is determined by the $\sqrt{\text{Root}}$. For instance, with a root $\sqrt{\text{ox}}$, an allomorph [en] is selected, giving rise to [oxen], while with a root $\sqrt{\text{sheep}}$, the zero [Ø] allomorph, resulting in the plural form [sheep]. These cases are instances of $\sqrt{\text{Root}}$ -driven number allomorphy, i.e., the number node is inward-sensitive to the $\sqrt{\text{Root}}$. An opposite direction in which the number conditions the allomorphy of the $\sqrt{\text{Root}}$ is also attested (see Moskal (2015b) for a typological survey in the nominal domain and Harley (2014) for the allomorphy of the $\sqrt{\text{Root}}$ conditioned by number in the verbal domain in Hiaki).

(16) ALLOMORPHS OF PLURAL IN ENGLISH (Gouskova & Bobaljik 2025)

[PL] $\leftrightarrow$ -en /]_N __, where $N \in \{\sqrt{\text{ox}}, \sqrt{\dots}\}$

[PL] $\leftrightarrow$ Ø /]_N __, where $N \in \{\sqrt{\text{sheep}}, \sqrt{\text{fish}}, \sqrt{\dots}\}$

[PL] $\leftrightarrow$ [-back, +tense] /]_N __, where $N \in \{\sqrt{\text{foot}}, \sqrt{\text{goose}}, \sqrt{\dots}\}$

[PL] $\leftrightarrow$ -i /]_N __, where $N \in \{\sqrt{\text{alumn}}, \sqrt{\text{syllab}}, \sqrt{\dots}\}$

[PL] $\leftrightarrow$ -z /]_N __

In the pronominal system, the three forms [we, you, they] do not differ in the morphological realization of plurality, as it is not the case that for one of them we may identify an explicit exponent of plural not shared by the others. Therefore, a zero [Ø] exponent of plural suggests itself as a natural explanation. Since all three forms include the feature [PERSON] in their representation, specifying the context for [PERSON] yields the correct distribution of the Ø plural allomorph. Zero exponent of plural is followed by allomorphy in the first- and third-person pronouns, leaving the second person unaffected.

	SG	PL
1	I	we
2	you	you
3	he/she	they

(a) English

Table 2.19: Pronouns in English

It is important to note that the syntactic environment must be kept intact in allomorphic conditioning. The environments in these examples differ in that the **trigger** of allomorphy, i.e., the node indicated in the context specification, is a $\sqrt{\text{Root}}$ in (16), while the trigger of allomorphy in (17) is PERSON. This means that we may differentiate between different *sets of allomorphs* of the *same morpheme* based on the *trigger*. In that sense, the zero allomorph in (16), triggered by roots such as $\sqrt{\text{sheep}}$ and $\sqrt{\text{fish}}$, is independent of the zero allomorph conditioned by the presence of PERSON in (17).

An often noted remark in the literature is that pronouns do not employ the same plural morphology as nouns, which some authors consider a property showing that plurality in the nominal domain fundamentally differs from plurality in pronouns (Ghomesi & Massam 2020), while others propose that this morphological difference does not carry over to semantics (Harbour 2016). In other words, the question is why **Is* is not the first person plural pronoun in English.

(18) HYPOTHETICAL FIRST PERSON PRONOUN IN ENGLISH

*Is

Returning to the set of allomorphs of plural in English, (16) shows that an argument which relies on non-identity of the plural exponent is not sufficient and, in fact, misleading in approaching morphological realization of plurality on pronouns. I argue that the absence of a hypothetical first-person pronoun in (18) demonstrates that pronouns in English ***do not select the elsewhere exponent***. As will be discussed extensively in the continuation, the elsewhere exponent of plural is attested in the pronominal systems in a number of typologically distinct languages. An answer that would resort only to semantic reasons would leave it unexplained why, in many languages, this is a legitimate option. These data show why studying the minimal syntactic environments that give rise to allomorphy is instrumental in uncovering the structure and nature of pronominal forms. The long-standing puzzle about the unavailability of **Is* in English receives a straightforward explanation once revisited from the perspective of allomorphy and relations that the person morpheme establishes in its syntactic environment.

Across languages, pronouns surface with discrete exponents of plural. This fact suggests that pronominal forms involve an elaborate internal structure with a dedicated terminal hosting plurality, on par with the nominal domain, and should not be considered feature bundles. In the following tables, I present illustrative examples of transparent plural exponents from the dataset that I investigated in this study. These data cover a large scale of typologically distinct languages,

demonstrating that exponing the plural morpheme via a discrete exponent is a natural possibility in the realizational component of grammar cross-linguistically. The exponent of the plural is separated by a hyphen and is marked in bold in the following tables. Furthermore, the respective language family information is provided for each language.

LANGUAGE	π	SG	PL
Akha Sino-Tibetan Pertsova (2022b)	1 _{INCL}	ŋá	àdu mà
	1 _{EXCL}		ŋá mà
	2	ŋó	ŋó mà
	3	àjǝ	àjǝ mà
Apatani Sino-Tibetan Pertsova (2022b)	1	ngo	ng- unu
	2	no	n- unu
	3 _{ANIM}	mo	mo- lu
	3 _{INANIM}	hii	hii- ka atang
Atakapa isolate; extinct Pertsova (2022b)	1	wi	ju- kit
	2	na	na- kit
	3	ha	ha- kit
Cayapa Barbacoan Pertsova (2022b)	1	i	la- la
	2	ñu	ñu- lla
	3	ja	ja- la
Chinantec Oto-Manguean Pertsova (2022b)	1	'hnaɿɿ	hn- o ɿɿ
	2	'ʔnuɿ	ʔn- o ɿɿ
	3	'uɿ	'uɿ
Hamtai Papuan; Trans-New Guinea Pertsova (2022b)	1	ni	n- ai
	2	nti	h- ai
	3 _M	əqo	əqu- oa
	3 _F	i	iʔj- oa
Jingpo Sino-Tibetan Pertsova (2022b)	1	ŋaiɿ	an ɿt ^h eɿ/j- an ɿt ^h eɿ
	2	naŋɿ	n- an ɿt ^h eɿ
	3	k ^h jɿɿ	k ^h - an ɿt ^h eɿ
Kuna Chibchan Pertsova (2022b)	1	an	an- mar
	2	be	be- mar
	3	we	we- mar
Luiseño Uto-Aztecan Pertsova (2022b)	1	no	tʃa:m
	2	o(m)	o- mom
	3	po	po- mom
Mangghuer Altaic; Mongolic Slater (2003)	1	bi	da-si
	2	qi	ta-si
	3	gan	gan-si
Mapuche Araucanian Smeets (2008)	1	ɪntʃe	ɪntʃ- in
	2	ejmi	ejm- in
	3	fey	fejeŋ- in

Table 2.20: Plural exponents in personal pronouns: A cross-linguistic perspective

LANGUAGE	π	SG	PL
Meithei Sino-Tibetan Chelliah (2011)	1 2 3	əy nəŋ má	əy- khoy nə- khoy mə- khoy
Nambikwara Nambikwara Pertsova (2022b)	1 2 3	tʔai]li] ʔwǎi]na] te]na]	tʔai]li]- nǎu] ʔa] ʔwǎi]na]- nǎu] ʔa] te]na]- nǎu] ʔa]
Nuosu Sino-Tibetan Gerner (2013)	1 2 3	nga ne cy	ngop wox nop wox cop wox
Pipil Uto-Aztecán Campbell (1985)	1 2 3	naha taha yaha	tehe- met /amehe- met anehe- met yehe- met
P'urhépecha isolate Pérez (2006)	1 2 3	ji cha indé	ju- chá cha / cha- jtsi indé- chá
Siroi Papuan; Trans-New Guinea Pertsova (2022b)	1 2 3	je ne nu	si- le ta- le na- le
Tlingit Na-Dene; Tlingit Pertsova (2022b)	1 2 3	ɣád waʔé hú	ʔuh- á·n ýi'h ^(w) - á·n h- ás
Totonac Totonac-Tepehua Pertsova (2022b)	1 2 3	'kit 'wiʃ 'ʔut	ki'- nan wi'f- in ʔu't- un
Tübatulabal Uto-Aztecán Voegelin (1935)	1 2 3	mik imbi in	ɪŋgil- u:ts imbu:- mu in- da
Udmurt Uralic Edygarova (2022)	1 2 3	mon ton so	m- i t- i so- os
Vaeakau-Taumako isolate Næss & Hovdhaugen (2011)	1INCL 1EXCL 2 3	iau, au koe ia	tha- tou , tha- tu mha- tou , mha- tu khou- tou , kho- tou lha- tou , lha- tu

Table 2.21: Plural exponents in personal pronouns: A cross-linguistic perspective (continued)

2.3.2.1 Elsewhere plural exponents

Serbian. In Slavic languages, the exponent [i] can be considered an elsewhere plural realization. In Serbian, it surfaces with both masculine (19) and feminine nouns (20), as well as those that take the formant [ov/ev] before combining with plural morphology (21). The same plural exponent is employed in the pronominal paradigm, in all person specifications (Table 2.22). Note that in the third-person feminine pronoun, gender triggers allomorphy of the number (22), which is not the case in all Slavic languages, such as Russian.

- | | | | | | | | | |
|------|----|---|------|----|---|------|----|---|
| (19) | a. | <i>prozor-i</i>
window.M-PL

'windows' | (20) | a. | <i>noć-i</i>
night.F-PL

'nights' | (21) | a. | <i>slon-ov-i</i>
elephant.M-OV-PL

'elephants' |
| | b. | <i>automobil-i</i>
car.M-PL

'cars' | | b. | <i>peć-i</i>
stove.F-PL

'stoves' | | b. | <i>miš-ev-i</i>
mouse.M-EV-PL

'mice' |
| | c. | <i>telefon-i</i>
phone.M-PL

'phones' | | c. | <i>ljubav-i</i>
love.F-PL

'loves' | | c. | <i>muž-ev-i</i>
husband.M-EV-PL

'husbands' |

	SG	PL
1	ja	m- i
2	ti	v- i
3 _M	on	on- i
3 _F	ona	on- e

(a) Serbian

- ## (22) VIs for PL in Serbian

[PL] $\leftrightarrow$ e / FEM

$$[\text{PL}] \leftrightarrow \mathbf{i}$$

Table 2.22: Pronouns in Serbian

Korean. The exponent of plural *[tul]* is selected for third-person pronouns, also being optionally present in first- and second-person plural pronouns. Importantly, the exponent *[tul]* surfaces as a regular plural marker on nouns and a plural agreement marker on verbs (Koopman 2005), as illustrated in (23). The alternation *[ka~i]*, observable in the realization of the nominative case, is an instance of phonologically-conditioned allomorphy (Kim 2006).

	SG	PL
MSC	ku-ka	ku- tul -i
FEM	kunye-ka	kunye- tul -i

Table 2.23: Elsewhere plural exponent in Korean (Smith et al. 2019)

- (23) *Ai-tul-i* *cywusi-lul masi-ess-ta-tul*.
 child-PL-NOM juice-ACC drink-PAST-DECL-PL

‘The children drank juice.’

Koopman (2005)

Ket (Yeniseian). The same plural exponent appearing in nouns [(V)n; -(V)ŋ] (Georg 2007), illustrated in (24-25), is attested in all three person specifications in pronouns, transparently combining with person exponents. The tone change, evident in both nouns and pronouns, is a general property of Ket phonology, present elsewhere in the language. Namely, when the plural suffix attaches to a monosyllabic noun stem, the tone changes (24), which is not the case otherwise (25). As pronouns are monosyllabic in Ket, the same tone change is observable.

	SG	PL		
1	ād	ótŋ	(24)	a. àŋ ~ áŋen <i>rope</i>
2	ū(k)	ókŋ		b. ōb ~ óbaŋ <i>father</i>
3	bū	būŋ	(25)	a. hómka ~ hómkam <i>Evenki person</i>
				b. ájkon ~ ájkon-iŋ <i>polar fox</i>

(a) Pronouns in Ket (Georg 2007)

Table 2.24: Elsewhere plural exponent in Ket

French. Kayne (2003) points out that the third-person pronoun surfaces with the same plural exponent present on nouns [s]. Note that [s] can in principle be recognized in first- and second-person pronouns. However, it surfaces with an additional formant [ou] in [n-ou-s; v-ou-s]. I analyze [ous] as a plural allomorph triggered by the presence of the [PARTICIPANT] feature in the person node, while the exponent [s] in the third person is an elsewhere, with [es] in the accusative being phonologically-driven.

	1		2		3			
	SG	PL	SG	PL	M.SG	F.SG	M.PL	F.PL
NOM	je	nous	tu	vous	il	elle	il-s	elle-s
ACC	me	nous	te	vous	le	la	l-es	l-es

Table 2.25: Person exponents in French (Kayne 2003)

Turkic. In Turkish, Uzbek, and Kazakh, the same plural marker [lar] that surfaces in nouns (27) is attested in third-person pronouns and agreement affixes on the verb. The first- and second-person pronouns do not employ the same plural marker, being realized as [biz] and [siz], respectively. However, the plural exponent [lar] may optionally appear in the first- and second-person pronouns (28), in which case it has semantic effects. Namely, when a pronoun that already realizes plurality in the form of the exponent [iz], shared between the first and second person, surfaces with an additional plural marker [lar], the obtained reading is ‘we people/you people’, lit: ‘we as a community’ (p.c., Rümeyşa Bektaş). This is consistent with the claim made in the

literature that [lar] can give rise to associative plurality, being syncretic with the regular additive plural marker (Görgülü 2011; Cinque 2018; Dikmen 2021).

- (26) *kitap-lar*
book-PL
‘books’

- (27) *biz-lar :: siz-lar*
1_{PL}-PL :: 2_{PL}-PL
‘we people/you people’

	SG	PL
1	ben	biz
2	sen	siz
3	o	on- lar

(a) Turkish (Csató & Johanson 2021)

	SG	PL
1	men	biz
2	sen	siz
3	u	u- lar

(b) Uzbek (Straughn 2011)

	SG	PL
1	men	biz
2	sen	siz
3	ol	o- lar

(c) Kazakh (Straughn 2011)

Table 2.26: Elsewhere plural exponents in Turkic

Cabécar (Chibchan). The plural exponent [wá] that combines with the nouns (28) is present only in the third-person plural pronoun, where it transparently combines with the third-person singular form [jié]. Crucially, this exponent is absent in the first-person inclusive, exclusive, and second-person pronouns. Number marking is not obligatory on nouns, and nominals lacking the overt exponent [wá] are not necessarily interpreted as singular (Verhoeven 2012). Therefore, the observed split in the realization of plurality can be explained in terms of the presence of the [PARTICIPANT] feature that requires a different plural marking from the elsewhere.

	SG	PL
1 _{EXCL}	yís~s	sá
1 _{INCL}		sé
2	bá	bás
3	jié~i	jié- wá

(a) Pronouns in Cabécar (Verhoeven 2012)

- (28) *míchi* (*dalóna*) *wá*
cat (black) PL
‘the (black) cats’

Table 2.27: Realization of plural in pronouns and nouns in Cabécar (Verhoeven 2012)

Modern Yucatec (Mayan). The plural exponent [o’ob] surfacing on nouns (29) combines only with third-person pronouns, accompanied by the phonological change that gives rise to [o’b], triggered by the preceding glottalized vowel. This plural marker is not obligatory in pronouns, consistent with the general optionality of plural marking in the language, proposed to merge as an adjunct (Butler 2011; Yu 2024; for an adjunction analysis of plural in Halkomelem, see Wiltschko 2008). Importantly, it cannot be combined with the first and second person pronouns, replicating the pattern present in Cabécar.

	SG	PL
1	teen	to'n
2	teech	te'x
3	leti'	leti'-o'b

(29) *ba'alche'-ob*
animal-PL

‘animals’

Yu (2024)

(a) Pronouns in Yucatec (Lehmann 2020)

Table 2.28: Elsewhere plural exponents in Modern Yucatec

Kulina (Arawan; Amazonian). An exponent of plural [*deni*] is attested in the second- and third-person pronouns (Table 2.29) and common nouns (30). In addition, the same exponent may give rise to an associative plurality (31). This plural exponent shows an animacy restriction and originates from the nominal for referring to Deni people (Dienst 2014). On the basis of the pronominal system given in Table 2.29, I propose the following set of Vocabulary Items (32), with [*deni*] figuring as an elsewhere exponent, and zero (as the pronoun [*ia*] does not provide substantial evidence for segmenting of an overt plural exponent, I consider the plural realized as $\emptyset$ in this case) being the realization of plurality in the context of the first person, i.e., AUTHOR. Note that only the first person undergoes the number-driven person allomorphy. In the second and third person, the presence of the plural number leaves the person exponents in the singular intact.

(30) *amonehe-deni*
woman-PL
‘women’

(31) *Zowi-deni*
Zowi-PL
‘Zowi and others’

	SG	PL
1	owa	ia
2	tia	tia- deni
3M	powa	powa- deni
3F	poni	poni- deni

(32) **VIs for PL in Kulina**

[PL] $\leftrightarrow \emptyset$ / $___ \pi_{\text{AUTHOR}}$

[PL] $\leftrightarrow$ *deni*

(a) Pronouns in Kulina (Dienst 2014)

Table 2.29: Elsewhere plural exponents in Kulina

In the preceding discussion, I made two points. Pronouns surface with explicit plural exponents, suggesting the presence of the number terminal in the narrow syntax. The unavailability of the hypothetical first-person plural pronoun in English [**Is*] is a consequence of the fact that pronouns often do not select an elsewhere exponent. However, this possibility is readily available in a number of typologically distinct languages. These data demonstrate why allomorphy provides a window into the structure of pronouns. When observed from the perspective of allomorphic alternations, it becomes evident that the realization of the plural has to take into consideration the PERSON morpheme, which is not the case in nouns, where it interacts with a $\sqrt{\text{Root}}$.

2.3.3 Dual and paucal exponents

In the same way as plural, the dual number receives an overt realization in pronouns. In Lavukaleve, dual pronouns are transparently built on top of plurals by merging the exponent of dual [*l*]. The exponent [*l*] surfaces elsewhere in the language, marking dual on nouns (Terrill 2011), in which case it attaches to the singular form of a noun, contrasting in this respect with pronouns.

	SG	PL	DU
1 _{EXCL}	ngai	e	e- l
1 _{INCL}		me	me- l
2	inu	imi	imi- l

Table 2.30: Dual exponent in Lavukaleve (Terrill 2011)

Across languages, the dual number receives an overt morphological realization on pronouns, as illustrated in Table 2.32. The dual exponents are marked in bold.

LANGUAGE	π	SG	PL	DU
Qiang LaPolla & Huang (2003)	1	qa	tɕi-le	tɕi- zzi
	2	ʔũ	ʔi-le	ʔi- zzi
	3	the:/qupu	them-le	thi- zzi
Meithei Chelliah (2011)	1	əy	əykhoy	i- bani
	2	nəŋ	nəkhoy	nə- bani
	3	má	məkhoy	mə- bani
Wardaman F. C. Merlan (2011)	1 _{EXCL}	ngayugu	yirrug	yirrug- guya
	1 _{INCL}		ngarrug	yawung- guya
	2	yinyang	nurrug	nurrug- guya
	3	narnaj	narnaj-bulu	narnaj- guya
Oksapmin Pertsova (2022b)	1 _{INCL}	nox	ⁿ dil	ⁿ di- t
	1 _{EXCL}		nuxul	nuxu- t
	2	go	gul	gu- t
	3	ox.m/ux.f	ixil	ixi- t
Mongsen Ao Coupe (2007)	1 _{INCL}	nì	ìsa/ìsəŋla	i- nət
	1 _{EXCL}		ìla/ìkhəla	kə- nət
	2	nəŋ	nəŋla/nəŋkhəla	nə- ŋət
	3	pa	túŋla/túŋəla/túŋkhəla	pa- nət /tuŋ- nət /tuŋ- ət

Table 2.31: Dual exponents: A cross-linguistic perspective

The same transparent morphological realization holds for other number specifications. This is exemplified here with an overt paucal morphology [*sɪ*] in Daakaka and an overt trial morphology [*het*] in Lihir.

	SG	PAUCAL		SG	TRIAL
1 _{EXCL}	nye	kinyem- si	1 _{EXCL}	yo	ge- het
1 _{INCL}		an- si	1 _{INCL}		kita- het
2	ngok	kam- si	2	wa	go- het
3	nge	nyo- si	3	e	die- het

(a) Daakaka (Von Prince 2015)

	SG	TRIAL
1 _{EXCL}	yo	ge- het
1 _{INCL}		kita- het
2	wa	go- het
3	e	die- het

(b) Lihir (Corbett 2000)

Table 2.32: Paucal and trial exponents

2.3.4 Minimal and Augmented exponents

In addition to the singular~plural distinction, the literature on cross-linguistic realization of number, and number on pronouns in particular, proposes the existence of minimal~augmented systems (McKay 1978; Noyer 1992; Corbett 2000, 2005; Harbour 2016). What qualifies a language to be considered as one surfacing with the minimal~augmented distinction? The existence of a pronoun whose referent consists of *exactly* two individuals, the AUTHOR and the ADDRESSEE, hence [1+2], paved the way for proposing the MINIMAL feature in natural language. It is important to note that the first-person minimal exclusive [1_{MIN.EXCL}], second person minimal [2_{MIN}] and third-person minimal [3_{MIN}] neatly correspond to the first-, second-, and third-person singular. The addition to the system is precisely the first-person minimal inclusive pronoun [1+2_{MIN.INCL}], often referred to as the ‘fourth person’, as it morphologically behaves more like singular and not plural or dual, as argued in the literature above, which led to the proposal that it should be considered together with other singular forms, i.e., as minimal. The system of features and the notational conventions are presented in Table 2.34. The schema of feature distribution is followed by data from Limbum (Bantu; Eastern Grassfields; Ndi 2017), a language which makes the MINIMAL~AUGMENTED distinction.

1 _{MIN.EXCL}	1 _{AUG.EXCL}	MIN	AUG
1+2 _{MIN.INCL}	1+2 _{AUG.INCL}	1 mɛ̀, m̀	wɛ̀r
2 _{MIN}	2 _{AUG}	1+2 sò	sèe
3 _{MIN}	3 _{AUG}	2 wɛ̀, à	wèe
		3 ye, e	woowèe, a

Table 2.33: Minimal~Augmented system

In addition to this general system, MINIMAL~AUGMENTED languages may involve UNIT AUGMENTED forms, understood as involving one more participant to the minimal pronoun, i.e., the resulting number of referents is two in the case of [1_{MIN.EXCL}], [2_{MIN}], and [3_{MIN}] and three in the [1+2_{MIN.INCL}]. In that sense, unit augmented corresponds to the dual in languages with the SINGULAR~PLURAL.

In the following data from Gaagudju (Australian; Aboriginal), the augmented exponent is uniformly realized as [mba] across all person specifications. Interestingly, this language in-

volves gender-driven allomorphy of augmented and unit augmented exponents. While [mba] is the gender augmented exponent selected in cases when the referent of the pronoun involves both masculine and feminine participants, if the group included in the referent of the pronoun consists of masculine participants *only*, the augmented morpheme is realized as [da]. Therefore, I analyze these cases as gender-driven number allomorphy, as will be discussed in Chapter 5. Furthermore, the unit augmented number distinguishes between the feminine unit augmented [FUA] and the masculine unit augmented [MUA]. As illustrated in the table below, two distinct exponents, [njdja] for feminine and [mana] for masculine, are chosen.

	MIN	AUG	MA	FUA	MUA
1	ngaayi~ngaanj-ma	ngaa- mba ~ngee- mba	ngaa- da	ngaa- njdja	ngaa- mana
1+2	maneerra	manee- mba	manaa- da	manaa- njda	manaa- mana
2	ngiinja	nginjee- mba	nginjaa- da	nginjaa- njdja	nginjaa- mana
3M	naawu	nowoo- mba	nowoo- da	nowoo- njdja	nowoo- mana
3F	ngaayu			ngoyoo- njdja	

Table 2.34: Pronouns in Gaagudju (Harvey 2011)

Although minimal morphology is typically silent across languages, on par with singulative, overt morphological realization of the augmented morpheme is common. This is exemplified in the following table on the basis of Bardi (Nyulnyulan; Australian), Kemtuik (Nimboran; Papuan), and Maranungku (Western Daly; Australian Aboriginal). Interestingly, in Bardi, number distinctions are not encoded in nouns. However, the pronominal system draws minimal~augmented number marking. All augmented pronouns surface with the formant [rr]. These data show that natural language has plural/augmented exponents whose context of insertion involves reference to PERSON, hence its presence with pronouns only.

These data demonstrate that an overt morphological realization of number holds not only for the plural but extends to other number feature specifications, such as dual and trial. The same holds for languages surfacing with the minimal~augmented distinction.

2.3.5 Number morphology in the absence of an overt $\sqrt{\text{Root}}$

The realization of number in pronouns and nouns gives rise to the following generalizations (33). The shared plural morphology on pronouns and nouns is not precluded by the system, as corroborated by several languages discussed here. Namely, in languages where pronouns share the same exponent of plural as nouns, the *elsewhere* exponent surfaces in the two environments. This is the case in Serbian. Furthermore, if only a subset of pronouns surfaces with the elsewhere exponent, it is the third person (French, Korean, Turkic, Modern Yucatec) or the third and the second person pronouns (Kulina). An **unattested pattern** is the one where the first- and second-person pronouns would surface with the elsewhere realization of plural present on nouns, while the third-person triggers a special allomorph.

LANGUAGE	π	MIN	AUG
Bardi	1	ngay(oo)	a- rr -oodoo
Bower (2012)	1+2	ayoo	a- rr -idil
	2	joo	goo- rr
	3	giniyingg(i)	i- rr
Kemtuik	1	gnam	gnam- naŋ
Donohue & Smith (1998)	1+2	imot	imot- naŋ
	2	mot	mot- naŋ
	3	nmot	nmot- naŋ
Maranungku	1	ngany	nga- tya
(Tryon et al. 1970)	1+2	ngangku	ki- tya
	2	nina	ni- tya
	3 _M	nankuny	wi- tya
	3 _F	ngankuny	wi- tya

Table 2.35: Augmented exponents: A cross-linguistic perspective

(33) **PLURAL EXPONENTS IN PRONOUNS AND NOUNS: GENERALIZATIONS****G1:** If pronouns and nouns share a plural exponent, it is the elsewhere exponent.**G2:** If the elsewhere plural exponent is not spread across a paradigm, it is the third person or the third- and the second-person pronouns that surface with the elsewhere form.

Why do pronouns not have the complete set of plural allomorphs available to nouns? Considering the set of allomorphs in English (34), why, for instance, does a pronoun that has an antecedent the noun [*oxen*] not surface with the same plural exponent [*en*] that the noun would have? Why does the plural morphology of the antecedent not survive in the respective pronoun?

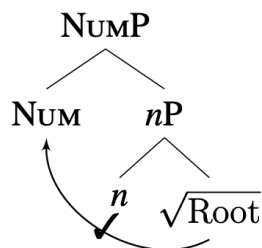
(34) **ALLOMORPHS OF PLURAL IN ENGLISH** (Gouskova & Bobaljik 2025)[PL] $\leftrightarrow$ -en /]_N __, where $N \in \{\sqrt{\text{ox}}, \sqrt{\dots}\}$ [PL] $\leftrightarrow$ $\emptyset$ /]_N __, where $N \in \{\sqrt{\text{sheep}}, \sqrt{\text{fish}}, \sqrt{\dots}\}$ [PL] $\leftrightarrow$ [-back, +tense] /]_N __, where $N \in \{\sqrt{\text{foot}}, \sqrt{\text{goose}}, \sqrt{\dots}\}$ [PL] $\leftrightarrow$ -i /]_N __, where $N \in \{\sqrt{\text{alumn}}, \sqrt{\text{syllab}}, \sqrt{\dots}\}$ [PL] $\leftrightarrow$ -z /]_N __

I propose that there is a principled explanation for these questions. At the point when the number node is targeted for insertion in pronouns, a $\sqrt{\text{Root}}$ cannot condition allomorphy of the plural node and give rise to one of the allomorphs, as it is **not present in the structure** when the Vocabulary Insertion into plural takes place. This explains why there is no language in which one of the allomorphs triggered by a particular set of $\sqrt{\text{Roots}}$ surfaces in pronouns.

The inventory of allomorphs in (34) corresponds to the structural representation in (35), where the $\sqrt{\text{Root}}$ triggers the allomorphy in the plural node. Crucially, when the $\sqrt{\text{Root}}$ is not present in the structure, it cannot act as a trigger. As a consequence, (i) $\sqrt{\text{Root}}$ -conditioned

allomorphs of plural are unexpected on pronouns, and (ii) only the elsewhere forms, which do not make reference to any particular $\sqrt{\text{Root}}$, are available.

(35) $\sqrt{\text{Root}}$ -DRIVEN NUMBER ALLOMORPHY



The two generalizations in (33) will have significant implications for the analysis of allomorphy in the next chapter. Consider the following pattern. In Udmurt (Uralic), the first- and second-person pronouns share the plural exponent $[i]$ to the exclusion of the third person, which surfaces with $[os]$.

	SG	PL
1	mon	m- i
2	ton	t- i
3	so	so- os

(a) Udmurt (Edygarova 2022)

(36) **VIs for Udmurt**

- [PL] $\leftrightarrow$ i / $___ \pi_{\text{PARTICIPANT}}$
 [PL] $\leftrightarrow$ os
 [1] $\leftrightarrow$ m
 [2] $\leftrightarrow$ t
 [3] $\leftrightarrow$ s

Table 2.36: Selective person-driven number allomorphy: Udmurt

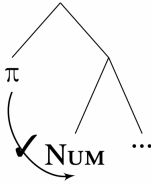
In principle, there are two ways to approach these data. In the present work, I argue for the following analysis. The PARTICIPANT feature as a whole, encompassing the first and second person, acts as a trigger of allomorphy, thus constituting a single unit when the choice of exponents is made. In the presence of the PARTICIPANT feature in the person node, the allomorph $[i]$ surfaces. When the context specification is not met, the elsewhere exponent $[os]$ is realized, which is the case in the third person (36). Most importantly, $[os]$ is an exponent of plural that surfaces on nouns, with the phonologically-conditioned version $[jos]$ in cases where the noun ends in a consonant (Edygarova 2022). This fact further corroborates the generalization introduced above, according to which the third person shares the exponent of plural with nouns and not the first and second.

An alternative analysis would propose that it is the third person that triggers allomorphy. That is, $[os]$ is inserted only in the context of the third person, while the exponent $[i]$, shared between the first and the second, is an elsewhere. This is where the generalizations in (33) become particularly relevant. If pronouns and nouns share plural morphology, it is always an elsewhere form. In addition, if the elsewhere form is not spread across the entire pronominal paradigm but present only with the subset of pronouns, it does not combine with the first and second person, while the third triggers a special allomorph. Instead, the opposite holds. The third person is

the one that surfaces with the elsewhere or the second and third, but crucially not the first and second, to the exclusion of the third. Therefore, empirical generalizations demonstrate that the alternative approach cannot be on the right track.

In determining the inventory of the exponents of plural in a particular language, one has to take into consideration that there are two potential triggers of allomorphy of the plural node in the nominal domain, (i) the $\sqrt{\text{Root}}$ (35) and (i) the PERSON head (37). The two morphemes act as sources of plural morphology in a language.

(37) PERSON-DRIVEN NUMBER ALLOMORPHY



I propose that the two syntactic configurations **generate two distinct sets of allomorphs**. When the Subset Principle, as a core component of the Vocabulary Insertion operation, inspects if the relevant structural conditions are met, it will consider the syntactic environment involving a $\sqrt{\text{Root}}$ as distinct from the one involving PERSON. I suggest that neither is more specific, that is, the presence of the $\sqrt{\text{Root}}$ should not be considered more specific than the presence of PERSON in the Vocabulary Insertion rule or vice versa. In fact, the two stand in a similar relation as different Lists. As a consequence, allomorphs of plural triggered by the $\sqrt{\text{Root}}$ **do not compete** for insertion with plural allomorphs triggered by PERSON. This proposal is exemplified on the basis of the hypothetical Vocabulary Insertion rules in (38), where both contexts are equally complex. Therefore, the allomorph $[a]$, triggered by the $\sqrt{\text{Root}}$ and the allomorph $[b]$, triggered by PERSON do not compete and the Subset Principle does not resolve the decision on the choice of an exponent.

(38) ALLOMORPHS OF PLURAL

$[\text{PL}] \leftrightarrow a / __ \sqrt{\text{Root}}$

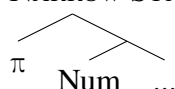
$[\text{PL}] \leftrightarrow b / __ \pi$

The question that arises is what would happen if the two triggers of allomorphy were simultaneously present in the structure, i.e., the $\sqrt{\text{Root}}$ and the PERSON feature. This is the case in DPs surfacing with the definite article, which realizes the third person (Bernstein 2008b; Longobardi 2008; Höhn 2016; Alexiadou et al. 2024). Why does the third-person feature, realized as the definite article, not trigger the allomorphy of number? I suggest that the answer lies in the derivational timing. Namely, the $\sqrt{\text{Root}}$ interacts with the plural node first, resulting in the $\sqrt{\text{Root}}$ -conditioned allomorphy of the plural. A subsequent merge of the definite article cannot affect the realization of the plural on the noun, as it has already undergone the Vocabulary Insertion.

2.3.6 Interim Summary

In this section, I demonstrated that pronouns employ dedicated plural, dual, trial, and augmented exponents across languages. I propose that these exponents realize the **number morpheme** in the underlying structure of pronouns. This morpheme is independent of the person node, instantiating a separate syntactic terminal. Therefore, pronouns are **multimorphemic**, consisting of terminals hosting PERSON and NUMBER (39). In the same way as the person morpheme, discussed earlier in this chapter, the number morpheme undergoes Vocabulary Insertion at PF, being supplied with a dedicated Vocabulary Item (VI), as illustrated using a hypothetical VI /b/ in the following representation.

(39) NARROW SYNTAX



(40) PF

M [π; / /] — M [NUM; /b/]

As the proposal that person and number instantiate separate nodes in the structure of pronouns clashes with a view that pronouns are unordered φ -bundles, I will take issue here with a recent proposal of Ghomeshi & Massam (2020), who argue that number is **formally structurally invisible** in pronouns, being bundled with person in a single φ head. In this squib, the authors suggest that the number on pronouns fundamentally differs from the number on nouns. Seven differences, argued to be central, are provided: (i) There are languages in which number is marked on pronouns but not on other nominals, (ii) Number in pronominals is usually different in morphological form from number in nominals, (iii) Associativity is often marked on nominals but rarely on pronominals, (iv) Number can instantiate differences in formality in pronominals, not nominals, (v) Number can be systematically marked twice on pronominals, while double marking on nominals tends to be idiosyncratic and/or limited to a particular noun class, (vi) Number contrasts in pronominals can differ from those in nominals, (vii) Pronominal number is compatible with the feature proper, whereas nominal number is not. Consequently, the structural representations of nouns and pronouns differ (41).

(41) a. Nominal structure

[_{DP} D [_{NUMP} Num [_{nP} n [_{√P} Root]]]]

b. Pronominal structure

[_{DP} D [_{φP} φ [_{nP} n [_{√P} Root]]]]

Ghomeshi & Massam (2020)

Let me start with the second point, which states that number differs in its morphological form in nouns and pronouns. In 2.3.2.1, I showed that the same plural exponents that appear in nouns are selected in the pronominal domain. My argumentation was based on a number of typologically different languages, covering the entire Slavic and Turkic families, Korean, Ket (Yeniseian), Modern Yucatec (Mayan), and Kulina (Arwan; Amazonian). Crucially, the comparison of the morphological realization of plural in nouns and pronouns cannot be investigated in simplistic terms, where a single morphological realization of plural in nouns is assumed, against

which the realization of plural in pronouns is evaluated. Natural languages employ several allomorphs of plural, with the elsewhere form being one of them. An in-depth investigation of the exponence of plural on pronouns has to go beyond the elsewhere forms, though they are readily available to pronouns across languages. Furthermore, looking at the internal composition of pronouns through the lens of allomorphy, which I argue for in this dissertation, straightforwardly explains why a hypothetical pronoun **Is* is ungrammatical in English. The answer is that the elsewhere exponent of plural is not selected in English in the context of PERSON. This contrasts with Serbian and other languages where the elsewhere is available. Given that allomorphy involves the alternation in the realizational component of grammar, such variation is expected across languages.

Moving to associativity, which I have already addressed in the context of Turkish, here I will provide an illustrative example of the associative plural on pronouns. In Hungarian, associative and additive plurality receive distinct morphological realizations. With first- and second-person pronouns, associative marker [i] is selected, while with the third person, additive [k] (Dékány 2012). Therefore, Hungarian data show that nothing prevents the associative plural from surfacing on pronouns. Given the fact that the associative inference arises with pronominal forms, since the semantics of the person features indicates that a particular person specification is ‘included in’ the referent of the pronoun (*x*), associative plural exponents are not only available, but in fact expected in pronouns (cf. Ackema & Neeleman (2018) for the claim that there is no language where pronouns employ an overt exponent of associative plurality).

	SG	PL
1	én	m- i
2	te	t- i
3	ő	ő- k

Table 2.37: Pronouns in Hungarian (Pertsova 2022b)

The next remark relates to honorification, and the first case study that the authors present. I suggest that in evaluating the honorification facts in Persian, we need to take into consideration the way honorification marking proceeds in general. In a large scale study based on 120 languages encompassing more than 35 genera, Wang (2023) shows that features selected for the realization of honorification involve plural, third person, and indefinite, precisely those surfacing with weakened presupposition projection (Sauerland 2008a), without invoking a special feature [HON]. Therefore, realizing honorification is rooted in a particular set of φ -features, showing the crucial interaction with pragmatics, not in the distinction between nouns and pronouns.

When it comes to double marking of number on pronouns, it is important to stress that the authors consider the co-occurrence of the plural exponent and the plural-triggered allomorphy of person as an instance of double plural marking. Therefore, it is crucial to make a terminological distinction between double plurals used in this paper and the phenomenon of double plurality in the sense of Alexiadou (2024), where a noun surfaces with two exponents of plural, one realizing

grammatical number and the other realizing lexical number. The co-occurrence of the trigger and the target of allomorphy is readily available in natural language (see Chapter 3) and goes beyond the pronominal domain, thus not being an exclusive property of pronouns.

I will address the sixth and first points together, as they involve a common underlying source. It is argued that in some languages, number is marked only on pronouns (point i) and that number distinctions present in pronouns are not marked on nouns within the same language (point vi). The first point was illustrated in data from Bardi that I discussed above, where pronouns realize overt augmented morphology, in contrast to nouns, which do not mark number distinctions. Variation of this type is a common cross-linguistic property. Namely, within the same language, the inclusive~exclusive distinction may be marked on agreement morphology but not on pronouns, or, vice versa, the clusivity distinction is marked in the pronominal system but not on agreement markers on the verb. Likewise, syncretism of person in a pronominal paradigm may be accompanied by an elaborate agreement morphology or morphological explicitness can go in the opposite direction, with elaborate pronouns but syncretic person marking on the verb (see Cysouw 2003; Sauerland & Bobaljik 2013; Bobaljik & Sauerland 2018; Harbour 2016). To account for these facts, there is no need to propose another type of person or number specialized for agreement morphology, which would be comparable to proposing different semantics of plural, specialized for the nominal and pronominal domains. In that sense, marking plural only on pronouns but not on nouns or expressing dual exclusively in the pronominal domain, does not appear as a convincing argument that there is a deep difference in the semantics of the plural or dual that is intrinsically tied to formal categories of nouns or pronouns.

The final point (vii) relates to the co-occurrence of definite articles and plural morphology on pronouns in Niuean. I argue that this interaction resides in the ways genericity and definiteness are realized both within and across languages (see Chierchia 1998; Dayal 2004; Alexiadou et al. 2007; Lazaridou-Chatzigoga & Alexiadou 2019), and does not depend on categorial distinction between nouns and pronouns. Therefore, the reasons for the peculiar restrictions on the co-occurrence of pronouns and plural morphology in Niuean pronouns should be sought in the way this language realizes genericity.

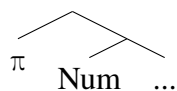
In this brief remark, my objective was to demonstrate that what appears to constitute a set of differences between pronouns and nouns, calling for two different structural representations of number, has independent explanations. When it comes to the representation of the pronominal structure provided in (41-b), where φ -features are subsumed under a single layer, I propose that the φ -component has an elaborate structural representation, neatly corresponding to compositional semantics. In fact, unpacking the φ -bundle is a central task for the theory of pronouns.

2.4 Summary

In this chapter, I introduced a **multi-morphemic** account of pronouns. In the narrow syntax, person and number are distinct morphemes (43). In the post-syntactic realizational component (44), the two morphemes are combined with the phonological features, enclosed in //. Through

detailed discussion and examination of the morphological realization of person and number features, this chapter demonstrated that both core components of pronouns give rise to a number of important generalizations.

(42) NARROW SYNTAX



(43) PF

$\mathbb{M} [\pi; /a/] \text{ — } \mathbb{M} [\text{NUM}; /b/]$

Individuation of person and number morphemes in the narrow syntax and the corresponding exponents in the post-syntactic Vocabulary serves as an empirical basis for the series of studies on allomorphy in pronouns discussed in the upcoming chapters.

Chapter 3

Allomorphy involving Number

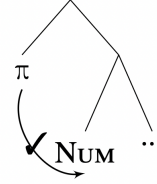
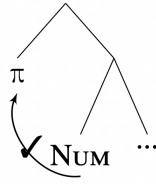
3.1 Introduction

The relation of person and number is one of the most challenging questions in the general theory of φ -features (Harley & Ritter 2002; Béjar 2003; Adger & Harbour 2008; Harbour 2016, among others), with implications for a number of central themes in morphology, syntax, semantics and their interfaces. The interaction of person and number features lies at the heart of investigating the structure of pronouns.

I approach this discussion from the perspective of allomorphy, proposing that it provides a key diagnostic for accessing the internal structure of pronouns. This chapter brings two studies: (i) Number-driven person allomorphy and (ii) Person-driven number allomorphy. The number node may trigger the realization of a person exponent y , which is otherwise realized in its elsewhere form as x . This pattern exemplifies **number-driven allomorphy of person** (1). In the dataset examined in the present study, the opposite direction is also possible. Namely, a specific exponent of plural $[y]$ is realized only in the context of the first person $[AUTHOR]$, but as $[x]$ elsewhere. In this case, **person-driven allomorphy of number** is at play (2). In structural terms, in representations where person surfaces above number (3-4), number-driven person allomorphy exemplifies outward-conditioning, i.e., the person morpheme is inward-sensitive to the number features. This contrasts with the person-driven number allomorphy, instantiating inward-conditioning, with number features being outward-sensitive to the person morpheme. The arrow indicates the directionality of allomorphy, going from the trigger to the target.

- | | |
|---|---|
| (1) NUMBER-DRIVEN PERSON ALLOMORPHY | (2) PERSON-DRIVEN NUMBER ALLOMORPHY |
| a. $[\pi] \leftrightarrow y / __ \#_{PL}$ | a. $[PL] \leftrightarrow y / __ \pi_{AUTHOR}$ |
| b. $[\pi] \leftrightarrow x$ | b. $[PL] \leftrightarrow x$ |

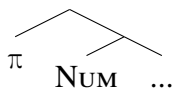
- (3) NUMBER-DRIVEN PERSON ALLOMORPHY (4) PERSON-DRIVEN NUMBER ALLOMORPHY



Number-driven person allomorphy has been discussed in the literature from different perspectives (Moskal 2015a, 2015b, 2018; Moskal & Smith 2016; Smith et al. 2019; Pertsova 2022a; Zompì 2023; Caha 2024). The opposite direction, person-driven number allomorphy, has not been empirically recognized. I show that both types of directionality involve (i) *Selective allomorphy* and (ii) *Across the board allomorphy*. In languages where allomorphy does not arise, the pattern is identified as the (iii) *Absence of allomorphy*.

On the basis of a number of empirical generalizations, I provide support for the central hypothesis of this thesis that allomorphy is a key tool for (a) accessing the inventory of morphemes that make up a pronoun across languages and (b) tracking the patterns of their interactions. Namely, allomorphic alternations involve an interaction of two morphemes, the *trigger* and the *target* of allomorphy. Identifying the trigger and the target in pronominal forms yields an *immediate result* - pronouns involve two morphemes that occupy distinct structural positions in their internal representation in the narrow syntax. This idea is at the core of the **multi-morphemic approach** to pronouns (5) proposed here, with person and number hosted on two separate nodes in the narrow syntax instead of being bundled on the same φ -head. Furthermore, I argue that the Vocabulary Insertion operation targets only terminal nodes. As a consequence, person and number are targeted for insertion separately.

- (5) MULTIMORPHEMIC APPROACH



The chapter is organized as follows. In the continuation of this section (3.1.1), I provide an overview of the data. Section 3.2 presents the study on the Number-driven person allomorphy, while Section 3.3 introduces the study on Person-driven number allomorphy. Section 3.4 summarizes.

3.1.1 Data

The studies discussed in this chapter are based on 210 languages from 54 language families, including 5 creoles and 11 isolates. Both types of directionality in allomorphic conditioning are investigated on the same data. The sample in the two studies comprises languages that employ a single morphological realization for the first-person plural pronoun, i.e., languages that do not make the INCLUSIVE~EXCLUSIVE distinction. This empirical domain will be independently

discussed in the next chapter, along with languages that surface with MINIMAL~AUGMENTED systems, as both involve additional considerations related to person features that need to be taken into account.

In the following table, I provide an overview of the data, which includes the list of language families with representative examples from each language family. As noted in the introductory chapter, in cases where individual language families, rather than branches, are parts of a larger language family, as is the case with Mongolic and Turkic within the Altaic, Dogon and Kordofanian within the Niger-Congo, or Yam, Engan, Sepik and languages of the Trans-New Guinea as part of the Papuan family, this information is included in the table. The complete set of data investigated in this study is included in Appendix A.

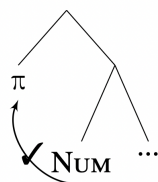
FAMILY	LANGUAGE
Afro-Asiatic	Goemai, Tamashek, Amharic, Hebrew, Hausa, Oromo, Pero, Dahalo
Altaic ^{Mongolic}	Mangghuer, Buryat
Altaic ^{Turkic}	Uzbek, Kazakh, Chuvash, Khakas, Turkish
Araucanian	Mapuche
Arawakan	Baure
Arawan	Kulina
Australian ^{Pama-Nyungan}	Pitjantjatjara, Warrongo
Austro-Asiatic	Pacoh, Sapuan
Austronesian	Madurese, Tukang Besi
Baining	Qaqet
Barbacoan	Cayapa, Awa Pit
Caucasian	Abkhaz, Kartvelian (Svan, Georgian), Nakh-Dagestanian (Hinuk)
Chibchan	Damana, Kuna, Tunebo
Chicham	Aguaruna
Chonan	Selknam
Dravidian	Kannada, Gondi, Gadaba, Brahui
Eskimo-Aleut	Aleut, Greenlandic, Central Alaskan Yupik
Guaicuruan	Pilagá, Toba
Hmong-Mien	Hmong Njua, Xong
Hokan	Chimariko, Salinan, Central Pomo, Eastern Pomo
Indo-European	Greek, Armenian, Germanic, Slavic, Celtic, Romance, Indic
Japonic	Japanese
Korean	Korean
Mayan	Modern Yucatec
Mixe-Zoque	Sierra Popoluca
Mosetenan	Mosetén
Muskogean	Choctaw
Nadahup	Hup
Na-Dene ^{Athapaskan}	Navajo
Na-Dene ^{Tlingit}	Tlingit
Nambikuaran	Nambikwara

FAMILY	LANGUAGE
Niger-Congo	Bantu (Eton, Chichewa, Kisi), Kwa (Idoma, Igbo), Fongbe
Niger-Congo Dogon	Yamsay, Tommo So
Niger-Congo Kordofanian	Orig
Nilo-Saharan	Ma'di, Kanuri, Lango, Masalit, Koyraboro Senni, Koyra Chiini
Oto-Manguean	Chinantec
Páezan	Páez
Panoan	Cashinahua
Papuan Yam	Nama
Papuan Engan	Kewa
Papuan Sepik	Ramu-Lower (Yimas), Lower (Kopar), Awtuw, Kwoma
Papuan Trans-New Guinea	Hua, Kuman, Coastal Marind, Asmat-Kamrau Bay (Asmat)
Penutian	Nisenan, Nez Perce, Takelma
Quechuan	Ecuadorian Kichwa
Salish	Bella Coola, Musqueam
Sino-Tibetan	Nuosu, Qiang, Meithei, Jingpho, Apatani, Cantonese
Takanan	Cavineña
Torricelli	Arapesh
Totonac-Tepehua	Totonac
Uralic	Hungarian, Udmurt, Estonian, Finnish, Khanty, Saami, Tundra Nenets
Uto-Aztecan	Hopi, Luiseño, Western Tarahumara, Tübatulabal, Yaqui, Pipil
Witotoan	Ocaina
Yeniseian	Ket
Yukaghir	Kolyma Yukaghir
creoles	Fa d'Ambô, Berbice Dutch Creole, Saramaccan Creole, Sango, MBM
isolates	Chitimacha, Haida, Maybrat, Sandawe, Tonkawa, Yuracare, Zuni

3.2 Number-driven person allomorphy

This study presents a detailed examination of the number-induced allomorphy of person in pronominal systems of natural languages. Number-driven person allomorphy in the structural configuration in (6) involves outward-conditioning, as the trigger of allomorphy is in a structurally lower position compared to its target. This means that the person morpheme is inward-sensitive to number features. As illustrated in the general schema for this direction in allomorphic conditioning, the person morpheme has two exponents, $[x]$ and $[y]$. In line with the Subset Principle, the more specific exponent $[y]$ is inserted if the required condition is met, that is, if the syntactic context contains the number node carrying the plural feature. If the plural feature is not present, the elsewhere exponent $[x]$ surfaces.

(6) NUMBER-DRIVEN PERSON ALLOMORPHY



NUMBER-DRIVEN PERSON ALLOMORPHY

$[1] \leftrightarrow y / __ \#_{PL}$

$[1] \leftrightarrow x$

LINEARIZATION STATEMENT

π -NUM

This study provides an inventory of attested patterns in number-driven person allomorphy. I distinguish between (i) *Selective allomorphy* and (ii) *Across-the-board allomorphy*, which both differ from the (iii) *Absence of allomorphy*. Selective allomorphy refers to cases where allomorphic conditioning is present in only one person feature specification. For instance, in a given language, only the first-person pronoun undergoes allomorphy in the plural, which is not the case in the second and third person, which preserve person exponents across both number specifications. This contrasts with the Across-the-board allomorphy, where all three person specifications undergo allomorphy. The exact opposite of this pattern is the Absence of allomorphy, where all person exponents remain intact in the presence of the plural number. Therefore, there is a gradient cross-linguistic scale, with the Selective allomorphy being in between the two poles. The obtained empirical landscape will serve as the basis for the discussion about the theoretical implications and modeling. I will spell out the proposal of pronouns as multi-morphemic complex syntactic objects, which stems from identifying the trigger and the target of allomorphy.

Before proceeding with the presentation of the data, I will provide an overview of the results from a quantitative perspective. Allomorphy in first-person pronouns is attested in 141 out of 210 studied languages. The second-person shows a slightly lower rate with 107 languages that exhibit the allomorphic alternation, while the third-person¹ figures as the least prone to allomorphy with

¹A brief remark on the different number of languages examined in the case of the third person is necessary. In 5 out of 210 languages in my sample for this study, there is no overt realization of a pronoun in the third person. This is the case in Choctaw, So/Soko, Zuni, Coastal Marind, and Eton. For instance, in Zuni (isolate), the third-person pronoun has an overt realization of the third person only in the object position, while the subject does not have an overt morphological form for the third-person pronoun (Newman 1965). Therefore, this language had to be excluded from the considerations of allomorphy in the third person. At the same time, the overt exponents of the first and second person are preserved in the plural. For this reason, Zuni is classified as a language without

87 languages. Therefore, a decline in the availability of allomorphy can be tracked from the first to the third person.

1	2	3
141/210	107/210	87/205

Table 3.1: Number-driven allomorphy: Summary

The question that arises is whether the observed result has implications for the representation of person and approaches to person features in terms of containment in particular (see Noyer 1992; Harley & Ritter 2002; Béjar 2003; Toosarvandani 2023; Sichel & Toosarvandani 2024, among others). Is it the case that person features have a *blocking effect* in allomorphic conditioning of other person features? For instance, could it be the fact that the second person undergoes allomorphy only if the first person undergoes allomorphy, too? I argue that the answer to these questions is negative. The support for this view comes from languages like Finnish (Table 3.2), where the second person exhibits an allomorphic alternation to the exclusion of the first and third. These data demonstrate that neither first- nor third-person features may prevent the number node from accessing the second person. This means that when a particular feature is the **target** of allomorphy, no reference is made to the containment. As will be addressed later in this dissertation, the distinction in frequency between the first and second person emerges as a consequence of a particular pattern unique to Indo-European languages (see Chapter 6, 6.6).

	1		2		3	
	SG	PL	SG	PL	SG	PL
NOM	minä	me	sinä	te	hän	he
ACC	minut	meidät	sinut	teidät	hänet	heidät
PART	minua	meitä	sinua	teitä	häntä	heitä

Table 3.2: Overt person morphology in Finnish (White 2006)

Furthermore, to check whether allomorphy shows regularity with respect to individual person specifications, I compared its appearance within pronominal paradigms. In Table 3.3, the left-most column indicates the person specification undergoing allomorphy, i.e., [1] stands for the first person undergoing allomorphy to the exclusion of the second and third, while [1,2] indicates that the first and second person undergo allomorphy to the exclusion of the third. A specific pattern of allomorphy is followed by information about its frequency and examples of languages in which it is attested. The observed data show that all logical possibilities are exhausted.

Within a pronominal system of a language, allomorphy can be limited to the first (37 languages), second (6 languages), or third-person (7 languages). Likewise, it can apply in the first

number-induced allomorphy of person.

and second person, to the exclusion of the third (35 languages), in the first and third to the exclusion of the second (11 languages), in the second and third to the exclusion of the first (10 languages). The most frequent patterns are those where allomorphy applies across the board, affecting all three person specifications (55 languages), or is absent altogether (47 languages).

PATTERN	FREQUENCY	EXAMPLES
1	37	Armenian, Icelandic, Hungarian, Luiseño
2	6	Chitimacha, Hamtai, Ocaina, Finnish, Fur
3	7	Iraqw, Kannada, Nez Perce, Pitjantjatjara, Warrongo
1, 2	35	Georgian, Italian, Korean, Latvian, Sandawe, Serbian
1, 3	11	English, Faroese, Haida, Manx, Pomo, Goemai, Igbo
2, 3	10	Baruya, Votic, Yale, Estonian, Tommo So
1, 2, 3	55	Abkhaz, Baure, Breton, Chichewa, German, Hua, Nupe
no allomorphy	47	Greenlandic, Japanese, E. Kichwa, Kuna, Mapuche

Table 3.3: Number-driven allomorphy and Person: Summary

Although frequency should not be considered particularly telling for the overall theory of allomorphy, given that even the least frequent patterns indicate that the observed pattern is possible in the natural language, certain asymmetries appear worth noting. Allomorphy limited to the first-person (37) shows a higher frequency than allomorphy limited to the second (6) or third person (7). Furthermore, allomorphy applying in the first and second person (35) is more frequent compared to the allomorphic pattern limited to the first and third (11) and second and third person (10). As noted above, I will address this fact later in this dissertation, providing a view from a broader Indo-European perspective.

3.2.1 The patterns

I distinguish between (i) *Selective allomorphy*, where allomorphic alternation is restricted to a particular person specification, [e.g., AUTHOR], (ii) *Across-the-board allomorphy*, defined here to be attested in pronominal systems where allomorphy applies in all three person specifications, and the (iii) *Absence of allomorphy*, exemplified by pronominal systems in which all person specifications, i.e., the exponents of the first-, second-, and third-person, do not undergo allomorphy in the plural.

3.2.1.1 Selective allomorphy

First-person allomorphy. In the pronominal system of a language, allomorphy can apply only in the first person to the exclusion of the second and third. As illustrated on the basis of Hungarian (Finno-Ugric), exponents of the second and third person are preserved in the plural. The first-person pronoun, however, exhibits a clear morphological alternation. Furthermore, this language surfaces with discrete, transparently identifiable plural exponents, being realized as [i] in

the first- and second-person and $[k]$ in the third-person. This means that in this language, the allomorph of the first person co-occurs with the overt trigger of allomorphy.

	SG	PL
1	én	m-i
2	te	t-i
3	ő	ő-k

Hungarian (Pertsova 2022b)

Table 3.4: Selective allomorphy limited to the first person

The same holds for a number of typologically unrelated languages. For instance, in Cayapa (Barbacoan; northern Ecuador) and Luiseño (Uto-Aztecan; California), the exponents of the second and third person are transparently preserved in the plural, combining with the overt number morphology. However, morphological relatedness cannot be identified in the first-person pronouns. Patterning with the data discussed above, these languages surface with an overt plural morphology that combines with the exponents of person. In Cayapa, the exponent of the first person in the singular $[i]$ changes to $[la]$ in the presence of an overt plural exponent $[la]$. In contrast, the exponents of the second $[ju]$ and third-person $[ja]$ are preserved in the plural. A related pattern is attested in Luiseño, where the exponent of the first person in the singular $[no]$ exhibits a clear allomorphy in the plural $[tʃa:]$, which is not the case with the exponents of the second $[o(m)]$ and third-person $[po]$, combining with the plural exponent $[mom]$.

	SG	PL		SG	PL
1	i	la-la	1	no	tʃa:-m
2	ñu	ñu-lla	2	o(m)	o-mom
3	ja	ja-la	3	po	po-mom

(a) Cayapa (Pertsova 2022b) (b) Luiseño (Pertsova 2022b)

Table 3.5: Selective allomorphy limited to the first person

Second-person allomorphy. Allomorphy can be limited to the second person. This pattern is exemplified in Table 3.6 on the basis of Finnish (Finno-Ugric), Fur (Nilo-Saharan), and Hamtai (Trans-New Guinea). In Finnish, $[m]$ and $[h]$, the exponents of the first and third person, respectively, do not undergo allomorphy, being preserved in the plural. The second person, however, shows $[s] \sim [t]$ allomorphic alternation triggered by the plural. A similar pattern is present in Fur, where the exponents of the first and third person, $[k]$ and $[ie]$, do not go allomorphy, with observable phonological changes in tone in the case of the third-person pronoun. This contrasts with the $[j] \sim [b]$ exponence alternation, present in the second person. In Hamtai, person exponents in the first- and third-person pronouns in both masculine and feminine genders do not undergo allomorphy, in contrast to the second person, which exhibits $[nti] \sim [h]$ alternation.

	SG	PL
1	m -inä	m -e
2	s -inä	t -e
3	h -än	h -e

(a) Finnish (White 2006)

	SG	PL
1	k -a	k -i
2	j -i	b -i
3	ie	ìè-èḡ

(b) Fur (Pertsova 2022b)

	SG	PL
1	n -i	n -ai
2	nt -i	h -ai
3 _M	ə q -o	ə q -uoa
3 _F	i	i -ṵjoa

(c) Hamtai (Pertsova 2022b)

Table 3.6: Selective allomorphy limited to the second person

Third-person allomorphy. This pattern is rare, though not impossible, in natural language. In Warrongo (Australian Aboriginal), [ng] and [y], the exponents of the first and second person, respectively, do not undergo allomorphy, contrasting with the third-person pronoun [ny-ola], which is realized in the morphologically unrelated form [jana] in the plural. A related pattern is attested in Iraqw (Cushitic). As illustrated in Table 3.7, the second- and third-person singular masculine pronouns are syncretic in this language. Importantly, the formant [ku:ŋ] is preserved in the second-person plural, thus illustrating the absence of allomorphy. However, the third-person plural pronoun receives a different realization [ʔinoʔín]. This language illustrates a pattern where the person distinction is neutralized in the singular but visible in the plural. Therefore, it straightforwardly shows why allomorphy matters. Without the plural and inspecting only the singular, we would not be able to see that the language differentiates the second and third person, which both become visible in the plural. I consider this language as exemplifying the pattern where allomorphy applies only in the third person, given that in addition to the absence of allomorphy in the second person, the first-person pronoun also shows phonological relatedness, sharing the formant [ʔa].

SG	PL	SG	PL
1	ng -aya ng -ana	1	ʔa-níḡ ʔa-tén
2	y -inda y -orra	2	kú:ŋ.M/ku:ŋgáʔ.F ku:ŋgáʔ
3	ny -ola j -ana	3	kú:ŋ.M/kí:ŋF. ʔinoʔín

(a) Warrongo (Tsunoda 2011) (b) Iraqw (Pertsova 2022b)

Table 3.7: Selective allomorphy limited to the third person

First- and second-person allomorphy. A pattern in which the first- and second-person pronouns undergo allomorphy to the exclusion of the third person, which remains invariant, is frequent across languages, as exemplified below on the basis of Georgian (Kartvelian), Latvian (Baltic), and Serbian (Slavic). In Georgian, the exponent of the first person in the singular [m] alternates with number-induced [č], while the exponent of the second person [š] alternates with [tk]. In contrast, the third-person form [is/igi] is preserved in the plural. Latvian exhibits a similar pattern with the morphological unrelatedness of exponents across singular and plural in the first and second person. The third person differs in this respect, with the exponent [vin]

remaining intact. In Serbian, two different exponents are employed for the realization of the person morpheme in the singular and plural in both the first- and second-person pronouns. In contrast, the third-person exponent $[(o)n]$ remains unaffected by allomorphy.

			SG		PL
	SG	PL			
1	m-e	č-ven	1	es	m-ēs
2	š-en	tk-ven	2	t-u	j-ūs
3	is/igi	is-ini/igi-ni	3 _M	vin-š	vin-i
			3 _F	vin-a	vin-as

(a) Georgian (Hewitt 2005)

			SG		PL
	SG	PL			
1	ja	m-i			
2	t-i	v-i			
3 _M	on	on-i			
3 _F	on-a	on-e			

(c) Serbian

Table 3.8: Selective allomorphy limited to the first- and second-person

First- and third-person allomorphy. This pattern is attested across Germanic languages, such as English and Faroese, and in several typologically unrelated languages, such as Goemai (Afro-Asiatic). In English, the zero allomorph of the plural does not affect the second-person pronoun. This contrasts with clear allomorphy in the first- and third-person pronouns. The exponent of the second person $[t]$ is preserved in the plural in Faroese. In contrast, the first-person singular pronoun $[eg]$ alternates with $[v]$ and the exponent of the third person $[h]$ alternates with $[t]$ in the plural. Goemai differs from Germanic languages in making the gender distinction in the second, but not in the third person. As illustrated in the table below, the masculine form of the second person does not undergo allomorphy, with the exponent $[g]$ remaining intact. The gender distinction is neutralized in the plural, where the masculine form $[gwě̃n]$ wins out as a single morphological realization of the second-person plural pronoun. The exponent of the first person in the singular $[h]$ undergoes allomorphy being realized as $[m]$ in the plural, while the third person exhibits $[ní] \sim [m\bar{u}]$ alternation.

SG			PL		
1	I	we			
2	you	you			
3 _M	he	th-ey			
3 _F	she	th-ey			

(a) English

SG			PL		
1	eg	v-it			
2	t-ú	t-it			
3 _M	h-ann	t-eir			
3 _F	h-on	t-ær			

(b) Faroese (Lockwood 1977)

SG			PL		
1	h-ě̃n	m-ě̃n			
2 _M	g-ő̃e	g-wě̃n			
2 _F	y-ő̃e~yĩ	g-wě̃n			
3	ní	mụ-ě̃p			

(c) Goemai (Hellwig 2011)

Table 3.9: Selective allomorphy limited to the first- and third-person

Second- and third-person allomorphy. In several languages, the second and third person may undergo allomorphy to the exclusion of the first person. This is the case in Finno-Ugric languages, Estonian and Votic, and in a typologically unrelated language, Tommo So (Niger-Congo; Dogon), as exemplified in the table below. In Estonian, the exponent of the first person in the singular, $[m]$, is preserved in the plural. This is not the case in the second person, where

two distinct exponents $[s]$ and $[t]$ are employed, and in the third person, which shows the alternation between $[t]$ and $[n]$. The same set of exponents with identical patterns of allomorphy is attested in Votic. In Tommo So, $[m]$ can be considered as an exponent of the first person. The plural form shows a clear morpho-phonological relatedness, preserving the exponent $[m]$, which is why I analyze it as the absence of allomorphy. In contrast, the second and third person involve allomorphic alternations, as the morphological forms of these pronouns are clearly distinct, without any morpho-phonological resemblance.

SG	PL	SG	PL	SG	PL
1	m -ina	m -eie	1	m -iä	m -õ; m -ü
2	s -ina	t -eie	2	s -iä	t -õ; t -ü
3	t -ema	n -emad	3	t -ämä	n -ämä

(a) Estonian (Pool 1999) (b) Votic (Pertsova 2022b) (c) Tommo So (McPherson 2013)

Table 3.10: Selective allomorphy limited to the second- and third-person

3.2.1.2 Across-the-board allomorphy

Allomorphy spread across all person specifications is a common property of Germanic and Celtic languages, as exemplified in Table 3.11, on the basis of German (Germanic), Breton (Celtic), and Scottish Gaelic (Celtic). In German, the exponents of person in the singular are not traceable in the plural forms, as the system involves substantial allomorphy. An exception in this respect is the third-person singular feminine pronoun, which is syncretic with the third-person plural. The realization of gender features and their interaction with the plural number will be discussed in Chapter 5. Breton and Scottish Gaelic exhibit the same high level of allomorphy, with the neutralization of the gender distinction in the plural.

SG	PL	SG	PL	SG	PL
1	ich	wir	1	m -e	ni
2	d -u	euch	2	t -e	c'hwi
3 _M	er	sie	3 _M	eñ	int
3 _F	sie	sie	3 _F	hi	int

(a) German (b) Breton (Press 2011) (c) Scottish Gaelic (Gillies 2009)

Table 3.11: Across the board allomorphy

Beyond Germanic and Celtic, across the board allomorphy affecting all three person specifications is attested in several languages. In Table 3.12, pronominal forms involve an additional formant shared across all persons and both numbers. In Chichewa (Bantu), this is the initial $[i]$. In Baure (Arawakan), it is the formant $[itiʔ]$ which alternates with $[otiʔ]$ in the third-person

masculine singular form, being preserved in the third-person plural. In Abkhaz, it is the formant *[ara]*. Exponents of this type, shared across the pronominal paradigm, will be addressed in Chapter 6. Given that these formants are shared across the entire pronominal system, we can safely conclude that they are not the realization of person or number. For purposes of the present discussion, it is important to note that once these formants are excluded, the remaining person and number morphology exhibits substantial allomorphy.

SG PL			SG PL			SG PL		
1	i-ne	i-fe	1	n-iti?	v-iti?	1	s-ara	h-ara
2	i-we	i-nu	2	p-iti?	j-iti?	2 _M	w-ara	š°-ara
3	i-ye	i-wo	3 _M	r-oti?	n-oti?	2 _F	b-ara	š°-ara
(a) Chichewa			3 _F	r-iti?	n-oti?	3 _M	j-ara	d-ara
(Bentley & Kulemeko 2001)			(b) Baure (Pertsova 2022b)			3 _F	l-ara	d-ara
						(c) Abkhaz (Pertsova 2022b)		

Table 3.12: Across the board allomorphy

Similar in this respect is Hua (Papuan), which has two invariant formants, *[gai]*, spread across Papuan (see Chapter 2), and *[a]*. Once these exponents are excluded, the remaining *[d]*, *[k]*, and *[∅]*, the exponents of the first, second, and third person, respectively, provide evidence for allomorphy. The first person morpheme is realized as *[r]* in the context of the plural, while the second person morpheme as *[p]*, being syncretic with the third person. The same pattern of substantial allomorphy, though without shared morphology, is present in Koyraboro Senni (Nilo-Saharan; Songhay) and Wolaytta (Cushitic), too, given that the exponents of person in the singular do not share any morphological resemblance with their plural counterparts.

SG PL			SG PL			SG PL		
1	d-gai-a	r-gai-a	1	aj	ir	1	ta	nu
2	k-gai-a	p-gai-a	2	n; ni	war	2	ne	hi:nte
3	gai-a	p-gai-a	3	a	i	3 _M	a	e:ta
(a) Hua (Heiman 1980)			(b) Koyraboro Senni			3 _F	i	e:ta
			(Pertsova 2022b)			(c) Wolaytta (Pertsova 2022b)		

Table 3.13: Across the board allomorphy

These data demonstrate that languages originating in different language families exhibit the same properties in their morphological realization. At the same time, substantial allomorphy attested in one language typically closely correlates with allomorphy in the entire family, with a certain degree of micro-variation, as Icelandic, English, and German all differ in patterns of allomorphy in their pronominal paradigms. What unites these languages is the fact that allomorphy is frequent and substantial, despite the variation. The same can be argued for other language families, such as Slavic or Papuan, where allomorphy in a single language is indicative

of allomorphy across the family. This fact provides a window into the strategies that individual languages and language families employ in realizing syntactic terminals in pronouns.

3.2.1.3 Absence of allomorphy

This pattern is characterized by the complete resistance of person morphemes to be affected by the number. The exponents of both person and number are overtly realized, revealing the compositionality transparently. Morphological compositionality of this type is transparent in languages like Cantonese and Japanese, in which the plural pronouns are built on top of singular ones, thus containing them in their representation, given that the discrete exponents of plural attach to the singular pronominal forms, leaving them unaffected.

	SG	PL		SG	PL
1	ngo ²³	ngo ²³ dei ²²	1	watashi	watashi -tachi
2	nei ²³	nei ²³ dei ²²	2	anata	anata -tachi
3	keoi ²³	keoi ²³ dei ²²	3 _M	kare	kare -tachi
			3 _F	kanojo	kanojo -tachi

(a) Cantonese (Pui Yee Yuen, p.c.)

(b) Japanese (Pertsova 2022b)

Table 3.14: Absence of allomorphy

Exponents of person are unaffected by the plural in Ecuadorian Kichwa (Quechuan), Mapuche (Araucanian), and P'urhépecha, an isolate spoken in Mexico (Table 3.15). In Ecuadorian Kichwa, the exponents [ɲuka], [kan], and [paj], present in the singular, are preserved in the plural. The same holds for Mapuche, where the exponents [iñch], [eym], and [fey] are clearly identifiable in the plural. Likewise, [x], [tʃ], and [i'm], analyzed here as the exponents of the first, second, and third person in P'urhépecha, transparently combine with the plural morphology, remaining unaffected.

	SG	PL		SG	PL		SG	PL
1	ɲuka	ɲuka -ɲjik	1	iñché	iñch -iñ	1	ji	ju -chá
2	kan	kan -kuna	2	eymi	eym -ün	2	cha	cha / cha -jtsi
3	paj	paj -kuna	3	fey	fey -engün	3	indé	indé -chá

(a) Ecuadorian Kichwa (Pertsova 2022b)

(b) Mapuche (Smeets 2008)

(c) P'urhépecha (Pérez 2006)

Table 3.15: Absence of allomorphy

Extending the discussion to other languages, data from Apatani (Tibeto-Burman), Korowai (Awyu-Dumut; Trans-New Guinea), and Udmurt (Finno-Ugric) show that this pattern is widely attested across languages. In Apatani, [ng] and [n], the exponents of the first and second person, respectively, combine with the plural exponent [unu]. The same resistance to allomorphy of

the person exponent is present in the third person in both animate and inanimate specifications. Korowai replicates the pattern of the absence of allomorphy, with the exponents $[n]$, $[g]$, and $[j]$ being clearly present in the plural forms. The same holds for Udmurt, where the person exponents remain intact but trigger allomorphy of the number, given that the plural exponent $[i]$ is shared across the first and second person, while the third-person plural pronoun surfaces with $[s]$.

	SG	PL		SG	PL		SG	PL
1	ng-o	ng-unu	1	n-up	n-oxup	1	m-on	m-i
2	n-o	n-unu	2	g-up	g-exenep	2	t-on	t-i
3 _{ANIM}	mo	mo-lu	3	j-up	j-exenep	3	s-o	s-oos
3 _{INANIM}	hii	hii-ka atang						
(a) Apatani (Pertsova 2022b)			(b) Korowai (Pertsova 2022b)			(c) Udmurt (Edygarova 2022)		

Table 3.16: Absence of allomorphy

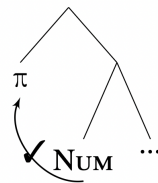
In the preceding discussion, I presented the inventory of alternations attested in number-driven person allomorphy in pronouns. In the next step, I turn to the implications of these empirical patterns for the structure of pronouns.

3.2.2 Towards the Analysis

The presentation of the data showed that person and number interact for purposes of allomorphy. I propose that in the narrow syntax, pronouns involve two syntactic terminals, one hosting PERSON [π] and one hosting NUMBER [NUM]. These morphemes are not bundled on the same head but are individuated, each occupying a separate head. Evidence for this stance comes from two empirical facts discussed here: (i) both person and number morphemes are mapped to discrete exponents at PF, and (ii) person and number morphemes interact for purposes of allomorphy.

I propose that the person node occupies the structurally higher position. The trigger (number) and the target (person) of allomorphy are immediately structurally adjacent, as illustrated in (7), followed by the general schema for number-driven person allomorphy, to underlie pronouns across languages. The Linearization mechanism, active at PF, operates on the two syntactic terminals, PERSON and NUMBER, giving rise to the Linearization statement in (7), in which the exponent of person linearly precedes the exponent of number, although both orders are attested.

(7) NUMBER-DRIVEN PERSON ALLOMORPHY



NUMBER-DRIVEN PERSON ALLOMORPHY

$[1] \leftrightarrow y / __ \#_{PL}$

$[1] \leftrightarrow x$

LINEARIZATION STATEMENT

π -NUM

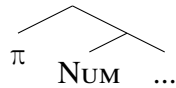
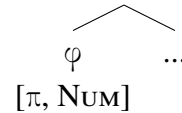
Considerations of directionality play an important role in the general theory of allomorphy. In (7), the person morpheme [π] shows inwards-sensitivity to number features [NUM], as it occupies a structurally higher position than the number. In that sense, the number-driven person allomorphy exemplifies outward-conditioning, given that the target of allomorphy [π] is in a structurally outer position compared to the trigger [NUM]. As noted earlier, the arrow in the tree indicates the directionality of allomorphy.

In the narrow syntax, the two morphemes combine via the standard structure-building operation *Merge*, resulting in the structural representation provided in (7). In the realizational component of grammar, the Vocabulary Insertion proceeds in a cyclic step-wise manner from the most deeply embedded morpheme outwards (Bobaljik 2000; Embick 2010; Moskal 2015b; Myler 2017; Kalin 2022a, 2022b; Adamson 2024). The Vocabulary Insertion operation first targets the number terminal, followed by the insertion into the person node.

3.2.2.1 Pronouns are multi-morphemic

The results of the previous chapter established that pronouns surface with dedicated number morphology, realized in the form of a discrete exponent. This empirical fact is accounted for here under the proposal that the exponent of the number realizes the number morpheme in the narrow syntax, which figures as a separate terminal, distinct from person. Therefore, the approach I am advancing here is MULTI-MORPHEMIC (8).

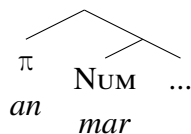
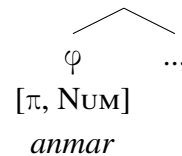
This view contrasts with an approach which considers φ -features on pronouns to constitute a bundle, hosted on a single terminal. I will call this **MONOMORPHEMIC** approach. As already noted in the course of the discussion, the bundling approach has been argued for in syntactic accounts, where person and number are subsumed under a φ P (Déchaine & Wiltschko 2002; Ghomeshi & Massam 2020, among others). As illustrated in (9), both person and number features are hosted on a single φ head.

(8) **MULTI-MORPHEMIC APPROACH**(9) **MONO-MORPHEMIC APPROACH**

Taking the pronominal system of Kuna (Chibchan) discussed above, where pronouns in the plural surface with overt exponents of both person and number, the two approaches assume the following representations (10-11). Under the Multi-morphemic approach, the plural exponent [*mar*], shared across all three person specifications, is the realization of the **NUMBER** terminal in the narrow syntax. Likewise, the exponent of the first person [*an*] realizes the **PERSON** (π) terminal. In the Mono-morphemic, i.e., bundling approach, both exponents correspond to a single φ morpheme.

	SG	PL
1	an	an-mar
2	be	be-mar
3	we	we-mar

Table 3.17: Pronouns in Kuna (Pertsova 2022b)

(10) **MULTIMORPHEMIC APPROACH**(11) **MONOMORPHEMIC APPROACH**

The two views differ with respect to the Vocabulary Insertion operation, which is defined as supplying the morphemes in the narrow syntax with the phonological content at PF. Under the Multi-morphemic view, both person and number terminals are targeted for insertion by two distinct insertion rules. Each Vocabulary Item targets exactly one terminal node, i.e., the Vocabulary Insertion operation applies twice, targeting each node separately and proceeding cyclically. This derives the correct individuation of the two distinct exponents.

If pronouns were mono-morphemic, however, the form [*anmar*] is realized at once. Therefore, person exponents would have to be stored in the memory and acquired by a child once in isolation [*an*], as the same exponent surfaces in the singular, and once together with the plural

exponent *[anmar]*, thus twice. Given that the decomposition is available in a straightforward manner, such an approach would have weak support, both on empirical and theoretical grounds. An account of Kuna in which *[an]* and *[anmar]* would be stored as single, separate Vocabulary Items, fails to capture significant empirical regularity in natural languages, which becomes immediately obvious once the person morpheme is individuated. Therefore, under this view, the mono-morphemic approach would share the view that Vocabulary Insertion targets terminal nodes with the multi-morphemic approach. However, while the multi-morphemic proposal recognizes two distinct exponents, the mono-morphemic one has to assume a single Vocabulary Item, which fails to explain the shared morpho-phonological material.

Alternatively, the mono-morphemic view may assume two distinct exponents for a single head, in which case one would have to posit recursive Vocabulary Insertion that applies iteratively to the same morpheme until it exhausts its features. That is, in (11), the Vocabulary Insertion would target φ , realizing *[mar]*. Proceeding iteratively, it would have to target the same φ morpheme again, realizing *[an]*. An additional problem for this view is that it remains entirely obscure whether the number or person should be realized first, and which process would be responsible for that if they were bundled on the same head. However, precluding recursive Vocabulary Insertion in pronouns tightly corresponds to the Uniqueness requirement of the Vocabulary Insertion operation (see Halle (1992) for ‘at most one Q’ requirement, where Q is a dummy position until realized with a phonological content, Embick (2015) for a discussion on the view that once Vocabulary Insertion applies to a node, the process of insertion terminates for this node, and Alexiadou (2024) for a concrete implementation of the Uniqueness requirement in the context of the Maximize Exponence account of double plurality). Under the Multi-morphemic approach developed here, each terminal, PERSON or NUMBER, is targeted only once. If a terminal undergoes Vocabulary Insertion, it cannot be targeted again.

A note on Fission seems to be called for at this point. This process refers to the creation of two morphemes at PF that correspond to a single morpheme in the narrow syntax (Halle & Marantz 1994). For instance, in (12), the morpheme $\mathbb{M}$ at PF corresponds to a single morpheme in the narrow syntax. In the post-syntactic component, this morpheme undergoes Fission, being split into two morphemes, m_1 and m_2 . Importantly, the target of the Vocabulary Insertion is not the original $\mathbb{M}$ morpheme, but the outcomes of Fission, m_1 and m_2 . Therefore, even in cases where Fission is assumed and required, Vocabulary Insertion does not apply recursively but targets individual outcomes of this operation separately.

(12) FISSION

$$\mathbb{M} \rightarrow m_1, m_2$$

In sum, in the proposal I developed so far, pronouns involve two distinct terminals in the narrow syntax, one hosting the person and one hosting the number. The decomposition extends to the PF component and the *Vocabulary* part of the grammar, as the two morphemes correspond to two exponents. In addition, Vocabulary Insertion proceeds cyclically, targeting only terminal nodes, which correctly derives a number of generalizations in the morphological realization and

provides further support for compositionality in natural language.

3.2.2.2 The role of the Vocabulary

The core idea of the present account is that allomorphy provides the key for accessing the internal compositionality of pronouns by identifying the *trigger* and the *target* of an allomorphic alternation. In this study, I have identified three types of allomorphic alternations in the pronominal domain, in cases when the number acts as a trigger and the person as a target: (i) *Selective allomorphy*, (ii) *Across-the-board allomorphy*, and (iii) *Absence of allomorphy*. What brings together German (Germanic), Chichewa (Bantu), Baure (Arawakan), Hua (Papuan), and Koyraboro Senni (Nilo-Saharan)? What makes them different from Greenlandic (Inuit), Ecuadorian Kichwa (Quechuan), Chinantec (Oto-Manguean), Apatani (Tibeto-Burman), and Udmurt (Finno-Ugric), which all share a unified property?

German and other languages listed above involve substantial number-induced allomorphy spread across all three person specifications. As a consequence, exponents of person employed in the singular are not traceable in the plural forms. This contrasts with Greenlandic and languages listed in the latter group, devoid of any allomorphic alternations, transparently preserving the exponents of person in the plural. Therefore, Germanic languages are revealing of the underlying structure and the inventory of features in pronouns, which is evidenced by allomorphy induced by the merge of any additional feature.

These empirical facts can be explained from the perspective of theories that argue for the dissociation of meaning and morpho-phonological form, as is the case in Distributed Morphology (Halle & Marantz 1994) and the Meaning First Approach (Sauerland & Alexiadou 2020). As noted in the introductory chapter, DM operates on the basis of three lists (Halle & Marantz 1994; Harley & Noyer 1999; Bobaljik 2017): (i) *The Syntactic Terminals*, a list which includes $\sqrt{\text{Roots}}$, categorizing heads (n , v , a), and functional morphemes (π , $\#$, γ , D , among others), (ii) *The Vocabulary*, a list consisting of Vocabulary Items, i.e., objects that supply syntactic terminals (morphemes) with the phonological material, and (iii) *The Encyclopedia*, a list which includes non-compositional semantic information, that emerges only in particular local contexts (Marantz 2001, 2007; Arad 2003; Alexiadou & Anagnostopoulou 2008b; Anagnostopoulou & Samioti 2014; Oikonomou & Alexiadou 2022; Alexiadou & Anagnostopoulou 2023).

The *Vocabulary* is relevant here. German employs two distinct exponents of the first-person feature (Table 3.18-a), one surfacing in the singular and one in the plural, while Ecuadorian Kichwa employs only one in both environments (Table 3.18-b). This means that the *Vocabularies* of the two languages differ. Namely, the German list of Vocabulary Items is larger than the one found in Ecuadorian Kichwa, as it contains the allomorph of the person feature, alongside the elsewhere form. Importantly, this difference does not emerge unpredictably. The larger set of Vocabulary Items in German is a consequence of allomorphy. This raises a specific prediction for language acquisition. Namely, a German-acquiring child has to store two Vocabulary Items for person, while a Kichwa-acquiring child only has one. These differences are placed here in the post-syntactic component of grammar and do not carry over back to the narrow syntax. Identity

in the form attested in Ecuadorian Kichwa is considered here to be the absence of allomorphy. I take this morphological identity to suggest that exactly the same feature in the narrow syntax is realized.

SG PL			SG PL		
1	ich	wir	1	ɲuka	ɲukanjik
2	du	euch	2	kan	kankuna
3 _M	er	sie	3	paj	pajkuna
3 _F	sie	sie			

(a) German

(b) Ecuadorian Kichwa (Pertsova 2022b)

Table 3.18: Allomorphy and its absence: German vs. Ecuadorian Kichwa

Further evidence for the multi-morphemic, i.e., multi-terminal approach argued for here comes from the fact that the trigger and the target of allomorphy freely co-occur in pronouns. I turn to this aspect of the analysis in the next step (3.2.2.3).

3.2.2.3 The trigger and the target co-occur

In the following data, discussed above in the context of the distribution of number-driven allomorphy with respect to person specifications, an allomorph of person co-occurs with the exponent of plural that triggers allomorphy within the same pronoun. This is exemplified here on the basis of languages in which the first-person morpheme undergoes allomorphy to the exclusion of the second and third. The exponent of the trigger of allomorphy, i.e., the number, is separated by the hyphen in the tables below.

SG PL			SG PL			SG PL		
1	es	me-k'	1	én	m-i	1	bi	da-si
2	du	du-k'	2	te	t-i	2	qi	ta-si
3	na	nran-k'	3	ő	ő-k	3	gan	gan-si

(a) Armenian (Ajello 2015)

(b) Hungarian (Pertsova 2022b)

(c) Mangghuer (Slater 2003)

Table 3.19: Co-occurrence of the trigger and target of allomorphy

Co-occurrence of the trigger and the target is something that *any* theory of allomorphy has to account for. These data have significant implications for the analysis of pronouns in which the trigger and the target are not transparently traceable to two distinct overt exponents. In these cases, it becomes tempting to assume Fusion or bundling in the syntax in most approaches. The simplest example would be the *I~we* alternation in English. This is the point where the theories that allow zero exponents, such as Distributed Morphology, and theories that explicitly rule it out, as is the case in Nanosyntax, diverge. Under the DM account, in the same way the trigger can receive an overt realization, as was the case above, it can be realized as zero.

	SG	PL		SG	PL
1	i	la-la	1	no	tʃa:-m
2	ñu	ñu-lla	2	o(m)	o-mom
3	ja	ja-la	3	po	po-mom

(a) Cayapa (Pertsova 2022b) (b) Luiseño (Pertsova 2022b)

Table 3.20: Co-occurrence of the trigger and target of allomorphy

Simultaneous presence of the trigger and the target of allomorphy, typically referred to as the **double marking** in the DM tradition (Halle & Marantz 1994 and subsequent literature), straightforwardly follows from the three points I have argued for above: (i) pronouns involve **two distinct morphemes** in the narrow syntax, (ii) the **two morphemes** correspond to **two exponents**, (iii) **Vocabulary Insertion targets only terminals/morphemes**. Under this proposal, the overt co-occurrence of the trigger and the target is expected. In fact, the only cases where the trigger and the target of allomorphy are not simultaneously *overtly* present are those involving a zero exponent.

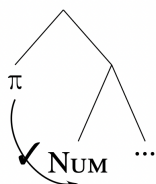
Crucially, these data demonstrate that the absence of allomorphy, where the two morphemes co-occur in an agglutinative manner without triggering any allomorphic alternations, as was the case in languages like Japanese, Kuna, and Udmurt, is not an alternative to the Selective- and Across-the-board allomorphy. That is, languages are not either agglutinative or have substantial allomorphy. The key evidence for this view is that the trigger and the target of allomorphy freely overtly co-occur in pronouns, as exemplified by the data discussed in this section. Double marking will be discussed again in Chapter 6, with special reference to realizational theories such as Nanosyntax and Spanning. In the next section (3.3), I move to the study of person-driven number allomorphy.

3.3 The Puzzle: Person-driven number allomorphy

In this section, I present the study on person-driven number allomorphy in pronouns. Number-driven allomorphy of person morphemes is well-attested across languages, as demonstrated in the previous discussion. This directionality of allomorphy is the one that has been approached from various perspectives in the literature (Moskal 2015a, 2015b, 2018; Moskal & Smith 2016; Harbour 2016; Smith et al. 2019; Zompì 2023). The **opposite direction**, in which realization of plural exponents would be sensitive to person feature specifications, has not been investigated in the literature. I demonstrate that, in addition to the number-driven person allomorphy, pronominal systems provide evidence for the person-driven number allomorphy. This creates a puzzle for the theory of allomorphy and the question of directionality (for considerations of directionality see Carstairs-McCarthy 1987, 2001; Bobaljik 2000, and Rolle (2023) for a recent overview). The question that arises is whether the two directions in allomorphic conditioning may receive a principled explanation. What is the rationale behind their availability? More broadly, how does the fact that both directions are found inform the overall theory of allomorphy?

The study is based on the same dataset used for the study on the number-driven person allomorphy. In the syntactic configuration where person surfaces above number (13), this means the number node is outwards-sensitive to person features, i.e., there is an inward-conditioning of number morphology by the person, as indicated in the following structural and linearized representations.

(13) PERSON-DRIVEN NUMBER ALLOMORPHY



PERSON-DRIVEN NUMBER ALLOMORPHY

$[PL] \leftrightarrow y / __ \pi_{AUTHOR}$

$[PL] \leftrightarrow x$

LINEARIZATION STATEMENT

$\pi-Num$

The first study demonstrated that identifying the trigger and the target of allomorphy in the pronominal domain provides evidence for a complex multi-terminal internal structure of pronouns. The present study further substantiates this finding. Importantly, it raises a different question, which is highly relevant for the overall theory of allomorphy: Is there a principled explanation for both directionalities in allomorphic conditioning? I demonstrate that the distinction in directionality cannot be pinned down to the difference between suppletion, i.e., morpho-syntactically induced allomorphy and phonologically-conditioned alternations.

Furthermore, is there a difference in frequency between the two types of directionality? Given that a large body of literature investigates different aspects of the number-driven person allomorphy, it seems that the former should be more widely attested, while the latter should be sporadically attested.

The question of which directionality of allomorphy is more frequent is tricky. Recall the discussion on the allomorphs of the plural in English from the previous chapter. English surfaces with several plural allomorphs, each specified for a particular list of $\sqrt{\text{Roots}}$, and the elsewhere

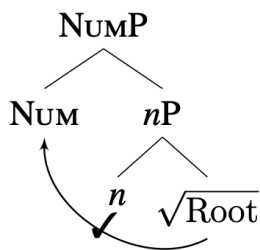
exponent $/-z/$ $[s]$. Notably, the Vocabulary Insertion rules in (14) first specify that the context of insertion for every allomorph is a noun, hence $[/]_N _]$. Only in the next step of specifying the context is the reference to particular $\sqrt{\text{Roots}}$, i.e., lists of $\sqrt{\text{Roots}}$, made.

(14) ALLOMORPHS OF PLURAL IN ENGLISH (Gouskova & Bobaljik 2025)

- [PL] $\leftrightarrow$ $-en /]_N _$, where $N \in \{\sqrt{\text{ox}}\}$
 [PL] $\leftrightarrow$ $\emptyset /]_N _$, where $N \in \{\sqrt{\text{sheep}}, \sqrt{\text{fish}}, \sqrt{\dots}\}$
 [PL] $\leftrightarrow$ $[-\text{back}, +\text{tense}] /]_N _$, where $N \in \{\sqrt{\text{foot}}, \sqrt{\text{goose}}, \sqrt{\dots}\}$
 [PL] $\leftrightarrow$ $-i /]_N _$, where $N \in \{\sqrt{\text{alumn}}, \sqrt{\text{syllab}}, \sqrt{\dots}\}$
 [PL] $\leftrightarrow$ $-z /]_N _$

In that sense, these Vocabulary Insertion items in fact involve two conditions on insertion that stand in a dependency relationship. The first condition is the presence of a noun (nP in the structural representation in 15) and the second, a more specific one, is that this noun has to surface with a particular $\sqrt{\text{Root}}$. In the case at hand, the first condition that requires the presence of the noun is not specific enough to account for the distribution of plural allomorphs. For this reason, we need to resort to the ancillary condition which specifies which $\sqrt{\text{Roots}}$, which first combine with n , co-occur with a particular allomorph of plural.

(15) $\sqrt{\text{Root}}$ -DRIVEN NUMBER ALLOMORPHY



Consider now the pronominal system of English. A dedicated exponent of plural is not transparently identifiable, thus resorting to the zero $[\emptyset]$ allomorph. This is particularly evident in the case of second-person pronouns. In the nominal domain (14), the zero allomorph of plural is selected with nouns surfacing with $\sqrt{\text{Roots}}$ such as $\sqrt{\text{sheep}}$ and $\sqrt{\text{fish}}$.

	SG	PL
1	I	we
2	you	you
3	he/she	they

(a) English

(16) VI for PL in English

- [PL] $\leftrightarrow$ $\emptyset / _ \pi$

Table 3.21: Pronouns in English

Now, let us see how the two allomorphs of the plural differ. In (17-a), the context of insertion specifies a noun, i.e., the nominalizing n head which combines with certain $\sqrt{\text{Roots}}$. In (17-b),

the context of insertion, i.e., the syntactic environment, is different. In this case, the trigger of allomorphy is the PERSON $[\pi]$ head.

(17) ZERO ALLOMORPHS OF PLURAL IN ENGLISH

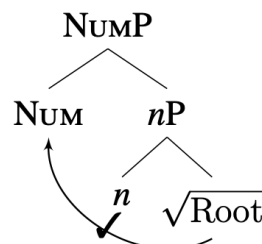
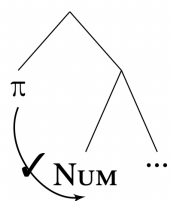
- a. $[\text{PL}] \leftrightarrow \emptyset /]_{\text{N}} ___$, where $\text{N} \in \{\sqrt{\text{sheep}}, \sqrt{\text{fish}}, \sqrt{\dots}\}$
- b. $[\text{PL}] \leftrightarrow \emptyset /]_{\pi} ___$

Therefore, if English is analyzed as having the zero allomorph of plural as in (17-b), this raises an important implication for all other languages. Namely, the prediction is that in all languages in which pronouns do not surface with the elsewhere plural realization for that language, the person node triggers allomorphy of the number. However, in the previous chapter, I proposed that $\sqrt{\text{Roots}}$ and **person** nodes **generate two distinct sets of allomorphs** that do not stand in a competition that can be resolved by the Subset Principle. In that sense, allomorphs of plural triggered by the person are independent of those conditioned by the $\sqrt{\text{Root}}$.

In Chapter 2, I argued against an accidental homophony analysis of Serbian plural morphology, where two syncretic exponents of plural $[i_1, i_2]$ would independently be stored in the Vocabulary, and separately selected with nouns [e.g., i_1] and pronouns $[i_2]$. I put forth an analysis where the plural exponent $[i]$ is a single exponent that takes part in two distinct syntactic derivations. In contrast, zero realization of plural in English provided in (17) necessarily need to be analyzed as a result of accidental homophony, as there is nothing that would bring close the person feature and the $\sqrt{\text{Roots}}$ such as $\sqrt{\text{fish}}$, $\sqrt{\text{sheep}}$, or $\sqrt{\text{feet}}$ to act as a single trigger of allomorphy. I argue that this difference between Serbian and English derives from the fact that Serbian employs the *elsewhere* exponent in the pronominal domain, which is not the case in English. The zero allomorph of plural in English is selected only with a highly limited set of $\sqrt{\text{Roots}}$. This prevents any conceivable uniform analysis with the allomorph triggered by the person node, sharply contrasting in this respect with Serbian. This analysis aligns with the two generalizations on the realization of number morphology in nouns and pronouns introduced in the previous chapter.

This discussion illustrates how the investigation of person-driven number allomorphy differs from the opposite direction. In determining the inventory of the exponents of plural in a particular language, one has to take into consideration that there are two potential triggers of allomorphy of the plural node, (i) the PERSON head (18) and (ii) the $\sqrt{\text{Root}}$ (19).

- (18) PERSON-DRIVEN NUMBER ALLOMORPHY (19) $\sqrt{\text{Root}}$ -DRIVEN NUMBER ALLOMORPHY



Therefore, the question is what is our criterion. If formalized from the perspective of the trigger, which is a $\sqrt{\text{Root}}$ or a person feature, there are two distinct sets of allomorphs. Formalized from the perspective of the target, however, which is in this case the number, all allomorphs of the plural are a single set whose members are specified for a particular structural environment. Importantly, when the Subset Principle, as a core component of the Vocabulary Insertion operation, inspects if the relevant structural conditions are met, it will consider the syntactic environment involving a $\sqrt{\text{Root}}$ as distinct from the one involving person. In Chapter 2, I suggested that neither is more specific, that is, that the presence of the $\sqrt{\text{Root}}$ should not be considered more specific than the presence of person in the Vocabulary Insertion rule or vice-versa. In fact, the two stand in a similar relation as different Lists. As a consequence, allomorphs of plural triggered by the $\sqrt{\text{Root}}$ *do not compete* for insertion with allomorphs of plural triggered by the person. This effectively explains accidental homophony in English and the availability of the *elsewhere exponents only* in nouns and pronouns, as the elsewhere form does not include the context specification.

Once allomorphs of plural triggered by the person are delineated from allomorphs of plural conditioned by the $\sqrt{\text{Root}}$, and only the former are considered in pronominal systems, a relevant **asymmetry** emerges. Namely, person-driven number allomorphy is less frequent than its directional counterpart, i.e., number-driven person allomorphy. I formalize this as a tendency toward an asymmetry.

(20) A TENDENCY TOWARD AN ASYMMETRY IN FREQUENCY

Person-driven number allomorphy is less frequent than the opposite direction.

Furthermore, particularly relevant in this respect are languages whose number morphology surfaces only on pronouns. Recall the data from Bardi, a language which exhibits the MINIMAL~AUGMENTED distinction. In this language, nouns do not surface with the overt number exponents, as this information is retrieved from the context. Pronouns, however, employ a dedicated exponent of augmented number, i.e., plural *[rr]*. To account for the distribution of the exponent *[rr]*, we would have to specify that it emerges only in the context of PERSON and not a particular noun or a nominal $\sqrt{\text{Root}}$, as was the case in English.

π	MIN	AUG
1	ngay(oo)	a- rr -oodoo
1+2	ayoo	a- rr -idil
2	joo	goo- rr
3	giniyingg(i)	i- rr

Table 3.22: Pronouns in Bardi (Bower 2012)

Before proceeding with the results of the present study, I will provide a brief remark on the realization of first-person plural pronouns in a comparative Germanic perspective. As the table below demonstrates, Germanic languages surface with the formant *[w/v]* in the nominative plural

and $[m]$ in the accusative singular. Focusing on the plural, if $[w]$ is considered the exponent of person in the plural, $[e]$ can be attributed the status of the plural exponent. The exponent with a similar phonological realization is present across other languages (compare *-it*; *-ir*; *-ið*; *-ij*; *-i*). An analysis along these lines would add one more exponent to the set of plural allomorphs in English. For individuation of $[m]$ as the exponent of the first person in Indo-European languages, see also the discussion in Chapter 6, 6.6.

	SG		PL	
	NOM	ACC	NOM	ACC
ENGLISH	I	me	we	us
FAROESE	eg	meg	vit	okkum
GERMAN	ich	mich	wir	uns
ICELANDIC	ég	mig	við	okkur
DUTCH	ik	mij	wij	ons
NORWEGIAN	jeg	meg	vi	oss
SWEDISH	jag	mig	vi	oss

Table 3.23: Shared allomorphs of PERSON in Germanic

Note that nothing would change in the discussion on the zero allomorph, as it is necessary in the second person. However, in the analysis of the second person, one can also resort to syncretism for explaining the obtained $[you \sim you]$ realization. This contrasts with $[sheep \sim sheep]$, where the analysis does not involve syncretism, but rather the $\sqrt{\text{Root}}$ and the zero allomorph of plural. The availability of this possibility shows how different analyses may be entertained depending on the nature of the trigger, namely whether it is a feature or a $\sqrt{\text{Root}}$, as the theory is more inclined to employ syncretism in the former but zero allomorphs in the latter.

In addition, if the formant $[th]$ is an exponent of the third person (Bernstein 2008a), then the third-person plural also provides evidence for the decomposition into $[th]$ and $[ey]$. In principle, the question of whether the remaining formants, $[e]$ in *we* and $[ey]$ in *they*, are the same exponent that underwent a morpho-phonological change raises a concern about which phonological rule of English would be responsible for this alternation.

3.3.1 The patterns

The same patterns identified in the number-driven person allomorphy are attested in the opposite direction. Namely, in (i) *Selective allomorphy*, the alternation is triggered by a particular person specification, [e.g., AUTHOR]. This contrasts with (ii) *Across-the-board allomorphy*, where all person specifications, i.e., the first-, second-, and the third-person, trigger a different realization of the plural morpheme. In the (iii) *Absence of allomorphy*, all person specifications combine with the same exponent of plural.

3.3.1.1 Selective allomorphy

AUTHOR-driven. The choice of plural exponents can be directly dependent on the presence of a particular person feature specification, such as *AUTHOR*, i.e., the first person. Ecuadorian Kichwa provides direct evidence for the existence of person-driven number allomorphy. While the second- and third-person pronouns share the plural exponent *[kuna]*, this is not the case in the first-person plural pronoun, which selects the plural exponent *[ɲjik]*. This pattern can be explained if the presence of the *AUTHOR* feature triggers the realization of the plural morpheme *[ɲjik]*. The more specific Vocabulary Item, specified for the presence of the *AUTHOR* feature, wins over the less specific elsewhere form (21). In the context of the second and third person, i.e., when the *AUTHOR* is not present in the structure and thus the context specification is not met, the elsewhere plural exponent *[kuna]* is realized. As elaborated in the previous section, this is a language in which person exponents do not undergo number-driven allomorphy, as singular forms are properly contained in plural ones. The set of VIs provided in (21) captures the Ecuadorian Kichwa data. The plural morpheme is sensitive to the presence of the *AUTHOR* feature in the structure, which is directly reflected in the morphology. This principle extends to all languages exhibiting *AUTHOR*-driven number allomorphy discussed here.

	SG	PL	(21) VIs for Ecuadorian Kichwa
1	ɲuka	ɲuka-ɲjik	[PL] ↔ ɲjik / ____ π _{AUTHOR}
2	kan	kan- kuna	[PL] ↔ kuna
3	paj	paj- kuna	[1] ↔ ɲuka
			[2] ↔ kan
			[3] ↔ paj

(a) Ecuadorian Kichwa (Pertsova 2022b)

Table 3.24: Selective person-driven number allomorphy: Ecuadorian Kichwa

A comparable pattern of allomorphy is attested in the Amazonian language Kulina. The exponent of the plural number surfacing with the second and third person, *[deni]*, is absent in the first-person plural pronoun. In this case, the distribution of plural morphology can be explained from the perspective of its co-occurrence with a particular person specification in the syntax. When the *AUTHOR* feature is absent, the elsewhere exponent *[deni]* surfaces. It is important to note that, in contrast to the situation in Ecuadorian Kichwa above, the person morpheme triggers allomorphy of the number morpheme, and the number morpheme also triggers allomorphy of the person morpheme, as the singular and plural forms do not share any material. This pattern exemplifies *bidirectional allomorphy*. To account for the Kulina data, I propose that the plural morpheme receives $\emptyset$ realization in the context of *AUTHOR* (22).

The sensitivity to the presence of the *AUTHOR* feature in the syntactic structure of a pronoun is found in the morphological realization of the plural morpheme in a number of unrelated languages. As illustrated in the table below, in Korowai (Awyu-Dumut; Trans-New Guinea), the second- and third-person pronouns surface with the formant *[exene]* in the plural, in contrast to

	SG	PL	
	1	owa	ia
	2	tia	tia- deni
	3M	powa	powa- deni
	3F	po-ni	po-ni- deni
(a) Kulina (Dienst 2014)			(22) VIs for Kulina
			[PL] $\leftrightarrow \emptyset / __ \pi_{\text{AUTHOR}}$
			[PL] $\leftrightarrow \text{deni}$
			[1] $\leftrightarrow \text{ia} / __ \#_{\text{PL}}$
			[1] $\leftrightarrow \text{owa}$
			[2] $\leftrightarrow \text{tia}$
			[3] $\leftrightarrow \text{po}$

Table 3.25: Selective person-driven number allomorphy: Kulina

the first person, which selects [*oxu*]. I analyze this exponent as an infix, given that the right-most [*p*] appears in singular forms too, especially in light of the fact that languages of the Trans-New Guinea family often surface with an invariant component of this type, as discussed in the previous chapter. Therefore, formants [*u*] and [*oxu*; *exenep*] can be considered the exponents of the singular and plural, respectively, which linearize as infixes between the exponents of person [*n*, *g*, *j*] and the invariant formant. Note that nothing in the present discussion would change if the parsing were different and the respective forms were [*oxuplexenep*]. In Luiseño, the second- and third-person pronouns surface with the plural exponent [*mom*], transparently combining with the person exponents without causing allomorphy. This contrasts with the situation in the first person, exemplifying bidirectional allomorphy, as both the person node triggers the allomorphy of the number, and the number triggers allomorphy of the person.

Modern Baba Malay (MBM), an Austronesian-based contact language, illustrates the same pattern of person-driven number allomorphy, showing that it should be considered a productive process in the natural language. Namely, Malay is the lexifier language, while Hokkien is the substrate language (Lee 2022). In Hokkien, the exponent of plural is in fact a noun with the meaning ‘people’. The Hokkien word for ‘people’ [*lang*] is replaced by the Malay word with the same meaning [*orang*]. Importantly for the present discussion, the first person in MBM does not surface with the plural exponent [*orang*]. In Hokkien, however, the first person takes [*lâng*], on par with the second and third, resulting in [*goá-lâng*]. Therefore, in this case, we may observe that the contact language, in contrast to its substrate, productively developed person-driven number allomorphy and bidirectional allomorphy.

	SG	PL		SG	PL		SG	PL
1	nup	n-oxu-p	1	no	tʃa:m	1	saya/gua	kita
2	gup	g- exene -p	2	o(m)	o- mom	2	lu	lu- orang /lu
3	jup	j- exene -p	3	po	po- mom	3	dia	dia- orang /dia

) Korowai (Pertsova 2022b)
(b) Luiseño (Pertsova 2022b)
(c) MBM (Lee 2022)

Table 3.26: Selective person-driven number allomorphy: A cross-linguistic perspective

PARTICIPANT-driven. Patterns of allomorphy demonstrate that the PARTICIPANT feature as a whole, shared by the first- and second-person, may act as a trigger of allomorphy, thus constituting a single unit at the level where the choice of exponence is made. As exemplified in the table below, on the basis of Faroese, Hungarian, and Udmurt, the first and second person pronouns share the plural exponent to the exclusion of the third person. In Faroese, the plural exponent *[it]* is selected with the first- and second-person pronouns, being absent in the third. Hungarian employs two realizations of plural in the pronominal system, *[i]* co-occurring with the first and second person, and *[k]*, surfacing with the third person. The same holds for Udmurt, where the exponent *[i]* is selected for the first and second person, and *[os]* for the third.

SG PL			SG PL			SG PL		
1	eg	v- it	1	én	m- i	1	mon	m- i
2	tú	t- it	2	te	t- i	2	ton	t- i
3 _M	hann	t- eir	3	ő	ő- k	3	so	so- os
3 _F	hon	t- ær						
(a) Faroese (Lockwood 1977)			(b) Hungarian (Pertsova 2022b)			(c) Udmurt (Edygarova 2022)		

Table 3.27: Selective person-driven number allomorphy: Participant trigger

More broadly, the same pattern is attested in Bella Coola (Salish), Domari (Indo-Aryan), and Mumuye (Savanna). The relevant exponents of plural are marked in bold in Table 3.28. Note that in Bella Coola, the exponent of the plural precedes the person. This is most transparently evident in the third-person pronouns, where the exponent of person *[ix]* combines with the gender exponents in the singular, being preserved in the plural. The exponent of plural, *[w]*, linearizes to the left of person *[ix]*.

SG PL			SG PL			SG PL		
1	ʔnc	ɬ-miɬ	1	ama	e- me	1	ṇ-ɬ	w- o -ɬ
2	ʔinu	ɬ-up	2	atu	it- me	2	mɔ-ɬ	n- o -ɬ
3 _M	tix	w-ix	3	pandži	pandž- an	3	wu	ji
3 _F	cix	w-ix						
(a) Bella Coola (Pertsova 2022b)			(b) Domari (Matras 2012)			(c) Mumuye (Pertsova 2022b)		

Table 3.28: Selective person-driven number allomorphy: Participant trigger

In some languages, the realization of plural shows interaction with other features, such as animacy. For instance, in Apatani (Tibeto-Burman), the number exponent *[unu]* surfaces with the first- and second-person pronouns, transparently combining with *[ng]* and *[n]*, the exponents of the first and second person, respectively. Therefore, the presence of the feature PARTICIPANT can be argued to underlie the distribution of this plural exponent. The third-person pronoun makes

the animacy distinction, which is transparently encoded in morphology. In the respective plural forms, singular exponents are preserved. Importantly, there is a clear distinction in the choice of the plural morphology. The third-person animate pronoun selects the exponent *[lu]*, thus contrasting with both the first- and second-person pronouns, while the third-person inanimate pronoun co-occurs with *[ka atang]*. In this case, it can be argued in principle that the animacy feature is the trigger of allomorphy, with *[unu]* being the elsewhere exponent. Given that animacy features are hosted together with gender on *n* (Adamson & Anagnostopoulou 2024) or with person on π (Toosarvandani 2023), the number morpheme can be sensitive to both structural positions. However, given the fact that first and second person are also animate, it is rather the PARTICIPANT that selects *[unu]*, while the animacy distinction drives the difference in exponence of plural in the third person.

	SG	PL
1	ngo	ng- unu
2	no	n- unu
3 _{ANIM}	mo	mo- lu
3 _{INANIM}	hii	hii- ka atang

Table 3.29: Pronouns in Apatani (Pertsova 2022b)

ADDRESSEE-driven. I found this exceptionally rare pattern in only three languages in my sample (Pero, P’urhépecha, and Warrongo). As the discussion below will elaborate in greater detail, the availability of this pattern of allomorphy is relevant not only for the structure of pronouns and allomorphy but also for the representation of the PERSON feature more broadly.

In Pero (West Chadic), the first- and third-person pronouns surface with the plural exponent *[ínù]*, in contrast to the second person, which lacks it. In this language, plural morphology is not marked on nouns. Number-related exponents appear only on pronouns and verbs (Frajzyngier 1989). Pronouns provided in Table 3.30 may occupy an argument position, surfacing as affixes on the verb. Alternatively, they may occur independently in appositional clauses and PPs, in which case they are preceded by a preposition. This distinction, affixal or independent, is orthogonal to the morphological shape, as it does not correlate with it, remaining constant in both cases. The table below shows that the second and third person draw gender distinctions. Furthermore, the two pronouns surfacing with the masculine gender in the second person differ in frequency, with *[kè]* being the more frequent form and *[kài]* appearing only sporadically.

	SG	PL
1	nè	m- ínù
2	kè/kài.M/cì.F	m-à
3	cákkà.M/tè.F	c- ínù

Table 3.30: Person-driven number allomorphy: Pero

Crucially, the formant [ínù], transparently identifiable in the first- and third-person pronouns, is absent in the second person. As the language lacks number morphology on nouns, the given exponent must be specified for syntactic contexts involving PERSON and precluded from the second-person environments. I argue that this distribution can be accounted for via the following set of Vocabulary Insertion rules (23). The plural exponent [à] is selected in the context of the person head, only if it contains the ADDRESSEE feature. In cases when the ADDRESSEE is absent, the exponent [ínù] surfaces. I include π in the context for insertion of the exponent [ínù] to indicate that this exponent surfaces only in pronouns, as the language lacks number morphology in nouns.

(23) **VIs for PL in Pero**

[PL] $\leftrightarrow$ à / ___ $\pi_{\text{ADDRESSEE}}$

[PL] $\leftrightarrow$ ínù / ___ π

These data are significant for the representation of features. Namely, the second-person plural pronoun can in principle be analyzed in two ways. Under one view, [mà] would be an exponent of the second person undergoing number-driven allomorphy, combining with the zero [Ø] allomorph of plural. Alternatively, it can be split into [m], the exponent of the second person, and [à], the exponent of the plural. The latter is the stance that I take here. Note that the person exponent is syncretic in the first- and second-person plural pronouns and realized as [m].

	SG	PL
1	nè	m -ínù
2	kè/kài.M/cì.F	m -à
3	cákkà.M/tè.F	c -ínù

Table 3.31: Person syncretism: Pero

How can two different allomorphs of plural co-occur with the same exponent of person? This question receives a straightforward explanation if the syncretic exponent of person [m] corresponds to **two different morphemes** in the narrow syntax, specified for different features. The features present in the morpheme are those relevant for allomorphy. The fact that person may act as a **trigger** of allomorphy, giving rise to the observed alternation, suggests that the syncretic Vocabulary Item in fact corresponds to different features in the syntax. In the next chapter (4.4), I will discuss comparable Algonquian data, further supporting this analysis.

With these considerations in place, I will now move to the discussion on two other cases of ADDRESSEE-driven plural allomorphy. In P'urhépecha, the exponent [chá] surfaces in the first- and third-person pronouns. Importantly, the plural exponents transparently combine with person exponents without triggering allomorphy. The same holds for the second person, where the singular pronoun [cha] is preserved in the plural. The second person, however, requires more attention. The form [cha] is used in both singular and plural, the exact interpretation depending on the context (Pérez 2006). If overt, the exponent of plural is realized as [jtsi].

These facts suggest that (i) the realization of plurality in P’urhépecha is optional in the second person, on par with optional plural exponents in the third-person plural in Yucatec Mayan and (ii) the second person triggers number allomorphy, as the resulting second person plural is not *[*cha-chá]* but *[cha-jtsi]*. The observed distribution can be explained if the **ADDRESSEE** feature is morpho-syntactically active, directly influencing the realization of the plural morphology.

	SG	PL
1	ji	ju- chá
2	cha	cha / cha- jtsi
3	indé	indé- chá

Table 3.32: Selective person-driven number allomorphy in P’urhépecha

An additional note is in order regarding the availability of the **ADDRESSEE**-driven allomorphy. The fact that a feature can act as a **trigger of allomorphy** provides direct evidence for its **morpho-syntactic activity** in the natural language. The status of the second person, i.e., the **ADDRESSEE** feature, is particularly controversial. Different approaches to semantics and morpho-syntactic representation of person features differ as to whether they allow and posit the existence of the **ADDRESSEE** feature in the general system of person (see Zwicky 1977; Plank 1985; Harley & Ritter 2002; Moskal 2018; Little 2018) or not (see Sauerland 2003; McGinnis 2005; Bobaljik & Sauerland 2023). In particular, the latter line of thought does not consider the **ADDRESSEE** feature a *primitive* of natural language. If **ADDRESSEE** is unavailable, how can it trigger allomorphy? I argue that the view that **ADDRESSEE** is not a semantic primitive, but a derived feature (see e.g., Bobaljik & Sauerland 2023), is not at odds with the finding that it can trigger allomorphy, given the timing of allomorphic alternations. At the point when the choice of exponents is made, the Vocabulary Insertion operation has available both primitives and derived features. That is, for a feature to condition allomorphy, it does not need to be a primitive of natural language, but it crucially has to be active in a given language. The results of the present study show that the **ADDRESSEE** feature is syntactically active. Namely, the sensitivity of the number node to the second person can explain the fact that a particular exponent of plural surfaces exclusively in the second person, while the first and third involve a different plural realization. These data demonstrate why and how allomorphic alternations can be informative for the general theory of features and their presence at different levels of representation.

3.3.1.2 Across-the-board allomorphy

A pronominal system in which the three person specifications all surface with distinct plural exponents is a possibility in natural language. This means that the realization of the number morpheme has to be sensitive to the presence of the first, second, and third person in the structure of a pronoun.

In Tübatulabal (Uto-Aztecan), the exponents of plural, *[u-ts]*, *[m-u]*, and *[da]*, transparently combine with the exponents of person. In this language, the first-, second-, and third-person

singular pronouns surface with the formant [ɪn], argued to be a meaningless² particle used in the formation of personal pronouns. The first person singular [nɪk] is obtained via transparent combination of the formant [ɪn] that undergoes metathesis [ɪn > nɪ] and the exponent of the first person [g], realized as [k] in the final position. In the plural, both the metathesis and the change [g > k] are absent. The shared exponent undergoes phonological change in the second person [ɪn > ɪm] when it combines with the second-person exponent [b]. Finally, the third person is realized as zero [Ø], combining with the shared formant which does not undergo any phonological changes in the context of the zero person exponent (Voegelin 1935).

As evident from the table below, there is no number-driven person allomorphy in Tübatulabal, as the exponents of person are preserved in the plural. Moving to the realization of plurality, the first person selects the exponent [u·ts]. The second person combines with an exponent [m·u], while the third with an exponent [da]. Therefore, the three person specifications combine with three different plural exponents. This can be explained if the realization of the plural is sensitive to the feature specification in the person node when it undergoes Vocabulary Insertion.

	SG	PL	
1	nɪk	ɪŋgil- u·ts	(24) VIs for PL in Tübatulabal
2	ɪmbi	ɪmb- u·mu	[PL] ↔ u·ts / ___ π _{AUTHOR}
3	ɪn	ɪn- da	[PL] ↔ u·mu / ___ π _{ADDRESSEE}
			[PL] ↔ da / ___ π

(a) Tübatulabal (Voegelin 1935)

Table 3.33: Across the board person-driven number allomorphy: Tübatulabal

To account for these data, I propose the following set of VI rules (24). At the point when the number node is targeted for insertion, if the person node contains the feature *AUTHOR* in the narrow syntax, the exponent [u·ts] is selected. If this is not the case and the person node has the feature *ADDRESSEE*, the exponent [u·mu] is inserted. When neither *AUTHOR* nor *ADDRESSEE* is present, the elsewhere exponent [da] is inserted, which is the case in the third person. This language does not employ plural morphology in the nominal domain, except in a limited number of syntactic environments such as the possessive (Voegelin 1935). In line with the data discussed above, I include π in the context of insertion for the exponent [j] to indicate that the given exponent surfaces only in the pronominal domain.

A comparable pattern is attested in Takelma (isolate; Native American). The exponents of the first, second, and third person are preserved in the plural, followed by phonological changes, which should not be considered as allomorphic alternations, given the transparent correspondence in realization. Instead, these alternations can be explained in terms of Readjustment rules (Halle & Marantz 1994) or a comparable process. However, the three pronominal forms surface with distinct plural morphology, marked bold in the following table.

²This formant can also be analyzed as an exponent of D (see the discussion in Chapter 6).

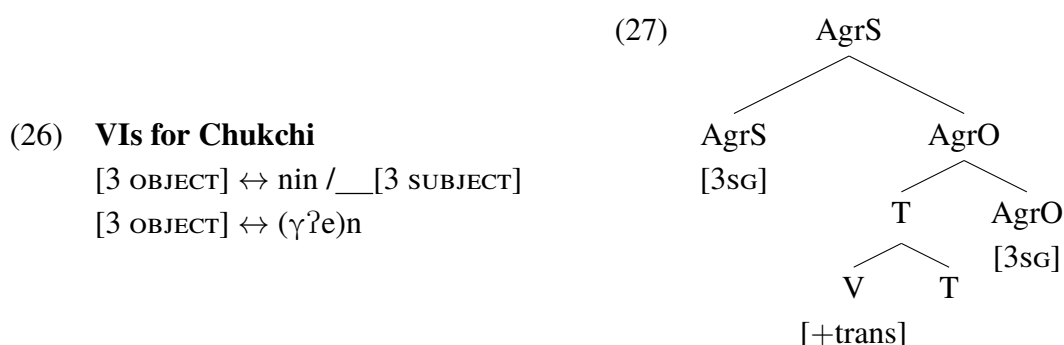
(25) **VI**s for **PL** in **Takelma**

[PL]	$\leftrightarrow$	m /	π_{AUTHOR}
[PL]	$\leftrightarrow$	p ^h /	$\pi_{\text{ADDRESSEE}}$
[PL]	$\leftrightarrow$	j /	π

Table 3.34: Across the board person-driven number allomorphy: Takelma

The empirical part of this chapter established that person-driven number allomorphy is well attested in natural language, existing alongside its directional counterpart. The availability of both directions is predicted to be impossible under Bobaljik's (2000) account of allomorphy. Namely, on the basis of the patterns of allomorphy in Chukchi agreement morphology, Bobaljik (2000) proposes that only **one direction** of allomorphy is **possible**, that is, an inward-conditioning allomorphy.

The proposal that only inward-conditioning is possible is based on two central assumptions: (i) the realization proceeds in a step-wise bottom-to-top manner targeting morphemes, (ii) when a morpheme receives phonological features in the course of the Vocabulary Insertion operation, its morpho-syntactic features are **deleted**, i.e., **overwritten**, in line with Halle (1990). Under this view, the Vocabulary Insertion is **replacive**. In the structure of the Chukchi verb (27), the subject morpheme (AgrS) merges higher than the object morpheme (AgrO). Crucially, subject morphemes trigger allomorphy of object morphemes, but not vice versa. If both the subject and the object surface with the third-person features, in the context of the third-person subject, the third-person object is realized as [nin]. When the subject is not realized with the third-person feature, the third-person object receives its elsewhere $[(\gamma?e)n]$ form (26).



As AgrO is lower in the structure, it undergoes Vocabulary Insertion prior to AgrS in the step-wise bottom-to-top model of the grammar. This means that at the point when AgrO is targeted for insertion, the features of AgrS are still available in the syntactic tree. After AgrO undergoes insertion, the morpho-syntactic features of AgrO are deleted. Therefore, when AgrS is targeted for insertion, the features of AgrO are no longer available. AgrO surfaces with the phonological features and can only lead to the phonologically-conditioned allomorphy, but not suppletion. This explains why AgrS can trigger the allomorphy of AgrO but not vice versa. The observed asymmetry motivated the **directionality condition** for allomorphy, which states that only the inward-conditioning is possible (Bobaljik 2000).

3.3.2.1 Directionality distinction makes reference to the structure

As noted above, the interaction of person and number in pronouns provides evidence for two directions of allomorphy: (i) Number-driven person allomorphy and (ii) Person-driven number allomorphy. As two morphemes occupy two distinct terminals in the underlying syntactic structure, whatever one's decision about their relative order, the two types of allomorphy necessarily give rise to outward-conditioning, i.e., inward-sensitivity and inward-conditioning, that is, outward-sensitivity. As a reminder, in outward-conditioning, the trigger of allomorphy is in a structurally lower position compared to its target. Thus, the target is inward-sensitive to the more deeply embedded trigger. This contrasts with the inward-conditioning, where the trigger is structurally outer, i.e., in a higher position than its target. In this case, the target is outward-sensitive

to its trigger.

As noted earlier in this chapter, the directionality of allomorphy has been a matter of debate in the literature (Carstairs-McCarthy 1987, 2001; Bobaljik 2000; Rolle 2023). Examples of **outward-conditioning**, i.e., inward sensitivity, figure prominently in the literature on allomorphy and include $\sqrt{\text{Root}}$ -driven allomorphy of English plural morphology and past tense. Rolle (2023) reports a set of outward-conditioning cases of allomorphy identified in the previous work, which include (i) [PERFECT] Aspect-driven allomorphy of agreement morphology in Latin (Carstairs-McCarthy 2001), (ii) gender-driven allomorphy of the definite article in Bulgarian (Gribanova & Harizanov 2017), (iii) [PROPER] feature, present on proper names and kinship nouns, triggering allomorphy of the accusative case in Moro (Jenks & Sande 2017), (iv) gender-driven allomorphy of number in Modern Standard Arabic (Winchester 2017), (v) [PERFECT] Aspect-driven allomorphy of the negation (Banerjee 2020). The opposite direction, that is, an **inward-conditioning** is exemplified by allomorphy in (i) Inuit subject and object agreement prefixes (Bobaljik 2000) and (ii) Hungarian plural morphology triggered by the outer possessive suffixes (Carstairs-McCarthy 2001).

I would like to point out that the distinction in directionality matters if formalized with reference to the hierarchical structure. Once linearized and concatenated, the inward-outward distinction vanishes. For this reason, I put forth the following correlation between the directionality and structure.

(28) DIRECTIONALITY AND STRUCTURE

Outwards- and inwards-directionality in allomorphic conditioning is a meaningful distinction **iff** the relevant condition is formalized in structural terms.

Rolle (2023) provides an important empirical generalization on directional (a)symmetries in allomorphy selection. Namely, both inward-conditioning and outward-conditioning are widely attested in grammatically-conditioned allomorphy, understood as allomorphy triggered by a morpho-syntactic feature (e.g., plural or tense), thus giving rise to a directional symmetry. This is consistent with the finding reported here on both number-driven person allomorphy and person-driven number allomorphy, as both involve feature triggers. The observed symmetry contrasts with examples of (i) lexically-conditioned allomorphy, understood as cases of $\sqrt{\text{Root}}$ -driven alternations, (ii) morphologically-conditioned allomorphy, in which a particular morphological property such as conjugation class leads to the alternation, and (iii) phonologically-conditioned allomorphy, which all show directional asymmetry, exemplifying outward-conditioning.

The trigger-target distinction and inwards-outwards conditioning tightly correlate with the Vocabulary Insertion operation. Namely, in a model where Vocabulary Insertion proceeds in a cyclic step-wise bottom-up manner (Bobaljik 2000; Embick 2010; Moskal 2015b; Kalin 2022a, 2022b; Adamson 2024), the *target* of allomorphy undergoes Vocabulary Insertion **prior** to its *trigger* under inwards-conditioning. In contrast, the *target* of allomorphy undergoes Vocabulary Insertion **after** the *trigger* in cases of outwards-conditioning.

3.3.3 Vocabulary Insertion constraining allomorphy

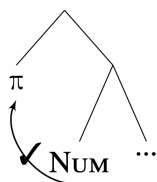
There are several ways in which Vocabulary Insertion may constrain the insertion of later nodes. The *replacive* view of Vocabulary Insertion, according to which phonological features of Vocabulary Items *replace* and *overwrite* features of the morpheme, thus being the only features available at PF (Halle 1990), yields a strong prediction for allomorphy. Namely, in a model of grammar where Vocabulary Insertion proceeds in a cyclic step-wise bottom-up manner (Bobaljik 2000; Embick 2010; Myler 2017; Kalin 2022a), outward-conditioning should be ruled out, given that the features of lower nodes that already underwent Vocabulary Insertion are no longer available to condition allomorphy of outer nodes.

The role of Vocabulary Insertion as a restrictor of allomorphy figures also in the Pruning operation (Embick 2010; see also the discussion in Marantz 2013), proposed to facilitate allomorphy by making two nodes sufficiently local. This is the case in English past tense and plural morphology, where a $\sqrt{\text{Root}}$ may condition allomorphy of the T and the Number node across categorizing heads *v* and *n*, respectively, in case they are realized as zero [$\emptyset$]. Vocabulary Insertion proceeds bottom-up from the most deeply embedded morpheme outwards, which is the $\sqrt{\text{Root}}$ in the case of verbs and nouns. After the $\sqrt{\text{Root}}$, the categorizing head is targeted. If it is realized as zero, pruning applies and leads to the recalculation of adjacency. After the categorizing head is removed, the $\sqrt{\text{Root}}$ and the T or the plural node become adjacent, enabling $\sqrt{\text{Root}}$ -conditioned allomorphy. One aspect of this operation that has not been discussed in the previous literature is that it makes a prediction on the directionality of allomorphy. Namely, pruning can make the two nodes more local only in the case of **outward-conditioning**, i.e., when the trigger of allomorphy is in a structurally lower position to its target, which is precisely the case in the past tense and plural allomorphy. Otherwise, a look-ahead problem emerges. That is, if the structurally outer node was able to condition allomorphy of the $\sqrt{\text{Root}}$, at the point when the $\sqrt{\text{Root}}$ undergoes Vocabulary Insertion, it would have to know that the categorizing head will be realized as zero. Therefore, this way of making the two nodes sufficiently local necessarily leads to a prediction on directionality when coupled with the Vocabulary Insertion operation.

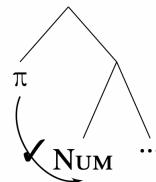
3.3.4 Why are both directions possible? Towards the analysis

The empirical part of this section established that person-driven number allomorphy is well-attested in natural language (30), existing alongside its directional counterpart (29). This fact supports the main hypothesis of this thesis that allomorphic alternations establish a window into the multi-morphemic structure of pronouns, proven by identifying the trigger and the target, i.e., two morphemes. The question that emerges is whether the two directionalities can receive a principled explanation. These data call for a theory of allomorphy that is able to accommodate allomorphic conditioning going in both directions and involving two same morphemes. Crucially, the directionality distinction cannot be reduced to the difference between grammatically-conditioned and phonologically-conditioned allomorphy (for phonologically-conditioned allo-

morphy see Paster 2006; Deal & Wolf 2017; Rolle & Bickmore 2022), as in both cases the trigger and the target are two features in the syntax, hence allomorphy is grammatically-conditioned in both of these directions.

(29) #-driven π allomorphy

outward-conditioning

(30) π -driven # allomorphy

inward-conditioning

I propose that for both directions of allomorphy to be available, it has to be ensured that at the point when number undergoes Vocabulary Insertion, person features are available to trigger allomorphy, and vice versa, when the person undergoes allomorphy, number features need to be visible. The computation of the form, i.e., the selection of the Vocabulary Item from the *Vocabulary* list of a particular language, has to take into consideration both morphemes participating in an allomorphic alternation for the two directions to emerge. I formalize this in the following terms.

- (31) Two morphemes, Ξ and Ω , may enter an allomorphic alternation, such that Ξ can trigger allomorphy of Ω and be a target of allomorphy triggered by Ω , if
- at the point when Ξ undergoes Vocabulary Insertion, features of Ω are visible
 - at the point when Ω undergoes Vocabulary Insertion, features of Ξ are visible

I propose that both types of directionality involving two same morphemes can be explained if Vocabulary Insertion makes reference to the features present in the narrow syntax, that is, features hosted on the trigger and the target nodes. If we have both the morpheme consisting of morpho-syntactic features prior to Vocabulary Insertion **and** the morpheme after it undergoes VI being supplied with the phonological content, we are armed with a larger set of possibilities to explain the mechanics of morpheme interaction at PF and the patterns of allomorphy.

This requires a particular view of the Vocabulary Insertion operation. I argue that both directions of allomorphy are explainable if phonological features are **adjoined to** morpho-syntactic and semantic features at PF. There is **no deletion** of features at PF, as the availability of both directionalities of allomorphy would be unexplainable. Person [π] and number [NUM] features are **not deleted** by the phonological features but **supplied** with them, in line with the initial idea of early DM (Halle & Marantz 1994).

Using the general schema for representing morphemes employed throughout this dissertation, I illustrate the adjoining of phonological features at PF in the following way (32). In the narrow syntax, prior to the Vocabulary Insertion operation, a morpheme consists of morpho-syntactic and semantic features (π or NUM) and an empty slot // for the phonological features (32-a), which can be inserted only once in line with the Uniqueness requirement for Vocabulary

Insertion (see Halle 1992, Embick 2015, Alexiadou 2024). Person and number morphemes are targeted in two separate cycles of Vocabulary Insertion. The Vocabulary Insertion adjoins the exponent $[x]$ to the person morpheme at PF. Likewise, it adjoins the exponent $/y/$ to the number morpheme. After Vocabulary Insertion, the features present in the narrow syntax are supplied with the phonological features, indicated by the filled slots in (32-b).

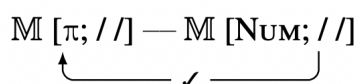
(32) *Adjoining of phonological features at PF*

- | | |
|---------------------------------------|--------------------------------------|
| a. $[\pi; / /] - [\text{NUM}; / /]$ | <i>prior to Vocabulary Insertion</i> |
| b. $[\pi; /x/] - [\text{NUM}; /y/]$ | <i>after Vocabulary Insertion</i> |

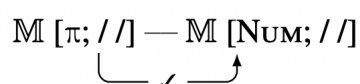
This idea can be stated in the following terms when coupled with the phonological material. In the representations below, I indicate the features present in the narrow syntax with π and NUM. The arrows indicate sensitivity. In (33), when the Number node is targeted for insertion, it looks into the person features π and not to the actual Vocabulary Item, as it carries only phonological features. The fact that I use flat representations to outline the point carries no theoretical significance. I discuss the Vocabulary Insertion in this way to indicate that the underlying hierarchical structures are not paired with the phonological material and there is no stage in the derivation when the phonological features are added to hierarchical representations in the narrow syntax, in line with the Universal Late Insertion argued for in the early DM (Halle & Marantz 1994) and the Meaning First Approach of Sauerland & Alexiadou (2020).

I propose that at the point when the number undergoes Vocabulary Insertion, person features are visible (33). The same holds for the opposite combination of trigger and target. When the person morpheme is targeted by the Vocabulary Insertion, number features are visible (34).

(33) NUM LOOKING INTO π



(34) π LOOKING INTO NUM



I focused here on the workings of the Vocabulary Insertion operation. In Chapter 6 (6.3), I develop a complete account of the availability of the two directionalities of allomorphy, taking into consideration **the structure of pronouns** for which the discussion in this and the upcoming chapters provided evidence. I make an argument that the availability of both inward-conditioning and outward-conditioning correlates with the structure, as pronouns involve a **single cycle of spell-out**. This argument is coupled with considerations about the nature of the morphemes that make up pronouns across languages and the workings of the Vocabulary Insertion operation.

3.4 Summary

This chapter discussed two studies: (i) Number-driven person allomorphy and (ii) Person-driven number allomorphy. Empirically, it recognized the following patterns in both directions: (i) *Selective allomorphy*, (ii) *Across-the-board allomorphy*, and (iii) *Absence of allomorphy*. These patterns provide an argument for the multi-morphemic approach to pronouns argued for here, as they indicate the presence of the two morphemes in the narrow syntax. In addition, this chapter brought the study on the person-driven number allomorphy in pronouns. The availability of both types of directionality raised a number of relevant questions for the theory of allomorphy.

Chapter 4

Allomorphy in Languages with Clusivity and Minimal~Augmented Distinction

4.1 Introduction

The inclusive~exclusive distinction has been a matter of an extensive debate in formal semantics (Sauerland 2008a; Bobaljik & Sauerland 2023; McGinnis 2005), general accounts of features and feature representations in natural language (Zwicky 1977; Noyer 1992; Harley & Ritter 2002; Bobaljik 2008a; Harbour 2016), typological investigations (Cysouw 2003; see also contributions in Filimonova 2005), and morphologically-oriented work, focused on the role of the containment-based representation of features and its implications for syncretism in pronominal paradigms (Moskal 2018, Pertsova 2022a).

The clusivity parameter is exemplified in Table 4.1 on the basis of Kuvi (Dravidian) and Wandala (Chadic). In these languages, the first-person plural pronoun has two different morphological realizations, depending on whether the second person is included in the referent of the pronoun. If it is, the *inclusive* form is realized. This contrasts with languages that do not mark clusivity, where both meanings are associated with a single form, i.e., the meaning of the inclusive is realized with the first-person plural pronoun. Therefore, from the perspective of an inventory of pronouns within a language, languages with clusivity have an additional pronominal form compared to those that do not mark this distinction in the morphology.

	SG	PL
1 _{EXCL}	nānu	māmbu
1 _{INCL}		māro
2	nīnu	mīmbu
3	tānu	tāmbu

(a) Kuvi (Steever 2020c)

	SG	PL
1 _{EXCL}	íyyà	ɲrè
1 _{INCL}		míyà
2	əkká	kùré
3	ɲánè	ítàrè

(b) Wandala (Frajzyngier 2012)

Table 4.1: The inclusive~exclusive distinction

The inclusive~exclusive distinction represents an empirical domain in which a number of

outstanding questions central to morphological and syntactic theory can be raised. Specifically, considerations relevant for clusivity pertain to questions about the size and internal composition of the morpheme in the narrow syntax, the mapping of form and meaning, the transparency of the correspondence between morphemes and exponents, the targets of Vocabulary Insertion operation, and the nature of the competition in grammar, to name a few.

In the present work, I address the following questions that have not been investigated on the basis of the inclusive~exclusive distinction in the literature before.

(1) QUESTIONS

- a. How does the clusivity parameter inform the structure of pronouns?
- b. How do the features involved in the clusivity parameter relate to the status of a morpheme as a basic unit of syntactic computation?
- c. Does the inclusive exponent relate to a single morpheme or two morphemes (terminals) in the narrow syntax?
- d. What are the targets of the Vocabulary Insertion operation in inclusive pronouns?

In essence, answering these questions can illuminate one of the central questions of morphological theory and the morphology-syntax and morphology-semantic interfaces, namely, what kinds of morphemes are available within and across languages.

This chapter presents two studies. The first study provides a detailed examination of the patterns of allomorphy and comparable morphological alternations in languages with the clusivity distinction. I present the formal analysis in terms of a **single morpheme** hosting both person features in the inclusive, thus placing the clusivity parameter in the narrow syntax without stretching the internal structure of a pronoun. Under this proposal, the **structure** of pronouns is **uniform**. That is, both inclusive and exclusive, as well as all the other pronouns in languages that do not mark clusivity, have a morpheme hosting person features. The featural content of this morpheme is the locus of the variation. In addition, the insertion targeting only terminal nodes is preserved without relying on any additional operations, thus replicating the same insertion mechanism that holds elsewhere in natural languages.

The second study focuses on languages with the so-called MINIMAL~AUGMENTED distinction. I propose that the *minimal inclusive* pronoun, which motivated the recognition of this system in natural languages, does not exist as a primitive but is compositionally derived from morphemes that exist independently in the natural language. Namely, it consists of the (i) **inclusive morpheme**, present in languages with clusivity, and (ii) the **singular** number.

The chapter is organized as follows. In Section 4.2, I provide a detailed discussion of the morphological realization of pronouns in languages with clusivity, covering both directions in allomorphic conditioning. In Section 4.3, I lay out the theoretical proposal in terms of the minimal inclusive morpheme. In the next step (4.4), I move to the **you* paradigmatic gap, revisiting it from the perspective of allomorphy. Section 4.5 presents the examination of analytic inclusives, understood here as forms where the first and second person are overtly realized in the in-

clusive. Section 4.6 focuses on the study of languages surfacing with the MINIMAL~AUGMENTED systems. Section 4.7 summarizes.

4.1.1 Data

The study is based on 118 languages that cover 32 language families and 6 isolates. An overview of the data is provided in the following table. As was the case in the previous two studies discussed in Chapter 3, when a particular language family can be divided into subfamilies, this information is included in the table. In the present dataset, this applies to Altaic (Tungusic), Australian (Bunuban, Iwaidjan, Pama-Nyungan, Tangkic, West Barkly, Wororan), and Papuan (Solomons East Papuan, Trans-New Guinea, Eleman). All languages investigated in this study are included in Appendix B.

FAMILY	LANGUAGE
Afro-Asiatic	Hdi, Mina, Wandala, Somali, Lamang
Algonquian	Ojibwe, Nishnaabemwin, Passamaquoddy
Altaic ^{Tungusic}	Udihe, Evenki
Arawak	Alto Perené
Australian	Kunbarlang, Ngandi; Wardaman
Australian ^{Bunuban}	Gooniyandi
Australian ^{Iwaidjan}	Maung
Australian ^{Pama-Nyungan}	Kuuk Thaayorre, Yanyuwa; Ngumpin-Yapa (Ngardi)
Australian ^{Tangkic}	Kayardild
Australian ^{West Barkly}	Wambaya
Australian ^{Wororan}	Worrorra, Ungarinjin
Austronesian	Oceanic (Belep; Daakaka), Kambera, Kilivila, Toqabaqita, Malagasy
Austro-Asiatic	Ho, Semelai, Temiar
Cariban	Pemón
Chibchan	Bribri, Ika
Choco	Embera
Dravidian	Betta Kurumba, Kolami, Konda, Kuvi, Malayalam, Tamil, Telugu
Kxa	Ju 'hoan
Mayan	Tojolabal
Misumalpan	Miskito, Sumu
Mixe-Zoque	Mixe (Ayutla)
Nilo-Saharan	Kunama
Sino-Tibetan	Mongsen Ao, Bunan, Dumi, Karbi, Limbu, Dolakha Newar, Akha
Panoan	Kakataibo
Papuan	Bunaq, Teiwa
Papuan ^{Solomons East Papuan}	Lavukaleve, Savosavo
Papuan ^{Trans-New Guinea}	Mian, Abui, Oksapmin
Papuan ^{Eleman}	Orokolo, Toaripi (Sepoe)
Peba-Yaguan	Yagua
Penutian	Sierra Miwok
Tucanoan	Cubeo, Siriano
Tupian	(Xingu) Asuriní, Sirionó
isolates	Vaeakau-Taumako, Movima, Nivkh (Amur), Tiwi, Kwaza, Urarina

4.2 The patterns

4.2.1 Absence of allomorphy

An exponent of the first-person morpheme employed in the singular may be shared with both exclusive and inclusive forms, thus resulting in the absence of allomorphy. In Kuuk Thaayorre (Pama-Nyungan), an exponent [*nga*] remains invariant. Urarina (isolate; spoken in North-western Peru) employs an exponent [*kan*]. Ngardi (Ngumpin-Yapa) surfaces with an exponent [*nga*], while Limbu (Sino-Tibetan), with [*aŋlan*]. Tamil (Dravidian) exemplifies a larger Dravidian pattern, where the first and second person share the initial formant [*n*], being differentiated on the basis of [*a/i*]. Notably, [*a*] remains intact. In Mian (Papuan), [*n*] is invariant.

	SG	PL
1EXCL	ngay	ngancn
1INCL		ngamp
2	nhunt	nhurr
3	nhul	peln

(a) Kuuk Thaayorre (Gaby 2017)

	SG	PL
1EXCL	kan _H	kanakaana
1INCL		kana
2	ii	ɟaara
3	aka	aka _H _H

(b) Urarina (Olawsky 2011)

Table 4.2: Absence of allomorphy

	SG	PL
1EXCL	ngaju	nganimpa
1INCL		ngali(m)pa
2	nyuntu	nyurrurla
3	nyantu	nyanungu

(a) Ngardi (Ennever 2021)

	SG	PL
1EXCL	aŋa	anige
1INCL		ani
2	khɛnɛʔ	khɛni
3	khunɛʔ	khɛŋhaʔ
3INANIM	khɛŋ	khɛŋhaʔ

(b) Limbu (Van Driem 2011)

Table 4.3: Absence of allomorphy

	SG	PL
1EXCL	nān	nānkaḷ
1INCL		nām
2	nī	nīnkaḷ
3M/F	avan/avaḷ	avar

(a) Tamil (Annamalai & Steever 2020)

	SG	PL
1EXCL	nē	nī
1INCL		nībo
2M	kēbo/kōbo	ībo
2F	ōbo	ībo
3M/F	ē/ō	ī

(b) Mian (Fedden 2011)

Table 4.4: Absence of allomorphy

4.2.2 Inclusive exponent

Allomorphy restricted to first-person inclusive pronouns is of particular interest. The emergence of the inclusive allomorph can be traced to two sources: (i) the person morpheme itself, and (ii) the number morpheme. As illustrated below on the basis of Adang (Papuan) and Udihe (Tungus; Altaic), the exponent of the first person morpheme [*n* in Adang; *b* in Udihe] remains constant in the exclusive. In contrast, inclusive forms involve a clear alternation. In Adang, the allomorph of person [*p*] transparently combines with the number-morphology [*i*], while in Udihe the allomorph [*min*] combines with the plural exponent [*ti*], which also surfaces with third-person pronouns.

	SG	PL
1EXCL	na	ni
1INCL		pi
2	a	i
3	sa	supi

	SG	PL
1EXCL	bi	bu
1INCL		minti
2	si	su
3	nuani	nuati

(a) Adang (Robinson & Haan 2014) (b) Udihe (Nikolaeva & Tolskaya 2011)

Table 4.5: Inclusive exponent

The same pattern is attested in other typologically distinct languages, such as Akha (Tibeto-Burman) and Nivkh (Amur dialect; Nivkh family). Akha data exemplify the transparent morphological compositionality, since singular forms of all person specifications are preserved in the plural, combining with the plural exponent [*mà*]. An exception in this respect is the inclusive form, realized as [*àdɯ*], thus clearly contrasting with the first person exponent [*ŋá*], present in the singular and the exclusive. Therefore, these data demonstrate that an observable morphological alternation in the language emerges in the inclusive. A comparable situation is present in Nivkh. The exponent of the first person [*nji*] is preserved in the exclusive, combining with the exponent of the plural [*ŋ*]. This patterning contrasts sharply with the inclusive form [*mer*; *mir*], which not only shows a different realization but also does not take the plural exponent [*ŋ*] present elsewhere in the paradigm.

	SG	PL
1EXCL	ŋá	ŋá mà
1INCL		àdɯ mà
2	ŋó	ŋó mà
3	àjɔ̌	àjɔ̌ mà

	SG	PL
1EXCL	nji	njiŋ
1INCL		mer; mir
2	chi	chiŋ
3	if	imŋ

(a) Akha (Pertsova 2022b)

(b) Nivkh (Pertsova 2022b)

Table 4.6: Inclusive exponent

Dumi (Sino-Tibetan) and Kakataibo (Panoan) exhibit the same morphological pattern. In Dumi, the exponent of the first person [*aŋ*], present in the singular, is preserved in the exclusive,

combining with the plural exponent [ki]. However, the inclusive form is realized as [iŋ], combining with the same plural exponent. In Kakataibo, the exponent of the first person is shared between the singular and the exclusive [‘ë], while the inclusive form [nu] shows a clear morphological alternation. Note that the formant [n] is an exponent of the case. Once removed, it becomes obvious that the exponents of the first-, second-, and third-person are preserved in the plural, transparently combining with the plural exponent [kama]. An exception in this respect is the realization of person in the inclusive.

	SG	PL		SG	PL
1 _{EXCL}	aŋ	aŋki	1 _{EXCL}	‘ë-n	‘ë-kama-n
1 _{INCL}		iŋki	1 _{INCL}		nu-kama-n
2	an	ani	2	mi-n	mi-kama-n
3	im	hammīl	3	a-n	a-kama-n

(a) Dumi (Van Driem 1993)

(b) Kakataibo (Zariquiey 2018)

Table 4.7: Inclusive exponent

Alto Perené (Kampa; Arawak) data is particularly illustrative for studying the ways morphological alternations emerge in languages with the inclusive~exclusive distinction. The exponent of the first person [naa] remains unaffected by the plural in the exclusive. However, in the inclusive, the person morpheme is realized as [aro]. It is important to note that all person specifications surface with the same exponent of plural [ite]. Furthermore, the first- and second-person surface with the formant [ka] in the singular, which is preserved in their plural counterparts. If the allomorph [aro] is triggered by the plural, then these data exemplify run-of-the-mill violation of the linear adjacency condition on allomorphy, as the alternation is present across an overt [ka]. Therefore, the allomorph [aro] can be explained either as a non-local allomorphy triggered by number or a result of a process different from allomorphy.¹

	SG	PL
1 _{EXCL}	naa-ka	naa-ka-ite
1 _{INCL}		aro-ka; aro-ka-ite; aro-r-ite
2	aviro-ka	aviro-ka-ite
3 _M	iri(ro)	iriro-ite
3 _{NON.M}	iro	iro-ite

Table 4.8: Inclusive exponent: Alto Perené (Mihás 2015)

¹Decomposition can be pushed even further. As evident from a closer inspection of the second and third person pronouns, a formant [ro] can be individuated. The same is preserved in the plural and can, in principle, be recognized in the inclusive form [aro] under a decomposition [a-ro]. However, individuation of this morpheme does not change the fact that the exponent [a] emerges in the inclusive, contrasting with [naa], present in the singular and exclusive forms. In addition, recognizing [ro] as a separate exponent increases the linear distance between [a] and the plural [ite], which would be the case of a long-distance allomorphy across two overt exponents.

4.2.3 Exclusive and inclusive exponents

The same allomorph of person, shared between the exclusive and inclusive, is attested in Dravidian languages Konda and Kuvi. In contrast to Tamil, where [nā] is preserved in the exclusive and inclusive pronouns, in Konda and Kuvi it changes to [mā]. Furthermore, in Kuvi, the first person exclusive, second, and third person share the same exponent of person [mbu], which is absent in the inclusive, thus exhibiting person-driven number allomorphy. The fact that the exclusive and the inclusive surface with two different allomorphs of plural suggests that the syncretic exponent of person realizes two different morphemes in syntax. The same syncretism of person in the two pronouns is attested in Mina (Chadic), Kwaza (Amazonian), Orokolo (Trans-New Guinea), and Lamang (Afro-Asiatic).

	SG	PL
1EXCL	nān(u)	māp (u)
1INCL		māṭ (u)
2	nīn(u)	mīr(u)
3PROX	i-di	i-vi
3REM	a-di	a-vi

(a) Konda

(Krishnamurti & Benham 2020)

	SG	PL
1EXCL	nānu	māmbu
1INCL		māro
2	nīnu	mīmbu
3	tānu	tāmbu

(b) Kuvi (Steever 2020c)

Table 4.9: Exclusive and inclusive allomorphs

	SG	PL
1EXCL	sə	na
1INCL		nók
2	ha	hi
3	∅, a	i

(a) Mina (Frajzyngier 2011)

	SG	PL
1EXCL	si	tsi'tsɛ
1INCL		txa'na
2	xyi	xyi'tsɛ
3	ĩ	ĩ

(b) Kwaza (Voort 2004)

Table 4.10: Exclusive and inclusive allomorphs

	SG	PL
1EXCL	ara	ela
1INCL		elavi:la
2	a	e
3	are	ere

(a) Orokolo (Pertsova 2022b)

	SG	PL
1EXCL	ìjò	nàjìn
1INCL		nàmwà
2	kàyà	kàyèni
3	nètʔè	nàxáj/nàxájí

(b) Lamang(Pertsova 2022b)

Table 4.11: Exclusive and inclusive allomorphs

4.2.4 Across-the-board alternation

A person morpheme in the first person singular, first person exclusive, and first person inclusive can be mapped to three different exponents. In Bunan (Sino-Tibetan), all pronouns surface with the same plural exponent [ɕi]. The exponents of person in the exclusive [hiŋ] and inclusive [eraŋ] are clearly identifiable and differ from each other and from the singular [gi], thus constituting three different allomorphs of the first-person morpheme. The same presence of the three allomorphs of person is attested in Wandala (Chadic; Afro-Asiatic), Daakaka (Oceanic; Austronesian), Belep (Oceanic; Austronesian), Vaeakau-Taumako (isolate; Polynesian outlier), and Kambera (Austronesian). In addition, Kambera involves a pattern that will be addressed in Chapter 6. All pronouns surface with the formant [nyu; and its phonologically-driven alternant nyi]. Being shared across the system, this formant cannot be an exponent of person and number features. Once removed, the resulting [ngga], [ma], and [ta] are allomorphs.

	SG	PL
1EXCL	gi	hin; hiŋ=ɕi
1INCL		eraŋ; eraŋ=ɕi
2	han	han=ɕi
3	tal	tal=ɕi

(a) Bunan (Widmer 2017)

	SG	PL
1EXCL	íyyà	ŋrè
1INCL		míyà
2	əkká	kùré
3	ŋánè	ítàrè

(b) Wandala (Frajzyngier 2012)

Table 4.12: Across the board alternation

	SG	PL
1EXCL	nye	kinyem
1INCL		ar; er
2	ngok	kimim
3	nge	nyoo

(a) Daakaka (Von Prince 2015)

	SG	PL
1EXCL	nao	ava
1INCL		ja
2	yo	ac
3	yer	la

(b) Belep (McCracken 2012)

Table 4.13: Across the board alternation

	SG	PL
1EXCL	iau	mha-tou
1INCL		tha-tou
2	koe	khou-tou
3	ia	lha-tou

(a) Vaeakau-Taumako (Næss & Hovdhaugen 2011)

	SG	PL
1EXCL	nyu- ngga	nyu- ma
1INCL		nyu- ta
2	nyu-mu	nyi-mi
3	nyu-na	nyu-da

(b) Kambera (Klamer 1998)

Table 4.14: Across the board alternation

4.2.5 Exclusive exponent

The final empirical possibility for realizing the person morpheme in the three cells under consideration is the one where a special allomorph of the first person is employed in the exclusive, while the singular and inclusive share the same exponent. This is the least frequent pattern in my sample, attested in only 5 out of 115 languages. This pattern of allomorphy is controversial, as under a containment-based approach where the exclusive is structurally contained by the inclusive (see Moskal 2018), it results in an *ABA pattern, which should be ruled out by the grammar (Bobaljik 2012). Moskal notes that two languages in her sample, Dolakha Newar and Mangarrayi, appear to give rise to *ABA. Furthermore, in a recent cross-linguistic study on the patterns of syncretism in clusivity, Pertsova (2022a) suggests that this pattern is not exceptional in any respect compared to other attested realizations and thus should be considered on par with others. In the theoretical part of this chapter, I will address why it is problematic to analyze the triple 1SG-1PL.EXCL-1PL.INCL on par with POSITIVE-COMPARATIVE-SUPERLATIVE, the approach entertained in Moskal (2018). At the same time, I will demonstrate that many examples argued to instantiate the *ABA in Pertsova (2022a) should not be analyzed this way. Before that, I will introduce two patterns attested in my sample that seem to provide evidence for allomorphy limited to the exclusive.

I provide data from two languages, Betta Kurumba (Dravidian) and Wardaman (Australian; F. Merlan 1994), which have not been discussed in the previous literature. In Betta Kurumba, the first person in the singular is realized as [na]. The same exponent is present in the first person inclusive. However, the exclusive form employs an allomorph [ya]. This pattern is particularly telling as both inclusive and exclusive forms combine with the same plural exponent [ŋgə]. The same plural marker is present in the second person, and arguably in the third [gə]. The same pattern emerges in a typologically unrelated language, Wardaman. While the first-person singular and the inclusive share the same exponent of person [nga], in the exclusive pronoun, person is realized as [yi]. Importantly, both inclusive and exclusive combine with the same transparent exponent of plural [rrug], shared with the second person, facilitating identification of person exponents.

	SG	PL
1EXCL	navə	ya-ŋgə
1INCL		na-ŋgə
2	niyə	ni-ŋgə
3PROX	idə	i-gə
3REM	adə	a-gə

(a) Betta Kurumba (Coelho 2020)

	SG	PL
1EXCL	ngayugu	yi-rrug
1INCL		nga-rrug
2	yinyang	nu-rrug
3	narnaj	narnaj-bulu

(b) Wardaman (F. C. Merlan 2011)

Table 4.15: Exclusive allomorph

Dolakha Newar data will be independently discussed in the upcoming section. I include it here for completeness, as this pattern was noted in Moskal (2018) as a case of *ABA. Namely,

the exponent of the first person [*ji*] emerges in the inclusive [*chiji*], being absent in the exclusive [*isi*]. Furthermore, additional data giving rise to *ABA include Mangarrayi, discussed in Moskal (2018) and Pertsova (2022b), and Pemón, noted in Pertsova (2022b). Starting with Pemón, note that the exponent of the first person [*(j)u*] is preserved in the inclusive, along with the formant [*re*], combining with the plural exponent shared with the second person [*nokon*]. This contrasts with the exclusive form [*ina*], which involves allomorphy. I suggest that the account of the Pemón data has to take into consideration the realization of the formant [*re*]. Namely, in the exclusive, it is not only the case that allomorphy of the person morpheme applies but [*re*] is affected too, as it disappears.

	SG	PL		SG	PL
1EXCL	ji	isi	1EXCL	(j)u-re	ina
1INCL		chiji	1INCL		(j)u-re-to; (j)u-re-nokon
2	chi	chipen	2	ama-re	ama-re-nokon
3PROX	u	upen	3	mue-re; mese-re	to; itʃanan; mesemonan
3DIST	ām(u)	āpen			

(a) Dolakha Newar (Genetti 2007)

(b) Pemón (Pertsova 2022b)

Table 4.16: Exclusive allomorph

The three exponents that give rise to *ABA in Mangarrayi [*ŋa-ŋi-ŋa*] deserve closer attention. First, they involve a morpho-phonological overlap, as evidenced by the shared formant [*ŋ*]. As noted in the introductory chapter and emphasized throughout, shared morpho-phonological material cannot be ignored. Phonologically-driven alternation can be excluded, as the same exponent of plural [*a*] is shared between the exclusive and the inclusive. Therefore, the question is why would it trigger a phonological alternation of the same vowel in the exclusive but not in the inclusive. If we isolate [*ŋ*] as the exponent of the first person, then it remains preserved across the three cells.

	SG	PL
1EXCL	ŋa -ja	ŋi- a
1INCL		ŋa - a
2	ŋa-ŋgi	ŋu- a

Table 4.17: Pronouns in Mangarrayi (Pertsova 2022b)

As noted above, Pertsova (2022a) argues that *ABA patterns are freely available in languages with clusivity. However, a closer examination of pronominal systems argued to involve this pattern may receive a different account. Seven languages are proposed to surface with this pattern: Worrorra, Puquina, Motuna, Ngandi, Nunggubuyu, Panare, and already discussed above Pemón and Mangarrayi.

In the analysis of Motuna, Pertsova (2022a) argues that the inclusive surfaces with an exponent of plural [ee], shared with the second person, while the exclusive is analyzed as a non-segmentable form [noni]. I propose a different analysis. Namely, [n] is the exponent of the first person, transparently identifiable in both singular and plural forms. As established in the previous chapter, the person-driven number allomorphy is present in natural language. Therefore, the realization of plurality in Motuna involves the competition of the two allomorphs [oni] and [ee]. There are, in principle, two ways to account for the trigger of allomorphy in this case. In one analysis, the ADDRESSEE feature, shared between the inclusive and the second person, triggers the allomorph [ee], while [oni] is an elsewhere. Alternatively, the AUTHOR feature triggers the allomorph [oni]. In the inclusive, the presence of the ADDRESSEE feature intervenes and blocks the AUTHOR-driven allomorphy of number. Consequently, the elsewhere form [ee] is realized both in the inclusive and second person. In both accounts, the exponent of person [n] remains intact. Thus, this pronominal system surfaces with the absence of allomorphy.

	SG	PL
1EXCL	n-i; n-ii	n-oni
1INCL		n-ee
2	ro; roo	r-ee

Table 4.18: Pronouns in Motuna (Pertsova 2022b)

4.2.6 Inclusive-driven number allomorphy

In the previous chapter, I established that natural language provides evidence for person-driven number allomorphy. As expected, languages with clusivity show the same patterns in this directionality in allomorphic conditioning. Consider the following example from Dumi (Sino-Tibetan), in which the first-person exclusive and inclusive pronouns surface with the plural exponent [ki]. This pattern can be explained if the presence of the first person, i.e., [AUTHOR] feature, conditions the insertion of the special allomorph of the plural. When the [AUTHOR] feature is absent, the context for insertion of the allomorph under consideration is not met. Hence, it is absent in the second and third person pronouns.

	SG	PL
1EXCL	aŋ	aŋ- ki
1INCL		iŋ- ki
2	an	an- i
3	im	hammɪl

(2) **VI for PL in Dumi**
 [PL] $\leftrightarrow$ ki / ___ π_{AUTHOR}

(a) Dumi (Van Driem 1993)

Table 4.19: Author-driven number allomorphy: Dumi

Of particular importance for the present discussion is a pattern in which the **inclusive person morpheme triggers allomorphy of the plural**. Namely, if there is an exponent of plural in the inclusive that is different from exponents of plural present with the exclusive and the second person, this means that the inclusive head consisting of both features simultaneously acts as a trigger of allomorphy. This empirical point has direct consequences for the theoretical formalization, demonstrating that the inclusive acts as a single head at the point when the choice of plural exponents is made. Furthermore, it shows that the inclusive head is a morpho-syntactically real object, as it triggers different morphological realizations of the plural morpheme than its subparts would have triggered in isolation. I now turn to the data attested in my sample.

INCLUSIVE-driven number allomorphy. In a pronominal system making the inclusive~exclusive distinction, the first-person inclusive pronoun may surface with a unique exponent of plural that is not employed by other person specifications. This is transparently exemplified in Kuvi (Dravidian). The plural exponent *[mbu]* surfaces in the first-person plural exclusive, second-person plural, and third-person plural pronouns. In contrast, the first-person plural inclusive pronoun selects the plural exponent *[ro]*. To account for these data, I argue that the inclusive form triggers allomorphy of the plural morpheme, giving rise to *[ro]*, while *[mbu]* is an elsewhere. In the Vocabulary Insertion rules (4), I indicate the inclusive head as $[1+2]$ for the ease and clarity of exposition. A closer examination of the inclusive~exclusive minimal pair, *[māmbu~māro]* shows that the person exponent is realized in the same way, *[mā]*, selecting for two distinct plurals. This pattern can be explained in terms of person-driven number allomorphy and underspecification of person exponents, as will be proposed for Algonquian.

	SG	PL
1EXCL	nānu	mā- mbu
1INCL		mā- ro
2	nīnu	mī- mbu
3	tānu	tā- mbu

(a) Kuvi (Steever 2020c)

(3) **VIs for PL in Kuvi**
 $[PL] \leftrightarrow ro / __ \pi_{1+2}$
 $[PL] \leftrightarrow mbu$

Table 4.20: Inclusive-driven number allomorphy: Kuvi

A comparable pattern is present in Kunbarlang (Australian; Aboriginal), where the inclusive surfaces with the plural exponent *[rrka]*, while the elsewhere form *[dbe]* is spread across all other person specifications. As in Kuvi, this distribution of number morphology can be explained if the inclusive person head acts as a trigger of allomorphy. Therefore, the realization of number morphology is sensitive to three different feature specifications in the person node.

	SG	PL
1 _{EXCL}	ngayi (nganj-)	nga- dbe
1 _{INCL}		nga- rrka
2	ngudda (nji-)	nuku- dbe (ngu- dbe -)
3 _(I)	nukka	be- dbe /kikka
3 _(II)	kikka	be- dbe /kikka

(a) Kunbarlang (Kapitonov 2021)

(4) **VIs for PL in Kunbarlang**[PL] $\leftrightarrow$ rrka / $__\pi_{1+2}$ [PL] $\leftrightarrow$ dbe

Table 4.21: Inclusive-driven number allomorphy: Kunbarlang

In Kwaza (isolate), the first-person inclusive pronoun surfaces with the plural exponent [na], while the exclusive and the second person combine with the plural exponent [tsɛ]. Note that the third person exhibits syncretism in both singular and plural numbers. Patterning with Kuvi and Kunbarlang, the distribution of plural exponents receives a straightforward explanation when approached from the perspective of allomorphy. **The combination of the first- and second-person features gives rise to the special allomorph of the plural.** When only one of them is present, i.e., the subset of the features of the inclusive, either first in the exclusive or the second in the second person plural, the elsewhere form is chosen.

	SG	PL
1 _{EXCL}	si	tsi'- tsɛ
1 _{INCL}		txa'- na
2	xyi	xyi'- tsɛ
3	ĩ	ĩ

(a) Kwaza (Voort 2004)

(5) **VIs for PL in Kwaza**[PL] $\leftrightarrow$ na / $__\pi_{1+2}$ [PL] $\leftrightarrow$ tsɛ

Table 4.22: Inclusive-driven number allomorphy: Kwaza

In Tamil (Dravidian), the inclusive head does not surface with the elsewhere exponent of plural [ka!], present both with the exclusive and the second person. Therefore, the realization of plurality in the inclusive again shows a different pattern than the rest of the paradigm.

	SG	PL
1 _{EXCL}	nān̄	nān̄- ka!
1 _{INCL}		nām
2	nī	nīn̄- ka!
3 _M	avan̄	avar
3 _F	ava!	avar

(a) Tamil (Annamalai & Steever 2020)

(6) **VIs for PL in Tamil**[PL] $\leftrightarrow$ $\emptyset$ / $__\pi_{1+2}$ [PL] $\leftrightarrow$ ka!

Table 4.23: Inclusive-driven number allomorphy: Tamil

In Miskito (Misumalpan), the first-person exclusive, second-person, and third-person pronouns share the same plural exponent, i.e., [*nani*], to the exclusion of the first-person inclusive that triggers the special allomorph [*won*]. Note that the exponents of person present in the singular are transparently preserved in the plural. The alternation in the realization of the first person [*jaŋ/ja:*] can be considered phonological in nature due to the substantial overlap.

	SG	PL
1EXCL	jaŋ	jaŋ nani
1INCL		ja:- won
2	man	man nani
3	witin	witin nani

(a) Miskito (Pertsova 2022b)

(7) **VIs for PL in Miskito**

[PL] $\leftrightarrow$ won / $__\pi_{1+2}$

[PL] $\leftrightarrow$ nani

Table 4.24: Inclusive-driven number allomorphy: Miskito

A slightly more complex realization is attested in Mina (Chadic; Afro-Asiatic). In this language, the exclusive, inclusive, and second-person plural pronouns surface with plural exponents [*a*], [*ók*], and [*i*], respectively. The analysis of the distribution of plural morphology in terms of allomorphy is provided in (9). In the context of the inclusive person morpheme, consisting of both the first- and second-person, the plural exponent [*ók*] is inserted. The exponent [*a*], which requires the presence of the first person, is inserted in the exclusive. Finally, the elsewhere [*i*] is chosen when the context specification is not met, which correctly derives its presence in the second- and third-person pronouns. Note that the third-person pronoun consists only of the overt plural exponent.

	SG	PL
1EXCL	sə	n- a
1INCL		n- ók
2	ha	h- i
3	∅, a	i

(a) Mina (Frajzyngier 2011)

(8) **VIs for PL in Mina**

[PL] $\leftrightarrow$ ók / $__\pi_{1+2}$

[PL] $\leftrightarrow$ a / $__\pi_1$

[PL] $\leftrightarrow$ i

Table 4.25: Inclusive-driven number allomorphy: Mina

Finally, Evenki illustrates the simultaneous presence of both PARTICIPANT-driven and inclusive-driven allomorphy. Namely, the formants [*b*] and [*s*] can be individuated as exponents of the first and second person, respectively, with [*u*] being an exponent of the plural. In line with the data discussed in the previous chapter, shared plural morphology between the first and second person is analyzed here as PARTICIPANT-driven number allomorphy. The inclusive form, however, behaves differently. The exponent [*u*] is absent, and the realization of the person shows allomorphy, as the default [*b*], shared between the singular and the exclusive, is absent. In principle, there are two ways to account for the realization of the inclusive form: (i) the person morpheme is realized

as $[m]$ combining with the plural allomorph $[it]$ or (ii) the person morpheme is realized as $[mit]$ combining with the $[\emptyset]$ allomorph of plural. Case paradigm suggests individuation of $[m]$ as an exponent of the first person (Table 4.26), thus supporting the first option, especially in light of the compositionality argued for in Chapter 2. Evenki paradigm is particularly important because it shows that the **inclusive** person head, that consists of the first and second person, is crucially **different** from the mere **PARTICIPANT** feature.

	SG	PL	
1EXCL	bi	bu	(9) VIs for PL in Evenki
1INCL		mit	
2	si	su	
3	nuŋan	nuŋartin	

(a) Evenki (Pertsova 2022b)

$[PL] \leftrightarrow it / __ \pi_{1+2}$
 $[PL] \leftrightarrow u / __ \pi_{PART}$
 $[PL] \leftrightarrow r(tin)$

Table 4.26: Inclusive-driven number allomorphy: Evenki

Allomorphy demonstrates that the realization of the plural morpheme recognizes the inclusive morpheme as different from the **PARTICIPANT** feature. Therefore, I propose the following set of VI rules for plural in Evenki pronouns (10). In the context of a more specific inclusive morpheme, the allomorph $[it]$ is inserted. When the inclusive morpheme is absent in the structure, the **PARTICIPANT** feature in the person morpheme triggers the realization of the allomorph $[u]$. Finally, the realization of plurality in the third person replicates the realization of plural in nouns with stems ending in $[n]$, that combine with the plural exponent $[r]$ followed by the formant $[tin]$, e.g., $[oron/oro-r]$ ‘deer’ (Bulatova & Grenoble 1999). Therefore, the plural exponent in the third person is an elsewhere.

	SG	PL.EXCL	PL.INCL
NOM	bi	bu	mit
ACC	minə	munə	mitpə
DAT	mindu:	mundu	mittu:

Table 4.27: Evenki first person pronoun (Bulatova & Grenoble 1999)

4.3 The Inclusive Morpheme $\mathbb{M}_{incl}$

From the perspective of morphological theory, one of the key questions is what kinds of morphemes are available. The clusivity distinction establishes an empirical area where it can be successfully examined. I suggest that the phenomenon of clusivity and the existence of an additional first-person inclusive pronoun primarily relate to the question of morphemes and what kinds of morphemes are possible in individual languages. Once the morpheme is carefully examined, relevant implications for the structure of pronouns arise. In the present work, I develop an argument for the following proposal (10).

(10) THE INCLUSIVE MORPHEME $\mathbb{M}_{incl}$

Languages with clusivity differ from languages that do not mark this distinction in the availability of a morpheme that hosts both first- and second-person features.

Under this proposal, the clusivity parameter is placed in the syntax (see Bobaljik (2024) for additional evidence that clusivity is a syntactic parameter on the basis of composed clusivity in Itelmen) and crucially accounted for from the perspective of a morpheme, as a basic unit of syntactic computation, that individual languages may generate. I define the inclusive morpheme as follows.

(11) THE INCLUSIVE MORPHEME $\mathbb{M}_{incl}$

def. The inclusive morpheme $\mathbb{M}_{incl}$ is an atomic unit of syntactic computation, consisting of two features

- a. $\mathbb{M}$ is a terminal node
- b. $\mathbb{M}$ is a target of the Vocabulary Insertion operation
- c. $\mathbb{M}$ may trigger allomorphy of the number morpheme

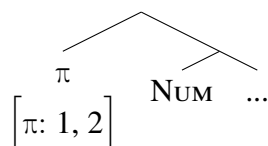
The core property of $\mathbb{M}$ is being a *morpheme*, that is, a basic atomic unit of syntactic computation that combines with other morphemes to build larger structures. Being a morpheme, $\mathbb{M}$ is a terminal node in the larger syntactic object in which it surfaces, i.e., in the inclusive pronoun. Crucially, **a morpheme difference is not a structural difference**. The combination of person features applies within the same morpheme, without interference of any other material in the composition of a pronoun, such as number or gender. The newly formed person morpheme then merges with other functional material, as illustrated below for Number. As evident in (13), the syntactic structure of inclusive pronouns is the same as the one proposed for all other pronouns, including languages without clusivity. Therefore, the proposal in terms of the inclusive morpheme derives the uniform pronominal syntactic structure.

(12) UNIFORM SYNTACTIC STRUCTURE

The syntactic structure of inclusive pronouns is not different from the structure of other pronouns.

That is, when the inclusive morpheme consisting of the two features merges in the structure, it establishes exactly the same terminal node as the person node hosting any other person feature specification, e.g., only the first or the second person. Under my proposal, **the clusivity parameter encoded in the syntax** and formalized in terms of the kinds of morphemes a language can generate **does not implicate a different syntactic structure** for inclusive pronouns.

(13) *An inclusive pronoun: The structure*



As noted above, the inclusive morpheme is the target of the Vocabulary Insertion operation. In line with the Subset Principle (Kiparsky 1973; Halle 1997) an exponent of person may realize all the features present in the inclusive morpheme or only the subset of the features.

(14) $\pi [1, 2] \leftrightarrow x$

As the upcoming discussion will demonstrate, exponents realizing inclusive morphemes may end up underspecified, thus being available across several cells in a pronominal system. Importantly, the underspecified Vocabulary Item applies to two different morphemes in the narrow syntax. Evidence for this comes from processes such as binding and agreement in languages with clusivity, where both features making up the inclusive morpheme are active.

4.3.1 Deriving the distribution of exponents

In this section, I show that the outlined proposal accounts for all the attested patterns of exponence realization. Starting with the absence of allomorphy, exemplified by the Urarina data, when the Vocabulary Insertion operation targets the person node, it realizes the subset of its features, that is, the first [AUTHOR] feature. In that sense, the exponent *[kan]* at PF is underspecified, and applies to two distinct morphemes in the narrow syntax: (i) the person morpheme consisting of the first-person feature only, present in the singular and the exclusive and (ii) the person morpheme consisting of the first- and second-person, realizing only a subset of them. Since no Vocabulary Item matches more features of the inclusive morpheme, the Vocabulary Item matching the subset of the features can apply to all three cells under consideration. In this case, the inclusive morpheme is syncretic with the first person. However, see the discussion on Algonquian later in this chapter, which shows that, in line with the Subset Principle, the second person can be expounded, too, yielding syncretism of the inclusive and the second person.

	SG	PL	
1 _{EXCL}	kan _H	kanakaana	(15) VI for person in Urarina [1, 2] $\leftrightarrow$ kan [1] $\leftrightarrow$ kan
1 _{INCL}		kana	
2	ii	ɲaara	
3	aka	aka _H _H	

(a) Urarina (Olawsky 2011)

Table 4.28: Deriving the distribution of exponents: Urarina

To account for the pattern in which the inclusive morpheme receives a different realization from the one attested in the singular and exclusive forms, I propose that the workings of the Subset Principle derive the distribution of the exponents. Namely, in Adang, there is a more specific Vocabulary Item $[p]$ that matches all the features present in the inclusive morpheme. This VI cannot be inserted in the person morpheme in the singular and the exclusive since it realizes more features than those present in the morpheme in the narrow syntax. The exponent $[n]$ is inserted in the singular, realizing the first person. In addition, it does not undergo number-driven allomorphy in the exclusive, which explains the formal identity of the two forms.

	SG	PL	
1 _{EXCL}	na	ni	(16) VIs for person in Adang [1, 2] $\leftrightarrow$ p [1] $\leftrightarrow$ n
1 _{INCL}		pi	
2	a	i	
3	sa	supi	

(a) Adang (Robinson & Haan 2014)

Table 4.29: Deriving the distribution of exponents: Adang

Moving to the next attested pattern, in languages like Mina, the realization of the person morpheme may also involve the competition of the two forms. This pattern contrasts with the one attested in Adang, as the exclusive form shows the number-driven allomorphy of the first person. In the inclusive, the number-induced allomorph of the subset of the features, i.e., the first person, is realized, which explains the identity in exponents in the two forms.

	SG	PL	
1 _{EXCL}	sə	na	(17) VIs for person in Mina [1] $\leftrightarrow$ n / $___$ # _{PL} [1] $\leftrightarrow$ s
1 _{INCL}		nók	
2	ha	hi	
3	∅, a	i	

(a) Mina (Frajzyngier 2011)

Table 4.30: Deriving the distribution of exponents: Mina

To account for three different exponents distributed across the three cells, attested in languages such as Vaeakau-Taumako, I propose that exponence realization can be traced to different sources within the same language. The person exponent attested in the inclusive [*tha*] is the most specific VI matching all the features of the inclusive morpheme. As noted above, it cannot be inserted in the exclusive and singular forms, as it matches more features than those present in the syntax in these two forms. The exponent [*mha*] is the result of the number-driven person allomorphy. As a consequence, it wins over the default exponent [*iau*] when the person morpheme in the exclusive undergoes insertion. Therefore, the complexity on both the left and right sides of a rule is consulted.

	SG	PL	
1 _{EXCL}	iau	mha-tou	(18) VIs for person in VT
1 _{INCL}		tha-tou	
2	koe	khou-tou	
3	ia	lha-tou	

(a) Vaeakau-Taumako
(Næss & Hovdhaugen 2011)

[1, 2] $\leftrightarrow$ tha
[1] $\leftrightarrow$ mha / ___ #_{PL}
[1] $\leftrightarrow$ iau

Table 4.31: Deriving the distribution of exponents: Vaeakau-Taumako

Finally, the controversial pattern of syncretism between the singular and the inclusive is derivable via underspecification and allomorphy. In the exclusive, the person morpheme is directly exposed to number, resulting in allomorphy. In the inclusive, however, the Vocabulary Insertion operation targets the subset of the features present in the inclusive morpheme, that is, the first person [*na*], which leads to the formal identity of person morphemes.

	SG	PL	
1 _{EXCL}	navə	ya-ŋgə	(19) VIs for person in Betta Kurumba
1 _{INCL}		na-ŋgə	
2	niyə	ni-ŋgə	
3 _{PROX}	idə	i-gə	
3 _{REM}	adə	a-gə	

(a) Betta Kurumba (Coelho 2020)

[1] $\leftrightarrow$ ya / ___ #_{PL}
[1] $\leftrightarrow$ na

Table 4.32: Deriving the distribution of exponents: Betta Kurumba

In what follows, I will provide further arguments that corroborate my proposal that the inclusive corresponds to a single morpheme in the narrow syntax.

4.3.2 How many exponents?

In Chapter 2, I discussed how an **exponent** at PF relates to a **morpheme** in the narrow syntax in pronouns. A discrete, clearly identifiable exponent of person or number was proposed to be a realization of a morpheme, i.e., a syntactic terminal. Across languages, the realization of the person morpheme in inclusive pronouns shows regularity that gives rise to a generalization in (20). Namely, while inclusive pronouns simultaneously consist of two person features, **AUTHOR** and **ADDRESSEE**, their morphological realization does not differ from other pronouns that surface with only one person feature specification. In fact, the realization of person in inclusive pronouns follows the same principles.

(20) **ONE INCLUSIVE EXPONENT**

The person morpheme in inclusive pronouns is realized with a single exponent.

Consider the following data from Kaera (Papuan), discussed earlier, and the corresponding set of VI rules. The first person morpheme is combined with the exponent $[n]$ in the singular, the second person receives zero realization, while the third person is realized as $[g]$. I analyze the formants $[a]$ and $[i]$ as exponents of singular and plural, respectively, while $[ng]$ is considered the realization of the D head. Inspection of the pronominal system reveals that person morphemes do not undergo number-driven person allomorphy. An exception in this respect is the first-person inclusive pronoun, which surfaces with the exponent $[p]$, contrasting with $[n]$ in the singular and exclusive forms (21).

	SG	PL	
1EXCL	n -ang	n -ing	(21) VI s for Kaera
1INCL		p -ing	$[1, 2] \leftrightarrow p$
2	ang	ing	$[1] \leftrightarrow n$
3	g -ang	g -ing	$[2] \leftrightarrow \emptyset$
(a) Kaera (Klamer 2014)			$[3] \leftrightarrow g$
			$[PL] \leftrightarrow i$
			$[SG] \leftrightarrow a$

Table 4.33: The inclusive exponent in Papuan

The same morphological realization present in Papuan holds for all the other languages discussed throughout this chapter. That is, the inclusive morpheme is realized with a single exponent. As noted in Chapter 2, if there is an exponent of person in pronouns, this exponent relates the form to a morpheme in the narrow syntax. If there is a person morpheme in the syntax, this syntactic terminal occupies a particular position in the pronominal structure. Therefore, I propose that the consistent appearance of one exponent is suggestive of one morpheme in the syntactic structure.

Crucially, the inclusive morpheme may trigger allomorphy of the plural, giving rise to the form that is present in neither the first-person exclusive nor the second-person. This suggests that

the inclusive morpheme as a whole, consisting of both of its features, triggers allomorphy and not one of its sub-components, thus pointing toward the inclusive morpheme being recognized as a unit.

4.3.3 Syncretism facts

Across languages, the realization of the person morpheme in the inclusive employs the same exponent present in the singular and exclusive forms. Recall data from Kuuk Thaayorre and Urarina. The exponent *[kan]* is spread across the three cells under consideration. These facts receive a straightforward explanation if the same exponent realizes the first-person feature across the paradigm. In the singular and exclusive, the person morpheme only consists of the first person [AUTHOR] feature. In the inclusive morpheme, which consists of the first and second person, following the Subset Principle (Kiparsky 1973; Halle 1997), the Vocabulary Item realizes the **subset** of the features of the inclusive morpheme, i.e., the first person. As a consequence, the exponent *[kan]* ends up **underspecified**.

	SG	PL
1EXCL	kan Ꞥ	kanakaana
1INCL		kana
2	ii	Ꞥaara
3	aka	akaꞤꞤ

Table 4.34: Absence of allomorphy in Urarina (Olawsky 2011)

The empirical fact that the person exponent is spread across the three cells is explainable through the workings of the Subset Principle and underspecification at PF if they apply to a **single morpheme** in the narrow syntax. In contrast, if the structure of inclusive pronouns differed from that of exclusives, involving more terminals for person features, why and how multiple terminals/morphemes may combine with an identical exponent would remain unexplained. The absence of allomorphy, that is, syncretism across the three cells, suggests that only one morpheme serves as an input for the Vocabulary Insertion operation.

4.3.3.1 Outlook

In the previous literature on the patterns of syncretism in clusivity, Moskal (2018) proposes that the person head in the inclusive consists of features AUTHOR and ADDRESSEE, noting that she remains agnostic as to whether portmanteau morphemes are (i) created via **Fusion** at PF prior to Vocabulary Insertion or (ii) they involve **insertion into non-terminals**.² This structure is adopted in Pertsova (2022a). Under this analysis, (i) there is a structural difference between

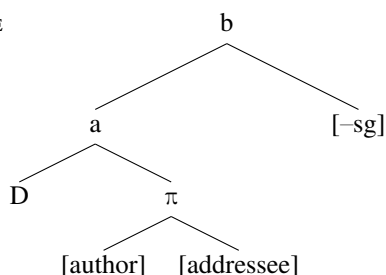
²Moskal (2018) proposes the following structures for exclusive and inclusive pronouns. The D head combines with the person node, while the number is higher in the structure. The two pronouns involve different syntactic structures.

inclusive and exclusive pronouns, (ii) fusion or non-terminal insertion obligatorily applies, and (iii) a mismatch between the number of morphemes in the narrow syntax and at PF arises.

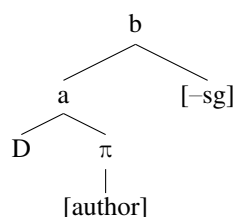
In the analysis I developed in terms of a **single morpheme** hosting both person features in the inclusive, the clusivity parameter is placed in the syntax without stretching the internal structure of a pronoun. Within the theoretical framework of Distributed Morphology, a morpheme hosting multiple features is a natural possibility. This view has several consequences. First, the **structure** of pronouns is **uniform**. That is, both inclusive and exclusive, as well as all the other pronouns in languages that do not mark clusivity, have a morpheme hosting person features. The exact featural content of this morpheme may differ. As argued above, the inclusive consists of both [AUTHOR] and [ADDRESSEE], while the exclusive surfaces with the [AUTHOR] feature only. From the syntactic structure and the computation perspective, the two pronouns are exactly the same, surfacing with one person morpheme. Second, the Vocabulary Insertion operation targets the person morpheme. When both features that make up an inclusive are hosted within a single morpheme, the **insertion into terminal nodes**, argued for throughout this dissertation based on empirical arguments beyond the clusivity distinction, is preserved. Therefore, one does not necessarily need to resort to the insertion into non-terminals to account for the clusivity facts.

Finally, an analysis that would preserve insertion into terminals but would resort to Fusion at PF, while operating with two morphemes in the narrow syntax, hosting [AUTHOR] and [ADDRESSEE] separately, would have to propose that Fusion always applies in inclusive pronouns. It is important to take into consideration what kind of operation Fusion is. It is highly language-specific and limited to PF interactions. On both empirical and theoretical grounds, it would be tentative to have a universal requirement on Fusion that applies precisely to these two morphemes in this particular syntactic environment. The existence of such a demand for PF would be mysterious. In addition, Fusion/creation of portmanteau morphemes is typically directly reflected in morphology. If Fusion obligatorily applies in inclusive pronouns, the question that arises is why the newly formed fused inclusive morpheme would be syncretic with the first person singular, a pattern attested in a number of unrelated languages (see 4.2.1). There is an additional problem with analysis in terms of Fusion. If Fusion is treated as contextual allomorphy (Bobaljik 2000; Trommer 2007; Fenger 2018), this would mean that the [AUTHOR] and [ADDRESSEE] features are the trigger and the target of allomorphy in the portmanteau inclusive morpheme, and not the number head. The allomorph arising in the inclusive is a consequence of the interaction of two person features, i.e., person-driven person allomorphy. The approach developed here does not

(22) INCLUSIVE



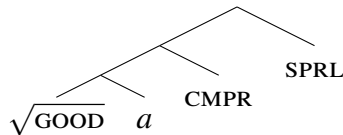
(23) EXCLUSIVE



need to rely on any of the additional assumptions. The Fusion operation is not invoked, the nature of the Vocabulary Insertion operation is preserved, and there is no mismatch between the narrow syntax and the PF.

In addition, Moskal's (2018) approach to clusivity aimed to test the predictions of the *ABA generalization in pronouns. Namely, hierarchical feature representation cashed out in terms of structural containment has been argued to figure as a condition on suppletive contextual allomorphy (Bobaljik 2012). According to the Containment Hypothesis, in a triple POSITIVE- COMPARATIVE - SUPERLATIVE, if a $\sqrt{\text{Root}}$ suppletes in the context of the comparative, then it suppletes in the context of the superlative too $[[[\sqrt{\text{Root}}] \text{CMPR}] \text{SPRL}]$. Thus, structural containment has a *blocking effect*, as the $\sqrt{\text{Root}}$ cannot surface in the superlative if the comparative is suppletive, hence *ABA.

(24) STRUCTURAL CONTAINMENT



(25) $\sqrt{\text{GOOD}} \leftrightarrow \text{be}(\text{tt}) / __] \text{CMPR}]$

$\sqrt{\text{GOOD}} \leftrightarrow \text{good}$

$[\text{SPRL}] \leftrightarrow \text{est}$

$[\text{CMPR}] \leftrightarrow \text{er}$

Why is the INCLUSIVE~EXCLUSIVE distinction fundamentally different from the containment pattern in comparatives and superlatives? **The target of allomorphy is different.** In the adjectival domain, the comparative node can trigger allomorphy of the $\sqrt{\text{Root}}$ (target). Likewise, the superlative node can trigger allomorphy of the same target, i.e., the $\sqrt{\text{Root}}$. Therefore, in **adjectives**, the **target remains constant** ($\sqrt{\text{Root}}$).

Let us see what happens in clusivity. In the exclusive, number can trigger the allomorphy of the person node (target). In the **inclusive**, the **target is not the same person morpheme**, but the person morpheme that consists of two features. Therefore, the clusivity distinction involves **increasing the complexity of the target** of allomorphy, which is not the case with $\sqrt{\text{Roots}}$ in adjectives. Therefore, when the number interacts with the person in the inclusive, its target is simply different from the target of the exclusive. For this reason, the two empirical domains do not constitute parallel environments for allomorphic conditioning.

4.4 **you* gap: The view from allomorphy

In languages that do not mark clusivity, such as English and German, the meaning of the inclusive, i.e., the pronoun whose referent includes both the first and the second person, is morphologically realized with the *first* person plural pronoun (*we*; *wir*) and not the second, giving rise to the prominent **you* gap noted in Zwicky (1977) and discussed in much of the subsequent work (see McGinnis 2005; Harbour 2016). In this section, I demonstrate that this paradigmatic gap extends to languages with clusivity.

As a reminder, the following patterns of morphological realization were discussed earlier in this chapter. In Urarina (isolate; spoken in Peru), the exponent of the person in the inclusive [*kan*] is syncretic with the singular and exclusive forms. This contrasts with Adang (Papuan), where the singular and exclusive exhibit syncretism, evidenced by the shared exponent [*n*], while the inclusive receives a different morphological realization [*p*]. In Mina (Chadic), the exclusive and inclusive are syncretic [*n*], while Vaeakau-Taumako (Austronesian) employs three distinct exponents for the three cells under consideration. Importantly, in none of the languages is the syncretism of the inclusive and the second person attested, corroborating the empirical validity of the **you* gap.

	SG	PL
1EXCL	kan _H	kanakaana
1INCL		kana
2	ii	ɟaara
3	aka	aka _H _H

(a) Urarina (Olawsky 2011)

	SG	PL
1EXCL	na	ni
1INCL		pi
2	a	i
3	sa	supi

(b) Adang (Robinson & Haan 2014)

Table 4.35: Syncretism in languages with clusivity

	SG	PL
1EXCL	sə	na
1INCL		nók
2	ha	hi
3	∅, a	i

(a) Mina (Frajzyngier 2011)

	SG	PL
1EXCL	iau	mha-tou
1INCL		tha-tou
2	koe	khou-tou
3	ia	lha-tou

(b) Vaeakau-Taumako
(Næss & Hovdhaugen 2011)

Table 4.36: Syncretism in languages with clusivity

4.4.1 The Algonquian puzzle

Algonquian data challenge this view. Namely, the exponent of the second person is found in the inclusive pronouns, as illustrated on the basis of Ojibwe and Nishnaabemwin data. As pointed

out in the literature (Harley & Ritter 2002; McGinnis 2005; Harbour 2016; Bobaljik & Sauerland 2023; Bobaljik 2024), the violation of the **you* gap is only apparent, as the inclusive and the second person plural are differentiated on the basis of distinct plural exponents [*awint/awaa* in Ojibwe and *win/waa* in Nishnaabemwin], thus not exhibiting full syncretism.

	SG	PL		SG	PL
1 _{EXCL}	niin	niin-awint	1 _{EXCL}	niin	niin-win
1 _{INCL}		kiin -awint	1 _{INCL}		giin -win
2	kiin	kiin -awaa	2	giin	giin -waa
3	wiin	wiin-awaa	3	wiin	wiin-waa

(a) Ojibwe (Harley & Ritter 2002) (b) Nishnaabemwin (Valentine 2001)

Table 4.37: Patterns of syncretism in Algonquian

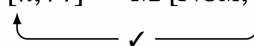
4.4.2 What does allomorphy show us?

I approach this discussion from the perspective of allomorphy. In the present work, I demonstrate that Algonquian data provide further evidence for the existence of **person-driven number allomorphy** in the natural language, introduced in the previous chapter.

As illustrated in Table 4.37, the **exclusive** and **inclusive** pronouns **share the plural exponent** [*awint* in Ojibwe; *win* in Nishnaabemwin]. Likewise, the second- and third-person pronouns surface with the same exponent of plural [*awaa*; *waa*]. Therefore, the realization of plurality involves the competition of two plural forms. I propose that this competition can best be accounted for in terms of **contextual allomorphy**, with the feature content of the person morpheme being crucial for explaining the distribution of plural morphology. Namely, the person morpheme in the exclusive carries the first person, i.e., the [AUTHOR] feature. The same holds for the inclusive, which, in addition to the first-person, surfaces with the second-person feature, too. I propose that the presence of the first person [AUTHOR] derives the distribution of the plural exponents. In the context of the [AUTHOR] feature, the allomorph of plural [*awint*; *win*] is selected. When the [AUTHOR] feature is absent, the elsewhere [*awaa*; *waa*] surfaces.

The computation of the plural exponent proceeds as follows. At the point when the plural node is targeted by the Vocabulary Insertion operation, the morpho-syntactic information about person, i.e., whether the AUTHOR is present, must be available. The Vocabulary of Ojibwe has two allomorphs of plural: (i) the more specific [*awint*], which requires a particular context specification to be met, and (ii) the elsewhere form [*awaa*]. In line with the Elsewhere Principle (Kiparsky 1973; Halle 1997), the presence of the [AUTHOR] feature in the inclusive and exclusive pronouns triggers the insertion of a more specific allomorph [*awint*]. In its absence, the elsewhere exponent [*awaa*] surfaces in the second and third person. The same principle extends to the Nishnaabemwin data. Therefore, the analysis formalized in terms of allomorphy correctly derives the distribution of plural morphology. In (26), I indicate a morpheme as a basic unit

of syntactic computation as ($\mathbb{M}$), features present in the narrow syntax with π and NUM, and an empty slot // for a Vocabulary Item (VI) to be inserted.

- (26) NUMBER MORPHEME LOOKING INTO PERSON PERSON-DRIVEN NUMBER ALLOMORPHY
- | | |
|--|--|
| $\mathbb{M} [\pi; / /] \text{ — } \mathbb{M} [\text{NUM}; / /]$
 | $[\text{PL}] \leftrightarrow \text{awint} / ___ \pi_{\text{AUTHOR}}$
$[\text{PL}] \leftrightarrow \text{awaa}$ |
|--|--|

At the same time, the person morpheme in the inclusive and the second-person plural pronouns is realized with exactly the same exponent, [*kiin*] and [*giin*], in Ojibwe and Nishnaabemwin, respectively, in violation of the **you* gap. How can two different allomorphs of plural co-occur with the same exponent of person? This puzzle receives a straightforward solution if the syncretic exponent of person, shared between the inclusive and the second person, corresponds to **two different morphemes** in the narrow syntax, specified for different features. The features present in the morpheme are those relevant for allomorphy. The fact that person may act as a **trigger** of allomorphy, giving rise to the observed alternation, suggests that the syncretic Vocabulary Item in fact corresponds to different features in the syntax. As a consequence, the violation of the **you* gap is an illusion that arises due to syncretism.

Therefore, the two pronouns, inclusive and the second-person plural, result from two different syntactic derivations, since the feature composition of the person node in the inclusive differs from the feature composition in the second person. In the realizational component of the grammar, however, a single underspecified VI [*kiin*; *giin*] applies to two distinct morphemes in the syntax, giving rise to syncretism. This is possible since the second-person feature is a **subset** of the features present in the inclusive (27). Note that these exponents can be segmented into person [*k/g*] and an additional formant [*ii*] that does not disrupt allomorphy (for further discussion, see Chapter 6 and the analysis of Serbian definiteness morphemes that linearize as infixes; these data receive the same analysis).

(27) **VIs for PERSON in Ojibwe**

[1, 2] $\leftrightarrow$ k

[1] $\leftrightarrow$ n

[2] $\leftrightarrow$ k

Syncretism obtained in morphology does not carry over to narrow syntax and semantics, which explains the existence of two different allomorphs of plural which co-occur with a single exponent of person, resulting in the **run-of-the-mill** **you* gap, observed elsewhere in natural languages. The results of allomorphy further support its status as a **universal gap** in pronominal paradigms and the crucial role of the first [AUTHOR] person feature that acts as a trigger of allomorphy and underlies the observed gap, raising implications for its prominent position in the representation of person features.

These facts further support the idea that the inclusive person head consists of two features, AUTHOR and ADDRESSEE, in the narrow syntax. Interestingly, applying an artificial language

learning methodology Maldonado & Culbertson (2022) demonstrate that language learners do not have difficulties in acquiring the second-inclusive forms (**you*), as the statistical difference between the first- and second-inclusive forms was marginal, concluding that they are both generated by grammar, with the second-inclusive being dispreferred over its widespread first-inclusive counterpart. Therefore, a pressing question that emerges is how to explain the simultaneous presence of the **you* gap in the natural language and its obvious learnability evidenced in an experimental setting.

4.5 Analytic inclusives: Creoles and beyond

As discussed above, the first-person inclusive morpheme surfaces with two features mapped to a single exponent. However, a logical possibility for the realizational component of natural language is employing two discrete exponents for the two features. Interestingly, these forms have been reported for Tok Pisin, an English-based creole, and Bislama, a pidgin, both spoken in Vanuatu, (Harbour 2016) and Dolakha Newar (Genetti 2007; see the discussion in Moskal 2018). As illustrated in the table below, first-person inclusive pronouns in Bislama and Tok Pisin transparently consist of first- and second-person singular pronouns, [*mi*] and [*yu*], respectively, in both plural and dual. Furthermore, these creoles show that the dual forms are built on top of plural ones, as exponents of plural [*fala*; *pela*] are contained in the dual, co-occurring with the dual exponent [*tu*]. I refer to these patterns as **analytic inclusives**.

π	SG	PL	DU	π	SG	PL	DU
1EXCL	mi	mi-fala	mi-tufala	1EXCL	mi	mipela	mitupela
1INCL		yu-mi	yu-mi-tu(fala)	1INCL		yumi(pela)	yumitupela
2	yu	yu-fala	yu-tufala	2	yu	yupela	yutupela
3	hem	ol(geta)	tufala	3	em	ol	tupela

(a) Bislama

(b) Tok Pisin

Table 4.38: Analytic inclusives in creoles (Harbour 2016)

The fact that transparent compositionality of this type is attested precisely in languages with a creole/pidgin origin cannot be accidental. Namely, these facts are highly reminiscent of the phenomenon of double plurality in the sense of Alexiadou (2024). On the basis of data coming from Ewe English, Hiaki Spanish, Bantu English, Bantu French, and Greek Turkish, Alexiadou (2024) demonstrates that double plural marking is associated with two distinct semantic primitives in the nominal extended projection. The two distinct plural exponents, present in a number of language mixing environments, are exponents of a lower lexical plurality, hosted on the nominalizing head *n*, and the higher grammatical plurality, DivP. In the data provided in (28), plural exponents coming from English and Spanish, [*s*], are hosted on the *n* head, realizing lexical plurality. This contrasts with number exponents [*wó*] and [*im*], originating in Ewe and Hiaki, respectively, which are hosted in the higher Div layer, thus realizing grammatical plurality.

(28) DOUBLE PLURALITY (Alexiadou 2024)

- a. *nye younger brother-s wó kata*
 1SG younger brother-PL PL all

‘all my younger brothers’

Ewe English

- b. *ume waka-s-im*
 the.PL COW-PL-PL

‘the cows’

Hiaki Spanish

The fact that two exponents of person are attested in creoles, Bislama and Tok Pisin, is consistent with the argument that language mixing pairs and language acquisition are highly inclined toward analyticity. Importantly, Alexiadou (2024) demonstrates that the same analysis in terms of Maximize Exponence can be extended to double plural marking outside language mixing pairs, such as Amharic and Breton.

In the preceding discussion, I proposed that inclusive pronouns involve a single morpheme in the syntax and a single exponent at PF. How to account for analytic inclusives? There are multiple ways to explore. Analytic inclusives can be the result of Fission, that splits the single morpheme into two morphemes at PF, it can be radical decomposition at the level of person features, comparable to double plurality in the domain of number (Alexiadou 2024), in which case one of the exponents would correspond to a lower head in the structure, or it may involve composed clusivity (Bobaljik 2024). More work in this domain is necessary to provide a conclusive account of analytic inclusives, taking into consideration both their morphological and interpretive properties.

The co-occurrence of the first- and second-person exponents has been reported for a language that does not have a mixing origin, Dolakha Newar, discussed in Moskal (2018). In this study, I present three other languages that surface with the same pattern that have not been addressed from this perspective in the previous literature. In my sample, the co-occurrence of the first and second person is attested in four languages: Ika (Chibchan), Mian (Papuan), Sumu (Misumalpan), and Dolakha Newar (Tibeto-Burman). In Ika, the first-person inclusive pronoun consists of the first [nən] and the second person [ma] pronouns, combining with the plural exponent [ka?].³ Likewise, in Sumu, the inclusive pronoun is composed out of the first person [jan] and second person [ma] pronouns, without requiring an additional plural exponent, contrasting in this respect with Ika. Mian data was introduced earlier in this chapter.

³Jana Bajorat (p.c.) suggests that both exclusive and inclusive forms are in active use in Ika, based on her original fieldwork data and corpus searches. In addition, during her fieldwork recordings, native speakers used the first-person inclusive pronoun in contexts where the exclusive was expected (e.g., *we as Ika people*), which suggests that the inclusive expands its distribution to non-inclusive environments. I thank Jana for the extensive discussion and clarifications about the data.

	SG	PL		SG	PL
1 _{EXCL}	nən	ni-wi	1 _{EXCL}	jaŋ	jaŋ-na
1 _{INCL}		ma-nən-kaʔ	1 _{INCL}		ma-jaŋ
2	ma	mi-wi	2	man	man-na
3	a	ikən-kaʔ	3	witiŋ	witiŋ-na

(a) Ika (Pertsova 2022b)

(b) Sumu (Pertsova 2022b)

Table 4.39: Analytic inclusives across languages

The observed data yield a relevant empirical generalization (29), which raises implications for languages without clusivity. Namely, transparent compositionality in the pronominal domain, where a plural pronoun consists of overt first- and second-person exponents, is attested *only* in languages with the exclusive~inclusive distinction.

(29) ANALYTIC INCLUSIVES: GENERALIZATION

Exponents of both the first- and second-person are simultaneously realized within the same plural pronoun only in languages that make the exclusive~inclusive distinction.

The fact that the co-occurrence of the first and second person emerges only in languages with the exclusive~inclusive distinction, being restricted to *inclusive* pronouns in these languages, supports the proposal that clusivity is a parameter encoded in the syntax, as argued here (see also Bobaljik & Sauerland 2023; Bobaljik 2024) and not the consequence of underspecification at PF (see Ackema & Neeleman (2013a) for the view that all languages surface with the clusivity distinction, the difference being located in the realizational component of the grammar).

I will dedicate more attention here to Dolakha Newar as it raises implications for the theory of allomorphy. In this language, the first-person inclusive pronoun [*chiji*] consists of the first [*ji*] and second person [*chi*] exponents, on par with the data discussed above. I propose the analysis in terms of the set of Vocabulary Items in (30). The first-person exclusive form undergoes number-driven allomorphy [*ji~isi*]. However, the first-person inclusive shows a sharp difference, as the obtained form is not [**chiisi*] but the one lacking allomorphy [*chiji*]. Furthermore, the second- and third-person combine with the plural exponent [*pen*]. I approach this morphological alternation in terms of person-driven number allomorphy. Namely, zero allomorph of the plural is selected in the context of the first person, i.e., the *AUTHOR* feature, while [*pen*] figures as the elsewhere exponent, hence its presence in the second and third person.⁴

⁴For an alternative analysis in terms of an Impoverishment Rule for plural in Dolakha Newar see Moskal (2018), who notices that the Dolakha Newar pattern exemplifies an *ABA pattern, under an analysis where inclusive properly includes exclusive in its structural representation. Thus, the triple *ji-isi-chiji* unambiguously gives rise to the *ABA as the first person exponent [*ji*] (A) undergoes allomorphy in the exclusive [*isi*] (B) and reverts to the (A) form [*ji*] in the inclusive [*chiji*]. As this example constitutes an obvious problem for the Containment-based approaches to allomorphy and to (number) features, it is proposed that the Impoverishment Rule has an effect of deleting the plural in the inclusive form. As a consequence, the exclusive is no longer in a containment relationship with the inclusive, as the inclusive consists solely of personal features. The same rule does not hold for Tok Pisin as the language employs an overt plural marker [*pela*]. At the same time, Tok Pisin facts are not problematic for the Containment hypothesis as the exclusive has no allomorphy.

π	SG	PL
1 _{EXCL}	ji	isi
1 _{INCL}		chiji
2	chi	chipen
3 _{PROX}	u	upen
3 _{DIST}	ām(u)	āpen

(a) Dolakha Newar (Genetti 2007)

(30) **VI_s for Dolakha Newar**[1] $\leftrightarrow$ isi / ___ #_{PL}[1] $\leftrightarrow$ ji[2] $\leftrightarrow$ chi[PL] $\leftrightarrow$ $\emptyset$ / ___ π_{AUTHOR} [PL] $\leftrightarrow$ pen

Table 4.40: Allomorphy in Dolakha Newar

What these data show is that the first-person morpheme is exposed to the number morpheme in the exclusive, which is directly reflected in the emerged allomorphy, but the same first-person feature is not affected by the same number feature in the *inclusive*. Therefore, the question is what blocks allomorphy in the inclusive. I propose that the apparent absence of allomorphy can be explained if the additional feature that inclusive pronouns have but exclusive ones do not, namely *ADDRESSEE*, blocks allomorphy. Recall that the difference between the inclusive and the exclusive person morphemes is in the *ADDRESSEE* feature, present in the former but absent in the latter.

In the exclusive, the number morpheme freely triggers allomorphy of the person morpheme, giving rise to the form [isi].

(31) **EXCLUSIVE: isi**

$$\mathbb{M} [\pi; / /] - \mathbb{M} [\text{NUM}; / /]$$

Moving to linearization facts, the plural exponent in Dolakha Newar linearizes to the right of the person exponents (32-a), as evident from the linearization of the plural marker [pen]. Crucially, the exponent of the second person, argued here to block allomorphy, does not linearize to the right of the first person (32-b), in which case it could be argued that it disrupts the linear adjacency between the first person and number, thus blocking allomorphy. Instead, it linearizes to the left of the first person, leaving the first-person linearly exposed to the number, i.e., the first-person and the plural are linearly adjacent.

(32) **LINEARIZATION STATEMENT**

- a. $\mathbb{M} [\pi; /chi/] - \mathbb{M} [\#] pen$
- b. $* \mathbb{M} [\pi; /ji/] - \mathbb{M} [\pi; /chi/] - \mathbb{M} [\#; / /]$
- c. $\mathbb{M} [\pi; /chi/] - \mathbb{M} [\pi; ji] - \mathbb{M} [\#; / /]$

The minimal pair in (33-34) demonstrates that allomorphy is blocked in the presence of an additional feature, which, however, does not disrupt the linear adjacency of the first person and

the number morpheme. The availability of allomorphy in (33) and its absence in (34), creates a problem for the linear adjacency condition on allomorphy (Embick 2010).

(33) EXCLUSIVE: *isi*

$$\mathbb{M} [\pi; /isi/] \text{ — } \mathbb{M} [\text{NUM}; / /]$$

↖ ↗
✓

(34) INCLUSIVE: *chiji*

$$\mathbb{M} [\pi; /chi/] \text{ — } \mathbb{M} [\pi; /ji/] \text{ — } \mathbb{M} [\text{NUM}; / /]$$

↖ ↗ ↗
✗

The question for more relaxed approaches to allomorphy, such as the one in terms of Accessibility Domain (Moskal 2015a; Moskal & Smith 2016), which allows allomorphy across an overt morpheme, is how to explain the observed blocking. In the case at hand, the Accessibility Domain approach wrongly predicts that the first-person morpheme is equally available to undergo allomorphy conditioned by the number, as the two surface in the same phase domain.

4.6 Minimal-augmented systems

The existence of a pronoun whose referent consists of exactly two individuals, AUTHOR and ADDRESSEE, motivated the proposal that natural languages, in addition to the SINGULAR~PLURAL distinction, may also draw the so-called MINIMAL~AUGMENTED distinction (McKay 1978; Noyer 1992; Corbett 2000, 2005; Harbour 2016). As noted in Chapter 2, the first-person minimal exclusive [1_{MIN.EXCL}], second-person minimal [2_{MIN}] and third-person minimal [3_{MIN}] neatly correspond to the singular first, second, and third person. The addition to the system is precisely the first-person minimal inclusive pronoun [1+2_{MIN.INCL}], often referred to as the ‘fourth person’, as it patterns more closely with the singular and not plural or dual, as reported in the literature above, which led to the proposal that it should be considered together with other singular forms, i.e., as minimal. Limbum (Bantu, Eastern Grassfields; Ndi 2017) exemplifies a language that makes the MINIMAL~AUGMENTED distinction. As illustrated in Table 4.41, in addition to the two pronouns in the augmented (plural) number, [wèr] and [sèe], corresponding to the first-person exclusive and inclusive, respectively, there is a third pronoun [sò], whose referent consists only of the first- and second-person. If there is any additional individual in the referent of a pronoun, the conditions for this pronominal form are not satisfied.

In contrast to the widespread clusivity distinction, languages with minimal inclusive pronouns are very rare. Although there is no consensus on which languages should be considered MIN~AUG, the existence of the first-person dual pronoun, especially in cases when (i) the dual form is present only in the first person but absent in the second- and third-person and (ii) the language has dual forms for each person surfacing with identical dual exponents and an additional dual pronoun which employs different exponents, qualify a language as a candidate. These two

1 _{MIN} .EXCL	1 _{AUG} .EXCL		
1+2 _{MIN} .INCL	1+2 _{AUG} .INCL		
2 _{MIN}	2 _{AUG}	MIN	AUG
3 _{MIN}	3 _{AUG}	1	mè, m wèr
		1+2	sò sèe
		2	wè, à wèe
		3	ye, e woowèe, a

Table 4.41: Minimal~Augmented system

considerations may serve as guiding factors in deciding whether a language exhibits the MINIMAL~AUGMENTED distinction or not.

In what follows, I introduce 13 case studies of pronominal systems surfacing with the minimal inclusive pronouns. The patterns of morphological realization are discussed in detail to scrutinize the regularities in exponence selection. In the next step, I discuss the implications for the structural representation of pronouns. Before proceeding with the obtained patterns and the discussion, I introduce the dataset that serves as a basis for this study.

4.6.1 Data

The study is based on 13 languages that exhibit the MINIMAL~AUGMENTED system. The selected sample covers a diverse set of language families, including Niger-Congo, Afro-Asiatic, Native American, Papuan, and Austronesian, and several groups and branches within the Australian family. The list of languages investigated in this study, along with information on their genetic affiliation and data source, is provided in Table 4.42.

LANGUAGE	FAMILY	SOURCE
Dii	Niger-Congo	Bohnhoff (2010)
Limbum	Bantu; Niger-Congo	Ndi (2017)
Gude	Chadic; Afro-Asiatic	Hoskison (1983)
Winnebago	Siouan; Native American	Noyer (1992)
Kemtuik	Nimboran; Papuan	Donohue & Smith (1998)
Ilocano	Malayo-Polynesian; Austronesian	Corbett (2000)
Maranao	Austronesian	Cysouw (2003)
Gaagudju	Australian Aboriginal	Harvey (2011)
Malakmalak	Northern Daly; Australian Aboriginal	Birk et al. (1976)
Maranungku	Western Daly; Australian Aboriginal	Tryon et al. (1970)
Limilngan	non-Pama-Nyungan; Australian Aboriginal	Harvey (2001)
Gurinja	Pama-Nyungan; Australian Aboriginal	Meakins & McConvell (2021)
Bardi	Nyulnyulan; Australian Aboriginal	Bowern (2012)

Table 4.42: Minimal~Augmented system: Language sample

4.6.2 The patterns

Limbum (Bantu; Niger-Congo). The first-person minimal exclusive pronoun [*mɛ̀, m̀*], surfaces with an overt exponent of person [*m̀*]. In the minimal inclusive pronoun, however, the person exponent is realized as [*s*]. Importantly, the exponent [*s*], present in the minimal inclusive, is preserved in the inclusive augmented pronoun. I propose the following set of VIs for Limbum. The exponent [*s*] is inserted into a morpheme consisting of both first- and second-person features, while the exponent [*m*] undergoes insertion when the person morpheme consists only of the first person.

	MIN	AUG	
1	mɛ̀, m̀	wɛ̀r	(35) Some VIs for Limbum
1+2	sò	sèe	
2	wɛ̀, à	wèe	
3	ye, e	woowèe, a	

[1, 2] ↔ s
[1] ↔ m

(a) Limbum (Ndi 2017)

Table 4.43: Pronouns in Limbum

Ilocano (Austronesian). Patterning with Limbum, the first person undergoes allomorphy in the augmented exclusive. The minimal inclusive pronoun is realized as [*ta*], being preserved in the augmented number. Therefore, three different exponents of person are distributed across the four cells under consideration, with the absence of allomorphy between the minimal inclusive and the minimal inclusive augmented.

	MIN	AUG	
1	ko	mi	(36) VIs for Ilocano
1+2	ta	tayo	
2	mo	yo	
3	na	da	

[1, 2] ↔ ta
[1] ↔ m / ___#_{AUG}
[1] ↔ k

(a) Ilocano (Corbett 2000)

Table 4.44: Pronouns in Ilocano

Maranao (Austronesian). The same pattern arises. The minimal inclusive pronoun is realized as [*ta*]. The exponent of person is preserved in the augmented inclusive [*tano*], as [*ta*] transparently combines with the plural exponent [*no*], shared with the second person augmented [*kano*].

Dii (Niger-Congo). The same pattern of morphological realization as the one discussed above emerges. The minimal inclusive pronoun [*ba*] remains unaffected by the augmented number, transparently combining with the augmented exponent. In line with the data discussed above, MIN.EXCL-AUG.EXCL-AUG.INCL employ three different exponents. Note that the dots in augmented inclusive [*ba...vɪ*] indicate that intervening material is allowed between the two components.

	MIN	AUG
1	ako	kami
1+2	ta	tano
2	ka	kano
3	sekanian	siran

(a) Maranao (Cysouw 2003)

- (37) **VIs for Maranao**
 [1, 2] ↔ ta
 [1] ↔ ka(mi) / ___#_{AUG}
 [1] ↔ ako

Table 4.45: Pronouns in Maranao

	MIN	AUG
1	mí	vó
1+2	ba	ba...ví
2	mó	ví
3	∅	v _H

(a) Dii (Bohnhoff 2010)

- (38) **Some VIs for Dii**
 [1, 2] ↔ ba
 [1] ↔ mí
 [2] ↔ mó

Table 4.46: Pronouns in Dii

Kemtuik (Nimboran; northern Irian Jaya, Papuan). The person morpheme is realized as [i] in the minimal inclusive, remaining intact in the augmented inclusive. In contrast to the languages discussed above, there is no allomorphy in the augmented exclusive, since the exponent of the augmented number [naŋ] transparently combines with the first person [g_{nam}]. As evident in the table below, all pronouns, with the exception of 1_{MIN.EXCL} and 1_{AUG.EXCL} surface with the formant [mot]. This is a common pattern across Papuan, which I analyze as the exponent of D (see Chapter 6 for details). Likewise, I analyze [naŋ] as the allomorph of [mot] selected in the context of the first person, hence only [g] as the exponent of the first person in (39), though nothing would change if the form [g_{nam}] were the realization of the first person. Furthermore, second-person realized as a zero morpheme [∅] is common across Papuan languages. Importantly, patterning with the languages discussed above, the augmented number does not trigger allomorphy of the person morpheme in the minimal inclusive pronoun.

	MIN	AUG
1	g _{nam}	g _{nam} naŋ
1+2	i mot	i motnaŋ
2	mot	motnaŋ
3	nmot	nmotnaŋ

(a) Kemtuik (Donohue & Smith 1998)

- (39) **Vocabulary Items for Kemtuik**
 [1,2] ↔ i
 [1] ↔ g
 [2] ↔ ∅
 [3] ↔ n
 [AUG] ↔ naŋ

Table 4.47: Pronouns in Kemtuik

Gaagudju (Australian). The same pattern was observed for other languages discussed above.

The minimal inclusive pronoun surfaces with the exponent [*manee*] that is absent in the first-person singular, but preserved in the augmented number.

	MIN	AUG
1	ngaayi~ngaanj-ma	ngaa-mba~ngee-mba
1+2	maneerra	manee -mba
2	ngiinja	nginjee-mba
3M	naawu	nowoo-mba
3F	ngaayu	—

Table 4.48: Pronouns in Gaagudju (Harvey 2011)

Winnebago (Siouan, Native American). The subject pronoun affixes in Winnebago exhibit exactly the same morphological realization as the one attested in Kemtuik. The first person minimal inclusive is realized as [*hin*], being preserved in the inclusive augmented [*hin- -wi*]. Patterning with Kemtuik, there is no allomorphy in the first-person exclusive augmented, given that the form [*ha*] is transparently combined with the augmented exponent, resulting in [*ha- -wi*]. This is the first language discussed here in which the first-person minimal exclusive and the minimal inclusive pronouns show partial morpho-phonological overlap. As discussed in Chapter 2, relatedness of this kind cannot be ignored in the morphological theory, especially in cases where the two forms relate to syntactic objects that overlap in features. Therefore, we may segment the pronominal affix [*hin*] into the exponent of the first person [*h*] and the exponent of the minimal number [*in*]. The syncretism of [*ha*] and [*hin*] is accounted for under the underspecification analysis coupled with the Subset Principle. Namely, the exponent [*h*] applies to the minimal inclusive morpheme, realizing only the subset of its features, that is, the first person. At the same time, the exponent of the minimal number would be realized as [*a*] in the first and second person, alternating with the allomorph [*in*] in the context of the inclusive person head. As person-driven number allomorphy is present in natural language, the analysis in terms of two allomorphs of the minimal number is readily available. Patterning with the languages discussed above, the minimal inclusive does not undergo allomorphy triggered by the augmented number.

	MIN	AUG
1	ha-	ha- -wi
1+2	hin-	hin- -wi
2	ra-	ra- -wi
3	∅	-ire

(a) Winnebago (Noyer 1992)

(40) **VIs for PERSON in Winnebago**

- [1, 2] ↔ h
 [1] ↔ h
 [2] ↔ r
 [MIN] ↔ in / ___ π₁₊₂
 [MIN] ↔ a

Table 4.49: Pronouns in Winnebago

In all languages discussed above, the minimal inclusive pronoun remains preserved in the augmented inclusive. The question that arises is whether allomorphy of the minimal inclusive

person head is even possible. Can augmented number ever affect the realization of the person morpheme that consists of the two features? Or is there a morphological gap? In the remainder of this section, I will focus on the data that may shed light on this question. In contrast, the triple 1_{MIN.EXCL}-1_{EXCL.AUG}-1_{INCL.AUG} resulted in an across-the-board allomorphy, as the three pronominal forms involved three distinct exponents of the first person. Exceptions to this pattern are Kemtuik and Winnebago, which employ two distinct exponents given that the augmented exclusive pronoun in these languages does not involve allomorphy.

Gude (Chadic). This pronominal system may be analyzed as involving four different exponents for the four cells. The first person minimal exclusive [nyɪ] undergoes number-driven allomorphy, resulting in [inə] in the augmented number. Importantly, the minimal inclusive pronoun [anwu] does not seem to show morpho-phonological relatedness with the augmented form [amə], unless [a] is identified as an exponent of the inclusive person morpheme. When it comes to the exponent of the augmented number, it can be argued that [nə] is the exponent shared by the first and second person with an allomorph [mə] which is conditioned by the presence of minimal inclusive. If this is the case, then [a] in [amə] can be considered morpho-phonologically related to [anwu]. However, more phonologically focused empirical work is necessary to address this question. If number-driven allomorphy is absent in this case, too, then Gude data replicate the pattern observed above, where the inclusive person head consistently resists allomorphy.

	MIN	AUG
1	nyɪ/ii	inə
1+2	anwu	amə
2	hə	unə
3 _M	ci	(ə)tii
3 _F	ki	(ə)tii

Table 4.50: Pronouns in Gude (Hoskison 1983)

Malakmalak (Daly; Australian Aboriginal). The minimal inclusive and the augmented inclusive share the initial formant [y]. However, note that the same formant is present in the first-person augmented exclusive pronoun but also in the third-person minimal masculine pronoun. Therefore, the recognition of [y] as an exponent of the inclusive person head that remains unaffected in the presence of the augmented number remains questionable, which may pave the way for analyzing this morphological alternation as augmented-driven number allomorphy.

Limilngan (Australian Aboriginal). On par with many other typologically related languages, Limilngan employs the initial formant [ng], which is shared between the first and the second person (and often the third person in languages of this area). Once [ng] is excluded, we are left with [ayki] and [ami] in the first-person minimal exclusive and minimal inclusive, respectively. All pronominal forms in the augmented number surface with [guyi], which can be considered the augmented exponent. If this is the case, the augmented inclusive pronoun consists only of

	MIN	AUG
1	ŋa	yawot
1+2	yaŋki	yɛŋkit
2	waŋari	nukut
3 _M	yöntön	wöröntön
3 _F	nöntön	wöröntön

Table 4.51: Pronouns in Malakmalak (Birk et al. 1976)

[guyi]. Hence, the realization of person is zero [Ø]. Therefore, this would be an example of a language in which the minimal inclusive person head has an augmented-induced allomorph.

	MIN	AUG
1	ngaykgi	n-guyi
1+2	ngami	guyi
2	nginyi	wun-guyi

Table 4.52: Pronouns in Limilngan (Harvey 2001)

Bardi (Nyulnyulan, Australian). As noted in Chapter 2, Bardi is a language with an overt, transparently identifiable augmented exponent [rr]. In contrast to the first-person minimal exclusive pronoun, the minimal inclusive does not include the initial formant [ng]. The augmented inclusive shares the initial formant [a], as do all four cells involving the first person, including the minimal exclusive once [ng] is taken out of consideration. For this reason, this language does not provide conclusive evidence for the augmented-driven allomorphy of the minimal inclusive morpheme.

	MIN	AUG
1	ngay(oo)	arroodoo
1+2	ayoo	arridil
2	joo	goorr
3	giniyingg(i)	irr

Table 4.53: Pronouns in Bardi (Bower 2012)

Maranungku (Australian Aboriginal). This language has two sets of minimal and augmented pronouns for subject and object positions. As evident in the table below, argument structure considerations interfere with the realization of the augmented number, since the exponent [tya], shared across all augmented pronouns in the subject position, alternates with [nya] when it surfaces in the object position. Once the typical formant [nga] is removed, the minimal inclusive is left with [ngku]. However, the person morpheme in the inclusive augmented is realized as [ki]. Therefore, this can be a clear example of augmented-driven inclusive allomorphy.

	MIN	AUG		MIN	AUG
1	nga-ny	nga-tya	1	-ngany	-ngeri-nya
1+2	nga- ngku	ki -tya	1+2	-ng angku	- ki -nya
2	nina	ni-tya	2	-nimpe	-ni-nya
3 _M	nankuny	wi-tya	3 _M	-na	-wi-nya
3 _F	ngankuny	wi-tya	3 _F	-nga	-wi-nya

(a) Subject

(b) Object

Table 4.54: Pronouns in Maranungku (Tryon et al. 1970)

Gurinji (Ngumpin-Yapa; Australian). In this language, the minimal inclusive form shares the formant [nga] with the first-person singular and the inclusive augmented. This shared material is explainable in terms of the Subset Principle, which realizes only the subset of the features present in the inclusive morpheme, in the case at hand, the first-person.

	MIN	AUG
1	ngayu	ngantipa
1+2	ngali	ngaliwa
2	nyuntu	nyurrulu
3	nyantu	nyarrulu

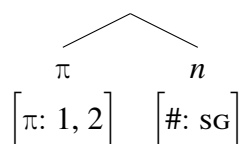
Table 4.55: Gurinji (Meakins & McConvell 2021)

4.6.3 Minimal inclusive: The structure

The distinguishing property of the minimal inclusive pronoun is the fact that it refers to exactly two individuals, *AUTHOR* and *ADDRESSEE*. The existing work on this phenomenon converges on the idea that this pronominal form should belong to the singular, and not to the plural system (McKay 1978; Noyer 1992; Corbett 2000, 2005; Harbour 2016). Examination of the morphological realization of minimal inclusive pronouns revealed that it typically surfaces with the exponent that is not present in the first-person singular. This exponent remains preserved in the augmented number.

I suggest that the minimal inclusive pronoun consists of two morphemes in its structural representation: the inclusive morpheme $\mathbb{M}_{incl}$ and the singular number, argued to be located on the *n* head (see Kramer 2015).

(41) *Minimal inclusive pronoun: The structure*



Therefore, the minimal inclusive is compositionally combined from the inclusive morpheme that exists independently in languages with the clusivity distinction and the morpheme carrying the singular feature.⁵ The exponent that arises in the minimal inclusive is the result of the Subset Principle. This form matches both features of the inclusive head [1, 2]. In the case of syncretism, that is, when the same formant is present in the first-person minimal, the minimal inclusive, and the minimal inclusive augmented, the underspecification derives the distribution of the exponents, since the given exponent corresponds to the subset of the features present in the inclusive head [1], as was the case in languages with clusivity and singular~plural distinction. Furthermore, the augmented number is located in the number (Num) morpheme, present in languages with the singular and plural.

4.7 Summary

In this chapter, I provided a detailed morphological and morpho-semantic examination of pronouns in languages that draw INCLUSIVE~EXCLUSIVE and MINIMAL~AUGMENTED distinctions. In the Introduction (4.1), I raised four questions: (i) How does the clusivity parameter inform the structure of pronouns?, (ii) How do the features involved in the clusivity parameter relate to the status of a morpheme as a basic unit of syntactic computation?, (iii) Does the inclusive exponent relate to a single morpheme or two morphemes (terminals) in the narrow syntax?, (iv) What are the targets of the Vocabulary Insertion operation in inclusive pronouns? After examination of the data, I proposed that inclusive pronouns in languages with clusivity do not have a different internal syntactic structure. Instead, the clusivity parameter is encoded at the morpheme level. The proposed $\mathbb{M}_{incl}$ is a morpheme that hosts both first- and second-person features, thus being the locus of the parameter. Therefore, the inclusive exponent relates to a single morpheme. As a consequence, the inclusive morpheme is targeted by the Vocabulary Insertion operation. This means that the realization of the structure proceeds in the same manner as elsewhere, with Vocabulary Insertion targeting a single morpheme. Furthermore, the minimal inclusive pronoun is proposed to be compositionally derived out of morphemes that independently exist in the natural language, that is, the inclusive morpheme, present in languages with clusivity, and the morpheme hosting the singular number.

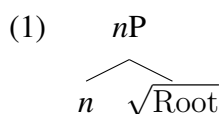
⁵Semantic reasons may require the singular/minimal number to surface above the person features in minimal inclusive pronouns, that is, the person must be in the scope of the minimal operator to obtain the correct semantic interpretation, as argued in Harbour (2016). I thank Uli Sauerland for bringing this fact to my attention and Artemis Alexiadou and Uli Sauerland for extensive discussions on this matter. The question that arises for future work is whether the singular/minimal number is base-generated above person, or it involves an operation comparable to quantifier raising applying within a word, in the case at hand, a pronoun. This important question awaits further research.

Chapter 5

Gender and Allomorphy

5.1 Introduction

An extensive research in nominal syntax brought arguments that gender does not head its own projection (Alexiadou 2004; Kramer 2010), but is located in the categorizing head n (Lecarme 2002; Lowenstamm 2008; Kramer 2015; Alexiadou, Lohndal, et al. 2015; Panagiotidis 2018; Deal 2016b; Adamson & Šereikaitė 2019; Adamson & Anagnostopoulou 2024). The n head (‘little n ’) is a syntactic terminal that is immediately adjacent to the $\sqrt{\text{Root}}$ (1). It has a twofold function: (i) nominalizing the acategorial $\sqrt{\text{Root}}$ and (ii) hosting gender features. As discussed in Chapter 1 in relation to the categorization assumption, a $\sqrt{\text{Root}}$ cannot surface as part of a more complex structure without being categorized. $\sqrt{\text{Roots}}$ are categorized by combining with category-defining functional heads, n , v , and a , which turn acategorial morphemes into nouns, verbs, and adjectives, respectively. On par with features such as person, number, Tense, Aspect, Voice, categorizing heads belong to functional morphemes. The nominalizing effects of little n are most transparently present in the case of Complex Event Nominals (CENs), since merging the n head closes the verbal domain of a nominalization, prevents verbal layers, and allows only the nominal layers (e.g., number, definiteness) to merge on top of it (Alexiadou 2020). Furthermore, the role of the n head is particularly important in the phenomenon of double plurality, where the two plural features are distributed across two nodes, NumP, hosting grammatical number, and the nominalizing head n , hosting lexical number (Acquaviva 2008; Alexiadou 2011, 2024; Kramer 2016). In addition to gender, the n head is the locus of features such as animacy (Kramer 2015; see Adamson & Anagnostopoulou (2024) for a view that animacy feature is directly dependent on gender and located on a and Panagiotidis (2018) for the view that animacy is located higher in the AnimP and Toosarvandani (2023) that is hosted together with person features in π). Being a categorizer, n is a phase head.



The study presented in this chapter builds on the line of work investigating the representation of gender features on pronouns (see Ritter 1995; Harley & Ritter 2002; Kratzer 2009; Panagiotidis 2018; Puškar-Gallien 2022). I focus on the **morphological realization** of gender features in pronouns across languages and discuss the principles of gender assignment and locus in the syntactic structure. The discussion focuses on the following set of questions.

QUESTIONS

1. How is gender morphologically realized in pronouns?
2. Do languages employ different means of realizing gender in pronouns?
3. Does the morphological realization of gender interact with person and number?
4. Are the same principles of gender assignment that apply in the nominal domain available to pronouns?
5. Where is the place of gender in the structure of pronouns in relation to person and number?

Studies on the patterns of allomorphy in pronouns discussed in previous chapters investigated the relation of person and number features and their sensitivity to each other. It has been demonstrated that both morphemes may act as triggers and targets of allomorphy. That is, the directionality in allomorphic conditioning goes in both directions. Adding gender features to this perspective increases complexity. Does gender interfere in allomorphy? Can it block an allomorphic alternation established between person and number? Which features does it interact with? The answers to these questions have important implications for the structure of pronouns.

Empirically, the data discussed in this chapter differ significantly from the interactions of person and number, since the gender distinctions reflected in morphology are attested in 80 out of the 350 languages investigated here. This fact is consistent with the argument made for the nominal domain that the number is a universal property of nouns, while gender is not (Alexiadou 2004). Cabécar, an indigenous Chibchan language spoken in the Talamanca Mountain Range of Costa Rica, has been demonstrated not to have the category of gender (Verhoeven 2012). The same applies to Turkish, which does not show the grammatical marking of gender features anywhere in the system (Panagiotidis 2024). Therefore, this study brings insights into the universality of building blocks making pronouns across languages.

The chapter is organized as follows. In the continuation of this section, I introduce the data (5.1.1), the Decision tree (5.1.2) for analyzing gender realization on pronouns, and evidence that gender features receive an overt realization in morphology (5.1.3). Section 5.2 presents a detailed discussion on the morphological realization of pronouns that make gender distinctions in both singular and plural numbers. In Section 5.3, the focus shifts to languages that exhibit systematic neutralizations of gender features in the plural. Section 5.4 discusses gender assignment in pronouns and identifies which types of gender are available in the pronominal domain. In the next section (5.5), I look at the $\sqrt{\text{Root}}$ domain more closely, revisiting the view that pronouns

involve a full silent noun in their representation. Section 5.6 presents implications of the obtained morphological, syntactic, and semantic results for the structure of pronouns, determining the place of gender with respect to person and number. Section 5.7 summarizes the chapter.

5.1.1 Data

This study is based on 80 languages that cover 25 language families and 4 isolates. All languages surfacing with gender distinctions in one of the person-feature specifications of the total sample of 350 languages are included in the present study. As the upcoming discussion will demonstrate, gender distinctions typically emerge in the third person but are not limited to it. For example, gender marking in both the second and third person is attested in Abkhaz (Caucasian), Pero, Amharic, Hebrew, Hausa (Afro-Asiatic), and Mian (Papuan). An overt gender marking limited to the second person is spread in Afro-Asiatic languages such as Iraqw, Goemai, and Tamashek, and in a Papuan (Sepik) language, Kwoma. Likewise, gender distinctions limited to the first and second person are attested in Páez, while across-the-board gender marking in the first, second, and third person is present in Itonama. The following table provides an overview of languages and their respective language families. Languages investigated in this study are included in Appendix C.

FAMILY	LANGUAGE
Afro-Asiatic	Semitic (Amharic, Hebrew), Cushitic (Somali), Berber (Tamashek)
Arawakan	Baure, Alto Perené
Arawan	Kulina
Australian	Ngandi, Ngankikurungkurr
Australian Pama-Nyungan	Yanyuwa
Australian Wororan	Worrorra
Austronesian	Kilivila, Sahu
Baining	Qaqet
Caucasian	Abkhaz
Dravidian	Kannada, Kolami, Malayalam, Malto, Tamil, Telugu, Tulu
Indo-European	Greek, Germanic (German), Celtic (Breton), Slavic (Serbian)
Japonic	Japanese
Korean	Korean
Mosetenan	Mosetén
Niger-Congo	Defaka, Ijo, Supyre
Páezan	Páez
Papuan	Lavukaleve, Savosavo, Inanwatan
Papuan Solomons East Papuan	Bilua
Papuan Sepik	Awtuw, Kwoma
Papuan Torricelli	Arapesh
Papuan Trans-New Guinea	Hamtai, Mian, Oksapmin
Salish	Bella Coola
Tucanoan	Cubeo, Siriano
Tupian	(Xingu) Asuriní
Yeniseian	Ket
isolates	Maybrat, Sandawe, Trumai, Itonama

5.1.2 The decision tree

For this study, I created a decision tree, inspired by the decision trees for comparative syntactic investigations developed in Roberts (2010) and Kalin (2022a), that yields morphological patterns of gender realization and its interaction with person and number, traceable across languages. The first criterion for including a language in the present study is an **observable gender distinction** in one of the person feature specifications. Therefore, the question at the top of the tree is the first cut. As noted above, 80 languages in the sample collected for this dissertation provided a positive answer to this question.

The next relevant question is **whether the gender distinction is present in both singular and plural**. This criterion results in a robust split. In only 23 out of 80 languages, gender distinctions are present in both number specifications. In the remaining 57 languages, gender marking is attested only in the singular, being neutralized in the plural. That is, languages surfacing with gender marking in both numbers correspond to less than one-third of the total number of languages investigated in this study. Therefore, two independent sets of questions were applied to the two resulting groups. I will first discuss the patterns observed in the group of languages that draw gender distinctions in both singular and plural.

Since gender has two features to consider, person and number, the next question applied to the group of languages that make gender distinctions in both singular and plural is **whether number-driven allomorphy can be identified**. Allomorphy is understood as the absence of any morpho-phonological relatedness of the two forms. This is important as it has to be ensured that we are indeed observing the exponents of gender, person, and number. This criterion gave rise to a further split. The number-driven allomorphy is attested in 4 out of 23 languages. In the remaining 19 pronominal systems, the person morpheme remains insensitive to the presence of the number.

The next step in unpacking the morphological realization of pronouns realizing gender features was identifying the exponent of person that is distinct from gender. For instance, two third-person pronouns that surface with different gender features share the same formant, which is also present in the plural number. This qualifies it as an exponent of the third person, as discussed throughout this dissertation. As elaborated before, a shared morpho-phonological material cannot be ignored and calls for the individuation of person exponents that relate the given form to the person morpheme in narrow syntax. The decision tree in (2) shows that in all languages that lack number-driven allomorphy, a person exponent is identifiable, i.e., all the languages responded positively to this question.

The final question asked with the aim to provide the full compositional analysis is **whether both gender and number are present as distinct exponents in the plural**. As noted in Chapter 2, in contrast to the singulative number, which typically stays silent, the plural receives an overt morphological realization. Therefore, a relevant question to ask is whether gender exponents will co-occur with number exponents in the environment where number is expected to be overt. As provided in the tree, languages fall into two groups with respect to this criterion.

Finally, among languages that draw gender distinctions in both singular and plural, there are

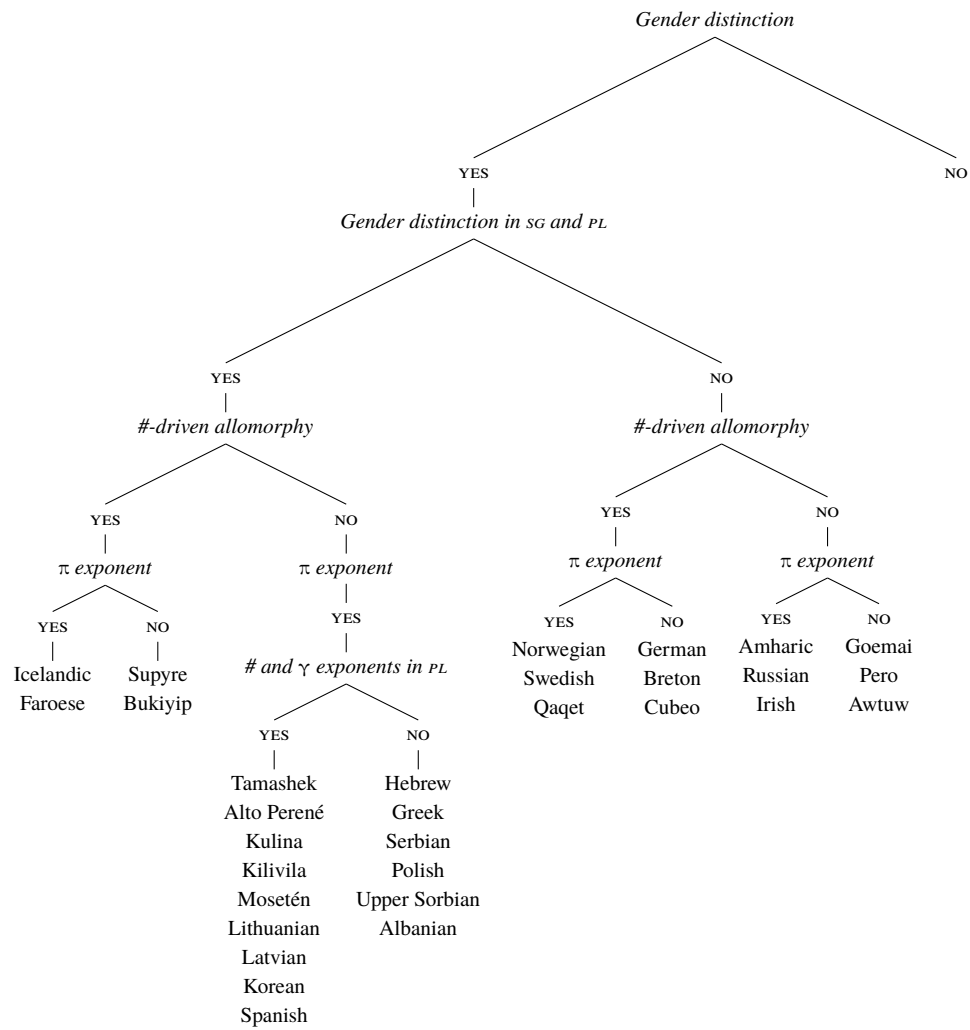
only four languages surfacing with the number-driven allomorphy. The same question raised for languages that do not surface with allomorphy was asked, namely **whether there is an identifiable exponent of person**. While Icelandic and Faroese respond positively to this test, Supyre and Bukiyip do not provide evidence for further segmentation of person exponents.

As noted above, 57 of the 80 languages investigated in this study exhibit gender neutralization in the plural. The same set of questions was asked. Languages falling into this group can be further differentiated depending on whether there is **number-driven allomorphy**. The crucial point becomes delineating person from gender exponents and correctly identifying both. The next applied criterion was **identifying the person exponent**, understood as a shared material that pronouns surfacing with the same person feature specification, but different gender features surface with. This resulted in four groups, as illustrated in the tree in (2).

In sum, what unifies all languages that respond positively to the **criterion π exponent** is the presence of a **shared morpho-phonological material** between the two (or more) variants of a pronoun that differ only in gender features.

The decision tree shows that individual languages employ different strategies for realizing gender features in pronouns. A question that arises is how this parameterization in the realizational component informs the investigation of the pronominal structure. In the upcoming sections, I will discuss the morphological realization of gender features and their relation to person and number in each branch of the tree. Before that, I will demonstrate that gender receives an overt realization in morphology (5.1.3), on par with person and number. In the next step, I will provide an analysis of languages that make gender distinctions in both singular and plural (5.2), moving to languages where gender distinctions are limited to the singular number only in Section 5.3.

(2) THE DECISION TREE



5.1.3 Point of departure: Exponents of gender

Before delving into patterns of allomorphy involving gender features, the natural question that arises is whether gender is realized via discrete, identifiable exponents in pronouns. In Chapter 2, I discussed the exponents of person and number in pronominal systems as evidence for their presence as morphemes in narrow syntax. In this section, the empirical perspective extends to cover gender features in natural languages.

In two Indo-European languages, Serbian and Lithuanian, one of the pronouns constituting the feminine~masculine minimal pair surfaces with an additional exponent. In Serbian, the third-person feminine pronoun surfaces with an exponent $[a]$, which is absent in the masculine form. Once this exponent is eliminated, the two pronouns are identical, consisting only of the exponent of person $[(o)n]$. Lithuanian employs a different strategy, marking the third-person masculine pronoun with an additional exponent $[s]$.

	SG	PL
1	ja	mi
2	ti	vi
3 _M	on	oni
3 _F	on- a	one

(a) Serbian

	SG	PL
1	àš	mēs
2	tù	jūs
3 _M	jì- s	jiẽ
3 _F	jì	jõs

(b) Lithuanian (Ramonienė et al. 2019)

Table 5.1: Exponents of gender in pronouns

Comparable patterns are attested in a range of typologically unrelated languages. Hebrew patterns with Lithuanian in employing an additional exponent $[a]$ in the second-person masculine form. This contrasts with the third person, where the two forms share the formant $[h]$, pointing toward its status as an exponent of person, and combining with the formants $[u]$ and $[i]$ in the masculine and feminine variants, respectively. In Korean, the feminine variant surfaces with an additional formant $[nye]$, transparently combining with $[ku]$, shared with the masculine form, and the exponent of the nominative case $[ka]$.

	SG	PL
1	ani	anachnu
2 _M	at- a	atem
2 _F	at	aten
3 _M	h- u	hem
3 _F	h- i	hen

(a) Hebrew (Ritter 1995)

	SG	PL
1	nay-ka	wuli-ka
2	ne-ka	nehuy-ka
3 _M	ku-ka	ku-tul-i
3 _F	ku- nye -ka	ku- nye -tul-i

(b) Korean (Smith et al. 2019)

Table 5.2: Exponents of gender in pronouns

In Sahu (Austronesian), the masculine, feminine, and neuter third-person pronouns surface with exponents $[u]$, $[mu]$, and $[e]$, respectively. Note that all third-person pronouns across both

number specifications share the exponent *[naŋa]*. As it is arguably present in the second person, too, this formant cannot be an exponent of the third person, but rather an exponent of D (see Chapter 6). The feminine third-person pronoun may potentially be further segmented into *[m]*, an exponent of gender, and *[u]*, an exponent of person shared with the masculine form. Alto Perené (Arawakan) makes a distinction between masculine referents, in which case the form *[iri(ro)]* is chosen, and all the other non-masculine referents *[iro]* (Mihás 2015). As evident from the data in Table 5.3, the masculine form surfaces with an extra exponent *[ri]*.

	SG	PL		SG	PL
1EXCL	ŋoi	ŋomi	1EXCL	naaka	naakaite
1INCL		ŋene	1INCL		aroka/arokaite/arorite
2	ŋana	ŋini	2	aviroka	avirokaite
3M	u -naŋa	a -naŋa	3M	i-ri -(ro)	i-ri -ro-ite
3F	mu -naŋa	a -naŋa	3NON-M	i-ro	i-ro-ite
3N	e -naŋa	e -naŋa			

(a) Sahu (Pertsova 2022b)

(b) Alto Perené (Mihás 2015)

Table 5.3: Exponents of gender in pronouns

These data demonstrate that exponents of both the masculine and the feminine genders are transparently identifiable alongside the person in the morphological realization of pronouns across languages. As noted in Chapter 2, an overt exponent implicates the presence of a morpheme, i.e., a syntactic terminal, in narrow syntax. The question that arises is what morpheme in the narrow syntax do these exponents refer to. In addition, where is it located in the syntactic structure of a pronoun with respect to person and number? In the next sections, I discuss the morphological realization of pronouns realizing gender features with the aim of elaborating on these questions.

5.2 Gender distinctions in singular and plural

Languages in which gender distinctions are observable in both singular and plural numbers are referred to as **parallel gender systems** (Corbett 1991). This means that two singular pronouns surfacing with the masculine and feminine gender correspond to two plural pronouns, i.e., syncretism does not apply. In this section, I will first discuss languages in which the gender distinction is preserved in the plural and both gender and number exponents are identifiable.

5.2.1 Person, number, and gender: 1-to-1 mapping

In Mosetén (Mosetenan), the masculine and feminine forms of the third-person pronoun, *[mi']* and *[mö']*, respectively, are preserved in the plural, transparently combining with the plural exponent *[in]*, shared across the paradigm. Note that the language involves a high degree of

syncretism. Namely, the distinction between the second and third person masculine is phonologically marked [*mi* vs. *mi'*]. Likewise, the third-person feminine pronoun shares the initial formant [*m*], which can be analyzed as an exponent of person, while the feminine gender is realized with the exponent [*ö'*]. Kilivila (Austronesian) third-person pronouns exhibit a complex morphological pattern that calls for a decomposition. Namely, both third-person pronouns share an initial formant [*m*] and a final formant [*na*], which can be analyzed as exponents of person and D, respectively. Importantly, the difference emerges in the formant [*to*], present in the masculine and [*ina*] in the feminine, suggesting an analysis in terms of gender exponents. All exponents present in the singular are preserved in the plural, combining with the plural exponent [*si*], present also in the first person, which linearizes to the right of gender.

	SG	PL		SG	PL
1	yäe	tsin	1EXCL	yegu	yakamesi
2	mi	mĩ'in	1INCL		yakidasi
3M	m-i'	m-i'-in	2	yokwa	yokwami
3F	m-ö'	m-ö'-in	3M	m-to-na	m-to-si-na
			3F	m-ina-na	m-ina-si-na

(a) Mosetén (Sakel 2011)

(b) Kilivila (Senft 1986)

Table 5.4: Preserving gender morphology

I propose the following analysis of Mosetén (3). In addition to person and number exponents, gender features are realized in morphology, with [*ö'*] being inserted when the feminine feature is present in a relevant node in narrow syntax and [*i'*] in the case of the masculine feature. Adopting the notational convention of Kramer (2015), I specify the presence of the feminine gender as [+FEM] and the masculine as [-FEM].

(3) VIs for Mosetén

- [3] ↔ m
- [PL] ↔ in
- [+FEM] ↔ ö'
- [-FEM] ↔ i'

The set of Vocabulary Items in (4) accounts for Kilivila. The exponent [*m*] realizes the third-person feature, while the exponent [*si*] realizes the plural number. Feminine [+FEM] and masculine [-FEM] gender features are mapped to discrete exponents [*ina*] and [*to*], respectively.

(4) VIs for Kilivila

- [3] ↔ m
- [PL] ↔ si
- [+FEM] ↔ ina
- [-FEM] ↔ to

A comparable pattern is present in Kulina (Arawan) and Alto Perené (Arawak). In Kulina, the exponent of the feminine gender *[ni]* is present in the plural, linearizing between exponents of person *[po]* and plural *[deni]*. Note that the formant *[wa]* in the masculine is also present in the first person, while the feminine exponent *[ni]* is not attested elsewhere in the singular. In contrast, Alto Perené draws a distinction between the masculine and all the other non-masculine referents, as already noted. In this language, it is the exponent of the masculine gender that receives an overt realization, being preserved in the plural, while the feminine remains zero. As illustrated in the table below, the masculine and non-masculine third-person pronouns show a significant overlap, differing precisely in the exponent *[ri]*, present in the former and absent in the latter.

	SG	PL
1	owa	ia
2	tia	tia-deni
3 _M	powa	powa-deni
3 _F	po- ni	po- ni -deni

(a) Kulina (Dienst 2014)

	SG	PL
1 _{EXCL}	naaka	naakaite
1 _{INCL}		aroka/arokaite/arorite
2	aviroka	avirokaite
3 _M	i- ri -(ro)	i- ri -ro-ite
3 _{NON-M}	i-ro	i-ro-ite

(b) Alto Perené (Mihas 2015)

Table 5.5: Preserving gender morphology

Korean third-person pronouns draw a gender distinction in the singular that is carried over to the plural. The third-person feminine pronoun surfaces with a formant *[nye]* that is absent in the masculine. On par with data discussed above, the overt exponent of gender linearizes between the person *[ku]* and number *[tul]* exponents. The remaining exponents, *[ka]* and *[i]*, are realizations of the nominative case. Their alternation is phonologically-driven (S. Cho 2016), as *[ka]* combines with preceding vowels and *[i]* with consonants.

	SG	PL
1	nay-ka	wuli-ka
2	ne-ka	nehuy-ka
3 _M	ku-ka	ku-tul-i
3 _F	ku- nye -ka	ku- nye -tul-i

(a) Korean (Smith et al. 2019)

(5) **VIs for Korean**

[3] ↔ ku

[PL] ↔ tul

[+FEM] ↔ nye

Table 5.6: Preserving gender morphology

Tamashek (Berber; Afroasiatic) exemplifies an interesting pattern where gender distinctions are made in the plural but not in the singular, thus violating Greenberg's Universal 45, according to which a language that has gender distinctions in the plural exhibits them in the singular, too (Greenberg 1963; 1990). This is the case in first- and third-person pronouns, where a single singular pronoun corresponds to two plural pronouns depending on the gender of the referent,

exhibiting one-to-many mapping from the singular to the plural. Note that the second person exhibits the parallel gender system with two distinct singular pronouns. As illustrated in the table below, all feminine plural pronouns surface with the formant [æ̣t], which can be analyzed as an exponent of the feminine gender. A departure from Greenberg’s Universal 45 is attested in several Berber languages, as discussed for the first-person pronouns in Ntifa spoken in Central Morocco (Plank & Schellinger 1997). Furthermore, note that in the first- and third-person pronouns, where gender distinctions are present in the plural but not in the singular, gender stacking arises under the analysis provided in (6). This is a possible outcome, as multiple *n* heads are readily available in natural language (see the discussion in Kramer (2015) on two adjacent *ns* in Somali). Furthermore, in the second person where stacking is absent, the exponent of gender that arises in the plural [æ̣t] co-occurs with the singular version of the pronoun [kæ̣m] that already marks gender, and can be decomposed into the exponent of person [k] and the exponent of gender and singular number [æ̣m]. These data are reminiscent of the phenomenon of double plurality (Acquaviva 2008; Alexiadou 2011, 2024; Kramer 2016) in the domain of gender features. In addition, one of the exponents can be the realization of animacy.

	SG	PL	
1M	næk	nækk-æn-ed	
1F	næk	nækk-æn-æ̣t-ed	(6) VIs for Tamashek
2M	kàyy	kæw-æn-ed	[PL] ↔ ed
2F	kàmm	kæm-æ̣t-ed	[+FEM] ↔ æ̣t
3M	ə̣nta	ə̣nt-æn-ed	[-FEM] ↔ æ̣n
3F	ə̣nta	ə̣nt-æn-æ̣t-ed	

(a) Tamashek (Heath 2005)

Table 5.7: Co-occurrence of masculine and feminine

In Romance languages, gender distinctions are preserved in the plural, with overt number morphology co-occurring with gender exponents. As noted earlier, Kayne (2003) pointed out that the plural exponent surfacing in nouns is present in third-person pronouns in French, where it transparently combines with singular forms differing in gender. Spanish employs a different strategy, as the formants [o] and [a] yield the difference between the two forms. Finally, Italian differs from both French and Spanish, as the feminine third-person pronoun is syncretic across singular and plural, which is not the case with the masculine.

The two plural forms, however, exhibit an important semantic difference. Taking Spanish as an example, in the nominal domain, the final vowels [o/a] are not exponents of gender, but realizations of the inflection class, as both can be used with the two gender specifications (Alexiadou 2004). In pronouns, while [ellos] can be chosen both in cases where the referent of a pronoun consists of male individuals and mixed groups consisting of both male and female, [ellas] can be used when the referent of a pronoun consists of female-only individuals, but not for mixed groups and cases when there is at least one male individual. Focusing on the two vowels that

SG PL			SG PL			SG PL		
1	je	nous	1	yo	nosotros.M/nosotras.F	1	io	noi
2	tu	vous	2	tú	vosotros.M/vosotras.F	2	tu	voi
3 _M	il	il-s	3 _M	él	ell-o-s	3 _M	l-ui	l- ei
3 _F	elle	elle-s	3 _F	ella	ell-a-s	3 _F	l-oro	l- oro
(a) French (Kayne 2003)			(b) Spanish			(c) Italian		

Table 5.8: Preserving gender morphology

make the distinction between the forms under consideration, $[o]$ and $[a]$, the insertion of $[a]$ triggers a stronger presupposition that the referent of the pronoun consists of female-only individuals. At PF, this is captured by the workings of the Subset Principle that inserts $[a]$ as more specific exponent of the n head when it surfaces with the feminine gender in narrow syntax and outcompetes $[o]$ which figures as an elsewhere realization.

In languages discussed so far, the person, gender, and number were realized as distinct exponents exhibiting the 1-to-1 mapping from syntax to morphology. This is not universally the case, as in several languages, number and gender are not identifiable as two discrete exponents in the realizational component of the grammar. I turn to these data in the next section.

5.2.2 Gender-driven number allomorphy

Slavic languages exhibit gender distinctions in both singular and plural, as illustrated in Table 5.9 on the basis of Serbian, Polish, and Upper Sorbian. In the three languages under consideration, the third-person feminine pronoun surfaces with the exponent $[a]$, transparently combining with person morphology, in contrast to the masculine version, which overtly realizes only person $[(o)n]$. The vowel $[a]$ in the third-person feminine pronoun, emerges in nouns surfacing with the feminine gender [*žen-a* ‘woman’; *stolic-a* ‘chair’] including nominals that surface with the masculine natural, i.e., conceptual gender [*vladik-a* ‘bishop’] (for an extensive discussion on this matter in Serbian see Puškar (2018); for a related cross-linguistic discussion on realization of inflection and gender in nouns denoting profession in Alexiadou (2004) for Spanish, and Kučerová (2018) for Italian). In pronouns, this vowel cannot be considered the realization of an inflection class, but the genuine exponent of the gender feature. Evidence for this comes from the fact that the pronoun $[ona]$ can be chosen *only* in cases when the referent surfaces with the feminine gender. Mismatches observed in the nominal domain do not carry over to pronouns. In the plural, masculine combines with the elsewhere exponent of the plural $[i]$, shared with the first- and second-person, but also with nouns, as argued in Chapter 2 for Serbian, while the feminine form is realized with $[e]$ in the plural. This morphological alternation calls for an explanation. As noted above for Romance, the feminine form $[one]$ can be chosen only when the referent of a pronoun consists of female-only individuals, thus triggering a stronger presupposition, while $[oni]$ is used for male-only individuals and mixed groups.

SG PL			SG PL			SG PL		
1	ja	mi	1	ja	my	1	ja	mi
2	ti	vi	2	ty	wy	2	ti	wi
3 _M	on	on-i	3 _M	on	on-i	3 _M	wun	won-i
3 _F	on-a	on-e	3 _F	on-a	on-e	3 _F	won-a	won-e

(a) Serbian (b) Polish (Sadowska 2012) (c) Upper Sorbian (Pertsova 2022b)

Table 5.9: Gender-driven number allomorphy

As evident from Table 5.9, the exponent $[e]$ in the plural is not further segmentable. Non-segmentable exponents that nevertheless implicate more than one feature are typically referred to as *cumulative exponence* or *portmanteau exponence* (for a recent overview on cumulative exponence see Fenger 2023). Analyses of several empirical domains involving portmanteau morphology converge on the idea that it is best captured in terms of contextual allomorphy, yielding a number of empirical and conceptual advantages over its alternatives (Bobaljik 2000; Trommer 2007; Fenger 2018). Now, we face two analytical possibilities. If considered an allomorph, the exponent $[e]$ can be (i) a number-driven allomorph of gender in an $[a \sim e]$ alternation or (ii) a gender-driven allomorph of number, giving rise to an $[e \sim i]$ alternation. This is a non-trivial question, as under the possibility (i) gender figures as a target of allomorphy, while under (ii) it is a trigger of allomorphy. Are there empirical facts that would help us discern between the two options? Note that the elsewhere exponent of the plural in Serbian $[i]$ disappears in the context of the feminine gender. The fact that it is the elsewhere that gets lost is a strong indication of a gender-driven number allomorphy. In addition, the newly emerged exponent $[e]$ is distinct both from the elsewhere plural $[i]$ and the feminine $[a]$. This fact can be captured in terms of *bidirectional allomorphy*. In the context of the feminine gender feature, the plural exponent is realized as $[e]$. At the same time, in the context of the plural, the feminine gender receives zero $[\emptyset]$ realization. When these conditions are not met, the elsewhere exponents are selected (7). In the following VI rules, γ stands for gender.

(7) **VI_s for Slavic**

- $[PL] \leftrightarrow e / _ \gamma_{+FEM}$
 $[+FEM] \leftrightarrow \emptyset / _ \#_{PL}$
 $[PL] \leftrightarrow i$
 $[+FEM] \leftrightarrow a$

In contrast to Slavic, Hebrew draws gender distinctions in both second- and third-person pronouns. The exponents of the second and third person, $[at]$ and $[h]$, respectively, remain unaffected by the number morpheme. In the plural, masculine forms surface with the exponent $[m]$, while feminine ones with $[n]$. The system once again implicates a closer interaction of gender and number to the exclusion of person. Observed regularity in the selection of exponents can be captured in terms of the VI rules in (8), where the plural exponent $[n]$ is selected in

the context of the feminine feature [+FEM], while [m] in the context of the masculine [-FEM]. When the context specification is not satisfied because the relevant features are not present, the gender-induced exponents [n/m] do not emerge, as evidenced by the first-person pronoun.

	SG	PL	
1	ani	anaxnu	
2 _M	ata	ate- m	(8) VIs for Hebrew
2 _F	at	ate- n	[PL] $\leftrightarrow$ n / ___ γ_{+FEM}
3 _M	hu	he- m	[PL] $\leftrightarrow$ m / ___ γ_{-FEM}
3 _F	hi	he- n	

(a) Hebrew (Ritter 1995)

Table 5.10: Gender-driven number allomorphy

The pattern in which gender distinctions are preserved in the plural but gender and number are not realized in a segmentable manner, resulting in two discrete exponents, is present in Greek (Table 5.11). Greek is a three-way gender language, as exemplified by the three distinct variants of the third-person pronoun. The exponents of person [t] and demonstrativity [af] remain invariant across the system. However, [os], [i], and [o] in the singular, alternate with [i], [es], and [a] in the plural. The analysis in terms of allomorphy suggests itself as an explanation, as the realization of number is sensitive to the presence of a particular gender feature specification. In the continuation of this chapter, I will provide a more elaborate analysis of Greek in relation to the first and second person and the exact type of gender assignment.

	SG	PL
1	ego	emis
2	esi	esis
3 _M	af-t- os	af-t- i
3 _F	af-t- i	af-t- es
3 _N	af-t- o	af-t- a

Table 5.11: Gender-driven number allomorphy

The central empirical insight of languages discussed here is a closer interaction of gender and number, directly reflected in the realizational component. While these features exhibit an interaction evidenced by non-segmentable exponents, the exponent of person is clearly identifiable. The question that arises is what implications for the structure of pronouns can we draw from this finding.

5.2.3 Multiple alternations

In a small number of languages that preserve gender distinctions in the plural, number-driven allomorphy of the person is simultaneously present. This is the case in Germanic languages such

as Icelandic and Faroese. The exponent of the third person, [h], alternates with [] in Icelandic and [t] in Faroese, but the plural forms are not syncretic, preserving the distinction between the masculine and female genders. In both languages, the exponent of person in the third-person plural is syncretic with the exponent of the second person, thus calling for an underspecification explanation of the realization of person. Crucially, for the present discussion, the person exponents are identifiable and segmentable in these languages. Furthermore, note that the first and second persons surface with shared plural (and case) exponents, realized as [i] in Icelandic and [it] in Faroese, which is not the case in the third person. The presence of the gender features makes the exponents that combine with the third-person distinct both from each other and from exponents in the first and second person. Again, the exponent of person is segmentable in the morphological realization of pronouns making a gender distinction, which is not the case with gender and number, thus further pointing toward a close connection of the two features. Icelandic and Faroese data contrast with Serbian, Hebrew, and Greek in exhibiting allomorphy of the person morpheme in the plural. Importantly, this does not affect the segmentability of person exponents.

	SG	PL		SG	PL
1	ég	v-ið	1	eg	v-it
2	þú	þ-ið	2	tú	t-it
3 _M	h -ann	þ -eir	3 _M	h -ann	t -eir
3 _F	h -ún	þ -ær	3 _F	h -on	t -ær

(a) Icelandic (Neijmann 2021)

(b) Faroese (Lockwood 1977)

Table 5.12: Multiple alternations

The last pattern attested in languages that preserve the gender distinction in the plural is the one where (i) number-driven allomorphy applies and (ii) it is not possible to identify a shared person exponent, thus contrasting in this respect with Icelandic and Faroese. In Supyre (Niger-Congo), third-person pronouns distinguish between three genders, typically referred to as gender 1, 2, and 3 (Carlson 1994). The three pronouns in the singular all differ from each other, without the possibility of further segmentation into a person exponent and an additional material. In the plural, they undergo number-driven allomorphy, combining with the exponent of plural [i]. A comparable situation is present in Bukiyip (Torricelli; Papuan), where the two third-person pronouns are realized with distinct exponents without any shared material, undergoing number-driven allomorphy. In addition, the language surfaces with a pronoun [echech/chachi], selected in cases when the referent of a pronoun is a mixed group, consisting of both masculine and feminine individuals. In contrast, the pronouns [omom/mami] and [owo/wawi] are chosen when the referent of a pronoun consists of masculine-only or feminine-only individuals, respectively. Therefore, masculine plural pronouns in this language contrast with Romance and Slavic, where only the feminine plural triggers a presupposition that female-only individuals are included in the referent of a pronoun, while masculine-only and mixed groups receive a uniform realization.

	SG	PL		SG	PL
1	mìi	wùu	1	yek	apak
2	mu	yìi	2	nyak	ipak
3 GEN 1	u	p-i	3 _M	énan/nani	omam/mami
3 GEN 2	ku	y-i	3 _F	okok/kwakwi	owo/wawi
3 GEN 3	li	c-i/k-i	3 _{MIXED}		echech/chachi

(a) Supyre (Carlson 1994)

(b) Bukiyip (Pertsova 2022b)

Table 5.13: Multiple alternations

A related pattern is attested in Hamtai (Papuan; Trans-New Guinea), where the masculine form remains preserved in the plural, thus patterning with languages discussed above where number does not trigger allomorphy, but the feminine form exhibits an allomorphic alternation [*i~iʔj*]. Both masculine and feminine combine with the same exponent of plural [*oa*].

	SG	PL
1	ni	nai
2	nti	hai
3 _M	əqo	əqu-oa
3 _{FEM}	i	iʔj-oa

Table 5.14: Pronouns in Hamtai (Pertsova 2022b)

Examination of the data discussed so far demonstrated that pronouns make gender distinctions in singular and plural numbers. Several relevant empirical generalizations emerged. Segmentation of person exponents is possible in languages where pronouns exhibiting gender marking undergo number-driven allomorphy and those that lack it. As exemplified on the basis of languages like Mosetén, the three features, that is, person, gender, and number, show 1-to-1 mapping between the narrow syntax and morphology with three distinct exponents (G1). Crucially, in languages where segmentation is not possible, gender and number features interact closely, while the exponent of person remains intact (G2). This pattern carries over to languages exhibiting number-driven allomorphy, as is the case in Icelandic and Faroese, where, despite allomorphy, the person exponent is identifiable and segmentable, which is not the case with gender and number.

(9) EMPIRICAL GENERALIZATIONS: GENDER IN BOTH SINGULAR AND PLURAL

G1: Person, gender, and number exponents are segmentable in the plural.

G2: In languages where segmentation is not possible, gender and number are realized via a non-segmentable exponent, while person remains visible.

5.3 Towards systematic neutralizations

This section presents an examination of languages in which gender distinctions are restricted to the singular number and neutralized in the plural, i.e., a single pronoun is used irrespective of the gender of its referents. As illustrated in the decision tree in (2), the two criteria applied, that is, (i) the presence of number-driven allomorphy and (ii) the possibility of segmenting the person exponent as distinct from gender and number, resulted in four groups. The central empirical contribution of these data is that **gender neutralization** in the plural is **systematic**, thus raising the question of its theoretical modeling. Before proceeding with providing an analysis for each group, I will present a detailed case study of Somali pronouns, which demonstrates how the post-syntactic feature-deletion Impoverishment operation accounts for the systematic neutralizations in a principled way.

5.3.1 Case study: Impoverishment blocks Concord in Somali

In Somali (Cushitic; Afro-Asiatic), personal pronouns may optionally co-occur with the non-remote definite determiner drawing the gender distinction [-*ka*.MSC / -*ta*.FEM] (Saeed 1999; Lecarme 2002; Özyıldız & Ivan 2017). The masculine determiner [*ka*] surfaces in its allophone form [*ga*], while the feminine [*ta*] surfaces as [*da*]. Note that the allophone [*ka*] is present in the second-person plural, where the pronoun ends in a consonant. All pronouns take the masculine determiner, except *iyáda* ‘she/her’. Therefore, the realization of the definite determiner shows sensitivity to the presence of the feminine gender features, on par with the nominal domain. Furthermore, the definite article linearizes to the right of a pronoun in both cases, consistent with the linearization of the definite article in nouns (10).

	SG	PL		
1EXCL	aní-ga	anná-ga	(10)	éy-ga dog-DET ‘the dog’ Lecarme (2002)
1INCL		inná-ga		
2	adí-ga	idín-ka		
3M	isá-ga	iyá-ga		
3F	iyá- da	iyá-ga		

(a) Pronouns (Saeed 1999)

Table 5.15: Distribution of the definite determiner in Somali

The alternation in the realization of the definite determiner can be analyzed in terms of (i) gender-driven allomorphy and (ii) agreement/nominal concord. Quite often, the two phenomena are not easily distinguishable (see Weisser (2019) for the set of diagnostics and discussion). In the former analysis, the definite determiner [*da*] instantiates a gender-driven definiteness allomorphy conditioned by the feminine gender feature in a pronoun. Analysis in terms of allomorphy is a plausible path to explore, especially in light of the fact that a comparable pattern of gender-driven allomorphy of definite articles is attested in other languages, as demonstrated

for Bulgarian (Gribanova & Harizanov 2017). Under the latter analysis, the determiner reflects DP-internal nominal agreement, on par with languages like French, German, and Greek. As pointed out in Lecarme (2002), a general property of Cushitic languages that sets them apart from Indo-European is that definite determiners do not show agreement for number with the noun but for gender instead. This is consistent with the argument that the inventory of characteristics that participate in the nominal concord is determined on a language-specific basis (Norris 2014). In line with the existing literature on nominal syntax in Somali (Lecarme 2002; Kramer 2015), I analyze gender-driven alternation in Somali definite determiners as a result of nominal concord. However, nothing in the present argument would change if the observed alternation in the realization of the definite article were an instance of allomorphy.

The following set of VI rules (11) accounts for the data. The competition between the two definite determiners is resolved in terms of the Subset Principle. Following the standard principles of nominal concord (Norris 2014; Landau 2016), the D head collects the feminine features of the pronoun. When the D head contains both definiteness and feminine gender feature, the more specific Vocabulary Item *[da]* is inserted. Otherwise, the elsewhere *[ga]* surfaces.

	SG	PL	
1EXCL	aní-ga	anná-ga	(11) VIs for the determiner in Somali
1INCL		inná-ga	
2	adí-ga	idín-ka	
3M	isá-ga	iyá-ga	
3F	iyá- da	iyá-ga	

[DEF, +FEM] ↔ da
[DEF] ↔ ga

(a) Somali (Saeed 1999)

Table 5.16: Gender-driven definiteness allomorphy in Somali

Three important points must be taken into consideration when proposing an account of Somali facts (12). The question is why nominal concord for gender features does not emerge with exactly the same realization of the third-person pronoun. In the plural, it reverts to the elsewhere form. The absence of the determiner *[da]* in the plural indicates that the exponent *[iyá]* in the plural does not realize a morpheme that contains a gender feature, i.e., gender features are not present in the plural, despite syncretism.

(12) GENDER-DRIVEN CONCORD OF THE DEFINITE DETERMINER IN SOMALI PRONOUNS

- a. feminine gender-driven concord is **present** in the **singular**
- b. feminine gender-driven concord is **absent** in the **plural**
- c. **plural** forms are **syncretic** with the **feminine singular**

In the previous chapter, I discussed the patterns of syncretism in Algonquian. In Ojibwe, the exponent of the second person *[kiin]* is attested in both second-person plural and first-person plural inclusive pronouns, thus giving rise to syncretism. At the same time, the exclusive and

inclusive pronouns share the plural exponent [*awint*]. The same holds for the second- and third-person pronouns that surface with the same exponent of plural [*awaa*].

	SG	PL	(13) VIs for Ojibwe
1EXCL	niin	niin-awint	[PL] $\leftrightarrow$ awint / π_{AUTHOR}
1INCL		kiin -awint	[PL] $\leftrightarrow$ awaa
2	kiin	kiin -awaa	[1, 2] $\leftrightarrow$ k
3	wiin	wiin-awaa	[1] $\leftrightarrow$ n
			[2] $\leftrightarrow$ k

(a) Ojibwe (Harley & Ritter 2002)

Table 5.17: Patterns of syncretism in Algonquian

To account for Algonquian data, I proposed an analysis in terms of **contextual allomorphy** and **underspecification**. In narrow syntax, the person morpheme in the exclusive carries the first person, i.e., the [AUTHOR] feature, while the inclusive morpheme surfaces with two person features, [AUTHOR, ADDRESSEE], thus constituting the superset of the exclusive. The presence of the first person is crucial for deriving the distribution of the plural exponents. I proposed that in the context of the first person, i.e., [AUTHOR] feature, the allomorph of plural [*awint*] is selected. In its absence, the elsewhere form [*awaa*] that does not require any context specification surfaces.

The fact that exactly the same exponent of person [*kiin*] combines with two different exponents of plural, [*awint*] and [*awaa*], is explained in terms of **underspecification** of the Vocabulary Item [*kiin*]. Namely, the syncretic exponent of person, shared between the inclusive and the second person, corresponds to **two different morphemes** in narrow syntax, specified for different features. The fact that person may act as a **trigger** of allomorphy, giving rise to the observed alternation, suggests that the syncretic Vocabulary Item in fact corresponds to different features in the syntax. In the realizational component of the grammar, i.e., at PF, a single underspecified VI [*kiin*] applies to two distinct morphemes in the syntax, giving rise to **syncretism**. This is possible since the second-person feature is a **subset** of the features present in the inclusive.

Somali exhibits a crucial difference. The syncretic exponent of person [*kiin*] triggers allomorphy of the number in the inclusive pronouns in Ojibwe, thus influencing the realization of other morphemes. The same does not hold for the syncretic form [*iyá*]. In the plural, the definite determiner cannot refer to gender features. That is, at the point when the determiner is targeted for Vocabulary Insertion, feminine-gender features are no longer present in a pronoun anymore. As a consequence, it resorts to the elsewhere form [*ga*]. The fact that in Ojibwe we have allomorphy and in Somali concord is orthogonal to this debate. The main point that these data make is that gender features in Somali cannot influence the realization of the definite determiner in the plural in the way they do in the singular. Therefore, an analysis in terms of underspecification of Vocabulary Items where the syntax contains the necessary feature accounts for the Algonquian syncretism, but it is simply not enough for Somali. In Somali, the syncretism is followed by the blocking of the concord that otherwise arises. This implies that we are in need of an analysis

that explains both the syncretism and blocking.

Syncretism, understood as mapping of two distinct feature bundles to the same morphological realization, can be explained in terms of (i) **Underspecification** and (ii) **Impoverishment** in Distributed Morphology. While underspecification does not invoke any specific operation, being a mere property of Vocabulary Items, Impoverishment is a post-syntactic operation with an effect of deleting a feature in a particular morpho-syntactic environment (Bonet 1991, 1995; Noyer 1992). In DM, Impoverishment takes place at PF prior to the Vocabulary Insertion operation. As a consequence, it does not affect narrow syntax but explains the patterns of syncretism in the realizational component. As a feature-deletion operation, Impoverishment should crucially be distinguished from Obliteration (Arregi & Nevins 2007, 2012; Kouneli 2021), which deletes the entire morpheme, and Pruning (Embick 2010). Being a post-syntactic operation, it is expected that Impoverishment would potentially interact with other operations located in this branch of the grammar, as discussed for Pruning (Calabrese 2012) and Local Dislocation (Watanabe 2013). In this study, I show how it interacts with the nominal concord, effectively blocking it.

Several distinct empirical domains motivated the view that this operation is closely tied with the morphological markedness (Noyer 1992; Nevins 2011a, 2012; Arregi & Nevins 2007, 2012; Nevins & Parrott 2010). Nevins (2011a) proposes that markedness can act as both a trigger and target of Impoverishment, thus distinguishing between (i) *markedness-targeted impoverishment*, in which case a marked category gets neutralized in a particular environment and (ii) *markedness-triggered impoverishment*, where it is the marked category that causes neutralization of other categories. It is demonstrated that the dual, argued to be the most marked number specification, can act in both neutralization directions, i.e., as a trigger and target. In addition to feature deletion, it has been proposed that Impoverishment may also involve feature insertion at PF. However, insertion is restricted to unmarked values (Noyer 1998) or opposite values of the same feature (Harbour 2003). In relation to markedness, Impoverishment rules have been proposed to be triggered when a language reaches and exceeds its complexity threshold (Noyer 1992). Thus, the obtained syncretism that arises as a result of Impoverishment involves neutralization of marked forms to lesser marked ones (Bobaljik 2002) at the level of morphology. At this point, we may observe a paradoxical situation. Namely, *the more complex the narrow syntax, the less complex the morphological realization*. Taking our main concern here, gender and number, if we decided to inspect only the plural realizations of pronouns in languages where neutralization of gender distinctions in the plural arises, we would not be in a position to detect the presence of gender in the narrow syntax in these languages.

Neutralization of gender distinctions in the plural is typically modeled in terms of Impoverishment (Noyer 1992; Bobaljik 2002; Harley 2008; Nevins 2011a). As illustrated above, syncretism in Somali is followed by the absence of the nominal concord in the plural. This can be explained if the feature that triggers the realization of a definite determiner marked for gender, which is, in the case at hand, feminine gender, is not available at the point when the determiner is targeted for Vocabulary Insertion. I propose that the application of an Impoverishment rule that

deletes feminine features in the plural successfully accounts for the distribution of the definite article in Somali.

	SG	PL
1 _{EXCL}	aní-ga	anná-ga
1 _{INCL}		inná-ga
2	adí-ga	idín-ka
3 _M	isá-ga	iyá-ga
3 _F	iyá- da	iyá-ga

(a) Somali (Saeed 1999)

(14) **VIIs for the determiner in Somali**

[DEF, +FEM] $\leftrightarrow$ da

[DEF] $\leftrightarrow$ ga

(15) **IMPOVERISHMENT RULE**

[+FEM] $\rightarrow \emptyset / __\text{PL}$

Table 5.18: Gender-driven definiteness allomorphy in Somali

If gender features are deleted in the plural as a result of Impoverishment, **no subsequent Vocabulary Insertion rule may refer to these features**. This is precisely what we find in Somali. As a consequence of deletion, the Vocabulary Insertion rule for the determiner cannot make reference to gender features in the plural, and the elsewhere realization of the definite article surfaces. Therefore, the set of VI rules introduced at the beginning of this section needs to be augmented with an Impoverishment Rule (15) that deletes the feminine feature in the plural.

Two desired effects arise. **First**, in the plural, the third-person pronoun cannot combine with the definite article [*da*] as there are no gender features that would trigger the nominal concord. **Second**, the resulting syncretic form [*iyá*] can be inserted in cases when the referent of a pronoun consists of male-only individuals and mixed groups, as feminine gender features are not present. These data demonstrate that Impoverishment rules derive patterns of syncretism that go beyond the properties of individual Vocabulary Items. Once deleted, a feature cannot be employed for subsequent operations. In this way, it not only accounts for syncretism but also explains the absence of the nominal concord in the plural.

5.3.2 The patterns

With these considerations in place, I will now move to the analysis of the four patterns attested in languages that make gender distinctions only in the singular and exhibit systematic neutralizations in the plural. As a reminder, the four patterns are yielded by the decision tree discussed at the beginning of this chapter.

5.3.2.1 Number-driven allomorphy: Segmentable person exponent

In Norwegian and Swedish, two closely related Germanic languages, the masculine and feminine variants of third-person pronouns share the formant [*h*]. In line with the criteria discussed earlier in this dissertation, shared morpho-phonological material calls for a decomposition. In that sense, the shared formant [*h*] can be analyzed as an exponent of the third-person. In the plural, both forms are neutralized in favor of a single third-person plural pronoun [*de*]. Following the

principles applied earlier, the alternation $[h \sim d]$ is the case of number-driven person allomorphy. Note, in addition, that the initial $[d]$ is present in the second-person plural, too, where it can be analyzed as an exponent of the second-person that does not undergo allomorphy. Such plural-induced syncretism of person exponents is a property that is spread across several Germanic and related Indo-European languages. As evident from Table 5.19, it applies in Norwegian but not in Swedish that exhibits the canonical case of number-driven person allomorphy evident in the $[d \sim n]$ alternation in the second person, thus showing that syncretism facts are subject to micro-variation even within a single branch of a particular language family. Crucially, both languages employ one plural pronoun that corresponds to two singular ones and does not carry any information about the gender of its referents. In this, Norwegian and Swedish differ sharply from Icelandic.

	SG	PL
1	jeg	vi
2	du	dere
3 _M	h-an	d-e
3 _F	h-un	d-e

(a) Norwegian
(Strandskogen & Strandskogen 1994)

	SG	PL
1	jag	vi
2	du	ni
3 _M	h-an	d-e
3 _F	h-on	d-e

(b) Swedish
(Hinchliffe & Holmes 2013)

Table 5.19: Identifiable person exponent

I propose the following set of Vocabulary Items (16) and an Impoverishment rule (17). With the deletion of the gender features in the plural, the Vocabulary Insertion can no longer make reference to gender. Therefore, person exponents found in the plural are plain exponents of the person morpheme in the context of the plural (or the default realization of the third person). In that sense, Norwegian and Swedish third-person plural pronouns do not differ from languages that do not draw gender distinctions anywhere in the system, discussed in the previous chapters.

(16) **VIs for PERSON in Norwegian/Swedish**

$[3] \leftrightarrow d / __ \#_{PL}$

$[3] \leftrightarrow h$

(17) **IMPOVERISHMENT RULE**

$[\gamma] \rightarrow \emptyset / __ PL$

The same holds beyond Indo-European, as illustrated on the basis of Qaqet (Baining) and Baure (Arawakan). In Qaqet, the third-person pronouns share the formant $[k]$, being neutralized to $[ta]$ in cases of human referents and $[nget]$ when the referent of a pronoun includes non-human referents. When it comes to Baure, it is important to note that all pronouns surface with the formant $[tiʔ]$. Therefore, it cannot be considered an exponent of person, number, or gender.

Once it is eliminated, $[n]$, $[p]$, and $[r]$, the exponents of the first, second, and third person, respectively, alternate with $[v]$, $[j]$, and $[n]$ in the plural, thus establishing a clear case of number-driven person allomorphy. Importantly, the vowel $[i]$ linearizes between person exponents and the formant $[tiʔ]$ in the first- and second-person pronouns across both numbers and the third-person feminine form. In the masculine, however, it alternates with the allomorph $[o]$, which is preserved in the plural. This calls for an analysis in terms of allomorphy, where $[o]$ is a more specific realization, conditioned by the masculine gender, while $[i]$ is the elsewhere form. At the same time, the person exponent undergoes number-driven allomorphy, as evident from the $[r \sim n]$ alternation. This is an example of a language in which person and number behave more closely to the exclusion of the gender, which is still available based on their adjacency in narrow syntax.

	SG	PL
1	ngua	ut
2	nyi	ngen
3 _M	ka	ta/nget
3 _F	ki	ta/nget

(a) Qaqet (Hellwig 2019)

	SG	PL
1	n-i-tiʔ	v-i-tiʔ
2	p-i-tiʔ	j-i-tiʔ
3 _M	r-o-tiʔ	n-o-tiʔ
3 _F	r-i-tiʔ	n-o-tiʔ

(b) Baure (Pertsova 2022b)

Table 5.20: Identifiable person exponent

5.3.2.2 Number-driven allomorphy: Non-segmentable person exponent

Languages that make a gender distinction in one of the person specifications, e.g., third person, often employ two entirely different morphological realizations for the two pronouns under consideration. These two forms do not share any morpho-phonological resemblance. Hence, the presence of a discrete exponent of person, shared between the two pronouns, cannot be identified. I propose that these data can best be captured in terms of gender-driven person allomorphy. This pattern is exemplified in Table 5.21 on the basis of Celtic languages, Breton, Scottish Gaelic, and Manx.

	SG	PL
1	me	ni
2	te	c'hwi
3 _M	eñ	int
3 _F	hi	int

(a) Breton (Press 2011)

	SG	PL
1	mi	sinn
2	t(h)u	sibh
3 _M	e	iad
3 _F	i	iad

(b) Scottish Gaelic
(Gillies 2009)

	SG	PL
1	mi	ʃin
2	u	ʃu
3 _M	e	ad ; əd
3 _F	i	ad ; əd
3 _{COMM}	ə	ad ; əd

(c) Manx (Pertsova 2022b)

Table 5.21: Gender-driven person allomorphy in Celtic

A way to account for Breton and comparable data is to assume that the feminine gender (i) triggers allomorphy of person in the singular and (ii) undergoes Impoverishment in the plural. I propose the following set of rules accounts for the data (18-19). Both third-person pronouns are realizations of the third-person feature in the context of the gender. The third-person masculine pronoun realizes only the person feature in the context of the gender specified as $[-FEM]$, i.e., masculine. Likewise, the third-person feminine pronoun is the realization of the third-person in the context of the feminine gender $[+FEM]$. Importantly, the gender distinction is not made in the plural. This fact can be captured by proposing an Impoverishment rule (19) which deletes feminine gender features in the context of the plural number, on par with languages discussed above. Now, with the deletion of gender, the third-person plural pronoun $[int]$ is the realization of the third person in the context of the plural. Note that the set of VI rules in (18) does not employ any rule ordering based on hierarchy. That is, it does not need to be stipulated that plural number features are higher than gender and therefore the VI rule making reference to number takes precedence. As a result of the Impoverishment rule in (19), gender features will simply never be present in the plural. Thus, there is no competition among the Vocabulary Items in (18).

(18) **VIs for Breton**

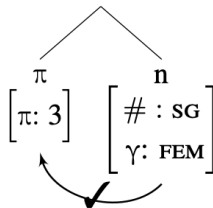
- $$\begin{aligned} [3] &\leftrightarrow int / __ \#_{PL} \\ [3] &\leftrightarrow hi / __ \gamma_{+FEM} \\ [3] &\leftrightarrow eñ / __ \gamma_{-FEM} \end{aligned}$$

(19) **Impoverishment Rule**

$$[\gamma] \rightarrow \emptyset / \#_{PL}$$

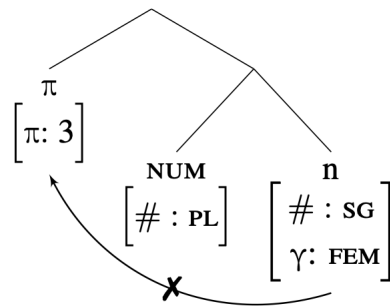
The question that emerges is whether the absence of gender distinctions in the plural correlates with the syntactic structure of pronouns, which may be different from the one present in the singular. As noted above, gender and singulative number are proposed to be hosted on n (Kramer 2015; Alexiadou 2019a, 2024). In other words, they constitute a single morpheme. This contrasts with the plural number, which is hosted on a higher layer encoding plurality, NumP. Now, consider the following structural representations for the third-person singular and plural pronouns (20-21). In the singular, the person morpheme, carrying the third-person feature, and n , hosting gender and the singular number, are immediately structurally adjacent (20). In this configuration, feminine gender features on the n head freely interact with the person features, resulting in gender-driven person allomorphy.

(20) **GENDER-DRIVEN PERSON ALLOMORPHY**



The plural pronoun, however, surfaces with a different structure (21). The number node [NUM] intervenes between gender and person. One may argue that the presence of the number morpheme has a *blocking effect*, preventing gender from triggering allomorphy of the person. As a consequence, the feminine-gender-induced allomorph of the third person [*hi*] cannot emerge in the plural. However, there is a reason not to pursue this analysis. Namely, this analysis would not explain why the plural forms systematically do not make any reference to gender features. On the basis of Greek, Hebrew, and Serbian data, we have seen that **number and gender closely interact to the exclusion of person** in languages that indeed preserve gender features in the plural. While the intervening number node may readily block allomorphy of the person, it remains unexplained why number and gender do not implicate any interaction. Instead, an inspection of the exponents in the third-person plural does not suggest gender specification of the pronoun's referent.

(21) GENDER-DRIVEN PERSON ALLOMORPHY BLOCKED



The same pattern of gender neutralization is attested in typologically related Germanic languages. English distinguishes masculine, feminine, and neuter in the singular, which is overridden in the plural. None of the forms in the singular morphophonologically correlates with the plural. Following the principle discussed above for Breton, third-person pronouns are realizations of the third-person feature in the context of gender. In case of the neuter [*it*], I assume the plain *n* head in the sense of Kramer (2015). In the plural, the elsewhere realization of the third-person arises. Namely, the pronoun [*they*] shares the formant [*th-*] proposed to act as an elsewhere realization of the third person in English (Bernstein 2008a; Alexiadou et al. 2024). As illustrated in (23), the definite article realizes the definiteness (ι) or kind operator ($\cap$) together with the third-person feature. Interestingly, the same formant arises in the third-person plural pronouns where gender distinctions are neutralized. This is readily explainable if the realization of the third-person resorts to the elsewhere exponent after Impoverishment, without making reference to the *n* head.

			(22) VIs for English
			[3] $\leftrightarrow$ he /___ γ_{-FEM}
			[3] $\leftrightarrow$ she /___ γ_{+FEM}
			[3] $\leftrightarrow$ it /___ γ
			(23) VI for the determiner (Alexiadou et al. 2024)
			$[\iota/\cap, 3] \leftrightarrow$ the
			(24) Impoverishment Rule
			$[\gamma] \rightarrow \emptyset / \#_{PL}$

	SG	PL
1	I	we
2	you	you
3 _M	he	th-ey
3 _F	she	th-ey
3 _N	it	th-ey

(a) English

Table 5.22: Gender-driven person allomorphy in Germanic

In Abkhaz (Caucasian), gender distinctions arise in the second and third person. Note that all person specifications across both numbers share the formant *[ara]*, hence it cannot be considered an exponent of person, number, or gender (see Chapter 6 on D morphemes). Once this component is eliminated, each pronoun surfaces with a single exponent that is not further segmentable. In (25), I offer an analysis in terms of contextual allomorphy. The first person does not make a gender distinction. In the plural, it alternates with the exponent *[h]*. Therefore, I analyze *[h]* as a number-driven allomorph of the first person while *[s]* as its elsewhere realization. In the second person, the exponent *[w]* is the realization of the second-person feature in the context of the masculine gender $[-FEM]$, while the exponent *[b]* is its realization in the context of the feminine gender $[+FEM]$. After Impoverishment, the exponent $[\check{s}^{\circ}]$ is the realization of the second person in the context of the plural number (or an elsewhere). The same principle extends to third-person pronouns.

			(25) VIs for Abkhaz
			[1] $\leftrightarrow$ h /___ $\#_{PL}$
			[1] $\leftrightarrow$ s
			[2] $\leftrightarrow$ $\check{s}^{\circ}$ /___ $\#_{PL}$
			[2] $\leftrightarrow$ b /___ γ_{+FEM}
			[2] $\leftrightarrow$ w /___ γ_{-FEM}
			[3] $\leftrightarrow$ d /___ $\#_{PL}$
			[3] $\leftrightarrow$ l /___ γ_{+FEM}
			[3] $\leftrightarrow$ d /___ γ_{-FEM}
			(26) Impoverishment Rule
			$[\gamma] \rightarrow \emptyset / \#_{PL}$

	SG	PL
1	s-ara	h-ara
2 _M	w-ara	$\check{s}^{\circ}$-ara
2 _F	b-ara	$\check{s}^{\circ}$-ara
3 _M	j-ara	d-ara
3 _F	l-ara	d-ara

(a) Abkhaz (Pertsova 2022b)

Table 5.23: Gender-driven person allomorphy in Abkhaz

Comparable patterns are attested in a number of typologically unrelated languages. In Cubeo

(Tucanoan), the masculine [ĩ] and feminine [õ] third-person variants are neutralized in favor of a single third-person plural form [dã], which does not share any morpho-phonological resemblance with the singular variants. The same holds for Defaka (Niger-Congo) and Wolaytta (Afro-Asiatic), where the singular variants surfacing with different gender features are morpho-phonologically distinct from each other but also from a plural third-person pronoun. Therefore, the formal analysis of Breton and Germanic can be extended to account for the wide range of typologically unrelated data.

	SG	PL		SG	PL		SG	PL
1 _{EXCL}	jĩ	jĩ-xã	1	i	wá	1	ta	nu
1 _{INCL}		bã-xã	2	í	ó	2	ne	hi:nte
2	bĩ	bĩ-xã	3 _M	o	iní/oní	3 _M	a	e:ta
3 _M	ĩ	dã	3 _F	á	iní/oní	3 _F	i	e:ta
3 _F	õ	dã	3 _N	je	iní/oní			

(a) Cubeo (Pertsova 2022b) (b) Defaka (Pertsova 2022b) (c) Wolaytta (Pertsova 2022b)

Table 5.24: Gender-driven person allomorphy across languages

Note that gender neutralization does not hold only for languages exhibiting singular~plural number system, but also for those with the minimal~augmented distinction, as illustrated in the following data from Gude and Malakmalak. In the context of the augmented number, gender features are neutralized.

	MIN	AUG		MIN	AUG
1	nyi/ii	inə	1	ŋa	yawot
1+2	anwu	amə	1+2	yaŋki	yɛŋkit
2	hə	unə	2	waŋari	nukut
3 _M	c-i	(ə)tii	3 _M	y-öntön	wöŋ-öntön
3 _F	k-i	(ə)tii	3 _F	n-öntön	wöŋ-öntön

(a) Gude (Hoskison 1983)

(b) Malakmalak (Birk et al. 1976)

Table 5.25: Gender neutralization in minimal~augmented systems

5.3.2.3 No number-driven allomorphy: Segmentable person exponent

In languages that make gender distinctions in both singular and plural, number-driven allomorphy of the person morpheme does not need to apply. In Amharic, gender distinction is made in both the second- and third-person pronouns, on par with other Semitic languages such as Hebrew. As illustrated in Table 5.26, the masculine singular forms are preserved in their entirety in the plural, transparently combining with the plural exponent [ənn]. Note that Amharic exhibits interesting linearization properties. The person exponent linearizes to the right of the

plural exponent, which has not been the case in the languages discussed so far. However, closer inspection of the linearization properties of definite determiners in Amharic (28-29) reveals that the linearization of the person mimics that of the definite article, which linearizes to the right, following both the root and plural exponents. In the nominal domain, gender features in Amharic are implicated by agreement facts observable on definite determiners, demonstratives, and verbs. There is no particular exponent for realizing masculine or feminine gender in the nominal domain (Kramer 2015). The two second-person pronouns distinguishing gender differ only in the final vowel, [ä] in the case of masculine, and [i] feminine (/j/ in the second-person feminine can be analyzed as phonologically-driven). As gender features typically do not receive an overt morphological realization, I analyze the exponents [ä/i] as exponents of gender, whose insertion is sensitive to the presence of the second person, i.e., ADDRESSEE, realized as [ant]. In the plural, gender features undergo neutralization to masculine variants of singular pronouns [antä/əss-u]. This is not surprising given the fact that masculine is the default in Amharic (Kramer 2015). Note that **neutralization** arises **in the context of an overt exponent of the plural**. Therefore, an **Impoverishment rule does not need to target a bundle deleting a feature of that bundle**. That is, the two features that act as a trigger and target of Impoverishment do not need to be on the same morpheme in order for deletion to apply. Instead, it deletes the feature in the morphology with an increase in the structural complexity of a pronoun.

	SG	PL	(27) Some VIs for Amharic
1	əne	əñña	
2 _M	ant-ä	ənn- ant-ä	
2 _F	anj-i	ənn- ant-ä	
3 _M	əss- u	ənn- äss-u	
3 _F	əss- ^w a	ənn- äss-u	

(a) Amharic (Leslau 1995)

[PL] ↔ ənn
 [2] ↔ ant
 [3] ↔ əss
 [+FEM] ↔ i / ____ π_{ADDRESSEE}
 [-FEM] ↔ ä / ____ π_{ADDRESSEE}

Table 5.26: Pronouns in Amharic: Analysis

(28) *bet-u*
 house-DEF
 ‘the house’

(29) *nigist-wa*
 queen-DEF.F
 ‘the queen’ Kramer (2010)

The realization of third-person pronouns exhibits different properties. Once the shared material [əss] is removed, the remaining exponents making the difference between the two third-person pronouns in fact correspond to definite determiners. In Amharic, definite determiners reflect the gender of the nominal they combine with. Masculine nouns combine with the definite determiner [u] and its allophone version arising after vowels [w], while feminine nouns combine with the definite article [wa]. Vocabulary Insertion rules for the definite article are provided in (30). The competition between the two determiners is resolved in terms of the Sub-

set Principle as the exponent $[wa]$ matches a greater number of features of the D head. However, as Kramer (2015) points out, further decomposition into the D exponent $[u]$ and a zero gender marker $/u+\emptyset/$ in the masculine and the D exponent $[u]$ and a gender marker $[a]$ $/u+a/$ surfacing as $[wa]$ is possible (Kramer 2010; 2015).

(30) **VIs for the definite determiner** (Kramer 2015)

D, [DEF], [+FEM] $\leftrightarrow$ wa

D, [DEF] $\leftrightarrow$ u

These formants are found in third-person pronouns: $[u]$ in the masculine and $[wa]$ in the feminine. On par with the second person, gender neutralization in the plural is followed by the insertion of the masculine forms. This is again consistent with the nominal domain, where definite determiners in the plural always take the masculine gender form, i.e., the default, irrespective of the actual gender of a noun (Kramer 2010). It is important to note that the set of Amharic independent pronouns provided in Table 5.26 corresponds to morphologically complex pronouns used only exceptionally (see also Chapter 6), while the arguments of a verb are typically realized as clitics concatenated with a verb. Despite this difference, Amharic data demonstrate that gender features on pronouns can be realized in various guises, including the strategies employed in the nominal domain, such as reflecting gender on the definite determiner that combines with the exponent of person. When it comes to its assignment, the gender of a pronoun is always dependent on the natural gender of the pronoun's referent.

In Hausa, a related Afro-Asiatic language, that patterns with Amharic in drawing gender distinctions in the second- and third-person, the second-person pronouns share the formant $[k]$, which can be analyzed as an exponent of person. It is preserved in the plural, transparently combining with the plural exponent $[uu]$, spread across the paradigm. Hausa contrasts with Amharic in not preserving one of the singular variants in its entirety, but only the person exponent. In the third person, I analyze $[sh]$ and $[s]$ as allomorphs. On par with data discussed above, the absence of gender distinctions in the plural is explainable in terms of Impoverishment (32).

	SG	PL	
1	n-ii	m-uu	(31) Some VIs for Hausa
2 _M	k-ai	k-uu	[PL] $\leftrightarrow$ uu
2 _F	k-ee	k-uu	[2] $\leftrightarrow$ k
3 _M	sh-ii	s-uu	[+FEM] $\leftrightarrow$ ee / $___ \pi_{\text{ADDRESSEE}}$
3 _F	?ita	s-uu	[-FEM] $\leftrightarrow$ ai / $___ \pi_{\text{ADDRESSEE}}$
(a) Hausa (Wolff 1993)			(32) IMPOVERISHMENT RULE
			$[\gamma] \rightarrow \emptyset / [\text{PL}]$

Table 5.27: Pronouns in Hausa: Analysis

In Russian, the exponents of person $[(o)n]$ are transparently identifiable in the plural, combining with the plural exponent $[i]$. Russian contrasts with Slavic languages such as Serbian,

Polish, and Upper Sorbian, discussed earlier in this chapter, in not preserving the gender distinctions in the plural, as the elsewhere exponent *[i]* is chosen in cases in which the referent of a pronoun consists of female-only individuals. In Irish, the formant *[s]*, shared between the masculine and feminine versions of the third-person pronouns, is preserved in the plural. Germanic languages exhibit a pattern where number induces syncretism of the person. Syncretism spreads throughout the paradigm in Irish. The fact that the remaining formants are morpho-phonologically distinct from each other suggests an underspecification analysis of the person exponent. Crucially, in both languages, gender distinctions vanish in the plural.

	SG	PL		SG	PL
1	ya	my	1	mé	s-inn
2	ty	vy	2	tú	s-ibh
3 _M	on	on-i	3 _M	s-é	s-iad
3 _F	on-a	on-i	3 _F	s-í	s-iad

(a) Russian

(b) Irish (Pertsova 2022b)

Table 5.28: Segmentable person exponents

In Pashto (Iranian; Indo-European), the masculine and feminine variants of the third-person share the formant *[d]* that is preserved in the plural. On par with other languages discussed here, gender distinction is neutralized through the form *[duy]* in the plural. Japanese variants of the masculine and feminine third-person pronouns can be analyzed as two independent forms, *[kare]* and *[kanojo]*, respectively, whose distinct morpho-phonological realization is conditioned by gender features. However, there is a certain amount of shared material, and the formant *[ka]* can be individuated as an exponent of the third person, with *[re]* and *[nojo]* realizing the masculine and feminine features, respectively. Importantly, *[kare]* is the form that is chosen in the plural, and the gender distinction is neutralized.

	SG	PL		SG	PL
1	zə	munɡ/munz	1	watashi-ga	watashi-tachi-ga
2	tə	tāse/tāsi	2	anata-ga	anata-tachi-ga
3 _M	day	duy	3 _M	kare-ga	kare-ra-ga
3 _F	dā	duy	3 _F	kanojo-ga	kare-ra-ga

(a) Pashto (David 2013)

(b) Japanese (Pertsova 2022b)

Table 5.29: Segmentable person exponents

The unifying property of languages discussed in this section is a shared exponent between the masculine and feminine variants of a pronoun. This suggests an analysis of the shared exponent as a realization of the person feature. In addition, this exponent is transparently traceable in the plural forms, thus showing that the number-driven allomorphy of the person does not apply. A pattern that emerges as a logical possibility is the one in which the masculine and feminine

variants do not share an exponent, but one of the exponents present in the singular is traceable in the plural. I turn to these data in the next section.

5.3.2.4 No number-driven allomorphy: Non-segmentable person exponent

Similarly to languages like Breton, in a subset of languages that lack number-driven allomorphy, the morphological distinction between the two pronouns can be argued to be a result of gender-driven allomorphy. The two variants, masculine and feminine, do not share a morpho-phonological material that can be considered a segmentable person exponent. However, in contrast to Breton and comparable data, there is no number-driven allomorphy of the person morpheme. Namely, the person exponent in one of the third-person variants spreads to the plural.

In Pero (Afro-Asiatic), discussed in Chapter 3, the third-person plural pronoun surfaces with the plural exponent [ínù], shared with the first person. The exponent of person [c] is present in the masculine singular form, too. Goemai (Afro-Asiatic) makes a gender distinction in the second person. The two variants, [gǝe] and [yǝe], can be further decomposed into [g/y], realizing person, and a formant [ǝe]. Importantly, the masculine form [g] spreads to the plural, combining with [w], that can be analyzed as an exponent of the plural in the context of the second person, and a formant [ǝn], present in the first person singular and plural.

	SG	PL		SG	PL
1	nè	m-ínù	1	h-ǝn	m-ǝn
2 _M	kè/kài	m-à	2 _M	g-ǝe	g-w-ǝn
2 _F	cì	m-à	2 _F	y-ǝe~yǐ	g-w-ǝn
3 _M	cákkà	c-ínù	3	ní	muǝp
3 _F	tè	c-ínù			

(a) Pero (Frajzyngier 1989)

(b) Goemai (Hellwig 2011)

Table 5.30: Non-segmentable person exponents

Awtuw (Papuan; Sepik) exhibits a similar pattern, with the masculine and feminine variants [rey] and [tey], being further decomposed into [r/t] and a shared formant [ey], a pattern typical of Papuan languages. The masculine form [r] is preserved in the plural, transparently combining with the plural exponent [om].

	SG	PL	
1	wan	n-om	(33) Some VIs for Awtuw
2	yen	om	
3 _M	r-ey	r-om	
3 _F	t-ey	r-om	

(a) Awtuw (Pertsova 2022b)

[3] ↔ r
[PL] ↔ om

Table 5.31: Gender-driven person allomorphy in Awtuw

5.3.3 Full syncretism: Neutralization to one of the singular variants

In a small number of languages in which a gender distinction is neutralized in the plural, this is achieved via syncretism with one of the singular forms. These data contrast with the absence of allomorphy of the person morpheme, in exhibiting the **full syncretism** with one of the singular variants. This is the case in Amharic and Somali, which have already been discussed in this chapter. As illustrated in Table 5.32, in Amharic, both second- and third-person pronouns employ the masculine variant of the singular, [antä] and [əssu]. In the plural, these forms transparently combine with the exponent of the plural number [ənn]. Somali employs a different strategy. Namely, the feminine variant of the third-person pronoun [iyá] spreads to the plural, optionally combining with the definite article [ga]. Plural number receives zero [Ø] realization.

	SG	PL		SG	PL
1	əne	əñña	1EXCL	aní-ga	anná-ga
2M	antä	ənn- antä	1INCL		inná-ga
2F	anči	ənn- antä	2	adí-ga	idín-ka
3M	əssu	ənn- ässu	3M	isá-ga	iyá -ga
3F	əss ^w a	ənn- ässu	3F	iyá -da	iyá -ga

(a) Amharic (Leslau 1995)

(b) Somali (Saeed 1999)

Table 5.32: Neutralization to singular

German patterns with Somali in spreading the feminine variant [sie] to the three cells in the plural, i.e., masculine, feminine, and neuter. It contrasts with Japanese, where the masculine variant [kare] is chosen in the plural. Together with Amharic and Somali, these languages demonstrate that there is no uniform strategy when it comes to syncretism with one of the singular variants, as both masculine and feminine are freely available to spread in the plural.

	SG	PL		SG	PL
1	ich	wir	1	watashi-ga	watashi-tachi-ga
2	du	ihr	2	anata-ga	anata-tachi-ga
3M	er	sie	3M	kare -ga	kare -ra-ga
3F	sie	sie	3F	kanojo-ga	kare -ra-ga
3N	es	sie			

(a) German

(b) Japanese (Pertsova 2022b)

Table 5.33: Neutralization to singular

Broadening the empirical perspective with languages originating in different families further corroborates this claim. In Páez, an indigenous Páezan language spoken in Colombia, the second person feminine pronoun [iʔkwe], which can be decomposed into [iʔ] and [kwe], shared with the first person feminine singular, is preserved in the plural, transparently combining with the plural exponent [ʔsʲ], spread across the paradigm. It appears tempting to analyze [kwe] as an

exponent of the feminine gender that additionally triggers allomorphy of the person, evident in alternations [ad^j~uʔ] and [id^j~iʔ] in the first and second person, respectively. Note that the decomposition can be pushed further. The first-person masculine form can be split into [a] and [d^j], while the first-person feminine into [ũ] and [ʔ]. Likewise, the second person masculine can be decomposed into [i] and [d^j], while the second person feminine into [i] and [ʔ]. From this perspective, masculine and feminine versions of the two person specifications share exponents. Namely, the exponent [d^j], shared between the first- and second-person masculine forms, can be analyzed as an exponent of the masculine gender. The exponents [a] and [i] are realizations of the first- and second-person features, respectively. Furthermore, the exponent shared between the first and second-person feminine forms, [ʔ], can be analyzed as an exponent of the feminine gender. In the first person, it triggers allomorphy of the person morpheme, resulting in the exponent [ũ] that emerges as a consequence of gender-driven person allomorphy. In the second person, allomorphy does not apply as the masculine and feminine variants share the exponent of person [i]. This opens a possibility to analyze [kwe] as an exponent of D or animacy. In that case, the person morpheme receives a zero [∅] realization in the plural.

Coming back to the main concern of this section, while Páez spreads the feminine form in the second-person plural, on par with German and Somali, Siriano, an indigenous Tucanoan language spoken in Colombia and Brazil, exhibits the opposite direction in spreading the masculine variant [ĩŋũ] to the plural. This exponent transparently combines with the exponent of number [sã], shared with the second person.

	SG	PL		SG	PL
1M	ad ^j	kweʔs ^j	1EXCL	j ^{HH}	mãrĩ
1F	ũʔkwe	kweʔs ^j	1INCL		g ^{HA}
2M	id ^j	ĩʔkweʔs ^j	2	mũũ	mũũ-sã
2F	ĩʔkwe	ĩʔkweʔs ^j	3M	ĩŋũ	ĩŋũ-sã
3	t ^j ã:	t ^j ã:weʔs ^j	3F	igo	ĩŋũ-sã

(a) Páez (Pertsova 2022b)

(b) Siriano (Pertsova 2022b)

Table 5.34: Neutralization to singular

These data demonstrate that languages employ both strategies in cases of full syncretism with one of the singular variants, as both masculine and feminine singular forms can be spread in the plural. In addition, Amharic, Japanese, Páez, and Siriano surface with an overt realization of the plural that freely co-occurs with the singular forms. These languages contrast with German and Somali, which exhibit full syncretism without additional plural morphology.

The question that arises is how to account for the observed patterns of spreading. In an approach to gender features where masculine, as less marked, would be a subset of the features of the feminine, feminine variants would never be able to be inserted in third-person masculine plural pronouns, as they contain features that are *not* present in the node. As a consequence, the Subset Principle would rule them out. Therefore, these data further substantiate an argument that

Impoverishment applies with an effect of deleting features. The two syncretic forms correspond to two different features in the narrow syntax, and both syncretic variants can spread to the plural.

5.4 Gender on pronouns: Locus and assignment

Natural language makes a distinction between conceptual and arbitrary gender (Kramer 2015, 2020; Alexiadou 2017; Panagiotidis 2018; Sudo & Spathas 2020; Adamson & Anagnostopoulou 2024). Namely, feminine gender in nouns such as *mother* exemplifies conceptual gender assignment (34-a), as it closely correlates with its semantics. In contrast to these cases, the arbitrary gender assignment reflects the fact that the gender of a noun typically does not interact with the semantics of a $\sqrt{\text{Root}}$ (34-b). For example, the noun *spoon* surfaces with the masculine gender in German [*der Löffel*], feminine in Serbian [*kašika*], and the neuter in Greek [*to koutáli*]. This fact shows that gender features are dissociated from the $\sqrt{\text{Root}}$, hosted in a separate projection, and arbitrarily assigned in individual languages. Therefore, conceptual gender corresponds to the natural gender, while arbitrary gender corresponds to the grammatical gender of a noun.

- (34) a. CONCEPTUAL GENDER
 mother [FEM]
 b. ARBITRARY GENDER
 spoon: *der Löffel* [MSC] in German, *kašika* [FEM] in Serbian, *to koutáli* [NEUT] in Greek

As noted in the Introduction, extensive research in this empirical domain led to a broad consensus in syntactic theory that in the nominal domain, gender does not head its own projection (Alexiadou 2004; Kramer 2010), but is located in the categorizing head *n* (Lecarme 2002; Lowenstamm 2008; Kramer 2015; Alexiadou, Lohndal, et al. 2015; Deal 2016b; Adamson & Šereikaitė 2019; Adamson & Anagnostopoulou 2024).

- (35) nP
 $\begin{array}{c} \diagup \quad \diagdown \\ n \quad \sqrt{\text{Root}} \end{array}$

There is a debate in the literature as to whether both conceptual and arbitrary genders are located on the same head. Kramer (2015) proposes the following possible specifications of the *n* head that aim to capture both types of gender (36). Namely, when gender features have semantic effects, as is the case with the conceptual gender, they are *interpretable* (*n i*). The distinction between masculine and feminine is captured through a binary feature, specified as [+FEM] in the case of feminine and [−FEM] in masculine nouns, while neuter nouns combine with a plain *n* that does not carry any gender features. This contrasts with an arbitrarily assigned gender that does not have semantic effects, hence uninterpretable (*n u*). The same binary feature accounts

for the distinction between masculine and feminine, as the categorizing head *n* is specified as [+FEM] when the arbitrarily assigned gender is feminine, as was the case with the noun ‘spoon’ in Serbian and [–FEM] when it is masculine in German. The idea that both types of gender are hosted on the same head figures prominently in accounts of so-called hybrid agreement nouns surfacing with different conceptual and arbitrary genders and the dual feature approaches thereof (Wurmbrand 2017; Smith 2021). However, to account for the same data, that is, hybrid nouns in Serbian, Puškar (2018) argues that natural gender and animacy are hosted on *n*, while grammatical (arbitrary) gender is on a higher layer, GenP. Panagiotidis (2018) proposes that conceptual gender is located on a higher layer encoding animacy (see also Distributed gender hypothesis in Steriopolo & Wiltschko 2010; Steriopolo 2018).

(36) Types of *n*

n *i* [+FEM]

n *i* [–FEM]

n

n *u* [+FEM]

n *u* [–FEM]

The representation of gender features has been a matter of debate in the literature (see Ritter 1995; Harley & Ritter 2002; Kratzer 2009; Panagiotidis 2018; Puškar-Gallien 2022). Ritter (1995) presents one of the first case studies looking at the representation of gender in Hebrew pronouns. In the feature hierarchy of Harley & Ritter 2002, gender is the projection located lower than both person and number, while Kratzer (2009) proposes that third-person pronouns consist only of gender features, lacking person. Panagiotidis (2018) pays a special attention to animacy features and its interaction with the conceptual (natural) gender, locating this kind of gender specification above the grammatical gender. Puškar-Gallien’s (2022) study on pronouns in Serbian presents the idea that, on par with nouns, pronouns surface with the lower *n* head, hosting natural gender and animacy, and a higher GenP, responsible for the grammatical gender. Furthermore, Bjorkman et al. (2022) provide the semantic account of Bjorkman et al. (2022), which takes as a criterion whether type conflicts arise when gender features combine with person and number. As noted in the Introduction (5.1), in the present work, I focus on the morphological realization of gender features, its role in allomorphy, and the implications that morphological facts have for the status of gender in pronouns.

Representation of gender in pronouns raises several important questions, relevant both for the syntactic theory and the structure of pronouns (1-3).

QUESTIONS

1. What types of gender are present in pronouns?
2. Where is gender in pronouns located?
3. What assignment principles hold for gender in the pronominal domain?

5.4.1 Types of gender in pronouns

In this section, I will provide an overview of the types of gender in pronouns, demonstrating that pronouns surface with the same kinds of gender available to nouns, i.e., (i) conceptual (natural) gender and (ii) arbitrary (grammatical) gender. Pronouns have an additional possibility of receiving (iii) the gender features determined by the referent.

5.4.1.1 Conceptual (natural) gender

The gender specification of a pronoun matches the conceptual, i.e., the natural gender of a noun that serves as its antecedent. This is exemplified here on the basis of English data and carries over to other languages (37). Namely, pronouns with an antecedent such as *mother* or *sister*, surfacing with the feminine conceptual gender, carry the same gender representation (37-a). Likewise, antecedents *father* and *brother*, exhibiting the masculine conceptual gender, match their respective pronouns surfacing with the masculine gender (37-b).

- (37) a. My **mother/sister** will support me. **She** always does.
 b. My **father/brother** will support me. **He** always does.

In this case, gender features have semantic effects, and in the system described above, they would be considered as surfacing with interpretable gender features. The question is where are these interpretable gender features hosted.

5.4.1.2 Arbitrary (grammatical) gender

The same matching of gender features extends to the arbitrary, i.e., grammatical gender. In German, the nouns *Stuhl* ‘chair’, *Kommode* ‘dresser’, and *Fenster* ‘window’ surface with the masculine, feminine, and neuter grammatical gender, respectively. Pronouns having these nouns as their antecedents exhibit the same gender specifications (38). As above, the question for pronouns is whether gender features are hosted on the same categorizing head *n* that combines with $\sqrt{\text{Roots}}$ in the nominal domain, that is, the *n* present in their antecedents.

- (38) a. *Der Stuhl ist rosa lackiert. Er war braun.*
 the._[MSC.SG] chair._[MSC.SG] is pink painted _[MSC.SG] was brown
 ‘The chair is painted pink. It was brown.’
 b. *Die Kommode ist rosa lackiert. Sie war braun.*
 the._[FEM.SG] dresser._[FEM.SG] is pink painted _[FEM.SG] was brown
 ‘The dresser is painted pink. It was brown.’
 c. *Das Fenster ist rosa lackiert. Es war braun.*
 the._[NEUT.SG] window._[NEUT.SG] is pink painted _[NEUT.SG] was brown
 ‘The window is painted pink. It was brown.’

5.4.1.3 Gender of a referent

In the data above, gender on pronouns was discussed in relation to gender specifications of the antecedent. However, in a number of environments and with first- and second-person pronouns in particular, the gender of a pronoun is not related to the gender features, conceptual or arbitrary, of any particular noun. Instead, the gender of a pronoun matches the natural (conceptual) gender of the referent. That is, if a first person, i.e., *AUTHOR*, surfaces with the masculine or feminine natural gender, this will be the gender of a pronoun. The same holds for second- and third-person referents. As noted above, gender marking on first- and second-person pronouns is considerably less frequent compared to gender distinctions in the third person. Nevins (2011a) proposes that the absence of gender distinctions in the first- and second-person pronouns can be explained in terms of an Impoverishment rule that deletes the gender features in the context of the marked first and second person features, but not in the unmarked third person. As already noted, Semitic languages such as Amharic and Hebrew, in addition to the third person, draw gender distinctions in the second person, too, depending on whether the addressee is male or female.

	SG	PL		SG	PL
1	əne	əñña	1	ani	anachnu
2 _M	antä	ənn-antä	2 _M	ata	ate-m
2 _F	anči	ənn-antä	2 _F	at	ate-n
3 _M	əssu	ənn-ässu	3 _M	hu	he-m
3 _F	əss ^w a	ənn-ässu	3 _F	hi	he-n

(a) Amharic (Leslau 1995)

(b) Hebrew (Ritter 1995)

Table 5.35: Gender distinctions in second- and third-person pronouns

Gender distinctions limited to the first-person pronouns are attested in Tengger Javanese, while other varieties of Javanese do not make comparable gender distinctions in the first person, which makes this pattern unique and rare (p.c. Jozina Vander Klok). In addition, pronouns are the only environment where Tengger Javanese overtly realizes gender features, with *(r)eyang* being used for first-person male referents and *isun* for first-person female referents. At the same time, Javanese employs a single pronoun *aku* for both contexts (Connors 2008).¹

Among Indo-European languages, the now extinct Tocharian distinguished masculine and feminine in the first person (Jasanoff 1989), while modern IE languages do not overtly mark gender distinctions in the first- and second-person pronouns. However, the presence of gender features can be **diagnosed indirectly** by looking at agreement facts, for example. As illustrated in (39) for Greek, the first (39-a) and second (39-b) person pronouns may co-occur with adjectives surfacing with both masculine and feminine gender in the predicative position. The choice of the gender specification of the adjective crucially depends on the gender of the referent. Now, the pronoun has a single morphological realization in both cases, i.e., the gender is not morphologically encoded.

¹I thank Jozina Vander Klok for introducing me to these data and providing the relevant reference.

- (39) a.
- Egó eímai éxypn-os/éxypn-i.*

1SG PRS.1SG smart-MSC.SG/smart-FEM.SG

‘I am smart.’

- b.
- Esí eísai ómorf-os/ómorf-i.*

2SG PRS.2SG beautiful-MSC.SG/beautiful-FEM.SG

‘You are beautiful.’

Greek

The same holds for a wide range of languages. Consider the following data from Serbian, where the single first-person pronoun freely co-occurs with the masculine and feminine genders on the participle in the past-tense formation. Comparable patterns are discussed for Brazilian Portuguese (Nevins 2011a), Czech (Parrott 2015), and Russian (Bobaljik 2017). In the same manner as in Greek, the morphological realization of the pronoun does not suggest the presence of gender features. However, they can be inferred on the basis of syntactic operations such as agreement.

- (40)
- Ja sam poseti-o/poseti-la Atinu.*

1SG.NOM AUX.1SG visit-MSC.SG/visit-FEM.SG Athens

‘I visited Athens.’

Serbian

In contrast to the conceptual (natural) and arbitrary (grammatical) gender features discussed above that are tightly connected with the gender specifications of their antecedents, gender assignment in this case does not provide evidence of correlation with any particular $\sqrt{\text{Root}}$. Therefore, the question that arises is which principles derive gender assignment in these cases and where gender features should be located.

In the nominal domain, it has been argued that gender specification on some nouns is dependent on the natural gender of a referent (Kučerová 2018). Such *valuation from the context* is possible only when the gender feature on *n* has not been valued in narrow syntax. Instead, gender features are valued at the syntax-semantics interface following the standard principles of the interpretation of referential indices. Once the features are valued from the context, they become available for the lower *n* head. Kučerová (2018) assigns the key role to the PERSON feature in the nominal structure hosted in D as, in contrast to other φ -features, it shows a tight connection with the referential index (Longobardi 2008; Landau 2010; Sudo 2012). Importantly, this way of gender feature valuation in the nominal domain is the **last resort** option, triggered only in syntactic configurations in which gender cannot be valued in the standard way on the *n* head. Once gender features are valued on D, i.e., the head that hosts person features in the nominal domain, gender features on the lower *n* head are valued through the matching Agree link between D and *n*. Pronouns differ from nouns in important ways. While valuation from the context is a last resort in the nominal domain, gender features on pronouns, and on first- and second-person pronouns in particular, are regularly valued from the context.

5.4.2 Do pronouns have roots?

The distinction between functional morphemes and roots becomes particularly relevant in the case of pronouns. Functional terminals, that is, features, are precisely those terminals that are overtly realized in the pronominal forms, while roots are terminals that are, at least morpho-phonologically, absent. The pronouns in (41-b) can have the roots in (41-a) combined with the categorizing head *n* as their antecedents (42). Thus, what makes a pronoun a pronoun morphologically is the absence of an overt root, where the root is understood as contributing the ‘lexical’ meaning.

(41) ROOTS AND PRONOUNS

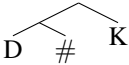
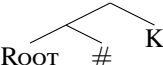
- a. $\sqrt{\text{girl}}$, $\sqrt{\text{boy}}$, $\sqrt{\text{balloon}}$, $\sqrt{\text{sea}}$, $\sqrt{\dots}$
- b. *she*, *he*, *it*

- (42) a. **The girl** arrived at the party. **She** is wearing a beautiful dress.
- b. **The balloon** flew away. **It** was red.

The apparent absence of overt roots in the sense of (41-a), motivated the proposal put forth in Moskal (2015a), according to which pronouns have less structure compared with nouns, building on this view in syntax (Postal 1969, Longobardi 1994, Déchaine & Wiltschko 2002). Under this view, in contrast to nouns, pronouns do not involve two structural components: (i) a root and (ii) the categorizing head *n*.²

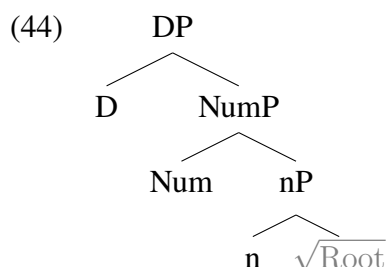
Roots and categorizing heads, however, do not exhibit a co-dependency (see Panagiotidis 2015). While roots require a categorizing head in order to surface in the structure, thus crucially depending on it, the categorizing head does not need a root to be a part of a complex head. This means that the dependency relationship is only **unidirectional**. An implication of this fact for the structure of pronouns is that the absence of the root does not immediately rule out the categorizer *n* and its morpho-syntactic and semantic contribution.³ Thus, nothing prevents the structure in (44), where the $\sqrt{\text{Root}}$ would be omitted but *n* projected. The ways in which roots merge in the morpho-syntactic representation are a matter of debate in DM. The literature provides arguments for four positions: (i) roots merged as complements, (ii) roots merged as modifiers, (iii) roots

²Moskal (2015a) proposes the structure of a pronoun in (43-a). The pronoun involves the pronominal base [D], the layer encoding ϕ -features abbreviated as [#], and the morphological case layer [K]. Building on this view, the representation of a pronoun in Smith et al. (2019) is given in (43-b). Smith et al. (2019) note that they treat the person formant as the root of the pronoun, where root is understood as the most deeply embedded morpheme that additionally undergoes allomorphy, without committing to a particular stance as to whether pronouns indeed contain roots in their structural representation in the technical sense.

- (43) (a) Moskal 2015a  (b) Smith et al. 2019 

³See also *give* in Koasati (Kimball 1994), discussed in Svenonius (2014) as realizing only little *v*.

merged either as complements or modifiers, and (iv) roots merged in a unique way (Alexiadou & Lohndal 2017b).



Entertaining a zero root is certainly a theoretical possibility. The empty light verb *go* has been proposed by Van Riemsdijk (2002) to account for the data in (45), widely attested in Germanic. The emptiness of this ‘super-light’ verb is argued to be licensed by a modal verb in these construals.

(45) *Ich muss nach Hause.*

I must PREP home

lit: ‘I must home’ ~ ‘I must go home.’

Van Riemsdijk (2002)

Natural language provides evidence for the existence of zero roots and stems, whose nullness does not need to be licensed and is not a result of any deletion processes. This is the case in Menomini (Bloomfield 1962) and Nimboran (Inkelas 1993). As illustrated in (46-a), the Menomini (Algonquian) verb ‘use’ is null, while the functional morphology realizing external and internal arguments receives overt realization. As a result, the entire construal conveys the meaning ‘He uses it’, despite the zero realization of the root.⁴ This contrasts with (46-b), where the verb ‘bring’ employs an overt root [*pi*], combining with the same set of person markers. These data provide convincing empirical support for the existence of zero roots and zero morphemes in general. Similarly, Inkelas (1993) discusses the set of verbs that have zero roots (*be*, *become*, *bring*, *dream*, *extend*, *go*, *hear*, *kiss*, *laugh*, *make cat’s cradles*, *say*, *sleep*) in Nimboran (Papuan). Patterning with Menomini data, only functional morphology receives an overt realization (47), which does not prevent the interpretation of the root, thus indicating its presence in the structure.

(46) a. Ø-*a-w*.

‘He uses it.’

⁴In Menomini, nullness of stems extends to other environments. Namely, certain stems are found only as part of secondary derivatives but never as an inflected form independently of a secondary derivation, which resulted in treating these stems as deficient or unused and the respective secondary derivations as unbased, i.e., lacking the transparent stem formant in Bloomfield (1962).

b. *pi'ta'w.*

‘He brings it.’

Menomini, Bloomfield (1962)

(47) *Ø-rár-be-d-u*

bring-PART-6_{LOC}-FUT-1

‘I will bring from here to above.’

Nimboran, Inkelas (1993)

A potential nullness of roots in pronouns differs from null roots discussed above in important ways. *Menomini* and *Nimboran* simply do not have exponents that would be realized as overt for certain verbs, similar to the non-existence of definite articles in some languages. While the absence of roots is quite obvious in pronouns, the question is whether we could provide a syntactic or morphological test proving that the root is really present or not in the internal composition of a pronoun. This is where empirical generalizations on the realization of number morphology discussed in Chapter 2 become particularly relevant.

5.4.3 Revisiting silent nouns

The idea that pronouns involve silent nouns has initially been entertained for various types of pronouns falling under the umbrella of E-type pronouns (Cooper 1979, G. Evans 1977, 1980, Heim 1982, 1990) and subsequently argued to hold for bound variable (Sauerland 2007, 2008b), and referential pronouns (Sauerland 2004; Elbourne 2005, 2013). Furthermore, it has been shown that virtually any pronoun can be replaced with a noun if the required conditions are met, as is the case with disambiguation (Schlenker 2005, Elbourne 2013). The proposals based on the idea that pronouns involve silent nouns differ in the size of the elided material. Sauerland (2003) proposes that pronouns are φ Ps that take full DPs as arguments, while Elbourne (2005) argues that pronouns are definite articles that take a noun phrase (NP) and a situation pronoun (s_i) as arguments. In both views, the silent material, DP or NP, undergoes deletion (ellipsis).

In this section, I will first introduce empirical arguments that motivated the proposal that pronouns involve a silent noun. In the next step, I will revisit this data from the perspective of allomorphy.

5.4.3.1 Gender

Based on gender facts, Sauerland (2003, 2007, 2008b) proposes that pronouns are φ heads that take full silent DPs as their arguments. The empirical evidence for this view comes from the matching of gender features between the pronoun and its antecedent. In German (48), nouns *spoon*, *fork*, and *knife*, which surface with the masculine, feminine, and neuter gender, respectively, act as antecedents of pronouns surfacing with the same gender specifications. According to the silent DP approach, this can be explained if pronouns inherit gender features from silent nouns via an agreement relationship that transfers the features of the noun. The silent elided DP

with which the pronoun establishes the agreement relationship is marked gray in the following examples, while $\triangle$ standardly indicates ellipsis.

- (48) a. *Tim hat **einen** Löffel gestohlen. Er* *[der Löffel_{MASC}] $\triangle$*
 Tim has a.ACC.MASC.SG spoon stolen. pro.NOM.MASC.SG the spoon
war aus Gold.
 was of gold
 ‘Tim has stolen a spoon. It was made of gold.’
- b. *Tim hat **eine** Gabel gestohlen. Sie* *[die Gabel_{FEM}] $\triangle$*
 Tim has a.ACC.FEM.SG fork stolen. pro.NOM.FEM.SG the fork
war aus Gold.
 was of gold
 ‘Tim has stolen a fork. It was made of gold.’
- c. *Tim hat **ein** Messer gestohlen. Es* *[das Messer_{NEUT}] $\triangle$*
 Tim has a.ACC.NEUT.SG knife stolen. pro.NOM.NEUT.SG the knife
war aus Gold.
 was of gold
 ‘Tim has stolen a knife. It was made of gold.’ Sauerland (2007)

Elbourne (2013) provides evidence for gender matching between pronouns and their antecedents in other languages with overt gender marking, such as French (49-50). Note that while German data in (48) involve an explicit linguistic antecedent, French examples involve the so-called *deep anaphora* (Hankamer & Sag 1976). Namely, the referent of a deep anaphor is present in a non-linguistic environment and understood as a strong visual clue in the immediate context (Elbourne 2005, 2013) that is sufficiently salient to license the ellipsis without explicitly realizing the antecedent in the sentence. This is the case in examples (49-50), where the referents *a large table* and *a desk* are present in the non-linguistic context, and the speaker may refer to them using a pronoun without overtly realizing the antecedent first.

- (49) Jean is trying to stuff a large table (*la table*, feminine) into the boot of his car. Mary says:
*Tu n' arriveras jamais à **la** / ***le** faire entrer dans la voiture.*
 you not arrive.FUT.2SG never to it.FEM / *it.MASC make enter into the car
 ‘You will never manage to get it into the car.’ Elbourne (2013)
- (50) Same scenario, with a desk (*le bureau*, masculine):
*Tu n' arriveras jamais à ***la** / **le** faire entrer dans la voiture.*
 you not arrive.FUT.2SG never to it.FEM / *it.MASC make enter into the car
 ‘You will never manage to get it into the car.’ Elbourne (2013)

Therefore, German and French data show that pronouns with both linguistic and non-linguistic antecedents match the gender features of their referents. Patel-Grosz & Grosz (2017) demonstrate that the same gender matching holds for **clitics** in Bavarian German (51).

- (51) a. *Da Tim hât an Leffl gstuin. Und i hâb'm gfundn.*
 the Tim has a.MASC spoon.MASC stolen and I have=*pro*.MASC found
- b. *Da Tim hât a Gåbl gstuin. Und i hâb's gfundn.*
 the Tim has a.FEM fork.FEM stolen and I have=*pro*.FEM found
- c. *Da Tim hât a Messa gstuin. Und i hâb's gfundn.*
 the Tim has a.NEUT knife.NEUT stolen and I have=*pro*.NEUT found
- ‘Tim stole **a spoon / fork / knife**. And I found **it**.’ Patel-Grosz & Grosz (2017)

5.4.3.2 Lexical plural

Elbourne (2013) provides evidence for the feature matching between pronouns and their referents from the realization of number. In a scenario where *John wants Mary to hand him his trousers*, only the plural form of the third-person pronoun can be used (52-a), while in the same scenario with a shirt, the opposite holds (52-b).

- (52) a. Could you hand **them**/*it to me, please?
- b. Could you hand **it**/*them to me, please?

Importantly, plural with nouns such as *trousers* falls under the notion of *lexical plurals*, which are not hosted in the syntactic terminal encoding plurality (NumP) but on the nominalizing head *n* instead (Acquaviva 2008; Alexiadou 2011, 2019b, 2019a, 2024; Kramer 2016). The question for the structure of pronouns is how to account for matching in the lexical plural features between the pronoun and its referent.

5.5 Roots are not morpho-syntactically active in pronouns

In Chapter 2, I presented two empirical generalizations on the realization of number morphology in pronouns. Pronouns and nouns may share the same exponent of plural, only if this exponent figures as an *elsewhere* form. Recall the data from Serbian, where masculine, feminine, and neuter nouns combine with the plural exponent [*i*]. The first, second, and third-person pronouns combine with the elsewhere [*i*], while the third-person feminine additionally shows an interaction with gender, giving rise to the allomorph [*e*], as illustrated in Table 5.36.

- (53) *prozor-i*
window.M-PL
'windows'
- (54) *noć-i*
night.F-PL
'nights'
- (55) *slon-ov-i*
elephant.M-OV-PL
'elephants'

	SG	PL
1	ja	m- i
2	ti	v- i
3 _M	on	on- i
3 _F	ona	on- e

(a) Serbian

(56) **VIs for PL in Serbian**

[PL] ↔ e / ____ FEM

[PL] ↔ i

Table 5.36: Pronouns in Serbian

Furthermore, if the elsewhere form is present only with a subset of pronouns in a given paradigm, it is the third person (French, Korean, Turkic, Cabécar, Modern Yucatec, Udmurt) or the second and third person (Kulina) that combine with the elsewhere realization, while the first person triggers a special allomorph, but not vice versa. Consider the following example from Udmurt, where the plural morphology surfacing on nouns (57) is present in the third-person pronoun.

	SG	PL
1	mon	m- i
2	ton	t- i
3	so	so- os

(a) Udmurt

- (57) *korka-os*
house-PL

'houses'

(Edygarova 2022)

Table 5.37: Selective person-driven number allomorphy: Udmurt

These facts gave rise to the following empirical generalizations.

(58) **PLURAL EXPONENTS IN PRONOUNS AND NOUNS: GENERALIZATIONS**

G1: If pronouns and nouns share a plural exponent, it is the elsewhere exponent.

G2: If the elsewhere plural exponent is not spread across a paradigm, it is the third person or the third- and second-person pronouns that surface with the elsewhere form.

Crucially, **pronouns do not have the complete set of plural allomorphs available to nouns**. The question that arises is why the plural morphology of the antecedent does not survive on the respective pronoun. The pronoun that has the noun *oxen* as an antecedent does not employ the allomorph [en] conditioned by $\sqrt{\text{Root}}$. Likewise, the pronoun with an antecedent *alumni* does not surface with the plural allomorph [i]. Therefore, the pressing question for approaches to

pronouns positing a silent noun in their representations is why the noun cannot trigger allomorphy of plural in the same way it does in canonical DPs. Why does the plural morphology of the antecedent not survive in the respective pronoun? **If a silent noun is always present, the same number morphology on nouns and pronouns is predicted.** The observed sharp asymmetry is unexpected.

(59) ALLOMORPHS OF PLURAL IN ENGLISH (Gouskova & Bobaljik 2025)

[PL] $\leftrightarrow$ -en /]_N __, where $N \in \{\sqrt{\text{ox}}, \sqrt{\dots}\}$

[PL] $\leftrightarrow$ $\emptyset$ /]_N __, where $N \in \{\sqrt{\text{sheep}}, \sqrt{\text{fish}}, \sqrt{\dots}\}$

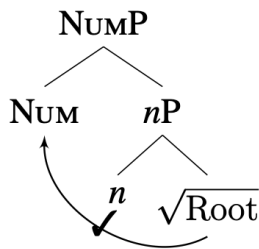
[PL] $\leftrightarrow$ [-back, +tense] /]_N __, where $N \in \{\sqrt{\text{foot}}, \sqrt{\text{goose}}, \sqrt{\dots}\}$

[PL] $\leftrightarrow$ -i /]_N __, where $N \in \{\sqrt{\text{alumn}}, \sqrt{\text{syllab}}, \sqrt{\dots}\}$

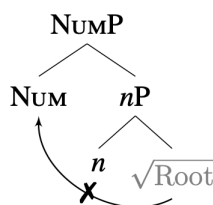
[PL] $\leftrightarrow$ -z /]_N __

The inventory of allomorphs in (59) is available in structural representations where the $\sqrt{\text{Root}}$ is present in the structure (60) and the number node can access it at the point when it undergoes Vocabulary Insertion.

(60) $\sqrt{\text{Root}}$ -DRIVEN NUMBER ALLOMORPHY



If there is a silent noun in pronouns, why is there no allomorphy of the plural conditioned by the $\sqrt{\text{Root}}$? I propose a principled explanation for this question. At the point when the number node is targeted for insertion in pronouns, a $\sqrt{\text{Root}}$ cannot condition the allomorphy of the plural node and give rise to one of the allomorphs, as it is **not present in the structure** when the Vocabulary Insertion into plural takes place (61). If the $\sqrt{\text{Root}}$ is not present, the number node cannot refer to it when it is targeted for Vocabulary Insertion. This explains the absence of a language in which one of the allomorphs triggered by a particular set of $\sqrt{\text{Roots}}$ would surface in pronouns. Therefore, (i) $\sqrt{\text{Root}}$ -conditioned allomorphs of plural are unexpected in pronouns and (ii) only the elsewhere forms, which do not make reference to any particular $\sqrt{\text{Root}}$ are available. Both predictions are corroborated by the data examined here.

(61) $\sqrt{\text{Root}}$ -DRIVEN NUMBER ALLOMORPHY BLOCKED

Therefore, allomorphy shows that $\sqrt{\text{Root}}$ s are not morpho-syntactically active in pronouns. This finding is consistent with the argument made in the literature that the representation of the n head is different in pronouns (see Panagiotidis 2002, 2003a, 2003b), as they do not involve the full lexical noun in the way n in nouns does. Integrating the findings from allomorphy and the insights reported in the literature, in this work I consider **n in pronouns** as a **complement/modifier-free head**, that does not combine with a $\sqrt{\text{Root}}$ in the syntax.

5.6 Implications for the structure of pronouns

Examination of the morphological realization of GENDER features on pronouns and their interaction with PERSON and NUMBER resulted in several empirical generalizations. Recall the set of generalizations in (9), repeated here as (62).

(62) EMPIRICAL GENERALIZATIONS: GENDER IN BOTH SINGULAR AND PLURAL

G1: Person, gender, and number exponents are segmentable in the plural.

G2: In languages where segmentation is not possible, gender and number are realized via a non-segmentable exponent, while person remains visible.

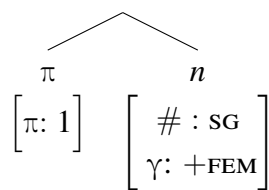
What implications for the structure of pronouns can we draw on the basis of these results? I propose that the first generalization (G1) tells us about the **inventory of syntactic terminals** while the second (G2) tells us about their **locality**. Namely, an overt realization of a gender exponent is suggestive of a morpheme in narrow syntax. However, the close interaction of gender and number, reflected in non-segmentable exponents for these features to the exclusion of person, suggests that the two should be adjacent. That is, a potential structure of a pronoun where person would be placed between number and gender, disrupting their adjacency, would not be able to explain the obtained close interaction of gender and number, directly reflected in morphology.

Crucially, the set of languages that make gender distinctions only in the singular further corroborates this stance. While gender features are readily available to be realized in morphology when they combine only with person features, adding number results in their neutralization. Therefore, the close interaction of number and gender holds across the board. I propose that the obtained facts strongly implicate structural adjacency of gender and number in pronouns, consistent with the nominal domain.

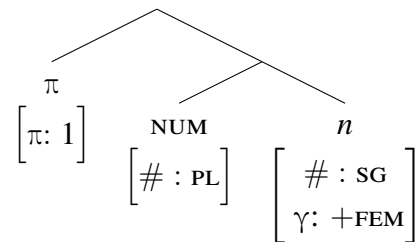
In light of the discussion about $\sqrt{\text{Root}}$ s and the dependency relation involving $\sqrt{\text{Root}}$ s and n , I propose that pronouns lack $\sqrt{\text{Root}}$ s, but surface with the n terminal. This terminal

hosts **gender** features, both conceptual (natural) and arbitrary (grammatical), and **singulative number**. In addition, it explains the availability of **lexical plurals** discussed above, in cases when the pronoun's referent involves a noun such as *trousers*. In the singular, the *n* terminal is immediately adjacent to the person $[\pi]$, while in the plural, the number node $[\text{NUM}]$ intervenes between the two. My focus here was on the examination of the morphological realization of gender features. Additional features that could be recruited on a language-specific basis, such as humanness and animacy, may appear together with these nodes, either immediately structurally adjacent or bundled on the same head, as proposed in the literature.

(63) FIRST PERSON FEMININE SINGULAR



(64) FIRST PERSON FEMININE PLURAL



In languages that lack the category of gender such as Cabécar (Verhoeven 2012), the *n* head does not need to surface with gender features, but with the singulative number and features such as animacy and humanness, which are typically associated with *n*. Likewise, Turkish is an example of a language in which grammatical gender features are not active, and the semantic gender is not encoded on any functional element. The same holds for pronouns that do not mark gender anywhere in the system (Panagiotidis 2024). Therefore, in these languages, pronouns will surface with fewer features compared to those that select gender for their inventory.

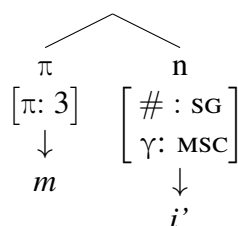
Taking Mosestén as an illustrative example that conforms to G1, I propose the following structural representations of third-person pronouns (66-69). The set of VI rules introduced earlier in this chapter can be augmented with *n*, the node hosting gender features.

(65) **VI**s for Mosestén

- $[3] \leftrightarrow m$
- $[\text{PL}] \leftrightarrow \text{in}$
- $n [+FEM] \leftrightarrow \text{ö'}$
- $n [-FEM] \leftrightarrow \text{i'}$

Structurally, third-person singular pronouns consist of two syntactic terminals. The person head $[\pi]$, surfacing with the third-person feature and the *n* head, hosting gender features and singulative number (Kramer 2015; Alexiadou 2019a, 2024). Both syntactic terminals receive an overt realization in morphology, with the gender distinction present in the narrow syntax being directly reflected in the form.

(66) THIRD-PERSON SINGULAR: MASCULINE



(67) THIRD-PERSON SINGULAR: FEMININE

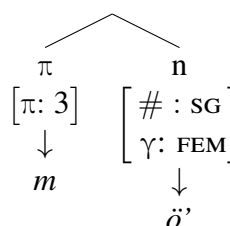
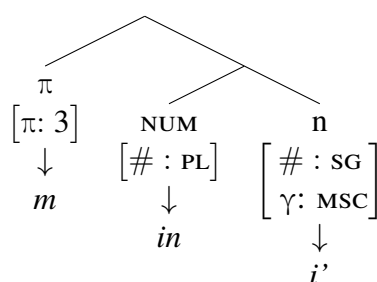


Table 5.38: Mosetén: Analysis of third-person singular pronouns

In the plural, the number node [NUM] structurally intervenes between the person [π] and *n*. All three syntactic terminals receive an overt realization in morphology. The ways in which the correct order of exponents is derived, either via head-movement in the syntax or a linearization mechanism at PF, will be independently discussed later in this dissertation.

(68) THIRD-PERSON PLURAL: MASCULINE



(69) THIRD-PERSON PLURAL: FEMININE

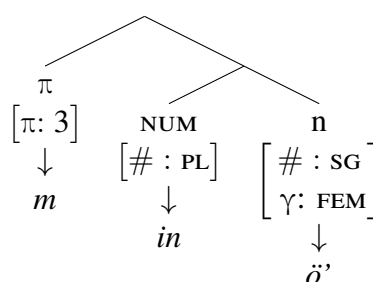


Table 5.39: Mosetén: Analysis of third-person plural pronouns

The locality of *n* and NUMBER terminals explains the patterns of allomorphy in languages that preserve gender distinctions in the plural, but do not realize all three features in an agglutinative manner like Mosetén. This was the case in Serbian, Greek, and Hebrew, among others. A single non-segmentable exponent can be analyzed as a gender-driven number allomorphy. At the same time, the person exponent consistently escapes the interaction of gender and number. I illustrate this with the third-person plural feminine pronoun in Serbian. In the context of the feminine feature on the *n* head, the plural is realized as [e], winning over the elsewhere [i].

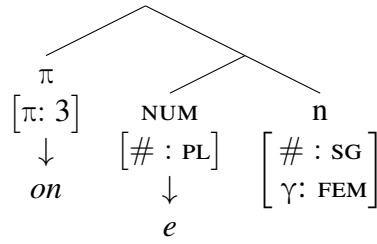
(70) **VIs for Slavic**[PL] $\leftrightarrow$ e / $___ \gamma_{+FEM}$ [PL] $\leftrightarrow$ i

Table 5.40: Serbian: Analysis of third-person plural pronouns

5.6.1 Impoverishment and Structural Complexity

Finally, languages that do not preserve gender distinctions in the plural demonstrate that the **Impoverishment** operation in the post-syntactic component of the grammar is invoked with the **increase of the structural complexity**. As noted in the overview of the literature, Impoverishment is understood as feature-deletion from a particular feature bundle, typically triggered by markedness (Noyer 1992; Arregi & Nevins 2007; Nevins 2011a). However, a legitimate question to ask is whether the observed neutralization may correspond to differences in the syntactic structure. That is, the question is whether syncretism can also correlate with the structure. Data from Amharic, Japanese, and Siriano demonstrate that the exponent of a morpheme that underwent Impoverishment freely co-occurs with the Impoverishment trigger, which is, in the case at hand, number.

	SG	PL		SG	PL
1	əne	əñña	1EXCL	jɨɨ	mãrĩ
2M	antä	ənn- antä	1INCL		gɨa
2F	anči	ənn- antä	2	mĩĩ	mĩĩ-sã
3M	əssu	ənn- ässu	3M	ĩŋĩ	ĩŋĩ -sã
3F	əss ^w a	ənn- ässu	3F	igo	ĩŋĩ -sã

(a) Amharic (Leslau 1995)

(b) Siriano (Pertsova 2022b)

Table 5.41: Neutralization to singular

This point further substantiates the hypothesis explored in this dissertation that pronouns are multi-morphemic, i.e., multi-terminal syntactic structures. In particular, I would like to propose the condition on Impoverishment cashed out in terms of structural complexity. In the data observed in this study, it arises with adding the number feature, that is, a new syntactic terminal. As a consequence, the trigger and the target of this operation do not need to be bundled on the same head. Instead, it involves an interaction of two syntactic terminals.

Furthermore, in a number of languages that exhibit gender neutralization in the plural, person exponents remain unaffected by number. If gender features were bundled with person in the same morpheme, we would expect comparable effects for the realization of person, too, or we would need to stipulate why the person feature, but not gender, remains unaffected by number.

Crucially, in Distributed Morphology, Impoverishment is a post-syntactic operation that applies before Vocabulary Insertion. This means that languages that exhibit neutralization in the plural do not need to have different syntax of pronouns. This points towards **universality in pronominal structure**. Languages that preserve gender distinctions in the plural and those that neutralize them differ only in the realizational post-syntactic component of the grammar. However, this distinction does not carry over to narrow syntax.

It is important to keep in mind that the universality thesis holds for languages that *do have* gender features. Furthermore, gender distinctions can emerge or get lost in the course of historical development. For example, old Indo-European pronouns did not distinguish gender, as the category of gender was the property of nouns and demonstratives (Sihler 1995). As noted in the overview of the data, gender distinctions are attested in 80 out of 350 languages studied here. This finding is in line with the argument that gender is not a universally available category (Alexiadou 2004). An important implication of this result for the overall theory of pronouns is that their internal structure and feature specification crucially depend on the inventory of features that individual languages select and realize in morphology.

5.7 Summary

In this chapter, I examined the morphological realization, assignment properties, and the locus of gender features on pronouns, a theme that has not been put under theoretical scrutiny in the previous literature. The observed data demonstrate that, in contrast to person and number, gender is not universally available to pronouns, yielding a finding consistent with the nominal domain. Languages split into two groups with respect to the availability of gender distinctions in the singular and plural. Crucially, both patterns yield a uniform result, which implicates a close interaction of gender and number. This empirical finding has direct consequences for the structure of pronouns. In addition, it has been demonstrated that the full set of gender assignment properties is available to pronouns, that is, (i) conceptual (natural) gender, (ii) arbitrary (grammatical) gender, and (iii) gender determined on the basis of a natural gender of the pronoun's referent.

Furthermore, in contrast to the interaction of person and number, gender introduces the need for additional explanations in terms of syncretism, as gender features undergo systematic neutralizations with the increase of the structural complexity in the plural. Neutralization patterns that invoke Impoverishment rules are only occasionally necessary in the relation of person and number, while gender features lead to robust patterns of deletion in the plural.

When it comes to allomorphy, there are two possibilities. Building on the insights from the previous chapters, gender may act as a trigger and as a target of allomorphy, on par with person and number features. Therefore, taking into account the three features results in the following four possibilities (71). When gender acts as a trigger, (i) gender-driven person allomorphy and (ii) gender-driven number allomorphy are expected. In contrast, when it figures as a target, (i) person-driven gender allomorphy and (ii) number-driven gender allomorphy may emerge.

(71) THE PLACE OF GENDER IN ALLOMORPHY

GENDER AS A TRIGGER

- gender-driven person allomorphy
- gender-driven number allomorphy

GENDER AS A TARGET

- person-driven gender allomorphy
- number-driven gender allomorphy

Data examined in this chapter demonstrate that gender features may trigger allomorphy of person in the singular number (e.g., Breton, Abkhaz) and allomorphy of number (Serbian). When it acts as a target, the situation is slightly different. Amharic provides evidence that a particular person feature specification (ADDRESSEE) may trigger allomorphy of gender, requiring a particular realization of gender features. The number, however, leads to widespread patterns of Impoverishment, i.e., neutralization of gender features, rather than generating a novel set of allomorphs of gender.

Chapter 6

The Structure of Pronouns

6.1 Central questions

The central question that this dissertation investigated is what the internal structure of pronouns is (1). I conducted a series of studies looking at the ways in which morphological realization reveals the internal syntactic complexity. Investigation of the internal structure of pronouns opened a set of additional questions related to the patterns of universality (2) and variation (3) in pronominal internal syntax, and broader generalizations about the structure present in the narrow syntax that is implicated by the morphological realization (4).

THE QUESTIONS

1. What is the internal structure of pronouns?
2. Are there universal patterns in the realization and the structure of pronouns?
3. Is there variation in the realization of pronouns? If there is, what is the source of variation?
4. Which broader generalizations about narrow syntax are derivable from observing the computation of the form?

In this chapter, I provide an overall account of the structure of pronouns, integrating the findings discussed in the previous chapters. The chapter is organized as follows. In Section 6.2, I provide the proposal on the structure of pronouns across and within languages. Before proceeding with the specifics of the analysis, I review the motivation for recruiting contextual allomorphy in investigating the internal structure of pronouns and the achievements of allomorphy applied as a tool in the pronominal domain. In Section 6.3, I discuss the implications of the study of pronouns for the theory of allomorphy. In 6.4, I focus on the extensions of the proposed analysis to case and definiteness features. Section 6.5 brings the comparison of the present proposal, couched within the Distributed Morphology, with other realizational approaches such as Nanosyntax and Spanning. Section 6.6 presents a puzzle for future work. Finally, in Section 6.7, I provide the overall conclusion of this dissertation and discuss the directions for future research.

6.2 The Structure of Pronouns: Proposal

In this dissertation, I proposed and investigated the following central hypothesis (1).

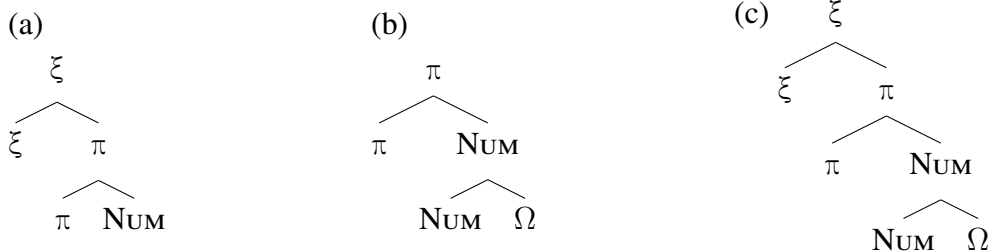
- (1) Contextual allomorphy is a key tool for
- a. *accessing* the inventory of morphemes that make up a pronoun across languages
 - b. *tracking* the patterns of their interactions

Why is allomorphy the tool? The motivation for recruiting allomorphy as a test for the internal structure was grounded on the idea that allomorphic alternations involve an interaction of two morphemes, the *trigger* and the *target* of allomorphy. Identifying the trigger and the target in pronominal forms provides us with an *immediate result* - pronouns involve two morphemes that occupy distinct structural positions in their internal representation in narrow syntax. Furthermore, allomorphy reflects the narrow syntax as it directly operates on the output of the syntax, i.e., the morphemes sent to PF for realization.

Identifying the trigger and the target of allomorphy corresponds to the *accessing* part of the hypothesis. Namely, it provides us with the inventory of morphemes in the narrow syntax that are compositionally combined via *Merge* to build the syntactic structure of a pronoun. Once we identify the morphemes that are present in the narrow syntax, we are in a position to examine the exact direction of allomorphy. This aspect corresponds to *tracking* the patterns of morpheme interactions. Which morpheme exactly acts as a trigger and which as a target? Natural languages provide evidence for number-driven person allomorphy. However, the opposite direction is also attested, as demonstrated in Chapter 3. Therefore, the examination of the patterns of allomorphy involves two steps, which include identifying the interacting morphemes and looking closely at the nature of the interaction that they establish.

What does this mean in the structure of a pronoun? The presence of the number-driven person allomorphy or the person-driven number allomorphy implicated two syntactic terminals in the structure of a pronoun, person (π) and number (NUM). Importantly, pronouns may involve more morphemes merged in different positions (2).

(2) BUILDING THE STRUCTURE



In (2-a), a morpheme ξ is merged on top of person [π] and number [NUM] morphemes, while in (2-b), a morpheme Ω is merged below the number morpheme. The presence of both ξ and Ω is an additional natural possibility (2-c). In the present work, I explored the hypothesis that the

structure of pronouns is revealed by inspecting series of **sequential interactions** of ξ with π and NUM (2-a), Ω with NUM and π (2-b), ξ and Ω with π and/or NUM (2-c).

In the preceding chapters, I presented a series of cross-linguistic studies on allomorphy and a case study on Algonquian plural morphology in pronouns. These facts motivate me to propose that pronouns involve a complex **multi-morphemic** internal structure. The inventory of morphemes that make up pronouns is drawn from the set of **functional morphemes**, as pronouns crucially lack $\sqrt{\text{Roots}}$ (see Chapters 2 and 5 and the discussion in the continuation of this chapter).

Crucially, the structure of pronouns exhibits patterns of **universality** and **variation** across languages. The first level of variation is located within a language, as singular pronouns do not have the morpheme encoding plurality, for example. An additional level of variation lies in the inventory of features that make up pronouns across languages. For example, some languages have a clusivity distinction. This phenomenon is accounted for in the present work in terms of a morpheme that hosts both first- and second-person features, in contrast to languages without clusivity that surface with the first person only. Therefore, the variation is encoded at the morpheme level without affecting or stretching the internal structure. Furthermore, gender features are not present in all languages. Therefore, the feature composition of the n head will differ. While in languages that have gender, it hosts gender features, singular number, and lexical plural, in languages without gender, it hosts only the singular and lexical plural. This variation is also encoded at the morpheme level. A significantly different kind of variation arises at PF. As noted in Chapter 3, languages originating in unrelated language families, such as German (Germanic), Chichewa (Bantu), Baure (Arawakan), Hua (Papuan), and Koyraboro Senni (Nilo-Saharan), may exhibit exactly the same patterns of allomorphy. This level of variation is not encoded in the morpheme, being the property of the Vocabulary of individual languages and the post-syntactic component of the grammar.

Some languages lacked allomorphy altogether, in which case individual morphemes were combined with discrete exponents in an agglutinative manner without showing any interaction. Crucially, this does not imply that in languages with allomorphy it is not possible to identify two distinct overt exponents. The trigger and the target of allomorphy freely overtly co-occur (see also 6.3.3). In languages like Mangghuer, the allomorphs $[da]$ and $[ta]$, triggered by the plural, co-occur with the overt realization of the trigger $[si]$.

			(3) VIs for Mangghuer	
	SG	PL		
1	bi	da-si	$[1] \leftrightarrow da / __\#_{PL}$	$[3] \leftrightarrow gan$
2	qi	ta-si	$[1] \leftrightarrow bi$	$[PL] \leftrightarrow si$
3	gan	gan-si	$[2] \leftrightarrow ta / __\#_{PL}$	
			$[2] \leftrightarrow qi$	

(a) Mangghuer (Slater 2003)

Table 6.1: Number-driven person allomorphy: Mangghuer

Why does allomorphy matter? In addition to revealing the internal structure by inspect-

ing the minimal sequential interactions, allomorphy proved instrumental in understanding the absence of $\sqrt{\text{Roots}}$. I proposed that the absence of the full inventory of the allomorphs of the plural triggered by a $\sqrt{\text{Root}}$ in a language is a consequence of the structural difference between nouns and pronouns. As $\sqrt{\text{Roots}}$ are not present in pronouns, when the Vocabulary Insertion targets the number node, the number cannot refer to the $\sqrt{\text{Root}}$ for purposes of allomorphy. Furthermore, the case study on Algonquian showed that the violation of the **you* gap is only apparent, as the first-person feature is present in the morpheme, reflected in the presence of the allomorph of number realized only in the context of the first person.

Beyond allomorphy, the present study investigated the ways in which syntactic complexity is implicated in morphological complexity. Going back to the Greek pronominal system that opened the discussion in this dissertation, the first-, second-, and third-person pronouns involve a significant morpho-phonological overlap, as the $[m]$, $[s]$, and $[t]$ consistently reappear in both singular and plural and across all case specifications. Individuating the person morpheme in the syntax that is combined with the phonological exponent at PF explains the shared morpho-phonological material in a principled manner. It is the first, second, or third person that compositionally combines with the remaining syntactic material.

	1		2		3					
	SG	PL	SG	PL	M.SG	M.PL	F.SG	F.PL	N.SG	N.PL
NOM	ego	<u>em</u> is	<u>es</u> i	<u>es</u> is	aftos	afti	afti	aftes	afto	afta
GEN	<u>em</u> ena	<u>em</u> as	<u>es</u> ena	<u>es</u> as	aftu	afton	aftis	afton	aftu	afton
ACC	<u>em</u> ena	<u>em</u> as	<u>es</u> ena	<u>es</u> as	afton	aftus	aftin	aftes	afto	afta

Table 6.2: Personal pronouns in Greek

The *Vocabulary* of a language, in the case at hand, Greek, consists of all its exponents. Now, under the approach argued for here, there is only one exponent realizing the first-person feature in pronouns, namely $[m]$. The same principle extends to the second- and third-person features, which are stored as $[s]$ and $[t]$, respectively. Therefore, there are only three exponents that need to be acquired by a child and stored in memory.

Let us imagine an alternative possibility, where Greek pronouns would not be further segmentable and stored in their entirety in the Vocabulary. The first problem that arises is that the shared morpho-phonological material remains unaccounted for. Most importantly, the provided pronominal system would consist of 30 Vocabulary Items. This means that the first-person, in two number (SG, PL) and three case (NOM, GEN, ACC) specifications, would have to be acquired and memorized multiple times, instead of only once. That is, a Greek acquiring child would have to repeat the acquisition of the second-person morphology with every instantiation of a pronoun in the six cells of the paradigm.

In contrast, under the approach pursued here, the second person is acquired and memorized as $[s]$ only once. As discussed earlier in this dissertation, the decompositional approach that recognizes multiple morphemes in the structure of a pronoun and several exponents that these

morphemes combine with at the PF, accounts for both the morpho-phonological overlap and also brings off economy. As such, it represents a cognitively preferable option.

As noted in Chapter 1, extensive research has collected significant evidence that words are not stored as such in memory, but decomposed. Evidence from this position comes from language-mixing varieties. Alexiadou & Lohndal (2018) demonstrate that bilingual speakers of several mixed varieties combine a root from one language with functional morphology from the other. Crucially, mixing arises below the word level, thus implicating that words are not stored as such in the memory, but decomposed into a root and several pieces of functional material. A line of experimental work provides evidence for the decompositional approach to words (Stockall & Marantz 2006; Solomyak & Marantz 2010; Fruchter & Marantz 2015; Flick et al. 2018; Gaston et al. 2021). In this dissertation, I proposed that pronouns consist of several functional morphemes. Therefore, the decomposition of the functional material reported in the existing experimental work is expected to carry over to pronouns. As pronouns lack roots, the question that arises for future experimental work is how the decomposition of the functional material in the absence of roots, that is, in the complex objects consisting of functional morphemes only, is reflected in processing. The expected decomposition would provide a strong novel argument in favor of the view that words are not stored and processed as such, but decomposed.

One additional remark in relation to acquisition and allomorphy is called for. Recall the general schema for allomorphic alternations (4). Allomorphy involves **departure** from **1-to-1 mapping** from the syntax to morphology, as a single morpheme ξ is mapped to two exponents, x and y . However, it is important to clarify in which way the notion of the 1-to-1 mapping is understood. The morpheme ξ belongs to the list of syntactic terminals (LIST 1), while the exponents x and y belong to the Vocabulary (LIST 2) of a particular language. Nothing in the theory requires these two lists to overlap. Considered from the perspective of allomorphy, the Vocabulary is necessarily larger than the list of syntactic terminals. In that sense, the two lists differ. This difference has implications for language acquisition. As children prefer 1-to-1 mapping (see Sauerland & Alexiadou (2020); Guasti, Alexiadou, & Sauerland (2023)), contextual allomorphy will require an effort in acquisition. This raises a prediction that the allomorph y , which requires the presence of a specific morpheme in its syntactic environment, has to be learned and stored in memory. As a consequence, children acquiring languages that have several allomorphs of the same morpheme are expected to acquire the full inventory of exponents later compared to languages where a given morpheme has only one exponent.

(4) *Contextual allomorphy*

- a. $[\xi] \leftrightarrow y / __ \Omega$
- b. $[\xi] \leftrightarrow x$

Importantly, the 1-to-1 mapping can be understood in one more way, preserving the asymmetry between the two lists. Namely, an exponent x realizes *only one morpheme at a time*. This idea is the central component of the view that the Vocabulary Insertion operation targets only

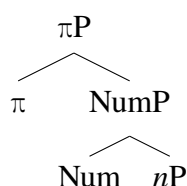
terminal nodes. The examination of the pronominal domain provided an argument in favor of this view, particularly in cases of an overt co-occurrence of a trigger and target of allomorphy (see 6.3.3 and the discussion in 6.5).

With these considerations in place, I now move to the central part of this chapter and the overall proposal of the structure of pronouns in the natural language (6.2.1) and implications for the theory of allomorphy (6.3).

6.2.1 Universality and Variation

On the basis of the results presented in the course of the discussion in previous chapters, I propose the following **internal structure of pronouns** (5). Personal pronouns consist of three morphemes, i.e., syntactic terminals in their structural representation: the ***n*** morpheme, the **number morpheme** (Num), and the **person morpheme** (π). The proposed morphemes show a number of close syntactic interactions, directly reflected in their morphological realization. The provided syntactic representation holds for all pronouns ‘proper’ across languages, that is, pronominal forms that do not appear in an adnominal position or comparable syntactic environments that allow ancillary nominal material, but as external and internal arguments of a verb in a clause.

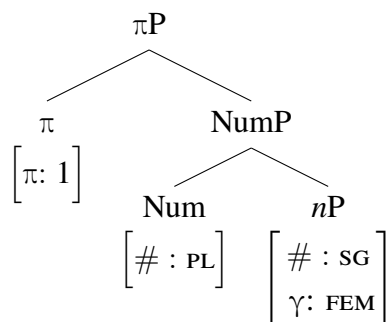
(5) *The structure of pronouns*



The syntactic structure in (5) answers the core question of this dissertation: *What is the internal structure of pronouns?* Abstract morphemes are basic atomic units of syntactic computations that figure as terminal nodes in the syntactic structure. In the composition of a pronoun, the nominalizing head *n* combines via *Merge* with the number (Num) node, which further combines with the person (π) node. In that sense, the pronoun with the proposed structure is a complete morpho-syntactic ‘word’, that is built compositionally following the same syntactic principles that apply to complex objects at the level of clausal syntax.

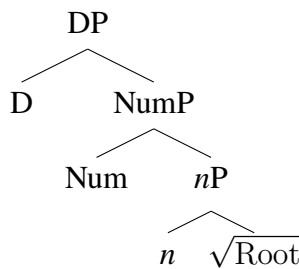
Furthermore, I propose the following feature distribution across the three morphemes. The *n* head is the locus of singular number and gender features in languages that include gender in their inventory. In addition, the lexical number interpretation arising with certain pronouns is located on the same head, on par with the nominal domain. The number node (Num) hosts plural features, while the person node (π) carries person features.

(6) *The structure of pronouns: Feature distribution*



The obtained structure of pronouns, proposed to hold universally, is consistent with the structure of the nouns. Extensive work in the syntactic theory on the internal structure of nominals converges on the idea that nouns have the following syntactic structure (see Alexiadou et al. 2007 for an extensive discussion and summary of the literature). The most deeply embedded morpheme in a noun is a $\sqrt{\text{Root}}$ (e.g., $\sqrt{\text{dog}}$, $\sqrt{\text{sea}}$, $\sqrt{\text{cake}}$, $\sqrt{\text{dress}}$, $\sqrt{\dots}$). The nominalizing head n categorizes the acategorical $\sqrt{\text{Root}}$ in a syntactic derivation. At the same time, n is the locus of gender features (Lecarme 2002; Lowenstamm 2008; Kramer 2015; Alexiadou, Lohn-dal, et al. 2015; Deal 2016b; Adamson & Šereikaitė 2019; Adamson & Anagnostopoulou 2024), singular number (Kramer 2015; Alexiadou 2019a, 2024), and lexical plural (Acquaviva 2008; Alexiadou 2011, 2024; Kramer 2016). Grammatical plural number is located on the NumP layer in the extended projection of a noun (Ritter 1992, 1995; Alexiadou 2011, 2019a, 2019b). Finally, D is the locus of person features in the nominal domain (Postal 1969; Abney 1987; Ritter 1991; Carstens 1991, 2000; Alexiadou et al. 2007; Baker 2008; Bernstein 2008a; Danon 2011; Landau 2016; Polinsky 2016; Höhn 2016, 2017, 2020; Kučerová 2018; Alexiadou et al. 2024).

(7) *The structure of nouns*



The proposed structure of pronouns (5) is similar to the structure of nouns. The position of the person head (π) in pronouns neatly corresponds to the locus of person features in the nominal domain, merging above the NumP in both representations. The realization of the number morphology, discussed in Chapter 2, and the series of case studies on allomorphy that have demonstrated that number acts as both the trigger and the target of allomorphy, motivated the establishment of the number (Num) terminal in the structure of pronouns. An extensive discussion in Chapter 5 on the assignment and realization of gender features in the pronominal domain implicated the presence of the n head in the structure of pronouns. In addition, the same effects that arise in the nominal domain, such as the lexical number with a noun *trousers*, for example, are present in the pronominal domain when the given noun serves as an antecedent of a pronoun *they* in English (Elbourne 2013). I proposed an account in terms of the lexical plurality located on the n head, on par with the nouns.

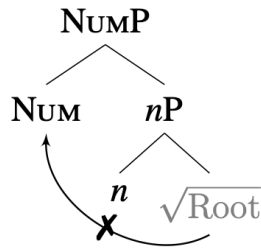
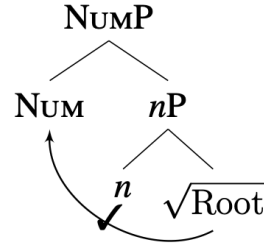
Furthermore, several arguments grounded in semantic evidence from different empirical domains demonstrate a tight connection of the person feature and the referential index (Longobardi 2008; Landau 2010; Sudo 2012; Kučerová 2018). Recent work in formal semantics brought arguments for the view that the index surfaces above φ -features (see Ahn (2019, 2022) for an explicit semantic account of definite descriptions, including pronouns, where the index (*idx*) is higher in the structure than the φ -features and Hanink (2021) for the view that the referential

index is below the D head but structurally adjacent to it). The exact position of the referential index, that is, whether it is bundled with the person head or above it, and its exact nature and semantic contribution are orthogonal to the present discussion. The relevant point to take into account is that the index shows a tight correlation with the person node and surfaces high in the structure, thus implicating the high locus of person features in the syntactic structure. The high structural position of person in pronouns is consistent with the extensive research in the nominal syntax, feature hierarchy developed in Harley & Ritter (2002), recent work on PCC, focused on pronouns and clitics in the clausal syntax (Toosarvandani 2023; Sichel & Toosarvandani 2024, among others), and findings obtained in formal semantics.

Going back to the proposed structure of pronouns in (5) and the standard account of nouns in (7), I proposed that the **structural difference** between pronouns and nouns is located in the $\sqrt{\text{Root}}$ domain, absent in the former but present in the latter. In addition to the apparent phonological silence of $\sqrt{\text{Roots}}$ in the pronouns, contextual allomorphy provided crucial evidence for the absence of $\sqrt{\text{Roots}}$ in pronouns. In Chapter 5, I provided a principled explanation for the absence of the *$\sqrt{\text{Root}}$ -driven number allomorphy* in pronouns.

Namely, I showed that pronouns do not have the complete set of plural allomorphs available to nouns. For example, the pronoun that has the noun *oxen* as an antecedent does not employ the allomorph [en] conditioned by the $\sqrt{\text{Root}}$. Furthermore, the pronoun with an antecedent *alumni* does not surface with the plural allomorph [i]. Therefore, the question raised for approaches to pronouns positing a silent noun in their representations was why the noun cannot trigger allomorphy of plural in the same way it does in canonical DPs. If the silent noun and its $\sqrt{\text{Root}}$ are present in the structure, we expect to find the same set of allomorphs conditioned by the $\sqrt{\text{Root}}$.

To explain the obtained asymmetry, I proposed that the inventory of allomorphs conditioned by the $\sqrt{\text{Root}}$ is available only when the $\sqrt{\text{Root}}$ is present in the structure and the number node can access it at the point when it undergoes Vocabulary Insertion. This is the case in the structure of nouns (9). In pronouns, however, at the point when the number node is targeted for insertion, a $\sqrt{\text{Root}}$ cannot condition the allomorphy of the plural node and give rise to one of the allomorphs, as it is not present in the structure when the Vocabulary Insertion into plural takes place (8). If the $\sqrt{\text{Root}}$ is not present, the number node cannot refer to it when it is targeted for Vocabulary Insertion. This explained the absence of a language in which one of the allomorphs triggered by a particular set of $\sqrt{\text{Roots}}$ would surface in pronouns. Therefore, (i) $\sqrt{\text{Root}}$ -conditioned allomorphs of plural are unexpected in pronouns and (ii) only the elsewhere plural forms, which do not make reference to any particular $\sqrt{\text{Root}}$ are available. Both predictions are corroborated by the data examined in this dissertation (see Chapters 2 and 5).

(8) *Pronouns*(9) *Nouns*

This study made the case that $\sqrt{\text{Roots}}$ are absent in pronouns on the basis of allomorphy. In addition, it demonstrated **why allomorphy serves as a key tool for explaining the structure of pronouns**. The pattern of allomorphy that otherwise arises is absent in the pronominal domain, since pronouns do not have the trigger ($\sqrt{\text{Root}}$) that is present in the nouns. Therefore, the most deeply embedded node in the structure of pronouns is the n head, while in the structure of nouns it is the $\sqrt{\text{Root}}$.

In recent work on pronouns, the person morpheme is argued to be below the number morpheme in the structure of pronouns (see Moskal 2015a, 2015b, 2018; Harbour 2016; Van Urk 2018; Smith et al. 2019; Stegovec 2019; Puškar-Gallien 2022). Smith et al. (2019) suggest that person is lower as it is the morpheme that undergoes allomorphy. In Chapter 3, we have seen that allomorphy goes in both directions. Empirical generalizations presented in Chapter 5, on the close interaction of gender and number to the exclusion of person show that placing the person feature on the n head (see Van Urk 2018) or in between the gender and number heads (see Puškar-Gallien 2022) would leave a number of discussed regularities in the realization of gender and number unexplained, suggesting that person cannot disrupt the adjacency of number and gender. Furthermore, if the person was located on n , it would remain mysterious why the number interacts with the gender, which is located on the same n head, but not with the person. In addition, Toosarvandani (2023) suggests that the proposal made in Harbour (2016), according to which person features must combine first before combining with number, does not depend on the hierarchical structure of the two nodes in the extended projection, which would require person to be below number. Instead, person features combine on their functional head, which is distinct from the head encoding number. Thus, number does not intervene in combinations of person features. This implies that Harbour's (2016) central proposal would not be affected if the person was above the number.

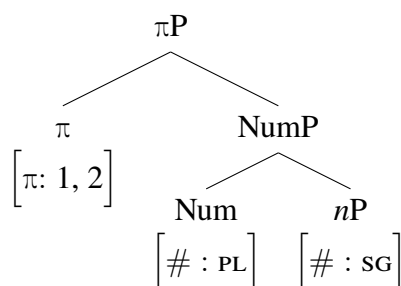
The proposed structural representation extends to languages with clusivity. In Chapter 4, I proposed that the inventory of pronouns in these languages involves an additional pronoun that morphologically signals the inclusion of the second person, i.e., an ADDRESSEE, in the referent of a pronoun. Careful examination of pronouns in languages with clusivity contributed a number of relevant empirical generalizations that motivated the view that languages with clusivity involve an **additional morpheme** ($\mathbb{M}_{incl}$) that hosts both first- and second-person features, defined as follows.

(10) THE INCLUSIVE MORPHEME $\mathbb{M}_{incl}$

def. The inclusive morpheme $\mathbb{M}_{incl}$ is an atomic unit of syntactic computation, consisting of two features

- a. $\mathbb{M}$ is a terminal node
- b. $\mathbb{M}$ is a target of the Vocabulary Insertion operation
- c. $\mathbb{M}$ may trigger allomorphy of the number morpheme

I proposed the following structure of the first-person inclusive pronoun (11). The inclusive morpheme ($\mathbb{M}_{incl}$), consisting of the first and second person [1, 2] occupies the same position in the internal structure of pronouns as the person morpheme discussed above, and merges with the number (Num) morpheme.

(11) *First person inclusive pronoun: The structure*

The central outcome of this proposal is placing the **clusivity parameter** in the syntax **without changing the internal structure** of a pronoun. The difference is encoded at the morpheme level. In that sense, the clusivity parameter is formalized in terms of the kinds of morphemes a language may generate. In languages with clusivity, both the first- and second-person features can be combined within the same morpheme. That is, these languages have an extra morpheme as a basic unit of syntactic computations. Crucially, the presence of an additional morpheme **does not implicate a different syntactic structure** for inclusive pronouns, as it merges in the same position as person morphemes with different feature specifications. This idea motivates the uniformity of the syntactic structure (12).

(12) UNIFORM SYNTACTIC STRUCTURE

The syntactic structure of inclusive pronouns is not different from the structure of other pronouns.

The structure of the first-person exclusive pronoun surfaces with the same set of morphemes, the only difference with the inclusive being located in the feature content of the person morpheme. The person morpheme in the exclusive consists of the first-person feature, on par with the first-person pronouns in languages without clusivity.

In the analysis in terms of a **single morpheme** hosting both person features in the inclusive, the clusivity parameter is placed in the syntax without changing the internal structure of

a pronoun. Within the theoretical framework of Distributed Morphology, a morpheme hosting multiple features is a natural possibility. Therefore, a portmanteau morpheme is not required, especially given the fact that the two features are simply two different specifications of the same feature, that is, **PERSON**. As a consequence, the **structure** of pronouns is **uniform**. That is, both inclusive and exclusive, as well as all the other pronouns in languages that do not mark clusivity, have a **morpheme** hosting **person features**. The exact **featural content of this morpheme differs**. From the perspective of the syntactic structure and the computation, the two pronouns are exactly the same, surfacing with one person morpheme. Second, the Vocabulary Insertion operation targets the person morpheme. With both features located in the same morpheme, it involves **insertion into terminal nodes** without resort to Fusion prior to PF that would turn two morphemes into one or the non-terminal insertion. Crucially, the proposed internal structure, coupled with the standard procedure of Vocabulary Insertion and the Subset Principle that takes into consideration both the left and the right sides of the rule of exponence, correctly derives all the patterns of the distribution of exponents, some of which I will briefly recapitulate here.

I identified six patterns in the morphological realization of pronouns in languages with clusivity. In Urarina, when the Vocabulary Insertion operation targets the inclusive person morpheme, the subset of its features is realized in morphology, that is, the first [AUTHOR] feature. The exponent [*kan*] at PF is underspecified, applying to two distinct morphemes in the narrow syntax: (i) the person morpheme consisting of the first person feature only, present in the singular and the exclusive and (ii) the person morpheme consisting of the first and second person, realizing the subset of them (13). In the absence of a more specific Vocabulary Item that matches more features of the inclusive morpheme, the Vocabulary Item matching the subset of the features is able to apply to all three cells under consideration. In this case, the inclusive morpheme is syncretic with the first person.

	SG	PL	
1EXCL	kan _H	kanakaana	(13) VI for person in Urarina
1INCL		kana	
2	ii	paara	
3	aka	aka _H _H	

[1, 2] ↔ kan
[1] ↔ kan

(a) Urarina (Olawsky 2011)

Table 6.3: Syncretism in languages with clusivity

Introducing clusivity data brought a new perspective, as it demonstrated that allomorphy is not the only way to achieve morphological alternation in pronouns. To account for the Adang data, I proposed that a more specific Vocabulary Item [*p*] matches all the features present in the inclusive morpheme. Being more specific, it takes precedence over the exponent [*n*] that realizes only the first person.

	SG	PL
1 _{EXCL}	na	ni
1 _{INCL}		pi
2	a	i
3	sa	supi

(14) **VIs for person in Adang**

[1, 2] ↔ p

[1] ↔ n

(a) Adang (Robinson & Haan 2014)

Table 6.4: Syncretism in languages with clusivity

In languages like Mina, the first person undergoes allomorphy in the exclusive, resulting in the form [n]. As the locality condition in the inclusive is the same, the number node freely accesses the person node and the subset of its features. In the absence of a more specific exponent, the form [n] is inserted in the inclusive.

	SG	PL
1 _{EXCL}	sə	na
1 _{INCL}		nók
2	ha	hi
3	∅, a	i

(15) **VIs for person in Mina**[1] ↔ n / ___ #_{PL}

[1] ↔ s

(a) Mina (Frajzyngier 2011)

Table 6.5: Syncretism in languages with clusivity

In Vaeakau-Taumako, the correct distribution of exponents is derived by employing competition on both the left and right sides of the rule. In the inclusive, the exponent [tha] matches both features [1, 2] present in the syntax. Hence, complexity arises on the left side of the rule. In the exclusive, the exponent [mha] is the result of the number-driven person allomorphy, as the first person is directly exposed to the number node. In this case, the complexity arises on the right side of the rule.

	SG	PL
1 _{EXCL}	iau	mha-tou
1 _{INCL}		tha-tou
2	koe	khou-tou
3	ia	lha-tou

(16) **VIs for person in VT**

[1, 2] ↔ tha

[1] ↔ mha / ___ #_{PL}

[1] ↔ iau

(a) Vaeakau-Taumako
(Næss & Hovdhaugen 2011)

Table 6.6: Syncretism in languages with clusivity

Now, does the existence of a morpheme such as $\mathbb{M}_{incl}$, where two person feature specifications, [AUTHOR] and [ADDRESSEE] are hosted in the same head clash with the view that person and

number are not hosted on the same head but establish two different morphemes in narrow syntax? As argued throughout this dissertation and demonstrated on the basis of the cross-linguistic data, pronominal systems in individual languages show **shared morpho-phonological material**, which calls for the individuation of person and number exponents. In the case of $\mathbb{M}_{incl}$, there is no shared material that would require a decomposition. The analytic inclusives were accounted for in terms of Fission at PF.

Introducing **gender** features into perspective led to a major split in the dataset. Namely, gender features in pronouns are overtly morphologically realized in 80 out of 350 languages investigated here. This result illustrated **variation** in pronouns across languages by showing that not all languages recruit gender features in their system. Those that do split into two groups: (i) languages with parallel gender systems, where gender distinctions are present in both singular and plural, and (ii) languages where the gender distinctions are present only in the singular. Both groups can be further differentiated on the basis of the patterns of their realization. However, three empirical generalizations (17) with particular significance for the structure of pronouns are worth noting again.

(17) *Empirical generalizations: Gender in relation to person and number*

G1: In a subset of languages that make gender distinctions in the singular and plural, person, gender, and number exponents are segmentable in the plural.

G2: In languages that make gender distinctions in the singular and plural, if segmentation is not possible, gender and number are realized via a non-segmentable exponent, while person remains visible.

G3: Gender neutralization arises with the addition of plural features to the structure.

I will illustrate each generalization with one representative example. In Mosetén, person [*m*], masculine [*i'*] and feminine [*ö'*] gender, and plural [*in*] features are realized with three distinct exponents in a 1:1 mapping, thus exemplifying G1. In Serbian, gender distinctions are preserved in the plural, but gender and number are not mapped to two overt exponents, consistent with G2. Breton exemplifies a language in which the gender distinction is neutralized in the plural.

SG		PL	SG		PL	SG		PL
3M	m-i'	m-i'- in	3M	on	on- i	3M	eñ	int
3F	m-ö'	m-ö'- in	3F	on-a	on- e	3F	hi	int
(a) Mosetén (Sakel 2011)			(b) Serbian			(c) Breton (Press 2011)		

Table 6.7: Gender in pronouns across languages

In Chapter 5, I proposed that the first generalization (G1) tells us about the **inventory of syntactic terminals** while the second (G2) and third (G3) are about their **locality**. The non-segmentable exponents of gender and number to the exclusion of person and neutralization of gender distinctions with the addition of the plural features are indicative of their close interaction

in the internal structure of pronouns. Serbian data were explained in terms of allomorphy, while Breton was explained in terms of Impoverishment. That is, both languages that draw gender distinctions in singular and plural and those with gender neutralization in the plural point toward a morpho-syntactic interaction of gender and number, as for these operations to arise, locality has to be established. This result is readily explainable under the structure proposed in (5), where gender features are located on the n head and number features on the Num head are structurally adjacent, while person surfaces higher in the structure.

Building on the discussion about $\sqrt{\text{Roots}}$, pronouns lack $\sqrt{\text{Roots}}$, but surface with the n terminal. As noted above, this terminal hosts **gender** features, both conceptual (natural) and arbitrary (grammatical), and **singulative number**, and explains the availability of **lexical plurals**.

Replicating the strategy proposed for clusivity, the presence of gender features does not immediately lead to a different internal structure of pronouns. Instead, the featural content of the n morpheme that is present in languages without gender, where it hosts the singular number, is the locus of the morpho-syntactic variation.

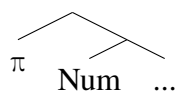
I propose that the **variation is located in a morpheme** as a basic unit of syntactic computations, both within and across languages. The syntactic variation in the structure of pronouns is manifested at **two levels**: (i) a morpheme can be completely absent from the structure, or (ii) its featural content may differ.

The first kind of variation is attested within a language. In the singular, pronouns will lack the Num morpheme in (5), consisting of the π morpheme and the n morpheme hosting the singular number. The second kind of variation accounts for phenomena such as clusivity and gender marking. In these cases, the variation is encoded within a morpheme in terms of features that it surfaces with. Languages with clusivity have an additional morpheme $\mathbb{M}_{incl}$, which hosts both the first [AUTHOR] and the second [ADDRESSEE] person. Therefore, the proposed syntactic structure is not modified. Instead, both features are located within the same π morpheme. The same principle extends to gender features, which are attested on a language-specific basis. If gender is present in a language, it is located on the n head that is independently available in the structure and hosts the singular number and the lexical plural. In the same manner as in the account of clusivity, the proposed syntactic structure is not modified. The difference lies in the featural content of the n head across languages.

6.2.2 Linearization

Syntactic operations that apply at the clausal level also hold for morphemes (Halle & Marantz 1994; Marantz 1995, 1997, 2001, 2007). The same applies to linearization that imposes the same mechanism on larger syntactic objects at the level of clausal syntax, and smaller syntactic objects, that is, syntactic terminals ($\sqrt{\text{Roots}}$, functional morphemes) within an M-word (Embick 2007). Abstracting from other layers in the structure of pronouns and focusing only on person and number, the proposed syntactic structure of pronouns, repeated here as (18), is linearized at PF in a language-specific way. In the linearization statement (19), I use the *-operator to indicate linear precedence (Embick & Noyer 2001), and [–] for the concatenation of the morphemes into a linear string. Under the multi-morphemic approach to pronouns, morphemes present in narrow syntax will undergo the linearization operation at PF.

(18) SYNTACTIC STRUCTURE



(19) LINEARIZATION STATEMENT

- a. $(\pi * (\text{Num} \dots))$
 b. $\pi - \text{Num}$

In the majority of languages discussed throughout previous chapters, the exponent of person linearly precedes the exponent of number, in line with (19). However, natural languages provide evidence for the opposite order of exponents, which is expected under the view that linearization applies at PF and can shift the order of morphemes in language-specific ways. In languages like Amharic and Ma'di, an exponent of person linearly follows an exponent of number. As discussed in Chapter 5, the exponents of the second [antä] and third [əssu] person masculine are preserved in the plural, where they transparently combine with the exponent of plural [ənn].

	SG	PL
1	əne	əñña
2 _M	ant-ä	ənn-ant-ä
2 _F	anj-i	ənn-ant-ä
3 _M	əss-u	ənn-äss-u
3 _F	əss- ^w a	ənn-äss-u

(a) Amharic (Leslau 1995)

(20) Some VIs for Amharic

- [PL] $\leftrightarrow$ ənn
 [2] $\leftrightarrow$ ant
 [3] $\leftrightarrow$ əss

Table 6.8: Linearization of person and number in Amharic

A related pattern is present in Ma'di (Central Sudanic; Nilo-Saharan), where the first [má] and the second [nɪ] person singular exponents are preserved in the plural, combining with the plural exponent [à], that is linearized in the initial position. Note that the third person exhibits syncretism in both singular and plural.

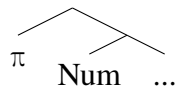
	SG	PL	
1	má	à-má	(21) Some VIs for Ma'di
2	ɲí	à-ɲí	[PL] $\leftrightarrow$ à
3	ká/kó	ká/kí'	[1] $\leftrightarrow$ má
			[2] $\leftrightarrow$ ɲí

(a) Ma'di (Blackings & Fabb 2003)

Table 6.9: Linearization of person and number in Ma'di

To account for Amharic and comparable data, the following linearization statement (23) is proposed, requiring that the number linearly precedes the person, while the syntactic structure remains the same.

(22) SYNTACTIC STRUCTURE



(23) LINEARIZATION STATEMENT: AMHARIC

- a. (Num * (π ...))
- b. Num – π

In this section, I provided a detailed elaboration on the multi-morphemic account of pronouns, discussing the patterns of universality and variation in syntactic structuring. The pronouns are composed out of three morphemes in their internal structure: the person morpheme (π), the number morpheme (Num), and the n morpheme. The inventory consists of functional morphemes only. Pronouns crucially lack the $\sqrt{\text{Root}}$ morpheme, which distinguishes them from nouns. In the next section 6.3, I turn to the results of allomorphy.

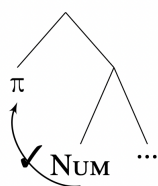
6.3 Allomorphy

Recruiting allomorphy as a test for the internal structure of pronouns yields a **twofold result**, as findings from the pronominal domain inform the theory of allomorphy in important ways. In this section, I will discuss the summary of the findings. In 6.3.1, I focus on the main empirical finding of the availability of allomorphic conditioning going in both directions, providing a theoretical explanation for the observed patterns. In the next step, I discuss the ways in which the addition of the features, gender in particular, interacts with person and number. In 6.3.2, I fully develop the account of the patterns of allomorphy in pronouns, arguing that the obtained findings are the result of the syntactic structure of pronouns, which involves a single cycle of spell-out, i.e., a single phase domain, and a particular view of Vocabulary Insertion. In the final part (6.3.3), I focus on the cases of the so-called double marking, understood as the overt co-occurrence of the trigger and the target of allomorphy. These data will serve as the basis for the comparison of the account of pronouns in Distributed Morphology with other realizational frameworks in the continuation of this chapter.

6.3.1 Directionality in allomorphic conditioning

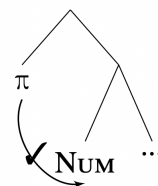
Pronouns provide evidence for both (i) number-driven person allomorphy and (ii) person-driven number allomorphy. In the structure where the person is above the number, the number-driven person allomorphy instantiates an **outward-conditioning** (24), as the trigger is in a structurally lower position compared to the target of allomorphy. In contrast, person-driven number allomorphy is the result of an **inward-conditioning** (25), given that the trigger is in a structurally higher position compared to the target.

(24) #-driven π allomorphy



outward-conditioning

(25) π -driven # allomorphy



inward-conditioning

The two directionalities of allomorphy in pronouns are illustrated here with data from Finnish and Ecuadorean Kichwa. In Finnish, the exponent of the second person [*s*] that is present in the singular alternates with [*t*] in the context of the plural number. In this, the second person differs from the first and third person, which do not undergo allomorphy in the context of the plural, being realized as [*m*] and [*h*], respectively, in both number specifications. Therefore, Finnish data exemplify the number-driven person allomorphy and outward-conditioning. This pattern contrasts with the one attested in Ecuadorean Kichwa, where the realization of the plural involves two allomorphs that compete for insertion. The presence of the first person triggers the realization of the plural exponent [nɟik]. When the first person is not present in the person

morpheme, the elsewhere plural form [*kuna*] surfaces, which is the case in the second- and third-person pronouns.

	SG	PL		SG	PL
1	m -inä	m -e	1	ɲuka	ɲuka-ɲjik
2	s -inä	t -e	2	kan	kan-kuna
3	h -än	h -e	3	paj	paj-kuna

(a) Finnish (White 2006)

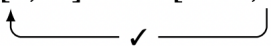
(b) Ecuadorian Kichwa (Pertsova 2022b)

Table 6.10: The two directions of allomorphy

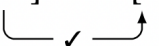
These data call for a theory of allomorphy that is able to accommodate **allomorphic conditioning** going **in both directions** and involving two same morphemes. As discussed earlier in this dissertation, the availability of both directions is predicted to be impossible under Bobaljik's (2000) approach to allomorphy. Examination of the pronominal systems revealed that **both directions of allomorphy** are **possible**. The results of the present study are consistent with the finding reported in the literature that both directions in allomorphic conditioning are available for grammatically-conditioned allomorphy (Rolle 2023), that is, when the trigger and the target involve morpho-syntactic features, which is the case with person and number.

Why are both directions possible? To explain the availability of both directions of allomorphy, I propose that at the point when the number undergoes Vocabulary Insertion, person features are visible (26). The same holds for the opposite combination of trigger and target. When the person morpheme is targeted by the Vocabulary Insertion, number features are visible (27).

(26) NUM LOOKING INTO π

$\mathbb{M} [\pi; / /] \text{ — } \mathbb{M} [\text{NUM}; / /]$


(27) π LOOKING INTO NUM

$\mathbb{M} [\pi; / /] \text{ — } \mathbb{M} [\text{NUM}; / /]$


I formulated this requirement in terms of (28). Allomorphy in both directions between two same morphemes is possible if the features of both morphemes are visible at the point when they are targeted by the Vocabulary Insertion operation.

(28) Two morphemes, Ξ and Ω , may enter an allomorphic alternation, such that Ξ can trigger allomorphy of Ω and be a target of allomorphy triggered by Ω , if

- at the point when Ξ undergoes Vocabulary Insertion, features of Ω are visible
- at the point when Ω undergoes Vocabulary Insertion, features of Ξ are visible

Furthermore, I suggested that both directions of allomorphy are explainable if phonological features are **adjoined to** morpho-syntactic and semantic features at PF. There is **no deletion** of features at PF, as the availability of both directionalities of allomorphy would be unexplainable.

Person [π] and number [NUM] features are **not deleted** by the phonological features but **supplied** with them, in line with the initial idea of early DM (Halle & Marantz 1994).

In the narrow syntax, prior to the Vocabulary Insertion operation, a morpheme consists of morpho-syntactic and semantic features (π or NUM) and an empty slot // for the phonological features (29-a), which can be inserted only once in line with the Uniqueness requirement for Vocabulary Insertion (see Halle 1992, Embick 2015, Alexiadou 2024 and the related discussion in Chapter 3; Section 3.2.2.1). Person and number morphemes are targeted in two separate cycles of Vocabulary Insertion. The Vocabulary Insertion adjoins the exponent [x] to the person morpheme at PF. Likewise, it adjoins the exponent / y / to the number morpheme. After Vocabulary Insertion, the features present in the narrow syntax are supplied with the phonological features, indicated by the filled slots in (29-b).

(29) *Adjoining of phonological features at PF*

a. [π ; //] – [NUM; //]

prior to Vocabulary Insertion

b. [π ; / x /] – [NUM; / y /]

after Vocabulary Insertion

This setup accounts for both directions of allomorphy presented here. Starting with Finnish, the number node that is lower in the structure triggers allomorphy of the person node, resulting in outward-conditioning. At the point when the person is targeted for insertion, it has available features of the number morpheme and can refer to them, which is directly reflected in the contextual allomorphy that arises (30).

	SG	PL
1	m -inä	m -e
2	s -inä	t -e
3	h -än	h -e

(a) Finnish (White 2006)

(30) #-driven π allomorphy

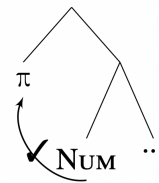


Table 6.11: Outward-conditioning: Finnish

In the post-syntactic component of the grammar, i.e., at PF, the person and number morphemes in Finnish pronouns are supplied with the phonological content (32). In the VI rules, I provide just the relevant points and do not provide the full paradigm.

(31) **VI for Finnish**

[2] $\leftrightarrow$ t / ___ #_{PL}

[2] $\leftrightarrow$ s

(32) *After Vocabulary Insertion*

[π ; / t /] – [NUM; / e /]

The same principle applies to Ecuadorian Kichwa. The structurally higher person node triggers allomorphy of the lower number node, giving rise to the inward-conditioning. At the point when the number morpheme is targeted for Vocabulary Insertion, the person features are available to condition allomorphy (33).

	SG	PL
1	ɲuka	ɲuka-ɲjik
2	kan	kan-kuna
3	paj	paj-kuna

(a) Ecuadorian Kichwa (Pertsova 2022b)

(33) #-driven π allomorphy

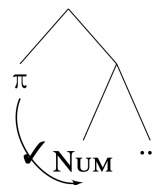


Table 6.12: Inward-conditioning: Ecuadorian Kichwa

At PF, the set of Vocabulary Items for the two morphemes is recruited (34), resulting in the adjoining of the phonological features (35).

(34) **VIs for Ecuadorian Kichwa**

[PL] $\leftrightarrow$ ɲjik / ___ π_{AUTHOR}

[PL] $\leftrightarrow$ kuna

[1] $\leftrightarrow$ ɲuka

(35) **After Vocabulary Insertion**

[π ; /ɲuka/] – [NUM; /ɲjik/]

The proposed system explains the observed patterns of allomorphy in a principled manner and provides the theoretical explanation thereof, combining the insights that both the trigger and the target are functional morphemes, hence allomorphy is grammatically-conditioned, and a specific view of Vocabulary Insertion. Crucially, these data raise several important questions about the syntax of these languages and its role in allomorphic conditioning.¹ What determines the directionality of allomorphy? That is, why is it inwards in some languages and outwards in others? Does a particular language *X* that shows inward-conditioning have different syntactic properties from a language *Y* that shows outward-conditioning? In addition, can one predict which properties are relevant for inward-conditioning and which properties lead to outward-conditioning? Furthermore, in languages like Faroese, discussed in Chapter 3, we find both directions in allomorphic conditioning simultaneously, as the first- and second-person surface with the person-driven number allomorphy, while the third-person exhibits number-driven person allomorphy. One potential way to explain these data is further examination of the features that take part in allomorphy, feature specifications, and their representation in syntax and semantics. Namely, as

¹I thank Artemis Alexiadou for raising these important questions and for extensive discussions on directionalities for finding the solutions to these puzzles.

discussed in Chapters 2 and 5, the third-person feature never triggers allomorphy of the number morpheme if the first- and second-person features surface with an elsewhere exponent present on nouns. The opposite holds. It is the third-person that consistently surfaces with the elsewhere. The third person may condition a specific allomorph of the plural morpheme, only if both the first and the second person trigger two distinct allomorphs of the plural, as was the case in Tübatulabal, which has three exponents of the plural for the three person specifications. These questions open a new direction for investigation in future research that goes beyond the structure of pronouns and focuses on the role of wider core syntactic properties and the nominal syntax in particular in contextual allomorphy.

6.3.1.1 What kinds of directionalities are there in pronouns?

In 6.3.1, I provided a theoretical proposal on the availability of the two directions of allomorphy, taking into account two features in the internal composition of pronouns: person and number. The question that arises is which patterns emerge when the structure increases its complexity with the addition of morphemes. The discussion in Chapter 5 introduced gender features into the perspective. Can gender act as both the trigger and the target of allomorphy? Can gender block an allomorphic alternation established between person and number? Does it interact with person only, with number only, or with both person and number simultaneously? I will discuss here the three patterns for which the cross-linguistic examination of gender in relation to person and number in pronouns provided evidence.

In languages with parallel gender systems, gender features may trigger allomorphy of the number morpheme in the plural. This is exemplified here on the basis of Serbian data. The gender distinction is made in the third person and preserved in the plural. The exponent of the person $[(o)n]$ transparently combines with the exponent of the feminine gender $[a]$. In the singular, the person morpheme and the gender hosted on n do not interact for purposes of allomorphy. In the plural, however, gender features interact with the plural features on the Num terminal. The elsewhere exponent of the plural $[i]$, present in the first- and second-person plural, the third-person plural masculine, and in the nouns, alternates with the exponent $[e]$ in the context of the feminine gender. Therefore, this pattern exemplifies gender- (n) driven number allomorphy.

	SG	PL
1	ja	m-i
2	ti	v-i
3 _M	on	on-i
3 _F	on-a	on-e

(a) Serbian

(36) n -driven # allomorphy

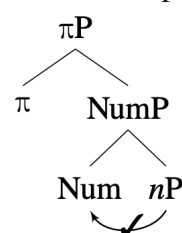


Table 6.13: Gender-driven number allomorphy: Serbian

Can gender trigger allomorphy of the person morpheme? In languages that make a gender distinction, but individuation of the person morpheme is not possible, i.e., there is no shared morpho-phonological material that would call for an analysis in terms of a dedicated exponent of person, the two distinct forms are explainable in terms of the gender-driven person allomorphy. Namely, the two third-person pronouns are realizations of the third-person feature in the context of gender. This is exemplified here on the basis of Breton data.

	SG	PL
1	me	ni
2	te	c'hwi
3 _M	eñ	int
3 _F	hi	int

(a) Breton (Press 2011)

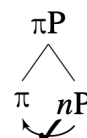
(37) *n*-driven π allomorphy

Table 6.14: Gender-driven person allomorphy: Breton

I proposed that the internal structure of pronouns in the singular consists of the person morpheme and the *n* morpheme, hosting singular number and gender features. In the syntactic structure consisting of only two morphemes, their features are readily available to enter an allomorphic alternation. Crucially, the gender distinction disappears in the plural. In Chapter 5, I provided a set of arguments, starting with the case study on the nominal concord in Somali pronouns, that gender features in languages that do not preserve the distinction in the plural are subject to the **Impoverishment operation**. Therefore, **gender** features are **deleted** in the plural and **cannot take part in allomorphy**. With gender absent in the plural as a consequence of deletion, the plural form [*int*] is the realization of the third-person in the context of the plural or, alternatively, an elsewhere third-person exponent. The decision between these two possibilities has no bearing on the structure of pronouns or on the account of allomorphy.

(38) **VIs for Breton**[3] $\leftrightarrow$ int / ___ #_{PL}[3] $\leftrightarrow$ hi / ___ γ_{+FEM} [3] $\leftrightarrow$ eñ / ___ γ_{-FEM} (39) **Impoverishment Rule**[+FEM] $\rightarrow \emptyset$ / #_{PL}

In 80 languages that morphologically mark gender features in pronouns, 57 exhibit systematic neutralization of gender in the plural. This systematicity receives a principled explanation under the Impoverishment account pursued here. Now, what does regularity in gender neutralization mean for allomorphy? **Impoverishment prevents allomorphy** as it removes the feature, in the case at hand, gender, that would serve as a trigger or a target. That is, as a consequence of

deletion, gender features can be neither triggers nor targets of allomorphy in the plural. Therefore, gender features can be entirely omitted from the computation of allomorphy in the plural in all languages with gender neutralization.

A potential interaction of gender with both the person and the plural number is possible only in languages with parallel gender systems. As exemplified on the basis of the Serbian data above, gender may interact with number for purposes of allomorphy in this case. These data demonstrate that one has to take into account an array of parameters in analyzing allomorphy in pronouns, as additional considerations, such as Impoverishment, may have a significant impact on whether allomorphy will apply or not.

If gender-driven person allomorphy is attested, the opposite direction, that is, person-driven gender allomorphy, may also arise, under the proposal made above that allomorphic conditioning goes in both directions in pronouns. It is important to note that the first requirement for this kind of allomorphy is that gender distinction is present in more than one person specification, which is a rare pattern. This is what we find in Amharic, a language that makes gender distinctions in the second and third person. In both person specifications, masculine singular forms are preserved in their entirety in the plural, transparently combining with the plural exponent [ənn]. The gender distinction in the second person is followed by the insertion of two exponents, [ä] for the masculine, and [i] for the feminine. As noted in Chapter 5, the realization of gender exponents in the third person [u/^wa] corresponds to definite determiners that reflect the gender of the nominal they combine with. Elsewhere in the language, gender features typically do not receive an overt morphological realization. The formants [ä/i] are different as they appear only with the second person, which motivates their analysis as person-driven gender allomorphy. With the Impoverishment rule in the plural, gender features are neutralized and cannot cause or affect allomorphic alternations.

	SG	PL
1	əne	əñña
2 _M	ant-ä	ənn- ant-ä
2 _F	anɿ-i	ənn- ant-ä
3 _M	əss- u	ənn- äss-u
3 _F	əss- ^w a	ənn- äss-u

(a) Amharic (Leslau 1995)

(40) π -driven n allomorphy

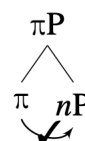


Table 6.15: Person-driven gender allomorphy: Amharic

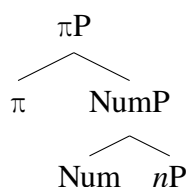
The examination of the gender data reveals that adding gender features does not generate a novel set of allomorphs of gender, but results in robust patterns of neutralization in the plural. If gender acts as a trigger or a target of allomorphy, this alternation is limited to adjacent nodes. In addition, while number-driven person allomorphy and person-driven number allomorphy are widely present across languages, this is not the case with allomorphic alternations involving gender features. The gender features have all the conditions to act as triggers and targets of

allomorphy. However, rather than trigger allomorphy in the way person and number do, gender features exhibit something else. They undergo systematic Impoverishment or interact only with immediately adjacent morphemes.

6.3.2 Locality

Allomorphic alternations examined here arise within a pronoun. Taking into consideration the structure of a pronoun and the nature of morphemes that make up pronouns is important for interpreting the results obtained for allomorphy. In (5), repeated here as (41), I proposed the internal structure of pronouns. Personal pronouns consist of three morphemes, i.e., syntactic terminals in their structural representation: the nominalizing head *n*, the **number morpheme** (Num), and the **person morpheme** (π). These morphemes belong to the inventory of functional morphemes. Pronouns crucially lack the second type of a morpheme, namely a $\sqrt{\text{Root}}$. Inspection of the pronominal structure reveals that pronouns involve the *n* head as the most deeply embedded morpheme. At the same time, the *n* head and all the nodes above it are present within the **same phase**, as there are no phase heads splitting the structure of a pronoun. In the nominal domain, phase heads are the categorizing head *n* and D (Alexiadou 2020). In pronouns, *n* is a complement/modifier-free head, since $\sqrt{\text{Root}}$ s are not present in the structure. The highest terminal node (π) in the structure of pronouns and the lowest *n* head are always in the same spell-out cycle.

(41) *The structure of pronouns*

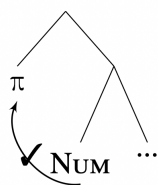


Under the idea that the spell-out of the syntactic structure proceeds in a cyclic² phase-based way (Marantz 2001, 2007 and subsequent work), cyclic spell-out is a delimiter of allomorphy (Embick 2010; Marantz 2013; Moskal 2015b; Moskal & Smith 2016; Kalin 2022a, 2022b; Adamson 2024), as it determines what is available at PF. Namely, only the morphemes that belong to the same phase are available and visible to each other for subsequent morphological operations at PF. A *cycle* in the phase-based spell-out is instantiated by sending a chunk of structure to the interfaces, which are then subject to further operations. The merge of a phase

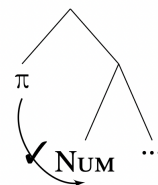
²In DM, the notion of *cyclicity* also stands for the idea that the Vocabulary Insertion operates on the inside-out basis, from the most deeply embedded morpheme outwards (Bobaljik 2000; Myler 2017). This procedure is referred to as the *inside-out* or *bottom-up* insertion. I do not focus on that aspect of the theory here, but the idea that larger pieces of syntactic structure are sent in phases to the interfaces, in accordance with the Minimalist theory (Chomsky 2000, 2001). As the same principles of structure building and realization that apply for complex objects at the level of clausal syntax hold for smaller objects, i.e., ‘words’, phase-based spell-out finds its natural extension to the word-level.

head triggers the spell-out of its complement, but not the spell-out of the phase head itself. In that sense, the phase heads (n , v , a , D) are *cyclic* heads.

I propose that pronouns (41) involve a **single cycle of spell-out**. All morphemes that make up a pronoun are available for interaction at PF. As pronouns do not involve a $\sqrt{\text{Root}}$ in their structure, the n head as the most deeply embedded morpheme does not block allomorphy and may act as both the trigger and the target. Furthermore, I propose that **availability of both directions of allomorphy**, widely attested in pronouns, is the result of the **syntactic structure** of pronouns and a specific view on the **Vocabulary Insertion**. All morphemes in a pronoun are sent to PF in one cycle of spell-out. Hence, they can freely interact in both directions within the same structure. In addition, the Vocabulary Insertion operation adjoins the phonological features to morphemes at PF without deleting them. As a consequence, the morpho-syntactic features present in the narrow syntax are not lost at PF. Therefore, **the two directions** of allomorphy arise **within the same phase** in the interaction of morphemes that are sent to interfaces in one cycle of spell-out.

(42) #-driven π allomorphy

outward-conditioning

(43) π -driven # allomorphy

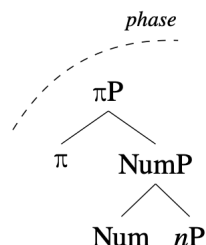
inward-conditioning

In relation to the theory of morphological phases, the main implication of the present proposal is that pronouns do not involve a phase head that splits their internal structure into several chunks sent to the interfaces in separate cycles. All morphemes that make up a pronoun are available for interaction within the same phase at PF. Importantly, the preceding discussion has shown that this does not lead to a more relaxed condition on contextual allomorphy, as only the adjacent heads are able to enter an allomorphic alternation in the pronominal domain. With the increase in complexity that arises with the addition of gender features, allomorphic alternations were not disrupted. As discussed above, gender establishes allomorphy with the number in the plural and the person in the singular, in which case it is immediately adjacent to both heads.

Crucially, the n head lacks a $\sqrt{\text{Root}}$. As there is no complement of n , this part of the derivation, typical of nouns, is not present in pronouns. In the first cycle of spell-out, the n head is sent to the interfaces together with all the functional material in the domain that it heads, that is, the functional morphemes number (Num) and person (π). As a consequence, all the layers in a pronoun, n , and the cyclic domain headed by n (Num, π) are available for the interaction at PF (44). In this configuration, the allomorphy freely arises in both directions. Taking the case of person-driven number allomorphy and number-driven person allomorphy, none of the heads [π , NUM] is a phase head. Therefore, they can interact for allomorphy in both directions within the

same phase.³ In addition to allomorphy going in both directions, the proposed syntactic structure of pronouns involving a single cycle of spell-out explains all the patterns of allomorphy discussed in this dissertation.

(44) **Available at PF**



Moreover, the directionality distinction in allomorphy is highly reminiscent of the directionality of the Agree operation.⁴ In the Downward Agree, the probe is in the structurally higher position than its c-commanding goal (Chomsky 2000, 2001). The literature has argued for the patterns of agreement that can be explained under the Upward Agree (Zeijlstra 2004, 2012; Wurmbrand 2012; Arregi & Hanink 2018; Bjorkman & Zeijlstra 2019; Bárány & Van der Wal 2022). In that sense, the inward-conditioning is similar to Downward Agree, while the outward-conditioning to Upward Agree, suggesting that the inward-outward distinction may figure as a more general property of natural language. Given that allomorphy and Agree are two distinct processes, the empirical question as to whether the availability of both directions requires the same locality conditions or exhibits additional uniform properties awaits further research.

6.3.3 Double marking: Co-occurrence of the trigger and target

Pronominal systems provide vast and profound evidence for so-called *double marking* (Halle & Marantz 1994). Double marking in this context is understood as an **overt co-occurrence of the trigger and the target of allomorphy**. These data are significant both for the theory of allomorphy and the Vocabulary Insertion operation. Two crucial implications arise: (i) allomorphy cannot be analyzed in terms of Fusion, and (ii) the Vocabulary Insertion targets only terminal nodes.

As discussed earlier in this dissertation, in Mangghuer (Altaic; Mongolic), the plural morpheme triggers the allomorphy of the first and second person. The exponent of the first person feature in the singular [*bi*] alternates with [*da*] in the plural. Crucially, the **trigger** of allomorphy, that is, the plural morpheme, realized in the form [*si*], **overtly co-occurs** with the **target** of allomorphy, i.e., the first person, realized in its allomorphic form [*da*]. The same extends to the second person. This pattern exemplifies a run-of-the-mill case of double marking, as the trigger and the target freely co-occur within the same form.

³The directionality asymmetry arising in Chukchi (Bobaljik 2000) may potentially stem from subjects and objects being located in two different phases.

⁴I thank Artemis Alexiadou for pointing out this idea to me.

	SG	PL			
	1	bi	da-si	(45) VIs for Mangghuer	
	2	qi	ta-si	[1] ↔ da / ___ # _{PL}	[3] ↔ gan
	3	gan	gan-si	[1] ↔ bi	[PL] ↔ si
				[2] ↔ ta / ___ # _{PL}	
				[2] ↔ qi	

(a) Mangghuer (Slater 2003)

Table 6.16: Number-driven person allomorphy: Mangghuer

The same holds for the opposite direction in allomorphic conditioning, as exemplified on the basis of person-driven number allomorphy in Ecuadorian Kichwa (Table 6.17). The trigger of allomorphy, which is in the case at hand the person morpheme, realized in the form [ɲuka] co-occurs with the target of allomorphy, that is, the plural morpheme, realized in its allomorphic form [ɲjik] in the context of the first person. An approach to pronouns that would consider them bundles of φ -features corresponding to a single Vocabulary Item would have to explain how the form [ɲuka] can appear in isolation, as is the case in the singular, and in the form [ɲukanɲjik], if it is not further decomposed.

	SG	PL		
	1	ɲuka	ɲuka-ɲjik	(46) VIs for Ecuadorian Kichwa
	2	kan	kan-kuna	[PL] ↔ ɲjik / ___ π _{AUTHOR}
	3	paj	paj-kuna	[PL] ↔ kuna
				[1] ↔ ɲuka

(a) Ecuadorian Kichwa (Pertsova 2022b)

Table 6.17: Person-driven number allomorphy: Ecuadorian Kichwa

Beyond person and number, double marking in pronouns, defined as the co-occurrence of the trigger and target of allomorphy, naturally arises in other combinations of features. As discussed above, in the alternation analyzed as person-driven gender allomorphy in Amharic, the trigger of allomorphy, i.e., the person morpheme realized as [ant], overtly co-occurs with the target of allomorphy, that is, the exponents of gender [ä/i].

The observed double marking stems from the very nature of contextual allomorphy. Namely, nothing prevents the overt co-occurrence of the trigger and the target. The core of the allomorphic process, which is the sensitivity of two morphemes to each other, directly reflected in morphology, does not impose any requirement on their overtness. In fact, the co-occurrence is consistently expected. It does not emerge only in the cases where one of the morphemes receives a zero realization. As will be discussed in the continuation of this chapter, the availability of double marking is a critical data point where Distributed Morphology and comparable realizational frameworks, such as Nanosyntax (Starke 2009; Caha 2009) and Spanning (Bye & Svenonius 2011; Merchant 2015; Svenonius 2016; Haugen & Siddiqi 2016) make different predictions.

6.4 Extensions

In this section, I will discuss the ways in which the proposed multi-morphemic analysis of pronouns can be extended to accommodate additional material present on a language-specific basis. The focus is on the morphological case (Section 6.4.1) and definiteness (Section 6.4.2). The discussion outlines the critical points that need to be taken into consideration in evaluating the contribution of these features and directions for future research.

6.4.1 Case

Investigation of case features in allomorphy requires a careful consideration of what kind of morpheme case is. The assignment of case features is crucially dependent on the argument structure, verbal or nominal (Marantz 2000; Alexiadou 2001; McFadden 2004; Bobaljik 2008b; Schäfer 2008; Wood 2011, 2017; Alexiadou, Anagnostopoulou, & Schäfer 2015; Baker 2015; Alexiadou & Anagnostopoulou 2021). Namely, assignment of the accusative to a nominal proceeds **configurationally**, depending on the presence of an established c-command relation with another nominal in a particular local syntactic environment. It is important to note that the type of case under consideration is the morphological *m*-case (Bobaljik 2008b) that should be distinguished from Case as abstract nominal licensing (Chomsky 2001; Pesetsky & Torrego 2012).

In the data examined in the previous chapters, allomorphic alternations involved person, number, and gender features as triggers and targets of allomorphy. Morphological case considerations introduce a different kind of syntactic node.⁵ As noted above, case is assigned configurationally, interpreting the syntactic structure and making reference to the argument structure of a verb in particular. In this, case differs from other features discussed here, such as person, number, and gender, whose presence does not rely on core syntactic principles, such as the c-command.

In Greek, [*m*], [*s*], and [*t*], the exponents of the first, second, and third person, respectively (Alexiadou & Stavrou 2000), remain unaffected by case. As evident from Table 6.18, morphological case shows an interaction with the number and gender exponents, that is, the material linearizing to the right of person exponents. It crucially does not interact with the person and definiteness [*af*] exponents (for detailed discussion of Greek inflection, see Alexiadou & Müller 2008, Markopoulos 2018). This pattern is expected if the case surfaces in a local environment with number and gender to the exclusion of person. This pattern is expected if the case features are inserted within the *nP* domain, traditionally associated with inflection, without extending the syntactic structure but being assigned to a morpheme that is already present.

⁵Thanks to Artemis Alexiadou for making me aware of this aspect of the analysis.

	1		2		3					
	SG	PL	SG	PL	M.SG	M.PL	F.SG	F.PL	N.SG	N.PL
NOM	ego	<u>e</u> mi <u>s</u>	e <u>s</u> i	e <u>s</u> i <u>s</u>	aft <u>o</u> s	aft <u>i</u>	aft <u>i</u>	aft <u>e</u> s	aft <u>o</u>	aft <u>a</u>
GEN	<u>e</u> me <u>n</u> a	<u>e</u> ma <u>s</u>	e <u>s</u> ena	e <u>s</u> a <u>s</u>	aft <u>u</u>	aft <u>o</u> n	aft <u>i</u> s	aft <u>o</u> n	aft <u>u</u>	aft <u>o</u> n
ACC	<u>e</u> me <u>n</u> a	<u>e</u> ma <u>s</u>	e <u>s</u> ena	e <u>s</u> a <u>s</u>	aft <u>o</u> n	aft <u>u</u> s	aft <u>i</u> n	aft <u>e</u> s	aft <u>o</u>	aft <u>a</u>

Table 6.18: Personal pronouns in Greek

Germanic languages, however, are particularly illustrative of allomorphic alternations involving case in both singular and plural. In English, German, and Icelandic, first-person pronouns undergo allomorphy triggered by case in both the accusative singular and plural.

	English		German		Icelandic	
	SG	PL	SG	PL	SG	PL
NOM	I	we	ich	wir	ég	við
ACC	me	us	mich	uns	mig	okkur

Table 6.19: Allomorphy in first person pronouns in Germanic languages

Germanic data present a puzzle. Namely, the person morpheme is consistently available to undergo allomorphy, both in the singular and plural.⁶ The question that arises is why does case have access to the person morpheme in the singular and plural in Germanic but not in Greek.

In this brief remark, I outlined the way case morphemes can be accommodated within the multi-morphemic account of pronouns. A detailed examination of the patterns of allomorphy involving case features and the observed asymmetry between Greek and Germanic is a topic for future research.

6.4.2 Definiteness

In this section, I present two case studies on the realization of definiteness in (i) third-person complex pronouns in Greek and (ii) first- and second-person complex pronouns in Serbian, showing how the multi-morphemic, multi-terminal approach to pronouns extends to accommodate additional features in the structure of pronouns.

In Greek, complex third-person pronouns surface with an additional exponent [*af*]. This exponent remains constant across all feature combinations. In this case, there are three features that have the potential to trigger allomorphy, namely, gender, number, and case. However, [*af*]

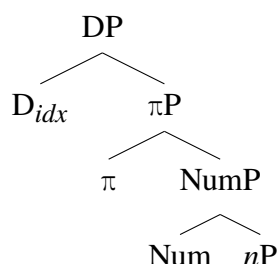
⁶Investigation of case morphology in pronouns motivated an approach to allomorphy that rejects adjacency and argues for a more relaxed locality condition stated in terms of an Accessibility Domain that encompasses the cyclic/phase head and one node above (Moskal 2015a, 2015b; Moskal & Smith 2016). However, the example from Tamil, where the case morpheme triggers allomorphy of the person morpheme across an overt number, that this proposal is based on, has been argued to be a result of the phonologically-conditioned allomorphy (Newell 2023), where adjacency assumed for suppletion is not required.

remains intact. As noted earlier in this dissertation (Chapter 2), in the absence of the formant [af], the remaining pronominal forms in the accusative and genitive correspond to clitics (for an extensive discussion on various aspects of the syntax of Greek clitics see Anagnostopoulou 1994, 1999; Philippaki-Warbuton & Spyropoulos 1998; Revithiadou & Spyropoulos 2008; Tsakali 2006; Alexiadou 2008; Alexiadou & Stavrou 2008; Alexiadou & Anagnostopoulou 2008a). In the present work, I focused on the morphological properties of pronominal forms. Clitics involve the exponent of the third-person in their realization, pointing towards the presence of the person (π) node in their internal structure. Importantly, the merge of the definiteness morpheme realized as [af] does not trigger the allomorphy of any of the existing morphemes and does not undergo allomorphy triggered by any of them.

	MSC		FEM		NEUT	
	SG	PL	SG	PL	SG	PL
NOM	<u>a</u> ftos	<u>a</u> fti	<u>a</u> fti	<u>a</u> ftes	<u>a</u> fto	<u>a</u> fta
ACC	<u>a</u> fton	<u>a</u> ftus	<u>a</u> ftin	<u>a</u> ftes	<u>a</u> fto	<u>a</u> fta
GEN	<u>a</u> ftu	<u>a</u> fton	<u>a</u> ftis	<u>a</u> fton	<u>a</u> ftu	<u>a</u> fton

Table 6.20: Third-person pronouns in Greek

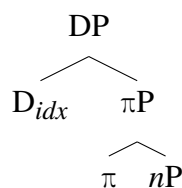
The approach to pronouns that proposes a complex, multi-morphemic, multi-terminal syntactic structure readily accounts for these data. I suggest that the exponent [af] is the realization of the definiteness (D) morpheme merged above person. In the structure of complex third-person pronouns in Greek, this morpheme is the highest functional node and is not dominated by any additional functional material. The demonstrative exponent [af] surfaces in the left-most position, linearly preceding both the person [t] and the remaining material encoding number, gender, and case. For complex pronouns in Greek, I propose the following syntactic structure (47), the VI rule for the D head (48), and the linearization statement in (49).

(47) *Complex pronouns in Greek*(48) **VIs for D in Greek**[D] $\leftrightarrow$ af(49) **LINEARIZATION STATEMENT**

a. (D * (π ...))

b. D – π

In the singular (50), the structure of complex pronouns lacks the NumP, and the D head combines with the person (π) and n morphemes.

(50) *The structure of pronouns*

These data contribute relevant implications for the discussion about the position of the person node in the hierarchical structure. If person $[\pi]$ was the lowest node in the structure of pronouns and definiteness (D), the highest one would have to explain how the two end up linearly adjacent.

I analyze the presence of the additional exponent $[af]$ as the realization of the strong definiteness (Schwarz 2009, 2013, 2019). Namely, natural languages distinguish between *weak* and *strong definiteness*⁷ associated with uniqueness and anaphoricity, respectively. While examination of Greek complex pronouns and clitics in the context of the theory of definiteness requires a close inspection of the syntactic and semantic environments in which the two are licit, it is important to note that clitics are already definite, evident from their availability to refer to salient referents in the linguistic or non-linguistic context. Definiteness in clitics also stems from individual φ -features in their structure being fully specified and interpretable (see Alexiadou & Anagnostopoulou 1998, 2021 for $[+D]$ on T in Greek and Roberts (2010) for the view that definiteness arises from interpretability of features). Importantly, merge of the higher definiteness morpheme, realized as $[af]$ comes with interpretive and syntactic effects. For these reasons, the presence of additional morphology in Greek complex pronouns is readily explainable under the syntactic account where the extra exponent realizes the definiteness head (see also Patel-Grosz & Grosz (2017) for the analysis of German *d*-pronouns and Bavarian clitics in terms of the *weak/strong* definiteness distinction).

An important remark on **definite articles** is called for. In some languages, pronouns involve definite articles in their realization. This was the case with Somali emphatic pronouns, as discussed in Chapter 5. Comparable examples in other languages motivated the view that some pronouns project a DP layer that hosts the definite article, exhibiting parallelism with the syntax of full DPs (see Déchaine & Wiltschko 2002). Two important points must be taken into consideration. In languages with definite articles (English, German, Greek), definite articles in DPs are obligatory (excluding bare plurals in Germanic). In addition, they are followed by a definite interpretation, but crucially, they are not followed by an emphatic or focused interpretation. In contrast, definite articles in pronouns are not obligatory, and their overt realization triggers a strong definite (emphatic, focus) interpretation. This suggests that the extra material in pronouns

⁷This distinction is empirically supported on the basis of a number of typologically different languages: German (Schwarz 2009) and several Germanic dialects (Schwarz 2019), Akan (Arkoh & Matthewson 2012), Hausa and Lakhota (Schwarz 2013), Thai (Jenks 2015), Korean (J. Cho 2016), Washo (Hanink 2018), Shan (Moroney 2021), Chuj (Royer 2022), Marka-Daring (Jenks & Konate 2022). The literature additionally debates whether weak and strong definiteness are realized in two different via distinct determiners (Schwarz 2009, Arkoh & Matthewson 2012, Jenks 2018, Jenks & Konate 2022) or strong definiteness involves compositional derivation out of weak definiteness (Hanink 2018, Ahn 2019, Royer 2022).

and definite determiners in full DPs should not be treated alike. However, some pronouns allow the overt co-occurrence of person features and definite articles, as is the case with third-person pronouns in Hebrew (Ritter 1995). This puzzle requires further research.

Greek data demonstrate that additional morphemes are readily accommodated under the syntactic approach to pronouns. I now turn to comparable examples from Serbian, a language without definite articles, extending the analysis to complex pronouns and clitics. As noted above, the empirical focus is on the first- and second-person pronouns. Specifically, I propose that complex first- and second-person pronouns in Serbian involve the infixation of a definiteness morpheme. An analysis in terms of infixation has not previously been proposed for complex pronouns, and for pronominal forms in Slavic languages in particular (see Stegovéc 2019; Ruda 2021b, 2021a; Puškar-Gallien 2022).

In Serbian, first- and second-person singular pronouns in non-nominative cases surface with formants *[en]* and *[eb]*, respectively, that are absent elsewhere in the pronominal system. On par with Greek, the presence of both complex forms and regular pronouns is limited to non-nominative cases. For completeness, I include the forms *[ja]* and *[ti]* in both tables, alongside zero $[\emptyset]$ in the set of regular pronouns, since Serbian is a *pro*-drop language. Motivation for an **infix-based analysis** comes from the fact that the person and case exponents constitute a unit on their own to the exclusion of the infix. Namely, when the infix is absent, the remaining exponents form (clitic) pronouns in Serbian. Moreover, clitics represent the default choice of a pronoun in Serbian, while the complex pronouns involving the infix are typically associated with the same set of environments as complex pronouns in Greek.

	1	2		1	2
NOM	ja	ti	NOM	ja/ $\emptyset$	ti/ $\emptyset$
ACC	m- en -e	t- eb -e	ACC	m<en>e	t<eb>e
DAT	m- en -i	t- eb -i	DAT	m<en>i	t<eb>i
GEN	m- en -e	t- eb -e	GEN	m<en>e	t<eb>e

(a) Complex pronouns

(b) Pronouns

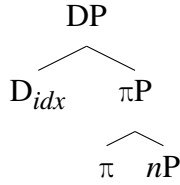
Table 6.21: Complex pronouns and pronouns in Serbian

On par with Greek, I analyze additional exponents present in complex pronouns as realizations of the higher definiteness head, corresponding to the strong index (Schwarz 2009, 2013, 2019), which linearize as infixes at PF. In the syntactic structure in (51), the target (D) of allomorphy is local to the trigger (π) being immediately adjacent to it, resulting in outwards-conditioned allomorphy. I propose the following set of VI rules for the D morpheme in Serbian (52). In the context of the first person, i.e., *AUTHOR*, the infix *[en]* is inserted, while in the context of the second person, *ADDRESSEE*, the infix *[eb]* is selected.⁸ During the linearization mechanism applying at PF, the *-operator requires the exponent of π to linearly precede the exponent of D

⁸The two allomorphs might presumably be further decomposed into *[e]* and *[n/b]*. I analyze the presence of *[e]* as phonologically-driven insertion triggered by the need to disrupt consonant combinations *[mn/tb]*.

(53-a), resulting in infixation (53-b).

(51) *The structure of pronouns*



(52) **VIs for D in Serbian**

[D] $\leftrightarrow$ en / $______ \pi_{\text{AUTHOR}}$

[D] $\leftrightarrow$ eb / $______ \pi_{\text{ADDRESSEE}}$

(53) **LINEARIZATION STATEMENT**

a. $(\pi * (\text{D} \dots))$

b. $\pi < \text{D} > n$

The exponent of the D morpheme exhibits two important properties. Infixation is inward, as the exponent of the D morpheme moves into a more embedded environment, linearizing between the exponent of person [*m*] and an exponent of the morphological case. As illustrated in Table 6.21, the relevant forms exist as clitics, and the infix disrupts the clitic. Furthermore, the infix that undergoes allomorphy in this case finds its trigger inwardly, as the π head is lower in the structure of pronouns. Both properties are consistent with the results of a large-scale cross-linguistic study on infixation and its interaction with allomorphy (Kalin 2022a). Furthermore, these data provide a relevant point for the general study of infixation and allomorphy. In contrast to lexically- and prosodically-conditioned suppletive allomorphy of infixes, allomorphy driven by morphological factors is reported to be rare across languages, attested in only 2 out of 51 languages studied in Kalin (2022a). Serbian data involve *morphologically conditioned allomorphy*, as the infix has two forms depending on the person specification it combines with.

The implication of the present analysis is that the proposed structure of pronouns remains intact. Namely, the first-person pronouns consist of the person morpheme (π) and the nominalizing morpheme (*n*) hosting gender and singular number. The structural relation of π and *n* remains the same, despite the presence of intervening [*en/eb*] exponents. The difference arises in the highest layer, which encodes strong definiteness. The infix does not disrupt the structural organization of the pronoun, but the linear order of the exponents. Any approach that would not analyze the additional exponent as an infix would have to resort to non-trivial adjustments of the structure and stretching of its internal composition. The present account achieves a uniform analysis of D morphemes in Greek and Serbian. The only difference lies in linearization. While in Greek the final landing site of the D morpheme is a prefix, in Serbian it is an infix that linearizes between person and case.

In previous chapters, I discussed several examples of invariant exponents in pronominal systems across languages that cannot be considered as exponents of person, number, or gender, as they consistently appear across the pronominal system. The analysis of these invariant formants as exponents of the D morpheme (or case) suggests itself as an explanation. More work on the realization of definiteness and the relation of person and definiteness in particular may provide important insights into this aspect of pronominal structure. The case studies focusing on Greek and Serbian data provide a starting point in this direction.

6.5 Comparison with other frameworks

In this section, I compare the account of pronouns developed in this dissertation and couched within Distributed Morphology, which argues for an (i) internal syntactic structure of pronouns and (ii) Vocabulary Insertion targeting terminal nodes of pronouns with comparable realizational frameworks that propose different models of realization. Pronouns represent an ideal testing ground for different theories of syntax, and realizational ones in particular, to effectively test their predictions. In Subsection 6.5.1, I introduce the theoretical frameworks in which the general architecture would be able to accommodate an elaborate internal structure of pronouns consisting of several terminals, but the realizational component significantly differs. Applying the Lexicalization algorithm of Nanosyntax (Starke 2018), I show that pronouns do not necessarily need to be realized in one cycle of spell-out. Importantly, Nanosyntax is a theory in which allomorphy does not exist. That is, the only available kind of allomorphy is phonologically-conditioned, while the grammatically- or lexically conditioned allomorphy, i.e., *suppletion*, is precluded by the architecture and several defining assumptions of the framework (Baunaz & Lander 2018). In the next step (Subsection 6.5.2), I will compare my account, developed within the Distributive Morphology framework, with the recent proposal modeled within Nanosyntax (Caha et al. 2024), demonstrating that the nanosyntactic account in terms of Fusion is not a satisfactory alternative to the context-sensitive rules of DM. Finally, in Subsection 6.5.3, I discuss *spanning* (Svenonius 2012) and *post-linearization spanning* (Haugen & Siddiqi 2016), demonstrating that both approaches preclude the existence of double marking discussed above.

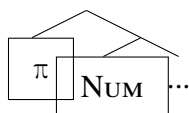
6.5.1 Allomorphy is not insertion into non-terminals

In the course of the discussion on the distinction between the Multi-morphemic and Monomorphemic approaches in Chapter 3, I focused on one realizational possibility for the Monomorphemic approach, namely, that both exponents correspond to a single terminal node in the syntax. There is an alternative possibility, however, which would employ the hierarchical structure that I am arguing for here, but it would maintain the single cycle of Vocabulary Insertion, either in form of the phrasal spell out of Nanosyntax (Starke 2009; Caha 2009) or the spanning view (Bye & Svenonius 2011; Merchant 2015; Svenonius 2016; cf. Ramchand 2008; Adger et al. 2009), where structurally adjacent terminals would be targeted for realization simultaneously.

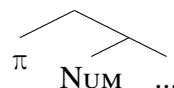
Under the approach that I am pursuing here, pronouns are not only multi-morphemic in the syntax but involve several exponents in their realization, as each morpheme is targeted by the Vocabulary Insertion separately. In that sense, **number** and **person** undergo **two separate cycles of Vocabulary Insertion**. In the representation in (54), I indicate the size of targets via boxes. The absence of a box in (55) indicates that both nodes can be simultaneously targeted under an approach where Vocabulary Insertion targets phrases or spans. Under the **phrasal spell out**, person would project a π P. The entire tree, consisting of person and number, stored in the post-syntactic lexicon, would be realized in one cycle of lexicalization. Under **spanning**, targets of Vocabulary Insertion are contiguous sequences of structurally adjacent heads which

establish head-complement relations. Much like trees, spans are stored as complex objects in the post-syntactic lexicon. As a consequence, spell-out targets nodes and the spans they dominate, adding phonological content to entire sequences of heads, rather than terminals of Distributed Morphology. In that sense, Vocabulary Insertion targeting π in (55) would also simultaneously realize Num as the head that π dominates. Therefore, the prediction of the spanning-based view is that all pronouns are stored as such in the lexicon, without further segmentation.

(54) TERMINAL NODE TARGETING



(55) PHRASE/SPAN TARGETING



I will dedicate more attention here to the predictions of Nanosyntax, as it does not necessarily need to intrinsically assume that all pronouns are stored as single objects in the post-syntactic lexicon. Recall data from Serbian, where the plural exponent $[i]$ surfaces on both pronouns (Table 6.22) and nouns (56-a). Gender-driven allomorphy of number observable in the third-person plural feminine pronouns $[e]$ carries over to the nominal domain (56-b).

	SG	PL
1	ja	m- i
2	ti	v- i
3 _M	on	on- i
3 _F	ona	on- e

(a) Serbian

- (56) a. *prozor-i*
window.M-PL
‘windows’
- b. *žen-e*
woman.F-PL
‘women’

Table 6.22: Plural exponents in Serbian

If $[i]$ in pronouns and nouns were treated as two distinct exponents, this would be the case of accidental homophony. Given that they [e.g., i_1 , i_2] would correspond to exactly the same feature in the narrow syntax, namely number, it appears highly unlikely that an analysis which would posit two syncretic exponents of plural, one combining with nouns and one combining with pronouns, would be on the right track. Instead, it is the single exponent that takes part in the realization of two distinct syntactic derivations. I argue that, under the nanosyntactic account, pronouns are not stored as single units in the post-syntactic lexicon.

I will provide here the Nanosyntactic model of realization, showing that, when it comes to pronominal domain and pronoun systems that employ overt exponents for individual features, Nanosyntax and Distributed Morphology do not part their ways. I assume here the lexicalization algorithm of Starke (2018), applied within the nanosyntactic framework for a number of unrelated phenomena (Caha 2023, 2024; Caha et al. 2024).

(57) LEXICALIZATION ALGORITHM (Starke 2018)

- a. Insert feature and spell out.
- b. If fail, try a cyclic (spec-to-spec) movement of the node inserted at the previous cycle and spell out.
- c. If fail, try a snowball movement of the complement of the newly inserted feature and spell out.
- d. If the spellout algorithm fails (a-c), go back to the previous cycle, and try the next option for that cycle.

Applied to the Serbian pronominal system provided above, the number feature is merged in syntax and sent for lexicalization. As the post-syntactic lexicon contains the exponent of plural [*i*], step (a) of the lexicalization algorithm is successful. In the next step, the person feature merges, undergoing the same spell-out procedure as was the case with the number. The lexicalization table for pronouns is provided in (58). I will not go into details of the way allomorphy should be modeled within the nanosyntactic framework (see 6.5.2) and the way the allomorph [*m*] emerges. For the purposes of the current discussion, it is relevant that the exponent of number can undergo lexicalization independently of the person.

(58) LEXICALIZATION TABLE: PRONOUN

	π	NUM
PRONOUN	m	i

The same procedure applies to nouns (59). The lexicalization of the noun [*prozori*] ‘windows’ proceeds as follows. The $\sqrt{\text{Root}}$ merged in the syntax is successfully lexicalized as [*proz*]. Spell-out of the $\sqrt{\text{Root}}$ is followed by two successive lexicalizations of the nominalizing head *n* and the plural head, which is mapped to exactly the same Vocabulary Item as was the case with pronouns. Therefore, Nanosyntax would have to realize the exponent of number in its own cycle of Vocabulary Insertion, separately from the rest of the material in the pronominal and nominal domains.

(59) LEXICALIZATION TABLE: NOUN

	$\sqrt{\text{Root}}$	n	NUM
NOUN	proz	or	i

6.5.2 Nanosyntax

Earlier in this chapter, I discussed instances of double marking in pronouns (6.3.3), where the trigger of allomorphy overtly co-occurs with its target. This empirical fact is readily explainable within Distributed Morphology and, in fact, expected. What does non-assuming double marking predict? That allomorphy does not even exist in natural language. I will take issue here with one such proposal, couched within the framework of Nanosyntax.

The fact that the overt trigger of allomorphy freely co-occurs with the target suggests that the phenomenon of allomorphy is an entirely distinct mechanism from Fusion. As noted earlier in this dissertation, the operation of Fusion stands for a post-syntactic mechanism that turns two different morphemes present in the narrow syntax (M_1, M_2) into a single morpheme (M) in the post-syntactic component of the grammar, as represented in (60). The Vocabulary Insertion operation targets the morpheme (M), which is the result of Fusion. Therefore, under Fusion, the two morphemes cannot be mapped to two distinct exponents, as the essence of the operation is a single exponent, which relates to two morphemes in the narrow syntax at the same time.

$$(60) \text{ FUSION} \\ M_1, M_2 \rightarrow M$$

In a recent Nonosyntactic account of allomorphy in Czech adjectives, Caha et al. (2024) propose a fusion-based approach to allomorphy as a theoretical alternative to the approach based on contextual rules employed in Distributed Morphology. In Czech, adjectivizing head *a* has two allomorphs: $[\emptyset]$ and $[n]$. Their distribution is entirely arbitrary and crucially dependent on the set of roots which combine with the $[\emptyset]$ allomorph, $[n]$ being the elsewhere form. Abstracting away from which roots exactly surface with the one or the other, I will represent this system schematically in the following way:

$$(61) \text{ a. } [a] \leftrightarrow \emptyset / __ \sqrt{\text{Root}_1}, \sqrt{\text{Root}_2}, \sqrt{\text{Root}_3} \\ \text{b. } [a] \leftrightarrow n$$

In a model without context-sensitive rules, the root is the sole controller of allomorphy. Roots combining with $[n]$ are not capable of lexicalizing the adjectivizing head *a*, thus both the root and *a* are realized by two distinct exponents. This is illustrated in Table 6.23, where the root *snad* ‘easy’ fails to lexicalize the adjectivizing head. Hence, the root and the adjectivizer are mapped to two exponents *snad* and *n*, respectively. In contrast, in adjectives with the zero adjectivizing head, roots are capable of lexicalizing both simultaneously: the $\sqrt{\text{Root}}$ and *a* (Caha et al. 2024). This is exemplified with the adjective *mlad* ‘young’. In this case, the form *mlad* is capable of lexicalizing both the root $\sqrt{\text{YOUNG}}$ and the zero adjectivizing head, resulting in Portmanteau realization (62-a). Under this account, there is only one step of lexicalization, and the adjectivizer *a* does not independently undergo spell-out. As a consequence, this analysis effectively rules out zero.

	$\sqrt{\text{Root}}$	a	(62) Portmanteau realization
EASY	<i>snad</i>	<i>n</i>	a. $\sqrt{\text{YOUNG}} + a \leftrightarrow \textit{mlad}$
YOUNG	<i>mlad</i>	$\emptyset$	b. $[a] \leftrightarrow n$

Table 6.23: Portmanteau realization (Caha et al. 2024)

Now, the following empirical problem emerges: **If all allomorphy is fusion, how can the overt trigger and the overt target ever co-occur?** Going back to the data discussed above, repeated here in Table 6.24, if the allomorph of person in the plural form is to be considered a result of fusion, which morphemes exactly are fused? The fusion approach leaves it unexplained why the number morpheme would receive an overt realization, if it has already undergone fusion and the Vocabulary Insertion together with its target, i.e., person.

	SG	PL	(63) Some VIs for Mangghuer
1	bi	da-si	$[1] \leftrightarrow \text{da} / __\#_{\text{PL}}$
2	qi	ta-si	$[1] \leftrightarrow \text{bi}$
3	gan	gan-si	$[\text{PL}] \leftrightarrow \text{si}$

(a) Mangghuer (Slater 2003)

Table 6.24: Number-driven person allomorphy: Mangghuer

Therefore, the cases of overt realization of triggers and targets of allomorphy, not unique to pronouns but widely spread in other domains of natural language, strongly favor cyclic insertion into both the trigger and the target, being realized as two distinct exponents. These data are informative for the study of fusion and portmanteau morphology, as they show that, if possible, the fusion operation must be distinguished from allomorphy.

Furthermore, let us imagine that Czech has 3 allomorphs of adjectivizing head a : in addition to $[\emptyset]$, $[n]$, there is also the third allomorph $[d]$. The question for the fusion-based theory of allomorphy is how it would distinguish between $[n]$ and $[d]$. In this case, the analysis would have to resort to lists as an integral part of context-sensitive rules of Distributed Morphology (64). Under the nanosyntactic approach outlined above, fusion would account for (64-c), but in both (64-a/b), the root would not be capable of lexicalizing the adjectivizing head $[a]$, as evident from two allomorphs, $[n]$ and $[d]$. Therefore, in these cases, the fusion cannot apply. As a consequence, the two allomorphs would have to be specified for individual sets of roots they combine with, either in the way given in (64) or a comparable approach which would transparently combine each root with its specific adjectivizer.

- (64) a. $[a] \leftrightarrow n / __\sqrt{\text{Root}_1}, \sqrt{\text{Root}_2}, \sqrt{\text{Root}_3}$
 b. $[a] \leftrightarrow d / __\sqrt{\text{Root}_4}, \sqrt{\text{Root}_5}, \sqrt{\text{Root}_6}$
 c. $[a] \leftrightarrow \emptyset$

In fact, natural languages often employ **several allomorphs** of the **same morpheme**. This situation was illustrated earlier in this dissertation in the realization of plurality in English, which employs at least five different allomorphs. An alternation between only two allomorphs, as is the case of the adjectivizing head [*a*] in the Czech data above, one of which is zero, is a possible allomorphic alternation. However, it is only one of the possible alternations. Languages employ several non-zero allomorphs of the same terminal simultaneously. For these reasons, I argue that the fusion-based approach cannot be considered a satisfactory overall theory of allomorphy, which rules out the context-based Distributed Morphology account. It is motivated on the basis of allomorphic alternation, which involves only two allomorphs, one of which is zero, and cannot account for the data involving two or more overt allomorphs. Crucially, it cannot account for the overt co-occurrence of the trigger and target, as fusion precludes two exponents. The task of the theory of allomorphy is to successfully account for both the arbitrariness in cases where there are several allomorphs of the same terminal and for subtle interactions which involve alternations between one allomorph that is selected under particular conditions and the elsewhere form.

6.5.3 Spanning

I will now examine the predictions of a model assuming that realization targets contiguous sequences of structurally adjacent heads, which establish head-complement relations and its specific predictions for pronouns. Under this view, pronouns would undergo a single cycle of Vocabulary Insertion, realizing all their morphemes at once. First, this view would assume that all pronouns are stored in the form in which they are realized in the lexicon. That is, there is no morphological segmentation of pronouns. This is in part the position of Cardinaletti & Starke (1999), who propose that weak pronouns and clitics are stored as such in the lexicon, while strong pronouns are derived in the syntax by merging an additional layer. Under the approach argued for here, no pronouns are stored in their entirety in the post-syntactic Vocabulary. Rather, what is stored are individual Vocabulary Items that relate to morphemes in the narrow syntax.

An approach which would assume a single cycle of Vocabulary Insertion for pronouns predicts that one and the same exponent of plural is realized in two different ways. This would be the stance of spanning (Svenonius 2012, 2016), where the person node would be targeted for insertion, simultaneously realizing the number node that it dominates. For instance, in Serbian data discussed above, the plural exponent [*i*] would be realized separately with nouns, unless one would subscribe to the view that the entire nominal, together with plural morphology, is stored as such in the lexicon. In contrast, if a single spell-out is pursued for pronouns, this means that the same exponent would be realized together with the person node in pronouns. This raises an apparent problem that the exponent [*i*] would be stored once in isolation (or even together with the noun) and once together with the pronoun, thus twice.

Furthermore, recall the data discussed in Chapter 3. In Kuna (Chibchan), plural pronouns consist of an exponent of plural [*mar*], and an exponent of person present in the singular. The person node would be targeted for insertion, together with the number node that it dominates. Therefore, the first-person plural pronoun is an unsegmentable form [*anmar*] realizing a span of

contiguous nodes (π , Num, n).

	SG	PL
1	an	an-mar
2	be	be-mar
3	we	we-mar

(65) THE STRUCTURE

π NUM n

(a) Kuna (Pertsova 2022b)

Table 6.25: Pronouns in Kuna and Spanning

This view implies that person exponents would have to be **stored in the memory** and **acquired** by a child **once in isolation** [*an*], as the same exponent surfaces in the singular, and **once together with the plural exponent** [*anmar*], thus twice. Given that the decomposition is available in a straightforward manner, such an approach would have weak support, both on empirical and theoretical grounds, putting a significant burden on the cognitive apparatus in requiring memorizing the same formant multiple times with each of its occurrences.

Haugen & Siddiqi (2016) develop a version of spanning called *post-linearization spanning* within their Restricted Realization Theory, proposed to be a preferable option over the insertion into terminal nodes standardly assumed in Distributed Morphology. The criticism is focused on cases handled by morpho-phonology and *Readjustment rules*, invoked in English past tense alternations such as *think~thought* and *bring~brought*, where the insertion of the zero [$\emptyset$] past tense exponent is followed by the phonological change of the stem. Crucially, the data examined here go **beyond the Readjustment rules**. None of the cases of co-occurrence of the trigger and target of allomorphy in pronouns involved Readjustment rules and thus an explanation in terms of morpho-phonology. Instead, the pronominal forms discussed here are true cases of **grammatically-conditioned** allomorphy, where the **overt trigger** co-occurs with the **overt target**. Recall the cases of double marking discussed in Section 6.3.3. In both the number-driven person allomorphy, attested in Mangghuer, and the person-driven number allomorphy, present in Ecuadorian Kichwa, the overt trigger and the overt target co-occur, without zero morphology and morpho-phonological effects that would call for readjustment (for additional data see Chapter 3 and the discussion throughout this dissertation, where the co-occurrence of triggers and targets is consistently noted).

	SG	PL
1	bi	da-si
2	qi	ta-si
3	gan	gan-si

(a) Mangghuer (Slater 2003)

	SG	PL
1	ɲuka	ɲuka-ɲjik
2	kan	kan-kuna
3	paj	paj-kuna

(b) Ecuadorian Kichwa (Pertsova 2022b)

Table 6.26: Double marking in both directions

This double marking involves allomorphy arising between two **functional morphemes**, the

hallmark of contextual allomorphy. Therefore, the criticism, taking into consideration morpho-phonological Readjustment rules, fails to predict and, in fact, does not discuss the double marking involving functional morphemes. The same remarks regarding storage in the memory and acquisition noted above for the standard version of spanning extend to post-linearization spanning.

This discussion shows why allomorphy resides at the heart of the morphological theory and the morphology-syntax interface. The competing theories differ precisely in their predictions about the inventory of objects that natural languages have in their post-syntactic components, i.e., in the Vocabulary. I showed that contextual allomorphy modeled in terms of context-sensitive rules in Distributed Morphology is an empirically preferable option and cannot be replaced with fusion proposed in Nanosyntax, as the range of allomorphic alternations would remain unexplained. Furthermore, several versions of spanning, which would lead to storing all pronouns as non-segmentable formants in a Vocabulary/post-syntactic lexicon of a language, would overlook a number of empirical generalizations such as the shared morpho-phonological material in singular and plural forms, for example, putting an additional burden on a child acquiring language and the memory, as all forms would have to be stored in each of their occurrences.

6.6 A Puzzle for Future Work: First person exceptionality

Throughout this dissertation, I used Greek pronouns to illustrate several central points. One aspect of the Greek pronominal system has not been addressed yet. Recall the Greek pattern, where $[m]$, $[s]$, and $[t]$ figure as exponents of the first, second, and third person, respectively. An obvious exception emerges in the first-person nominative pronoun, lacking the person exponent $[m]$. The absence of the person exponent only in the nominative singular has been argued to figure as a Non-Elsewhere Nominative Stem (NENS), with $[m]$ being the elsewhere exponent of the first-person in Greek (Christopoulos & Zompì 2023). Theoretical accounts of this phenomenon focus the analysis on the role of the nominative case (McFadden 2018; Christopoulos & Zompì 2023; Caha 2023). This pattern creates a puzzle for the containment-based approach to case (Caha 2009), where the accusative contains the nominative, as specifying an allomorph of the first person in the nominative would predict it in the accusative too.

	1		2		3					
	SG	PL	SG	PL	M.SG	M.PL	F.SG	F.PL	N.SG	N.PL
NOM	ego	<u>e</u> mi <u>s</u>	e <u>s</u> i	e <u>s</u> is	aftos	afti	afti	aftes	afto	afta
GEN	<u>e</u> mena	<u>e</u> ma <u>s</u>	<u>s</u> ena	e <u>s</u> as	aftu	afton	aftis	afton	aftu	afton
ACC	<u>e</u> mena	<u>e</u> ma <u>s</u>	e <u>s</u> ena	e <u>s</u> as	afton	aftus	aftin	aftes	afto	afta

Table 6.27: Personal pronouns in Greek

Staying in the realm of Greek, one may wonder why the non-elsewhere nominative stem is attested **only in the first person**, but not in the second and third. As demonstrated in the table

below, person morphemes are realized in the nominative singular in both second- and third-person pronouns. This pattern remains unexplained under an approach that makes reference only to case. I suggest that the successful explanation of the absence of person morphology in the first-person nominative has to take into account both person and case considerations. If the solution to this puzzle lies only in the nominative, it remains unexplained why Greek does not employ hypothetical pronouns *exi* and *afkos* (66) in second and third person nominative singular, respectively.

(66) HYPOTHETICAL GREEK PRONOUNS

- a. *exi* [2SG.NOM]
- b. *afkos* [3SG.NOM]

Exceptional behavior of first-person pronouns in the nominative case appears to be the reflection of a more general pattern, attested across individual branches of the Indo-European language family, such as Germanic, Slavic, and Baltic languages. In Chapter 3, I noted that the first-person pronouns show a higher rate of allomorphy compared to the second- and third-person pronouns. This result is precisely the effect of a number of Indo-European languages exhibiting this pattern.

As illustrated in the table below, the first-person pronoun in German surfaces with the exponent *[m]* only in non-nominative cases. The second-person pronoun, however, realizes the exponent *[d]* in both nominative and non-nominative cases. Icelandic exhibits a similar pattern. The exponent *[m]* is present in the first person only in non-nominative cases. Note that the absence of the person morpheme in the plural in German, i.e., the clear number-driven person allomorphy, remains orthogonal to the current discussion. The relevant point is the singular, where the exponent of the first person is consistently present across all non-nominative cases. In sharp contrast, the second-person pronoun is present both in the nominative and non-nominative cases.

	1		2	
	SG	PL	SG	PL
NOM	ich	wir	du	ihr
ACC	mich	uns	dich	euch
DAT	mir	uns	dir	euch

(a) German

	1		2	
	SG	PL	SG	PL
NOM	ég	við	þú	þið
ACC	mig	okkur	þig	ykkur
DAT	mér	okkur	þér	ykkur

(b) Icelandic (Neijmann 2021)

Table 6.28: First person exceptionality in Germanic

The same holds for the Slavic languages. As illustrated below, on the basis of Serbian, first-person pronouns surface with the exponent *[m]*, which is absent in the nominative. Second-person exhibits a different behavior, realizing the second-person exponent *[t]* in the nominative singular.

	1		2	
	SG	PL	SG	PL
NOM	ja	mi	ti	vi
ACC	mene	nas	tebe	vas
DAT	meni	nama	tebi	vama

Table 6.29: First person exceptionality in Slavic

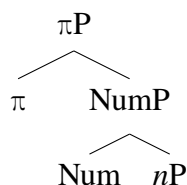
Integrating the discussion on Greek with a broader cross-linguistic Indo-European data indicates that the first person indeed behaves exceptionally across languages. This suggests that the explanation for the absence of person exponents in first-person nominative cases calls for an analysis that makes use of both the case and person features. This puzzle awaits further research.

6.7 Overall Conclusion

In this dissertation, I investigated the internal structure of pronouns. The focus of the inquiry was on the properties of pronominal internal syntax, that is, the inventory of morphemes that make up pronouns within and across languages. I approached the data from the perspective of allomorphy, proposing that it serves as a key tool for investigating the internal structural composition of pronouns.

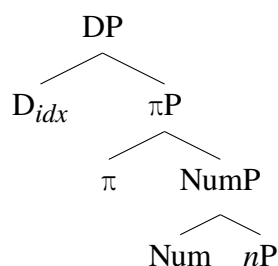
On the basis of a series of studies, I proposed a **multi-morphemic** structure of pronouns. Personal pronouns consist of three morphemes, i.e., syntactic terminals in their structural representation: the *n* morpheme, the **number morpheme** (Num), and the **person morpheme** (π). Evidence for the proposed structure comes from the observed close syntactic interactions of the three morphemes, directly reflected in their morphological realization. The provided syntactic representation (67) holds for all pronouns ‘proper’ across languages, that is, pronominal forms that do not appear in an adnominal position or comparable syntactic environments that allow ancillary nominal material, but as external and internal arguments of a verb in a clause. Pronouns crucially lack $\sqrt{\text{Roots}}$. I propose that there is no morpheme that can merge with the *n* head in pronouns, i.e., *n* in pronouns is universally complement/modifier-free. This structural part of a pronoun is exempt from variation and holds universally.

(67) *The structure of pronouns*



The **variation** in the proposed structure is **located within a morpheme** as a basic unit of syntactic computations, both within and across languages. This syntactic variation is manifested at **two levels**: (i) a morpheme can be completely absent from the structure, or (ii) its featural content may differ.

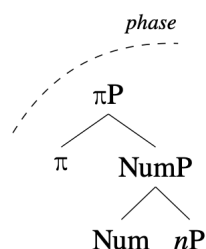
The first kind of variation is attested within a language. For example, in the singular, pronouns will lack the Num morpheme, consisting of the π morpheme and the *n* morpheme hosting the singular number. This kind of variation is also attested in the realization of definiteness in Greek and Serbian. Pronouns may surface with an additional higher layer encoding definiteness, which is followed by semantic effects. Therefore, the maximal structure that a pronoun may have is provided in (68), exemplified on the basis of Greek complex pronouns.

(68) *Complex pronouns in Greek*

The second kind of variation accounts for phenomena such as clusivity and gender marking, discussed in Chapters 4 and 5, respectively. To account for these data, I proposed an analysis in terms of variation encoded within a morpheme. Languages with clusivity surface with an additional morpheme $\mathbb{M}_{incl}$, which hosts the first [AUTHOR] and the second [ADDRESSEE] person. In that sense, the proposed syntactic structure (67) is not modified. Instead, both features are located within the same π morpheme. The same principle extends to the gender features, which are available on a language-specific basis. If present in a language, gender is located on the n head that is independently available in the structure and hosts the singular number and the lexical plural. In the same manner as in the account of clusivity, the proposed syntactic structure is not modified. The difference lies in the featural content of the n head across languages. Finally, variation applies at PF in the patterns of allomorphy that individual languages exhibit.

The question that arises for future work is whether additional layers need to be invoked in languages that surface with additional features, such as *humanness*, for example, or any other feature that a particular language may draw on a language-specific basis. Is there a need for any additional morphemes beyond the proposed maximal structure that consists of n , Num, π , and D, or will these potential features be located on one of the existing heads in the way clusivity and gender are analyzed here? Studies focused on individual languages may contribute important insights into these questions.

The results of the pronominal domain shed light on several aspects of the theory of allomorphy. Two central contributions were discussed: (i) availability of **inward-conditioning** and **outward-conditioning** and (ii) *double marking*, understood as an overt co-occurrence of a trigger and a target of allomorphy. I proposed that pronouns (69) involve a **single cycle of spell-out**. All morphemes that make up a pronoun are available for interaction at PF. As pronouns do not involve a $\sqrt{\text{Root}}$ in their structure, the n head as the most deeply embedded morpheme does not block allomorphy and may act as both the trigger and the target. Furthermore, I proposed that the finding on the **availability of both directions of allomorphy** is the result of the **syntactic structure** of pronouns and a specific view on the **Vocabulary Insertion**. Therefore, **the two directions** of allomorphy arise **within the same phase** in the interaction of morphemes that are sent to interfaces in one cycle of spell-out. As noted earlier in this chapter, the directionality finding opened a new path for future research focused on questions about the syntax of languages exhibiting one or the other direction in allomorphic conditioning.

(69) **Available at PF**

The aspect of the proposal that is relevant both for the structure and for allomorphy is that the inventory of morphemes that make pronouns across languages is drawn from the set of functional morphemes. The structure of pronouns is similar to that of nouns, the difference being in the $\sqrt{\text{Root}}$ domain.

I investigated the syntax and morphological realization of pronouns ‘proper’, that is, pronouns appearing in argument positions without ancillary nominal material. Natural languages have several other types of pronouns, such as adnominal pronouns in *we linguists* contruals, possessives, logophors, and demonstratives, to name a few. Which morphemes do these pronouns involve in their structure, and do they surface with an elaborate internal structure? Are there parallels with the syntax of pronouns studied here? In addition, in this chapter, I briefly touched upon the morphological case, providing one possible direction for the analysis. The assignment of case in pronouns and its tight correlation with the clausal syntax and argument structure opens several questions for investigation.

Furthermore, experimental work provides evidence that words are not stored in their entirety in memory, but are decomposed into $\sqrt{\text{Roots}}$ and functional material. If functional material is decomposed in the nominal and verbal domains, it is expected that individual components of pronouns, such as person and definiteness, are also independently stored. This important question and the investigation of processing in pronouns await further research.

In this dissertation, I presented a detailed study of the internal structure of pronouns and the patterns of allomorphy in the pronominal domain. The results contribute to the research on nominal and pronominal syntax and contextual allomorphy and open a range of questions and directions for future work.

Appendix A

This appendix includes languages investigated in the study on number-driven person allomorphy and person-driven number allomorphy. These languages do not have the clusivity distinction. The appendix is organized according to the source of the data: (i) Mouton Grammar Library [MGL], (ii) Mouton-CASL, (iii) Mouton Pacific Linguistics, (iv) Mouton Sinitic Languages of China, (v) Individual reference grammars, grammatical descriptions, and studies and the (vi) Database (Pertsova 2022b). Within each group, languages appear in the alphabetical order.

A.1 Mouton Grammar Library

LANGUAGE	π	SG	PL
Afrikaans Donaldson (1993)	1	ek	ons
	2	ju	julle
	3 _M	hy	hulle
	3 _F	sy	hulle
Aguaruna Overall (2017)	1	wi-/wii	ii.SPEC/hutii.NONSPEC
	2	amĩ	atum(i)
	3	nĩ-/nĩĩ	dita
Bangime Heath & Hantgan (2018)	1	mĩ	(n)nē(ē)
	2	á, áw	āā, àà-rú
	3	mĩ(i)	nĩ(i)
Basque Hualde & De Urbina (2011)	1	ni	gu
	2	zu	zuek
	2 _{FAM}	hi	
	3 _{PROX}	(h)au(r)	haue-
	3 _{MESIAL}	(h)ori	horie-
	3 _{DIST}	(h)ura	haie-

LANGUAGE	π	SG	PL
Berbice Dutch Creole Kouwenberg (1994)	1	$\varepsilon k\varepsilon$	$enfi/ i\dot{f}i, i\dot{c}i, ixi$
	2	ju	$j\acute{e}nd\varepsilon$
	3	ori	eni/ eni, ini
Breton Press (2011)	1	me	ni
	2	te	c'hwi
	3 _M	eñ	int
	3 _F	hi	int
Cavineña Guillaume (2008)	1	i-ke	e-kwana
	2	mi-ke	mi-kwana
	3	tu-ke	tu-na
Domari Matras (2012)	1	ama	eme
	2	atu	itme
	3	pandži	pandžan
Emai Schaefer & Egbokhare (2016)	1	i	ma
	2	u	vba
	3	o	yan
Eton Van de Velde (2008)	1	mà	bî
	2	wò	mîn
Fa d'Ambô Hagemeyer et al. (2020)	1	mu/mun/m/n/ma/mi	non
	2	bo	namin- sêdji/nam'sêdji/nam'sê
	3	ê, i	ineyn, êneyn, ineñi
Fongbe Lefebvre & Brousseau (2002)	1	nyè	mí
	2	hwè	mí
	3	é(yè)	yé
Goemai Hellwig (2011)	1	hěn	měn
	2 _M	gõe	gwěn
	2 _F	yõe~yĩ	gwěn
	3	ní	muěp
Hup Epps (2005)	1	ʔãh	ʔín
	2	ʔám	nín
	3	tíh	híd
Hinuk Forker (2013)	1	de	eli
	2	me	meži
	3	hado	had/had-be
Jamul Tiipay Miller (2001)	1	nyaach	nya'wach
	2	maach	menya'wach
	3 _{PROX}	peya	peya
	3 _{MED}	nyip	nyip
	3 _{DIST}	puu	puu

LANGUAGE	π	SG	PL
Jamsay Heath (2008)	1 2 3	mí ú wó	émé é bé
Khatso Donlay (2019)	1 2 3	ŋa ³³ nɛi ³³ i ³³	ŋa ³³ ts ^h γ ³³ nɛi ³³ ts ^h γ ³³ i ³³ ts ^h γ ³³
Kisi Childs (1995)	1 2 3	í à ò	ṛ là à
Kolyma Yukaghir Maslova (2003)	1 2 3	met tet tude(l)	mit tit titte(l)
Koyra Chiini Heath (2011)	1 2 3	ay ni a/ŋga~ŋa	yer wor~war i/ŋgi-yo
Kulina Dienst (2014)	1 2 3 _M 3 _F	owa tia powa po-ni	ia tia-deni powa-deni po-ni-deni
Lango Noonan (1992)	1 2 3	án yín én	wán wún gín
Lezgian Haspelmath (1993)	1 2 3	zun wun am	čun kün a-bur
Madurese Davies (2010)	1 2 3	sengko' ba'na aba'na	ba'-aba' ba'-aba' ba'-aba'
Ma'di Blackings & Fabb (2003)	1 2 3	má ɲí ká/kó	àmá àɲí ká/kí'
Mapuche Smeets (2008)	1 2 3	iñché eymi fey	iñchiñ eymün feye-ng-ün
Coastal Marind Olsson (2021)	1 2	nok oγ	nok yoγ

LANGUAGE	π	SG	PL
Meithei	1	əy	əykhoy
Chelliah (2011)	2	nəŋ	nəkhoy
	3	má	məkhoy
Modern Baba Malay	1	saya/gua	kita
Lee (2022)	2	lu	lu-orang/lu
	3	dia	dia-orang/dia
Mosetén	1	yäe	tsin
Sakel (2011)	2	mi	mi'in
	3 _M	mi'	mi'in
	3 _F	mö'	mö'in
Tundra Nenets	1	mən'°	mən'°aq
Nikolaeva (2014)	2	pidər°	pid°raq
	3	pida	pidoh
Nusu	1	nga	ngop wox
Gerner (2013)	2	ne	nop wox
	3	cy	cop wox
Pipil	1	naha	tehemet/amehemet
Campbell (1985)	2	taha	anehemet
	3	yaha	yehemet
Saramaccan Creole	1	mi, m	u
McWhorter & Good	2	i	ũ, únu
(2012)	3	a	de
Sierra Popoluca	1	ʔüch	ʔüch-tam
Boudreault (2018)	2	mich	mich-tam
	3	jeʔ	jeʔ-yaj
Qaqet	1	ngua	ut
Hellwig (2019)	2	nyi	ngen
	3 _M	ka	ta.HUM
			nget.NON-H
	3 _F	ki	ta/nget
Qiang	1	qa	tɕi-le
LaPolla & Huang (2003)	2	ʔũ	ʔi-le
	3	the:/qupu	them-le
Slave	1	sɪ̥	naxɪ̥
Rice (1989)	2	nɪ̥	naxɪ̥
	3	ʔedi̥	ʔegedi̥

LANGUAGE	π	SG	PL
Supyre Carlson (1994)	1	m̐i	w̐u
	2	mu	y̐i
	3 ^{GEN} 1	u	p̐i
	3 ^{GEN} 2	ku	y̐i
	3 ^{GEN} 3	li	ci/ki
Tamashek Heath (2005)	1	næk	nækk-æn-ed̐.M nækk-æn-æt-ed̐.F
	2 ^M	kàyy	kæw-æn-ed̐
	2 ^F	kæmm	kæm-æt-ed̐
	3	ənta	ənt-æn-ed̐.M ənt-æn-æt-ed̐.F
Tauya MacDonald (1990)	1	ya	sen
	2	na	ten
	3	ne	nen
Tommo So McPherson (2013)	1	mí	émmé
	2	ú	é
	3	wó	bé
Warrongo Tsunoda (2011)	1	ngaya	ngana
	2	yinda	yorra
	3	nyola	jana
Xong Sposato (2021)	1	wel	boub
	2	monx	manx
	3	beul	beul
Central Alaskan Yupik Miyaoka (2012)	1	wii ~ wiiŋa	waŋkuta
	2	ɪ̥p̥it	ɪ̥p̥ici
	3.RFL	ɪ̥mi	ɪ̥miŋ
	3.ABS	ɪ̥ii	ɪ̥ait
	3.REL	ɪ̥iin	ɪ̥aita

A.2 Mouton-CASL

LANGUAGE	π	SG	PL
Bangla David (2015)	1	am-i	am-ra
	2	tum-i	tom-ra
	3 _{PROX}	in-i	ẽ-ra
	3 _{DIST}	un-i	õ-ra
	3 _{NEUT}	tin-i	tã-ra
Dhivehi Gnanadesikan (2016)	1	ahareṅ/ma	aharumeṅ/mameṅ
	2	kalē	kalēmeṅ
	3	ēnā~ēn	emīhuṅ
Hindko Bashir & Connors (2019)	1	māẽ	assī
	2	tū	tussī
	3 _{PROX}	é	ẽ
Pashto David (2013)	1	zə	munḡ/munḡ
	2	tə	tāse/tāsi
	3 _M	day	duy
	3 _F	dā	duy

A.3 Mouton Pacific Linguistics

LANGUAGE	π	SG	PL
Kopar Foley (2022)	1	ma	e
	2	mi	o
	3	mu	mbu
Nama Siegel (2023)	1	yèndo(n)	yèndfem
	2	fèmo(n)	fèmofem/fèmafem
	3	yèmo(n)	yèmofem

A.4 Mouton Sinitic Languages of China

LANGUAGE	π	SG	PL
Caijia	1	ηo^{33}	$\eta o^{24}/\eta o^{24}\text{-xu}^{55}$
Lü (2022)	2	$n\eta u^{33}$	$n\eta u^{24}/n\eta u^{24}\text{-xu}^{55}$
	3	je^{33}	$je^{24}/je^{21}\text{-xu}^{55}$
Yichun	1	ηo^{34}	$\eta o^{34}li$
Li (2018)	2	ηi^{34}	$\eta i^{34}li$
	3	$ki\epsilon^{34}$	$ki\epsilon^{34}li$

A.5 Individual Grammars

LANGUAGE	π	SG	PL
Albanian	1	unë	ne
Newmark et al. (1982)	2	ti	ju
	3 _M	ai	ata
	3 _F	ajo	ato
Amharic	1	əne	əñña
Leslau (1995)	2 _M	antä	ənn-antä
	2 _F	anči	ənn-antä
	3 _M	əssu	ənn-ässu
	3 _F	əss ^w a	ənn-ässu
Armenian	1	es	mek'
Ajello (2015)	2	du	duk'
	3	na	nran-k'
Asmat	1	no	na
Corbett (2000)	2	o	ca
	3	a	a
Azeri	1	män	biz
Ragagnin (2021)	2	sän	siz
	3	o	onlar
Bashkir	1	min	bið
Berta (2021)	2	hin	hið
	3	ul	u-lar
Brahui	1	i	nan
Elfenbein (2020)	2	nī	num
	3	ō(d)	ofk

LANGUAGE	π	SG	PL
Cantonese	1	ngo ²³	ngo ²³ dei ²²
Pui Yee Yuen, p.c.	2	nei ²³	nei ²³ dei ²²
	3	keoi ²³	keoi ²³ dei ²²
Chichewa	1	ine	ife
Bentley & Kulemeka (2001)	2	iwe	inu
	3	iye	iwo
Chuvash	1	äbi	äbi-r
Agyagási (2021)	2	äzi	äzi-r
	3	vil	vi-zäm
Dutch	1	ik	wij/we
Donaldson (2008)	2	jij/je	jullie/je
	3 _M	hij	zij/ze
	3 _F	zij/ze	zij/ze
	3 _N	het	zij/ze
English	1	I	we
	2	you	you
	3 _M	he	they
	3 _F	she	they
Estonian	1	mina/ma	meie/me
Pool (1999)	2	sina/sa	teie/te
	3	tema/ta	nemad/nad
Faroese	1	eg	vit
Lockwood (1977)	2	tú	tit
	3 _M	hann	teir
	3 _F	hon	tær
Finnish	1	minä	me
White (2006)	2	sinä	te
	3	hän	he
French	1	je	nous
Kayne (2003)	2	tu	vous
	3 _M	il	ils
	3 _F	elle	elles
Gadaba	1	ān	ām
Bhaskararao (2020)	2	īn	īm
	3 _{M.PROX}	iyon	iyor
	3 _{M.DIST}	ōn	ōr
	3 _{NON-M.PROX}	id(d)u	iv(v)u
	3 _{NON-M.DIST}	ad(d)u	av(v)u

LANGUAGE	π	SG	PL
Scottish Gaelic Gillies (2009)	1	mi	sinn
	2	t(h)u	sibh
	3 _M	e	iad
	3 _F	i	iad
Gondi Steever (2020a)	1	nanna	mammat
	2	nima	nimat
	3 _{M.PROX}	ver	ver
	3 _{M.DIST}	or	or
	3 _{M.INTER}	bor	bor
German	1	ich	wir
	2	du	ihr
	3 _M	er	sie
	3 _F	sie	sie
	3 _N	es	sie
Georgian (Hewitt 2005)	1	m-e	č-ven
	2	š-en	tk-ven
	3	is/igi	is-ini/igi-ni
Greek Vasiliki Rizou, p.c.	1	ego	emis
	2	esi	esis
	3 _M	aftos	afti
	3 _F	afti	aftes
	3 _N	afto	afta
Hausa Wolff (1993)	1	nii	muu
	2 _M	kai	kuu
	2 _F	kee	kuu
	3 _M	shii	suu
	3 _F	ʔita	suu
Hebrew Ritter (1995)	1	ani	anaxnu
	2 _M	ata	atem
	2 _F	at	aten
	3 _M	hu	hem
	3 _F	hi	hen
Hindi McGregor (1995)	1	maimi	ham
	2	tū	tum
	3	vah/yah	ve/ye
Hopi Jeanne (1978)	1	niʔ	ʔitam
	2	ʔim	ʔima
	3	pam	pima
Hmong Njua Kunyt (1984)	1	ku	pé
	2	káo	mê
	3	nw	púa

LANGUAGE	π	SG	PL
Hua	1	dgai-a	rgai-a
Heiman (1980)	2	kgai-a	pgai-a
	3	gai-a	pgai-a
Icelandic	1	ég	við
Neijmann (2021)	2	þú	þið
	3 _M	hann	þeir
	3 _F	hún	þær
	3 _N	það	þau
Igbo	1	mí	ànyí
Georgi & Amaechi (2023)	2	gí	únù
	3	yá	há
Italian	1	io	noi
Erica Conti, p.c.	2	tu	voi
	3 _M	lui	lei
	3 _F	loro	loro
Itonama isolate	1 _M	os-ni	sih-ni-ʔ-ke
McGinnis (2005)	1 _F	os-ni-ʔka	sih-ni-ʔ-ke
	2 _M	oʔ-ni	dih-ni-ʔ-ke
	2 _F	koʔ-ni	dih-ni-ʔ-ke
	3 _M	oh-ni	oh-ni-tʸe-ʔ-ke
	3 _F	pi-ni	oh-ni-tʸe-ʔ-ke
Kannada	1	nānu	nāvu
Steever (2020b)	2	nīnu	nīvu
	3 _M	avanu	avaru
	3 _F	avaḷu	avaru
Kazakh	1	men	biz
Straughn (2011)	sen	siz	
	ol	olar	
Ket	1	āḍ	ótn
Georg (2007)	2	ū(k)	ókṽ
	3	bū	būṽ
Kirghiz	1	men	biz
Karakoç & Kalieva (2021)	2	sen	si-ler
	3	al	a-lar
Korean	1	nay-ka	wuli-ka
Smith et al. (2019)	2	ne-ka	nehuy-ka
	3 _M	ku-ka	ku-tul-i
	3 _F	kunye-ka	kunye-tul-i

LANGUAGE	π	SG	PL
Kuman Corbett (2000)	1	na	no
	2	ene	ene
	3	je	je
Latvian Kalnača & Lokmane (2021)	1	es	mēs
	2	tu	jūs
	3 _M	šis	šie
	3 _F	ši	šis
Lithuanian Ramonienė et al. (2019)	1	àš	mēs
	2	tù	jūs
	3 _M	jìs	jiẽ
	3 _F	jì	jõs
Luo Stafford (1967)	1	an	wan
	2	in	un
	3	en	gin
Mandarin Chinese Corbett (2000)	1	wǒ	wǒmen
	2	nǐ	nǐmen
	3	tā	tāmen
Mangghuer Slater (2003)	1	bi	da-si
	2	qi	ta-si
	3	gan	gan-si
Nandi Creider & Creider (1989)	1	áne:	áce:k
	2	ínye:	ókwe:k
	3	íne:	íce:k
Nez Perce Deal (2016a)	1	'iin	nuun
	2	'iim	'imé
	3	'ipí	'imé
Noghay Karakoç (2021)	1	men	biz
	2	sen	siz
	3	ol	o-lar
Norwegian Strandskogen & Strand- skogen (1994)	1	jeg	vi
	2	du	dere
	3 _M	han	de
	3 _F	hun	de

LANGUAGE	π	SG	PL
Pero Frajzyngier (1989)	1	nè	mínù
	2 _M	kè/kài	mà
	2 _F	cì	mà
	3 _M	cákkà	cínù
	3 _F	tè	cínù
Pite Saami Wilbur (2014)	1	mån	mij
	2	dån	dij
	3	sån	sij
Pokot Herreros Baroja et al. (1989)	1	aní	achá
	2	nyí	akwá
	3	nyínté, nyé	cháné/chá
Polish Sadowska (2012)	1	ja	my
	2	ty	wy
	3 _M	on	oni
	3 _F	ona	one
Central Pomo Corbett (2000)	1	ʔa·	ya
	2	ma	máya
	3	mu·l	mú·tuya
P'urhépecha Pérez (2006)	1	ji	ju-chá
	2	cha	cha / cha-jtsĩ
	3	indé	indé-chá
Russian	1	ya	my
	2	ty	vy
	3 _M	on	oni
	3 _F	ona	oni
Vlax Romani Hancock (1995)	1	me	amé
	2	tu	tumé
	3 _M	vo(v)	von
	3 _F	vój	von
Serbian	1	ja	mi
	2	ti	vi
	3 _M	on	oni
	3 _F	ona	one
Spanish Laura Ros García, p.c.	1	yo	nosotros.M/nosotras.F
	2	tú	vosotros.M/vosotras.F
	3 _M	él	ellos
	3 _F	ella	ellas

LANGUAGE	π	SG	PL
Svan Tuite (2023)	1	mi	näj
	2	si	sgäj
	3 _{PROX}	ala	alj-är
	3 _{DIST}	eža	ež-är
Swedish Hinchliffe & Holmes (2013)	1	jag	vi
	2	du	ni
	3 _M	han	de
	3 _F	hon	de
Tatar Berta (2021)	1	min	biz
	2	sin	siz
	3	ul	a-lar
Turkish Csató & Johanson (2021)	1	ben	biz
	2	sen	siz
	3	o	on-lar
Tübatulabal Voegelin (1935)	1	nık	ıngıl-u:ts
	2	ımbi	ımbu:mu
	3	m	ında
Udmurt Edygarova (2022)	1	mon	mi
	2	ton	ti
	3	so	soos
Uyghur Yakup (2021)	1	män	biz
	2	sän	si-lär
	3	u	u-lar
Uzbek Straughn (2011)	1	men	biz
	2	sen	siz
	3 _{PROX}	bu	bu-lar
	3 _{DIST}	šu	šu-lar
Modern Yucatec Lehmann (2020)	1	teen	to'n
	2	teech	te'x
	3	leti'	leti'o'b
Yoruba Bamgbose (2000)	1	mo	a
	2	o	ẹ
	3	ó	wón
Zuni Newman (1965)	1	ho [?]	hon
	2	to [?]	ton
	3	NA	NA

A.6 Database

LANGUAGE	π	SG	PL
Abkhaz	1	sara	hara
	2 _M	wara	š ^o ara
	2 _F	bara	š ^o ara
	3 _M	jara	dara
	3 _F	lara	dara
Aleut	1	θiŋ	tuman
	2	txin	txitʃi
	3	hiŋan	iŋakun
Apatani	1	ngo	ngunu
	2	no	nunu
	3 _{ANIM}	mo	molu
	3 _{INANIM}	hi	hiika atang
Arapesh Bukiyip	1	jek; eik	apak
	2	njak; njek	ipak
	3 _M	enan/nani	omom/mami
	3 _F	okok/k ^w ak ^w i	owo/wawi
Atakapa	1	wi	jukit
	2	na	nakit
	3	ha	hakit
Awtuw	1	wan	nom
	2	yen	om
	3 _M	rey	rom
	3 _F	tey	rom
Baruya	1	nimino	nemino
	2	gimino	sarimino
	3	gamino	kumino
Baure	1	niti?	viti?
	2	piti?	jiti?
	3 _M	roti?	noti?
	3 _F	riti?	noti?

LANGUAGE	π	SG	PL
Bella Coola	1	ʔnc	ʔmiʔ
	2	ʔinu	ʔup
	3 _M	tix	wix
	3 _F	cix	wix
Buryat	1	bi	bide
	2	shi	ta
	3	ene	ede
Cashinahua	1	e(a)n	nu(ku)n
	2	mi(a)n	ma(tu)n
	3	Ø; jatun	jabu; jabun
Cayapa	1	i	lala
	2	ñu	ñulla
	3	ja	jala
Chimariko	1	noʔot	nač ^h itot
	2	mamot	mamq ^h etot
	3	p ^h aʔmot	p ^h aʔmot
Chinantec	1	ʔhnaʔ	hnoʔ
	2	ʔnuʔ	ʔnoʔ
	3	ʔuʔ	ʔuʔ
Chitimacha	1	ʔiʃ	ʔuʃ
	2	him(ʔ)	was
	3	hus	huŋks
Choctaw	1	anoʔ	pishnoʔ/hapishnoʔ
	2	chishnoʔ	hachishnoʔ
	3	yamma(k)	NA
Dahalo	1	ʔápi	ɲápi
	2	ʔá:ta	ʔatta
	3 _M	ʔúdu	ʔummámu
	3 _F	ʔídi	ʔummámu
Damana	1	ra	nabind͡ʒina
	2	ma	mabind͡ʒa
	3	na	ihkənand͡ʒina
Dani	1	an	nit
	2	hat	hit
	3	at	it
Defaka	1	i	wá
	2	í	ó
	3 _M	o	iní/oní
	3 _F	á	iní/oní
	3 _N	je	iní/oní

LANGUAGE	π	SG	PL
Degema	1	mɛɾɛɾ	eniɾ
	2	wɔɾɔɾ	maɾaɾɲ
	3	ɔɿjɾ	ɓaɾaɾw
Ekari	1	ani	ini:
	2	aki	iki:
	3	okai	okei
Fur	1	k-a	k-i
	2	j-i	b-i
	3	ie	ìè-èɲ
Haida	1	ltha	talung
	2	dung	dalung
	3	la	tla
Hamtai	1	ni	nai
	2	nti	hai
	3 _M	əqo	əquoa
	3 _{FEM}	i	iʔjoa
Hungarian	1	én	mi
	2	te(INF)/ön(F)	ti(INF)/önök(F)
	3	ő	ők
Idoma	1	aɿmiɿ	aɿlɔɿ
	2	aɿmɔɿ	aɿaɿ
	3	aɿnuɿ	uɿwaɿ
Ijo	1	aɿrɪɿ	woɿniɿ
	2	aɿrɪɿ	ɔɿmɪɿnɪɿ
	3 _M	eɿrɪɿ	ɔɿmɪɿnɪɿ
	3 _F	aɿraɿʊɿ	ɔɿmɪɿnɪɿ
Iraqw	1	ʔaníɲ	ʔatén
	2 _M	kú:ɲ	ku:ɲgáʔ
	2 _F	kí:ɲ	ku:ɲgáʔ
	3 _M	kú:ɲ	ʔinoʔín
	3 _F	kí:ɲ	ʔinoʔín
Irish	1	mé	sinn
	2	tú	sibh
	3 _M	sé	siad
	3 _F	sí	siad

LANGUAGE	π	SG	PL
Japanese	1	watafi	watafi tatfi
	2	anata	anata tatfi
	3 _M	kare	kare tatfi
	3 _F	kanod ₃₀	kanod ₃₀ tatfi
Jingpho Kachim	1	ŋai↓	an↓t ^h e↓/jan↓t ^h e↓
	2	naŋ↓	nan↓t ^h e↓
	3	k ^h i↓	k ^h an↓t ^h e↓
Kanuri	1	wu↓	a↓ndi↓
	2	ɲi↓	na↓ndi↓
	3	fi↓	sa↓ndi↓
Kewa	1	ni↓	ni↓a↓a↓
	2	ne↓	ni↓mi↓
	3	ni↓pu↓	ni↓mu↓
Ecuadorian Kichwa	1	ɲuka	ɲukanjik
	2	kan	kankuna
	3	paj	pajkuna
Kisi	1	i↓	ŋ↓
	2	a↓	la↓
	3	o↓ (o); le↓ (le); i↓ (i)	a↓ (a); la↓ (la); ŋ
Kobon	1	(j)ad	hon
	2	ne; ni	kəl; kale
	3	nipe; ne	kəl; kale; kalipe
Koiari	1	da	no
	2	a	ya
	3	ahu	yabu
Korowai	1	nup	noxup
	2	gup	gexenep
	3	jup	jexenep
Koyraboro Senni	1	aj	ir
	2	ṇ; ni	war
	3	a	i

LANGUAGE	π	SG	PL
Kuna	1	an	anmar
	2	be	bemar
	3	we	wemar
Kwoma	1	ada/an	nota/no
	2 _M	mita/mi	kwota/kwo
	2 _F	ni ⁿ dza/nitɕa/ni	kwota/kwo
	3 _M	ɾita/ɾi	ɬetɕa/ɬe
	3 _F	sia/si	ɬetɕa/ɬe
Luiseño	1	no	tʃa:m
	2	o(m)	omom
	3	po	pomom
Manx	1	mi	ʃin
	2	u	ʃu
	3 _M	e	ad; əd
	3 _F	i	ad; əd
	3 _{COMM}	ə	ad; əd
Masalit	1	ama	mi:
	2	maŋ	ki:
	3	ti:	i;; wi:
Maybrat	1	tuo	amu
	2	nuo	anu
	3 _M	ait	ana
	3 _{NONM}	au	ana
Mumuye	1	n̩-ɭ	wo-ɭ
	2	mɔ-ɭ	no-ɭ
	3	wu	ji
Musqueam	1	ʔənθə	ɬniməɬ
	2	nəwə	ɬwələp
	3	t̪a	t̪aləm

LANGUAGE	π	SG	PL
Nambikwara	1	tʔai ʔli ʔ	tʔai ʔli ʔ-nāu ʔa ʔ
	2	ʔwāi ʔna ʔ	ʔwāi ʔna ʔ-nāu ʔa ʔ
	3	te ʔna ʔ	te ʔna ʔ-nāu ʔa ʔ
Navajo	1	shí	danihí
	2	ni	danihí
	3	bí	daabí
Nupe	1	mi	ji
	2	wo; o	je; e
	3	wũ; u	a
Nisenan	1	ni	ne:s
	2	mi	me:m
	3	mi:m	mise:m
Ocaina	1	xo	xaho; xa:
	2	o	monhto
	3 ^{ANIM}	nia	u:hia
	3 ^{INAN}	tia	u:hia
Ogbronuagum	1	aamɪ	əjo
	2	ja	epi
	3	jɔ	aba
Orig	1	ji ʔgə ʔn	ni ʔgə ʔn
	2	ɔ ʔgə ʔn	nɔ ʔgə ʔn
	3	tu ʔgə ʔn	ne ʔgə ʔn
Ormuri	1	(h)az	māx
	2	tu	tyōs, tōs
	3	afó	afáy, afóyn
Oromo	1	a(i)n(in)i(:)	nu:; nuwini:; nuja:
	2	at(in)i(:)	isani; isanu:
	3 ^M	inn(in)i(:)	isa:ni; worri:
	3 ^F	isi:ni(:)	isa:ni; worri:
Pacoh	1	kɪ:	hɛ:
	2	maj	ʔipɛ:
	3	dɔ:	ʔapɛ:; ɲa:j
Páez	1 ^M	adʲ	kweʔsʲ
	1 ^F	ũʔkwe	kweʔsʲ
	2 ^M	idʲ	iʔkweʔsʲ
	2 ^F	iʔkwe	iʔkweʔsʲ
	3	tʲã:	tʲã:weʔsʲ

LANGUAGE	π	SG	PL
Pilagá	1	hayem	qom'i
	2	am	am'i
	3 _{PROX}	ho'	
	3 _{PROX}	m'e	
	3 _{FAR}	č'a	čalo
Salinan	1	hɛ:ɾk ^h	ha:ɾk ^h
	2	moʔ	mo:m
	3	hɛ'joʔ	hɛ'jo:ʔt
Sandawe	1	ci	suŋ
	2	hapu	siŋ
	3 _{M/N}	hewe	heso
	3 _F	hesu	heso
Sango	1	mbiɫ	iɫ
	2	mɔɫ	aɫlaɫ
	3	loɫ	aɫlaɫ
Sapuan	1	ʔǎj	mu ɲa
	2	sǎj	mu sǎj
	3	ʔin	ʔin
Selknam	1	ya	ikuwa-y
	2	ma	ma-y
	3	ta	ta-y
Siroi	1	je	sile
	2	ne	tale
	3	nu	nale
So/Soko	1	mbá	ísó
	2	aje	ínó
	3	índé	NA
Takelma	1	gi:ɫɫ	go:uɫɫm
	2	maɫ	ma:ɫɫp ^h
	3	a:ɫɫk ^h	a:j
Western Tarahumara	1	né	ramué
	2	mué	ʔémi
	3	boʔné	aʔbóe
Tlingit	1	ɣád	ʔuháːn
	2	waʔé	ʔiːh ^(w) áːn
	3	hú	hás
Tonkawa	1	ʃaje	geuʃage
	2	naje	wenage
	3	ʔejelɐ	ʔewegɐ

LANGUAGE	π	SG	PL
Totonac	1	'kit	ki'n̄an
	2	'wiʃ	wi'f̃in
	3	'ʔut	ʔu'tun
Tunebo	1	as	is
	2	ba	ba:
	3	eri; ei	ein
Upper Sorbian	1	ja	mi
	2	ti	wi
	3 _M	wun	woni
	3 _F	wona	wone
Votic	1	miæ	møx; my:
	2	siæ	tøx; ty:
	3	tæmæ	næmæ/næmæt/nævæt
Yaqui	1	ʔínepo	ʔitépo
	2	ʔémpo	ʔeméʔe
	3	ʔáapo	bémpo
Wolaytta	1	ta	nu
	2	ne	hi:nte
	3 _M	a	e:ta
	3 _F	i	e:ta
Yagaria	1	dagaea	lagaea
	2	gagaea	lapagaea
	3	agaea	pagaea
Yale (Kosarek)	1	na	nu; nun
	2	an	ʔaun
	3	el	sin
Yuracare	1	se	tua
	2	me	pa
	3	na	now

Appendix B

This appendix includes languages that surface with the inclusive~exclusive distinction. It is split into three parts, according to the source of the data: (i) Mouton Grammars, (ii) Individual reference grammars, grammatical descriptions, and studies, and the (iii) Database (Pertsova 2022b). Within each group, languages appear in the alphabetical order. The analysis is provided in the column PATTERN. Standard notation (A, B, C) for indicating allomorphy is used. A is the basic (elsewhere) form, B indicates morphological alternation, and C a further alternation with respect to B. For each language, the first line in the PATTERN column includes the analysis for the 1 SINGULAR - 1 PLURAL EXCLUSIVE. The second line includes the analysis of the 1 PLURAL INCLUSIVE. For example, in Alto Perené, 1SG and 1PL.EXCL do not involve allomorphy, hence AA. The inclusive involves the morphological alternation, hence B in the second line.

The relevant forms are 1SG-1PL.EXCL-1PL.INCL. In that sense, Alto Perené is an example of an AAB pattern. With respect to the analysis in **Chapter 4**, AAA - **absence of allomorphy**; AAB - **inclusive exponent**; ABB - **exclusive and inclusive exponents**; ABC - **across the board alternation**; ABA - **exclusive exponent**.

B.1 Mouton Grammar Library

LANGUAGE	PATTERN	π	SG	PL
Alto Perené Mihás (2015)	AA B	1EXCL 1INCL	naaka	naakaite aroka/arokaite/aror- ite
Mongsen Ao Coupe (2007)	AB C	1EXCL 1INCL	nì	ìla/ìkhàla ìsa/ìsòŋla
Belep McCracken (2012)	AB C	1EXCL 1INCL	nao	ava ja
Bunaq Schapper (2022)	AA B	1EXCL 1INCL	neto	nei i
Bunan Widmer (2017)	AB C	1EXCL 1INCL	gi	hin~hiŋ=çi eraŋ~eraŋ=çi
Daakaka Von Prince (2015)	AB C	1EXCL 1INCL	nye	kinyem ar [TOP]; er (o)
Dumi Van Driem (1993)	AA B	1EXCL 1INCL	aŋ	aŋki iŋki
Hdi Frajzyngier (2011)	AB C	1EXCL 1INCL	íí	áŋní ámú
Kakataibo Zariquiey (2018)	AA B	1EXCL 1INCL	‘ën	‘ëkaman nukaman
Kambera Klamer (1998)	AB C	1EXCL 1INCL	nyungga	nyuma nyuta
Karbi Konnerth (2014)	AA B	1EXCL 1INCL	nè	nè-tūm e-tūm
Kayardild N. D. Evans (1995)	AA A	1EXCL 1INCL	ngada	nga-l-da nga-ku-l-da
Kilivila Senft (1986)	AB B	1EXCL 1INCL	yegu	yakamesi yakidasi

LANGUAGE	PATTERN	π	SG	PL
Kunbarlang Kapitonov (2021)	AA A	1EXCL 1INCL	ngayi (nganj-)	ngadbe ngarrka
Kuuk Thaayorre Gaby (2017)	AA A	1EXCL 1INCL	ngay	ngancn ngamp
Kwaza Voort (2004)	AB B	1EXCL 1INCL	si	tsi'tsɛ txa'na
Lavukaleve Terrill (2011)	AB C	1EXCL 1INCL	ngai	e me
Limbu Van Driem (2011)	AA A	1EXCL 1 INCL	aŋa	anige ani
Mian Fedden (2011)	AA A	1EXCL 1INCL	nē	nī nībo
Mina Frajzyngier (2011)	AB B	1EXCL 1INCL	sə	na nók
Neverver Barbour (2012)	AA B	1EXCL 1INCL	(i-)na	(i-)nam/(gu)mam (i)-git
Ngardi Ennever (2021)	AA A	1EXCL 1INCL	ngaju	nganimpa ngali(m)pa
Dolakha Newar Genetti (2007)	AB A	1EXCL 1INCL	ji	isi chiji/thiji
Savosavo Wegener (2012)	AA B	1EXCL 1INCL	a'gni/ai	ave mai
Seenku McPherson (2020)	AB C	1EXCL 1INCL	ń	íwí mí
Teiwa Klamer (2010)	AA B	1EXCL 1INCL	na('an)	ni('in) pi('in)
Toqabaqita Lichtenberk (2008)	AB C	1EXCL 1INCL	nau	kamiliqa, kamaliqa kulu
Udihe Nikolaeva & Tol- skaya (2011)	AA B	1EXCL 1INCL	bi	bu minti
Urarina Olawsky (2011)	AA A	1EXCL 1INCL	kanɐ	kanakaana kana
Vaeakau-Taumako Næss & Hovdhau- gen (2011)	AB C	1EXCL 1INCL	iau, au	mhatou, mhatu thatou, thatu
Wandala Frajzyngier (2012)	AB C	1EXCL 1INCL	íyyà	ɲrè míyà
Wardaman F. C. Merlan (2011)	AB A	1EXCL 1INCL	ngayugu	yirrug ngarrug

B.2 Individual Grammars

LANGUAGE	PATTERN	π	SG	PL
Adang Robinson & Haan (2014)	AA B	1EXCL 1INCL	na	ni pi
Betta Kurumba Coelho (2020)	AB A	1EXCL 1INCL	navə	yaŋgə naŋgə
Blagar Steinhauer (2014)	AA B	1EXCL 1INCL	na	ni pi
Cabécar Verhoeven (2012)	AB B	1EXCL 1INCL	yís~s	sá sé
Kaera Klamer (2014)	AA B	1EXCL 1INCL	nang	ning ping
Kolami Subrahmanyam (2019)	AA B	1EXCL 1INCL	ān	ām nēnd/nēm
Kamang Schapper (2014b)	AA B	1EXCL 1INCL	na	ni si
Konda Krishnamurti & Benham (2020)	AB B	1EXCL 1INCL	nān(u)	māp(u) māṭ(u)
Kurux Kobayashi & Oraon (2019)	AA B	1EXCL 1INCL	e:n	e:m na:m
Kuvi Steever (2020c)	AB B	1EXCL 1INCL	nānu	māmbu māro
Lihir Corbett (2000)	AB B	1EXCL 1INCL	yo	ge giet
Malagasy Keenan & Polinsky (2001)	AB C	1EXCL 1INCL	áho	izaháy isíka

LANGUAGE	PATTERN	π	SG	PL
Malayalam Sreekumar (2020)	AA A	1EXCL 1INCL	$\tilde{n}\tilde{a}\tilde{n}$	$\tilde{n}\tilde{a}\tilde{n}\tilde{a}\tilde{l}$ nammal/nām
Malto Steever (2020d)	AA B	1EXCL 1INCL	$\tilde{e}\tilde{n}$	$\tilde{e}\tilde{m}$ nām
Mokilese Corbett (2000)	AB C	1EXCL 1INCL	ngoah, ngoahi	kamai kisai
Nishnaabemwin Valentine (2001)	AA B	1EXCL 1INCL	niin	niin-win giin-win
Ojibwe Harley & Ritter (2002)	AA B	1EXCL 1INCL	niin	niin-awint kiin-awint
Passamaquoddy Bruening (2001)	AA B	1EXCL 1INCL	nil	nilun kilun
Western Pantar Holton (2014)	AA B	1EXCL 1INCL	nang	ning ping
Sawila Kratochvíl (2014)	AA B	1EXCL 1INCL	nannu	ninnu innu
Somali Saeed (1999)	AA B	1EXCL 1INCL	aní-ga	anná-ga inná-ga
Sursurunga Corbett (2000)	AB B	1EXCL 1INCL	iau	gim git
Tamil Annamalai & Steever (2020)	AA A	1EXCL 1INCL	$\tilde{n}\tilde{a}\tilde{n}$	$\tilde{n}\tilde{a}\tilde{n}\tilde{k}\tilde{a}\tilde{l}$ nām
Telugu Krishnamurti (2020)	AB C	1EXCL 1INCL	$\tilde{n}\tilde{e}\tilde{n}\tilde{u}$	$\tilde{m}\tilde{e}\tilde{m}(\tilde{u})$ $\tilde{m}\tilde{a}\tilde{n}\tilde{a}\tilde{m}(\tilde{u})$
Trumai Guirardello (1999)	AA B	1EXCL 1INCL	ha	ha wan ka wan
Tulu Bhat (2020)	AB C	1EXCL 1INCL	$\tilde{y}\tilde{a}\tilde{n}\tilde{u}$	$\tilde{e}\tilde{n}\tilde{k}\tilde{u}\tilde{l}\tilde{u}$ nama
Wambaya Nordlinger (1998)	AA A	1EXCL 1INCL	ngawurniji, ngawu	ngirriyani ngurruwani
Wersing Schapper & Hendery (2014)	AB C	1EXCL 1INCL	nenal	nyenal tenal
Worrorra Clendon (2014)	AA A	1EXCL 1INCL	ngayu	arri ngarri

B.3 Database

LANGUAGE	PATTERN	π	SG	PL
Abui	AA	1EXCL	na	nɪ
	B	1INCL		pɪ
Akha	AA	1EXCL	ŋá	ŋá mà
	B	1INCL		àdu mà
(Xingu) Asuriní	AB	1EXCL	dʒe	ure
	A	1INCL		dʒane
Atayal	AB	1EXCL	kun; kuziŋ	sami
	C	1INCL		ita
Bajau (Sama)	AB	1EXCL	aku	kami
	C	1INCL		kita
Bilua	AA	1EXCL	aŋa	aŋ-ŋe
	B	1INCL		aŋ-me
Bribri	AB	1EXCL	dʒě; ɲã; ɲĩ;	să
	B	1INCL		sě; sã
Chang	AB	1EXCL	ŋo	kən
	C	1INCL		sən
Cubeo	AA	1EXCL	jĩ	jĩ-xã
	B	1INCL		bã-xã
Embera	AB	1EXCL	mũ	tai
	B	1INCL		taŋʔi
Evenki	AA	1EXCL	bi	bɸ
	B	1INCL		mit
Gayo	AB	1EXCL	akɔ	kamɪ
	B	1INCL		kɪtə
Gooniyandi	AA	1EXCL	ŋaɲi	ɲidi
	B	1INCL		ja:di
Hani	AA	1EXCL	ŋaɿ	ŋaɿduɿ
	A	1INCL		ŋaɿjaɿ

LANGUAGE	PATTERN	π	SG	PL
Ho	AB C	1EXCL 1INCL	aŋ	ale abu
Iban	AB C	1EXCL 1INCL	ʔaku	kami kitaj
Ida'an	AB C	1EXCL 1INCL	aku	kəm̩mi kito
Ika	AA A	1EXCL 1INCL	nən	ni-wi ma-nən-kaʔ
Inanwatan	AA B	1EXCL 1INCL	n-ái-ti	n-íi-ti d-ái-ti
Ju 'hoan	AB C	1EXCL 1INCL	miɿ	eɿ-!aɿ mɿ-!aɿ
Kokota	AB C	1EXCL 1INCL	ara	ɣai ɣita
Kunama	AB C	1EXCL 1 INCL	aba	ame kime
Lamang	AB B	1EXCL 1INCL	ijò	nàjìn nàmwà
Manam	AB C	1EXCL 1INCL	ŋau; ŋa	ʔeʔa ʔita
Mangarrayi	AB A	1EXCL 1INCL	ŋa-ja	ŋi-ɭa ŋa-ɭa
Maranao	AB C	1EXCL 1INCL	sakən	səkami səktano
Maringe	AB C	1EXCL 1INCL	ia-ra	ɣe-pa ta-pa
Maung	AA A	1EXCL 1INCL	ŋa-b̥i	ŋar-i 'ŋar-wuri
Miskito	AA A	1EXCL 1INCL	jaŋ	jaŋ nani ja:won
Sierra Miwok	AB C	1EXCL 1INCL	kani	mahi ʔotiti
Mixe (Ayutla)	AA B	1EXCL 1INCL	ʔi ^h ts	ʔi:ts(t) ʔatəm
Motuna	AA A	1EXCL 1INCL	ni; nii	noni nee
Movima	AB C	1EXCL 1INCL	inɭa	iyɭi i:de

LANGUAGE	PATTERN	π	SG	PL
Ngandi	AB C	1EXCL 1INCL	ŋaja	ŋer ŋorkor
Ngankikurungkurr	AA A	1EXCL 1INCL	ŋa-ji	ŋa-gur na-jin nime
Nivkh (Amur)	AA B	1EXCL 1INCL	ŋji	ŋjiŋ mer; mir
Oksapmin	AA B	1EXCL 1INCL	nox	nuxul ⁿ dil
Orokolo	AB B	1EXCL 1INCL	ara	ela elavi:la
Paiwan	AB C	1EXCL 1INCL	ti-akən	ti-amən ti-tjən
Palauan	AB C	1EXCL 1INCL	nak	kəmám kið̃
Pemón	AB A	1EXCL 1INCL	(j)u-re	ina (j)u-re-to / (j)u-re-nokon
Pengo	AA A	1EXCL 1INCL	a:n-eŋ/a:n	a:p-eŋ/a:p a:h-eŋ/a:s
Sahu	AA A	1EXCL 1INCL	ŋoi	ŋomi ŋene
Semelai	AB C	1EXCL 1INCL	ʔəŋ	jɛ-ʔen hɛ-ʔen
Siriano	AB C	1EXCL 1INCL	jʰu	mãrĩ gʷa
Sirionó	AB C	1EXCL 1INCL	'se	'ure 'ɲaɖe
Sumu	AA A	1EXCL 1INCL	jaŋ	jaŋ-na ma-jaŋ
Taba	AB C	1EXCL 1INCL	jak; lak	am tit
Central Tagbanwa	AB C	1EXCL 1INCL	ako	kami kita
Temiar	AB C	1EXCL 1INCL	jʌjʔ	kanʌʔ/kandʌʔ ejʔ/ʌ:ʔ
Tiruray	AA B	1EXCL 1INCL	bɛgen; gu	(bɛ)gej (bɛ)tom
Tiwi	AB C	1EXCL 1INCL	ŋia	ŋawa ŋaɣa

LANGUAGE	PATTERN	π	SG	PL
Tobelo	AB	1EXCL	ŋo-hi	ŋo-mi
	C	1INCL		ŋo-ne
Tojolabal	AA	1EXCL	ke-ʔna	ke-ʔen-tik-on
	A	1INCL		ke-ʔen-tik
Toaripi (Sepoe)	AB	1EXCL	ara-va	ela-va
	C	1INCL		iauo-a
Ungarinjin	AB	1EXCL	ŋin	ŋjarun
	C	1INCL		ŋarun
Warembori	AB	1EXCL	iw-i	am-i
	C	1INCL		k-i
Yagua	AB	1EXCL	raj	nú:j
	C	1INCL		wũ:j
Yanyuwa	AA	1EXCL	ŋaŋa	ŋanu
	A	1INCL		ŋa ^m bala

Appendix C

This appendix includes 80 languages that make gender distinctions in one of the person feature specifications. Languages are alphabetically sorted according to the language family.

LANGUAGE	π	SG	PL
Amharic Afro-Asiatic Leslau (1995)	1	əne	əñña
	2 _M	antä	ənn-antä
	2 _F	anči	ənn-antä
	3 _M	əssu	ənn-ässu
	3 _F	əss ^w a	ənn-ässu
Hebrew Afro-Asiatic Ritter (1995)	1	ani	anachnu
	2 _M	ata	atem
	2 _F	at	aten
	3 _M	hu	hem
	3 _F	hi	hen
Somali Afro-Asiatic Saeed (1999)	1 _{EXCL}	aní-ga	anná-ga
	1 _{INCL}		inná-ga
	2	adí-ga	idín-ka
	3 _M	isá-ga	iyá-ga
	3 _F	iyá-da	iyá-ga
Goemai Afro-Asiatic Hellwig (2011)	1	hěn	měn
	2 _M	gõe	gwěn
	2 _F	yõe~yĩ	gwěn
	3	ní	muěp
Tamashek Afro-Asiatic Heath (2005)	1	næk	nækk- áen-əd. _M nækk-áen-æt-əd. _F
	2 _M	kàyy	kæw-áen-əd
	2 _F	kàmm	kæm-æt-əd
	3	ònta	ənt-áen-əd. _M ənt-æn-æt-əd. _F

LANGUAGE	π	SG	PL
Pero Afro-Asiatic Frajzyngier (1989)	1	nè	mínù
	2M	kè/kài	mà
	2F	cì	mà
	3M	cákkà	cínù
	3F	tè	cínù
Dahalo Afro-Asiatic Pertsova (2022b)	1	ʔápi	ɲápi
	2	ʔá:ta	ʔatta
	3M	ʔúdu	ʔummámu
	3F	ʔídi	ʔummámu
Iraqw Afro-Asiatic Pertsova (2022b)	1	ʔaníŋ	ʔatén
	2M	kú:ŋ	ku:ŋgáʔ
	2F	kí:ŋ	ku:ŋgáʔ
	3M	kú:ŋ	ʔinoʔín
	3F	kí:ŋ	ʔinoʔín
Oromo Afro-Asiatic Pertsova (2022b)	1	a(i)n(in)i(:)	nu:; nuwini:; nuja:
	2	at(in)i(:)	isani; isanu:
	3M	inn(in)i(:)	isa:ni; worri:
	3F	isi:ni(:)	isa:ni; worri:
Wolaytta Afro-Asiatic Pertsova (2022b)	1	ta	nu
	2	ne	hi:nte
	3M	a	e:ta
	3F	i	e:ta
Hausa Afro-Asiatic Wolff (1993)	1	nii	muu
	2M	kai	kuu
	2F	kee	kuu
	3M	shii	suu
	3F	ʔita	suu
Baure Arawakan Pertsova (2022b)	1	nitiʔ	vitiʔ
	2	pitiʔ	jitiʔ
	3M	rotiʔ	notiʔ
	3F	ritiʔ	notiʔ
Alto Perené Arawakan Mihás (2015)	1EXCL	naaka	naakaite
	1INCL		aroka/arokaite/aror-ite
	2	aviroka	avirokaite
	3M	iri(ro)	iroite
	3NON-M	iro	iroite
Kulina Arawakan Dienst (2014)	1	owa	ia
	2	tia	tia-deni
	3M	powa	powa-deni
	3F	po-ni	po-ni-deni

LANGUAGE	π	SG	PL
Ngandi Australian Pertsova (2022b)	1EXCL	ŋaja	ŋer
	1INCL		ŋorkor
	2	ŋugan; ŋuwan	ŋukar
	3M	ŋiwan	bawan
	3F	ŋawan	bawan
Ngankikurungkurr Australian Pertsova (2022b)	1EXCL	ŋa-ji	ŋa-gur
	1INCL		na-jin nime
	2	ŋipi	na-gur
	3M	nem	wirim
	3F	ŋajim	wirim
Yanyuwa Australian, Pama- Nyungan Pertsova (2022b)	1EXCL	ŋaŋa	ŋanu
	1INCL		ŋa ^m bala
	2	ji ⁿ da	jiru
	3M.AN	jiwa	alu
	3M.INAN	wali	alu
	3.F.AN	a ⁿ da	alu
Worrorra Australian, Wororan Clendon (2014)	1EXCL	ngayu	arri
	1INCL		ngarri
	2	ngunju	nyirri
	3M	awa	arrka
	3F	nyangka	arrka
Kilivila Austronesian Senft (1986)	1EXCL	yegu	yakamesi
	1INCL		yakidasi
	2	yokwa	yokwami
	3M	mtona	mtosina
	3F	minana	minasina
Sahu Austronesian Pertsova (2022b)	1EXCL	ŋoi	ŋomi
	1INCL		ŋene
	2	ŋana	ŋini
	3M	unaŋa	anaŋa
	3F	munana	anaŋa
	3N	enaŋa	enaŋa
Qaqet Baining Hellwig (2019)	1	ngua	ut
	2	nyi	ngen
	3M	ka	ta.HUM
			nget.NON-H
	3F	ki	ta/nget
Abkhaz Caucasian Pertsova (2022b)	1	sara	hara
	2M	wara	š ^o ara
	2F	bara	š ^o ara
	3M	jara	dara
	3F	lara	dara

LANGUAGE	π	SG	PL
Kannada Dravidian Steever (2020b)	1	nānu	nāvu
	2	nīnu	nīvu
	3M	avanu	avaru
	3F	avaḷu	avaru
Kolami Dravidian Subrahmanyam (2019)	1EXCL	ān	ām
	1INCL		nēnd/nēm
	2	nīv	nīr
	3M.DIST	am(d)	avr
	3M.PROX	im(d)	ivr
	3NON-M.DIST	ad	adav
	3NON-M.PROX	id	idav
Malayalam Dravidian Sreekumar (2020)	1EXCL	ñān	ñanṇaḷ
	1INCL		nammaḷ/nām
	2	nī	niṇṇaḷ
	3M.PROX	ivaṇ	ivar [+HUM]
	3M.DIST	avaṇ	avar [+HUM]
	3F.PROX	ivaḷ	ivar [+HUM]
	3F.DIST	avaḷ	avar [+HUM]
	3N.PROX	itə	iva [-HUM]
	3N.DIST	atə	ava [-HUM]
Malto Dravidian Steever (2020d)	1EXCL	ēn	ēm
	1INCL		nām
	2	nīn	nīm
	3M	āh	ār.DIST/īr.PROX
	3F	āḍ	
Tamil Dravidian Annamalai & Steever (2020)	1EXCL	nān	nānkaḷ
	1INCL		nām
	2	nī	nīnkaḷ
	3M	avaṇ	avar
	3F	avaḷ	avar
	3N	atu	avaikaḷ
Telugu Dravidian Krishnamurti (2020)	1EXCL	nēnu	mēm(u)
	1INCL		manam(u)
	2	nīwu/nuwwu	mīru
	3	wāḍu/atanu.M/adi.NON	māru/wāṇḍḍu.H/awī.NONH
Tulu Dravidian Bhat (2020)	1EXCL	yānu	eṅkulu
	1INCL		nama
	2	ī	nuwgulu
	3M.PROX	imbe/umbe	NA
	3M.DIST	āye	
	3F.PROX	imbolu/um- bolu/mōlu	
	3F.DIST	ālū	

LANGUAGE	π	SG	PL
Swedish Indo-European Hinchliffe & Holmes (2013)	1	jag	vi
	2	du	ni
	3 _M	han	de
	3 _F	hon	de
Norwegian Indo-European Strandskogen & Strand- skogen (1994)	1	jeg	vi
	2	du	dere
	3 _M	h-an	d-e
	3 _F	h-un	d-e
Afrikaans Indo-European Donaldson (1993)	1	ek	ons
	2	ju	julle
	3 _M	hy	hulle
	3 _F	sy	hulle
Dutch Indo-European Donaldson (2008)	1	ik	wij/we
	2	jij/je	jou/je
	3 _M	hij	zij/ze
	3 _F	zij/ze	zij/ze
	3 _N	het	zij/ze
Serbian Indo-European	1	ja	mi
	2	ti	vi
	3 _M	on	oni
	3 _F	ona	one
Russian Indo-European	1	ya	my
	2	ty	vy
	3 _M	on	oni
	3 _F	ona	oni
Polish Indo-European Sadowska (2012)	1	ja	my
	2	ty	wy
	3 _M	on	oni
	3 _F	ona	one
Upper Sorbian Indo-European Pertsova (2022b)	1	ja	mì
	2	tì	wì
	3 _M	wùn	woni
	3 _F	wona	wone
Lithuanian Indo-European Ramonienė et al. (2019)	1	àš	mēs
	2	tù	jūs
	3 _M	jìs	jiẽ
	3 _F	jì	jõs

LANGUAGE	π	SG	PL
Latvian Indo-European Kalnača & Lokmane (2021)	1	es	mēs
	2	tu	jūs
	3 _M	šis	šie
	3 _F	ši	šis
Breton Indo-European Press (2011)	1	me	ni
	2	te	c'hwi
	3 _M	eñ	int
	3 _F	hi	int
Scottish Gaelic Indo-European Gillies (2009)	1	mi	sinn
	2	t(h)u	sibh
	3 _M	e	iad
	3 _F	i	iad
Irish Indo-European Pertsova (2022b)	1	mé	sinn
	2	tú	sibh
	3 _M	sé	siad
	3 _F	sí	siad
Manx Indo-European Pertsova (2022b)	1	mi	jin
	2	u	ju
	3 _M	e	ad; əd
	3 _F	i	ad; əd
	3 _{COMM}	ə	ad; əd
Albanian Indo-European Newmark et al. (1982)	1	unë	ne
	2	ti	ju
	3 _M	ai	ata
	3 _F	ajo	ato
Vlax Romani Indo-European Hancock (1995)	1	me	amé(n)
	2	tu	tumé(n)
	3 _M	vo(v)	von
	3 _F	vój	von
Pashto Indo-European David (2013)	1	zə	munḡ/munṣ
	2	tə	tāse/tāsi
	3 _M	day	duy
	3 _F	dā	duy
French Indo-European Kayne (2003)	1	je	nous
	2	tu	vous
	3 _M	il	ils
	3 _F	elle	elles

LANGUAGE	π	SG	PL
Italian Indo-European Erica Conti, p.c.	1	io	noi
	2	tu	voi
	3 _M	lui	lei
	3 _F	loro	loro
Spanish Indo-European Laura Ros García, p.c.	1	yo	nosotros.M/nosotras.F
	2	tú	vosotros.M/vosotras.F
	3 _M	él	ellos
	3 _F	ella	ellas
Japanese Japonic Pertsova (2022b)	1	watashi-ga	watashi-tachi-ga
	2	anata-ga	anata-tachi-ga
	3 _M	kare-ga	kare-ra-ga
	3 _F	kanojo-ga	kare-ra-ga
Korean Korean Smith et al. (2019)	1	nay-ka	wuli-ka
	2	ne-ka	nehuy-ka
	3 _M	ku-ka	ku-tul-i
	3 _F	kunye-ka	kunye-tul-i
Mosetén Mosetenan Sakel (2011)	1	yäe	tsin
	2	mi	mĩ'ʔin
	3 _M	mi'	mi'in
	3 _F	mö'	mö'in
Defaka Niger-Congo Pertsova (2022b)	1	i	wá
	2	í	ó
	3 _M	o	iní/oní
	3 _F	á	iní/oní
	3 _N	je	iní/oní
Ijo Niger-Congo Pertsova (2022b)	1	a-ɾɾɿ	woɿniɿ
	2	aɿɾɿ	ɔɿmɿniɿ
	3 _M	e-ɾɿ	ɔɿmɿniɿ
	3 _F	aɿra-ɔɿ	ɔɿmɿniɿ
Páez Páezan Pertsova (2022b)	1 _M	adʲ	kweʔsʲ
	1 _F	ũʔkwe	kweʔsʲ
	2 _M	idʲ	iʔkweʔsʲ
	2 _F	iʔkwe	iʔkweʔsʲ
	3	tʲã:	tʲã:weʔsʲ

LANGUAGE	π	SG	PL
Lavukaleve Papuan Terrill (2011)	1EXCL	ngai	e
	1INCL		me
	2	inu	imi
	3NEUT.M	foina	foiva
	3PROX.M	fona	fova
	3DIST.M	feana	feava
	3UNSPEC.M	foana	foava
	3NEUT.F	foia	foiva
	3PROX.F	fo	fova
	3DIST.F	fehea	feava
	3UNSPEC.F	fohoa	foava
	3NEUT.N	foiga	foiva
	3PROX.N	foga	fova
	3DIST.N	feaga	feava
	3UNSPEC.N	foaga	foava
Savosavo Papuan Wegener (2012)	1EXCL	a'gni/ai	ave
	1INCL		mai
	2	no	me
	3M	lo/la	ze/zepo/za
	3F	ko/ka	ze/zepo/za
Inanwatan Papuan Pertsova (2022b)	1EXCL	n-ái-ti	n-íi-ti
	1INCL		d-ái-ti
	2	(Ø)-ái-ti	(Ø)-íi-ti
	3M	í-ti-gi	í-ti-ga
	3F	í-ti-gɔ	í-ti-ga
Bilua Papuan; Solomons East Papuan Pertsova (2022b)	1EXCL	aŋa	aŋ-ŋe
	1INCL		aŋ-me
	2	ŋo	me
	3M.DIST	vo	se
	3M.PROX	nei	ni
	3F.DIST	ko	se
	3F.PROX	komi	ni

LANGUAGE	π	SG	PL
Awtuw Papuan; Sepik Pertsova (2022b)	1	wan	nom
	2	yen	om
	3M	rey	rom
	3F	tey	rom
Kwoma Papuan; Sepik Pertsova (2022b)	1	ada/an	nota/no
	2M	m̐ita/mi	kwota/kwo
	2 F	ni ⁿ d̐za/nit̐a/ni	kwota/kwo
	3M	r̐ita/r̐i	̐et̐a/̐e
	3F	sia/si	̐et̐a/̐e
Bukiyip Papuan; Torricelli Pertsova (2022b)	1	yek	apak
	2	nyak	ipak
	3M	énan/nani	omam/mami
	3F	okok/kwakwi	owo/wawi
	3MIXED		echech/chachi
Hamtai Papuan; Trans New Guinea Pertsova (2022b)	1	ni	nai
	2	nti	hai
	3M	əqo	əquoa
	3FEM	i	iʔjoa
Mian Papuan; Trans New Guinea Fedden (2011)	1EXCL	nē	nī
	1INCL		nībo
	2M	kēbo/kōbo	ībo
	2F	ōbo	ībo
	3M	ē	ī
	3F	ō	ī
Oksapmin Papuan; Trans New Guinea Pertsova (2022b)	1EXCL	nox	nuxul
	1INCL		ⁿ d̐il
	2	go	gul
	3M	ox	ixil
	3F	ux	ixil
Bella Coola Salish Pertsova (2022b)	1	ʔɲc	ʔmiʔ
	2	ʔinu	ʔup
	3M	tix	wix
	3F	cix	wix
Cubeo Tucanoan Pertsova (2022b)	1EXCL	j̐i	j̐i-xã
	1INCL		bã-xã
	2	b̐i	b̐i-xã
	3M	ĩ	dã
	3F	õ	dã
Siriano Tucanoan Pertsova (2022b)	1EXCL	j̐u	mãrĩ
	1INCL		g̐a
	2	m̐ũũ	m̐ũũ-sã
	3M	ĩŋũ	ĩŋũ-sã
	3F	igo	ĩŋũ-sã

LANGUAGE	π	SG	PL
(Xingu) Asuriní ^{Tupian}	1EXCL	dʒe	ure
Pertsova (2022b)	1INCL		dʒane
	2	ene	pene
	3M	ẽ	ŋi
	3F	ŋa	ŋi
Maybrat ^{isolate}	1	tuo	amu
McKay (1978)	2	nuo	anu
	3M	ait	ana
	3NONM	au	ana
Sandawe ^{isolate}	1	ci	suŋ
Pertsova (2022b)	2	hapu	siŋ
	3M/N	hewe	heso
	3F	hesu	heso
Trumai ^{isolate}	1EXCL	ha	ha wan
Guirardello (1999)	1INCL		ka wan
	2	hi	hi wan
	3M	ine	inak wan
	3F	inatl	inak wan

	1SG	2SG	3SG.M	3SG.F	1PL	2PL	3PL.M/F/N
NOM	ād	ū(k)	bū	bū	ótn	ókŋ	būŋ
GEN	āb	ūk	bú-da	bú-di	ótn-na	ókŋ-na	búŋ-na
DAT	áb-aŋa	úk-aŋa	bú-daŋa	bú-diŋa	ótn-naŋa	ókŋ-naŋa	búŋ-naŋa
BEN	áb-ata	úk-ata	bú-data	bú-dita	ótn-nata	ókŋ-nata	búŋ-nata

Table C.1: Gender in non-nominative cases in Ket (Georg 2007)

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